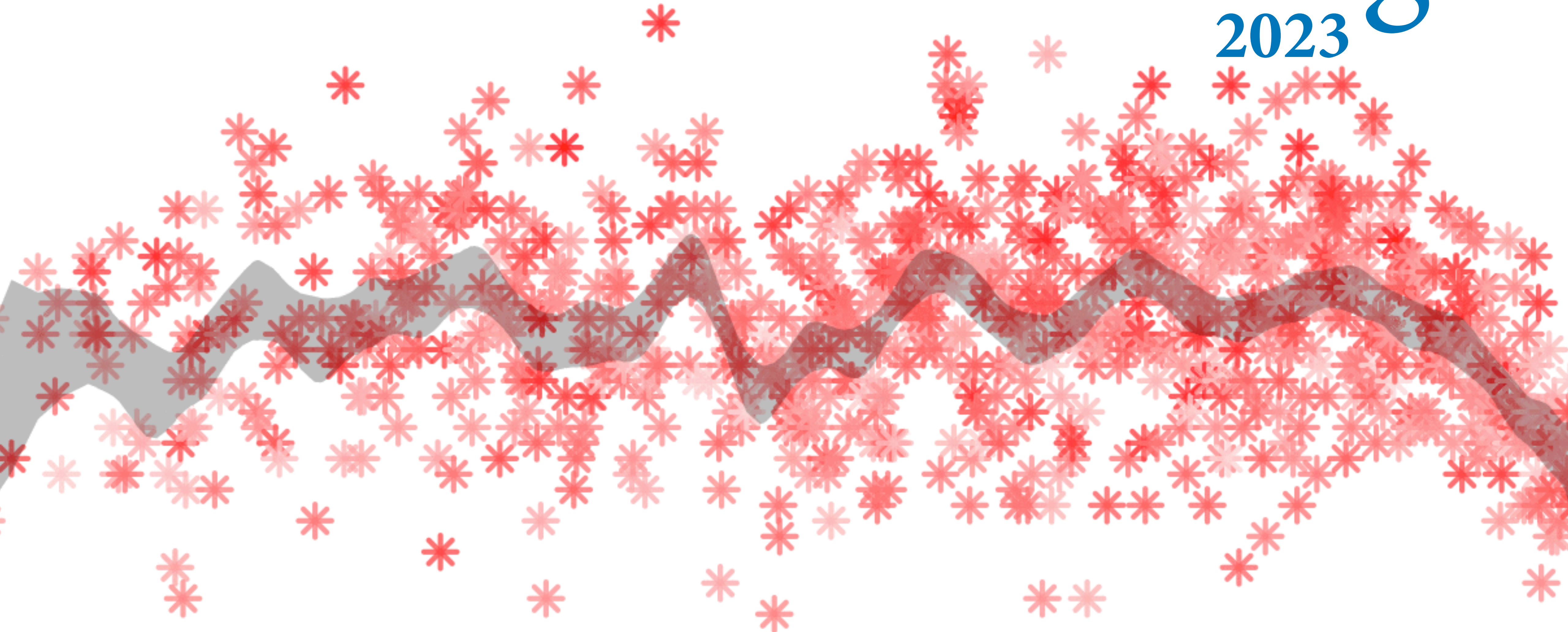


Statistical Rethinking

2023



6. Good & Bad Controls



The Fork

$X \leftarrow Z \rightarrow Y$

The Pipe

$X \rightarrow Z \rightarrow Y$



The Collider

$X \rightarrow Z \leftarrow Y$

The Descendant

$X \rightarrow Z \rightarrow Y$
 $\downarrow$
 A





The Fork

$$X \longleftarrow Z \longrightarrow Y$$

X and Y associated
unless stratify by Z



The Pipe

$$X \longrightarrow Z \longrightarrow Y$$

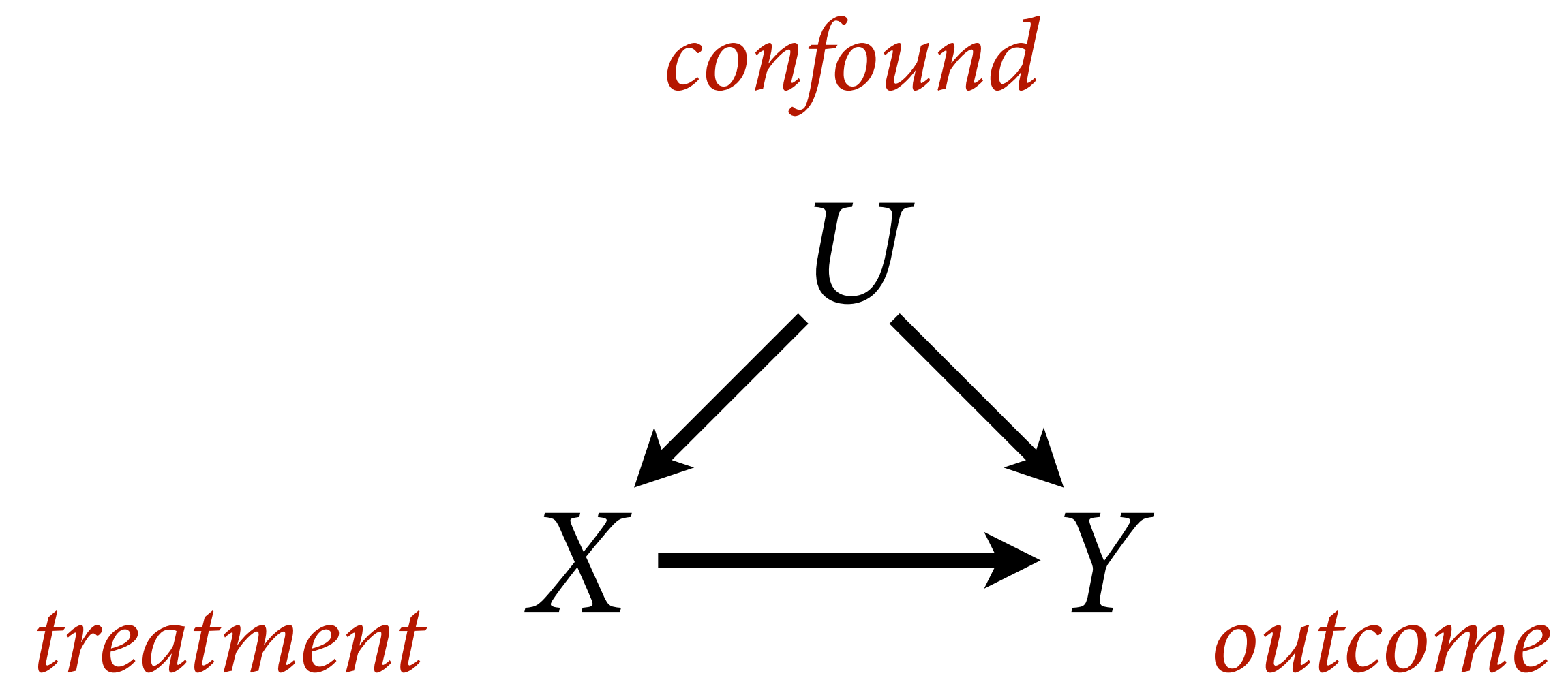
X and Y associated
unless stratify by Z

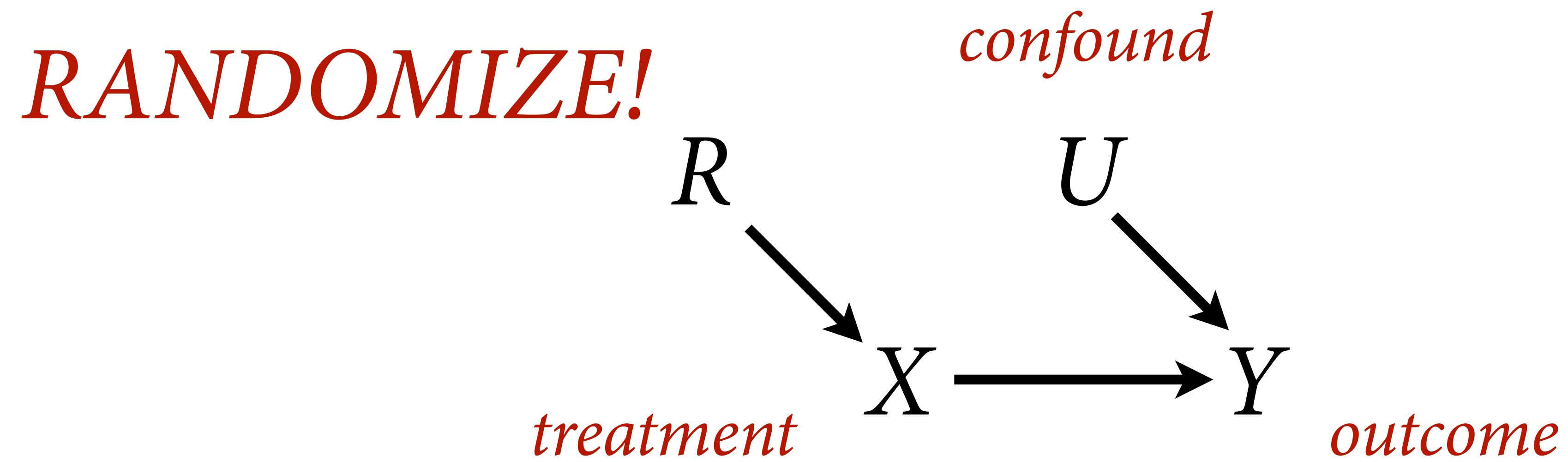


The Collider

$$X \longrightarrow Z \longleftarrow Y$$

X and Y not associated
unless stratify by Z





Causal Thinking

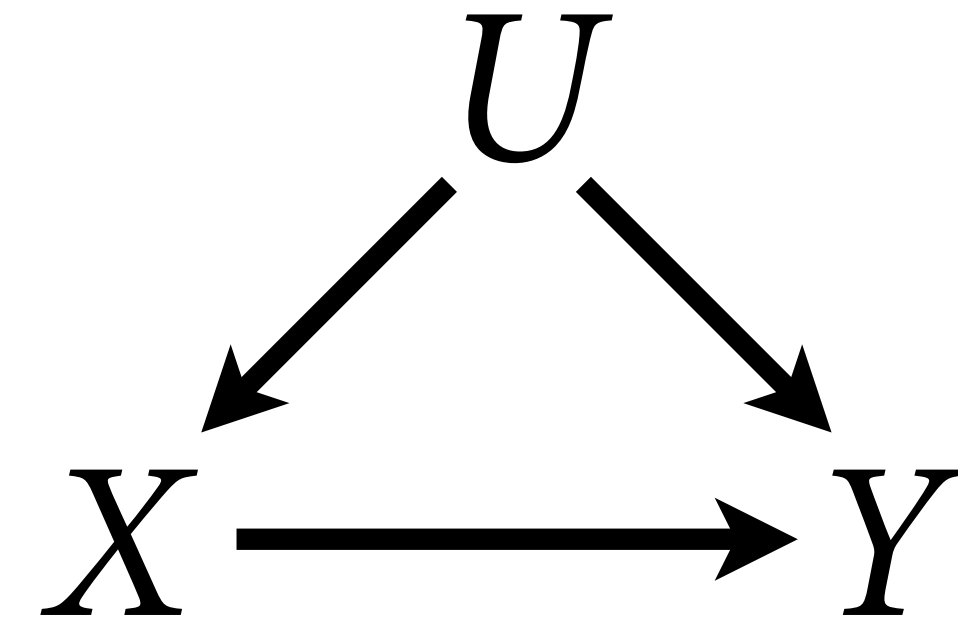
In an experiment, we cut causes of the treatment

We *randomize* (we try at least)

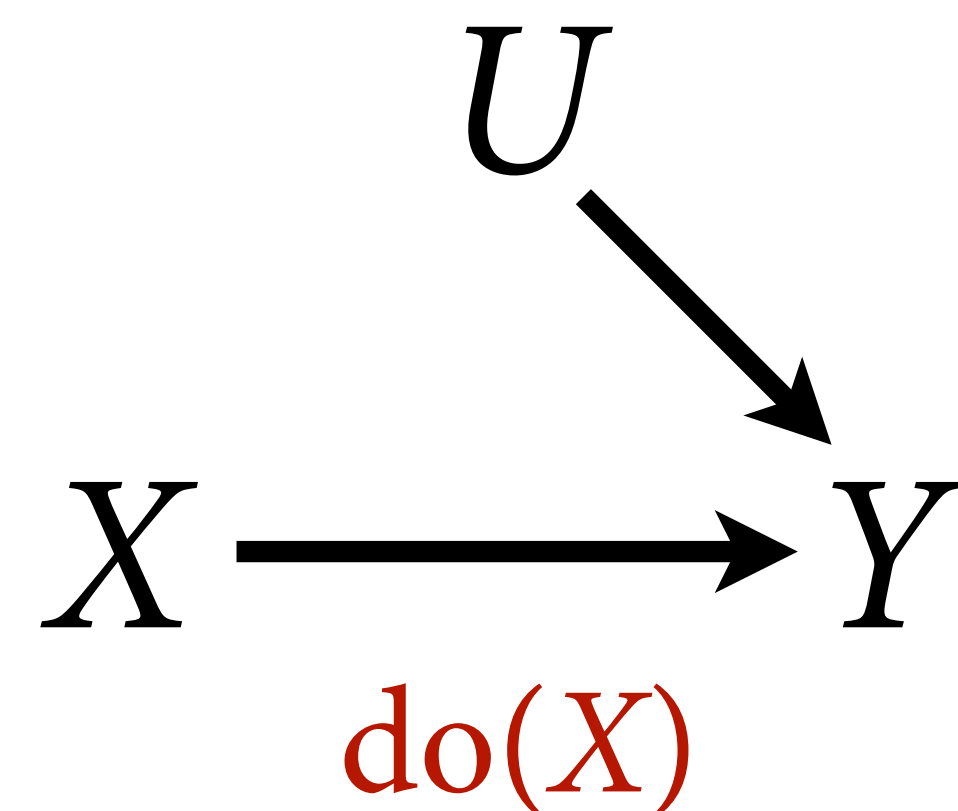
So how does causal inference without randomization ever work?

Is there a statistical procedure that mimics randomization?

Without randomization



With randomization



Causal Thinking

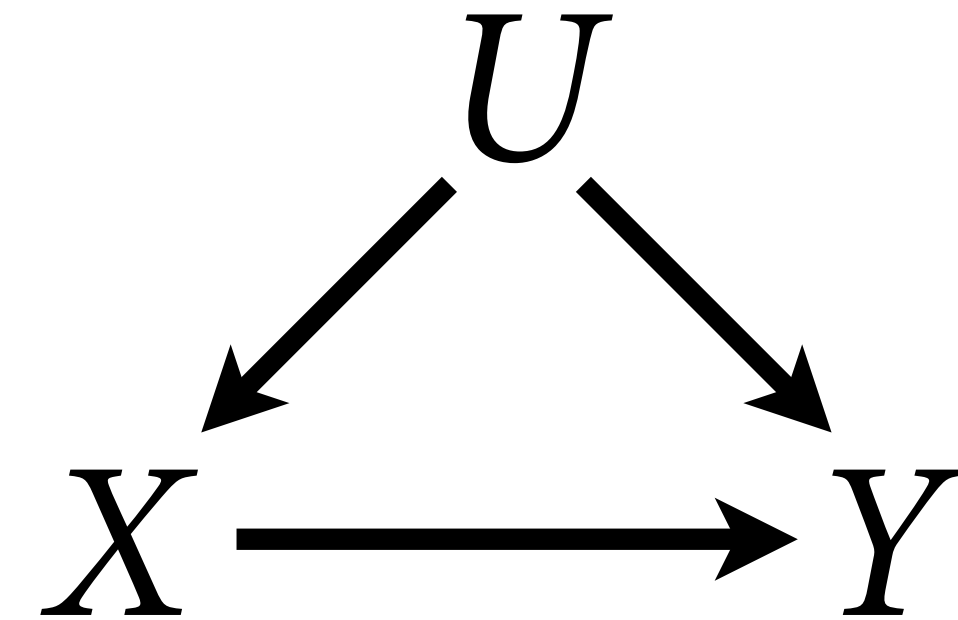
Is there a statistical procedure that mimics randomization?

$$P(Y | \text{do}(X)) = P(Y | ?)$$

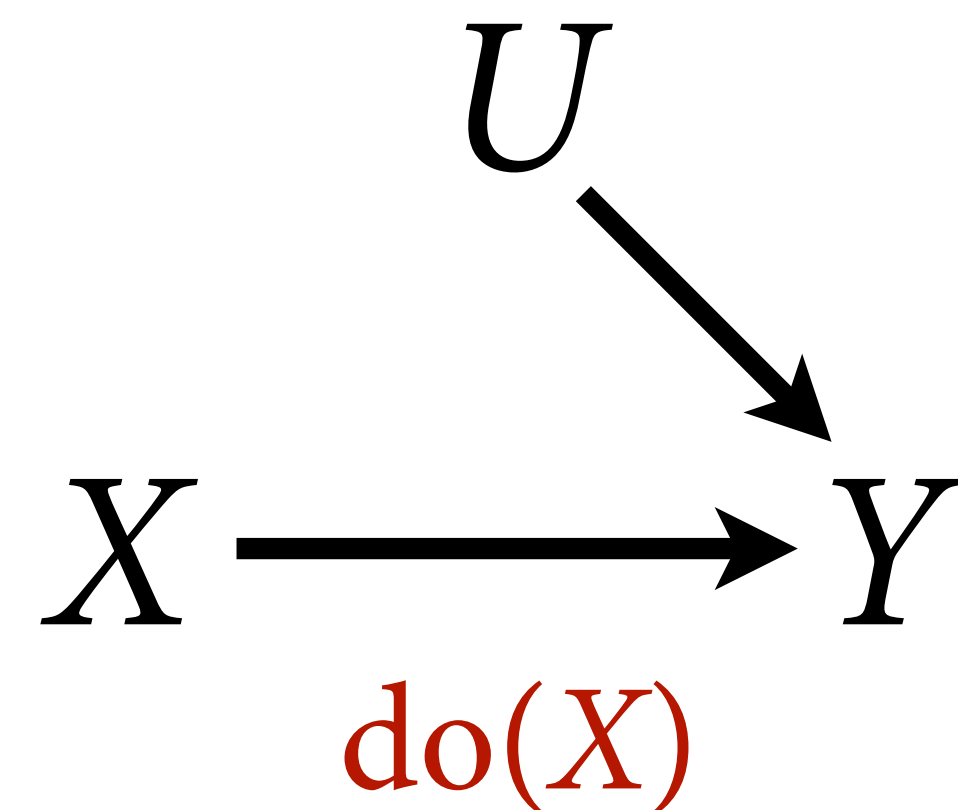
$\text{do}(X)$ means intervene on X

Can analyze causal model to find answer (if it exists)

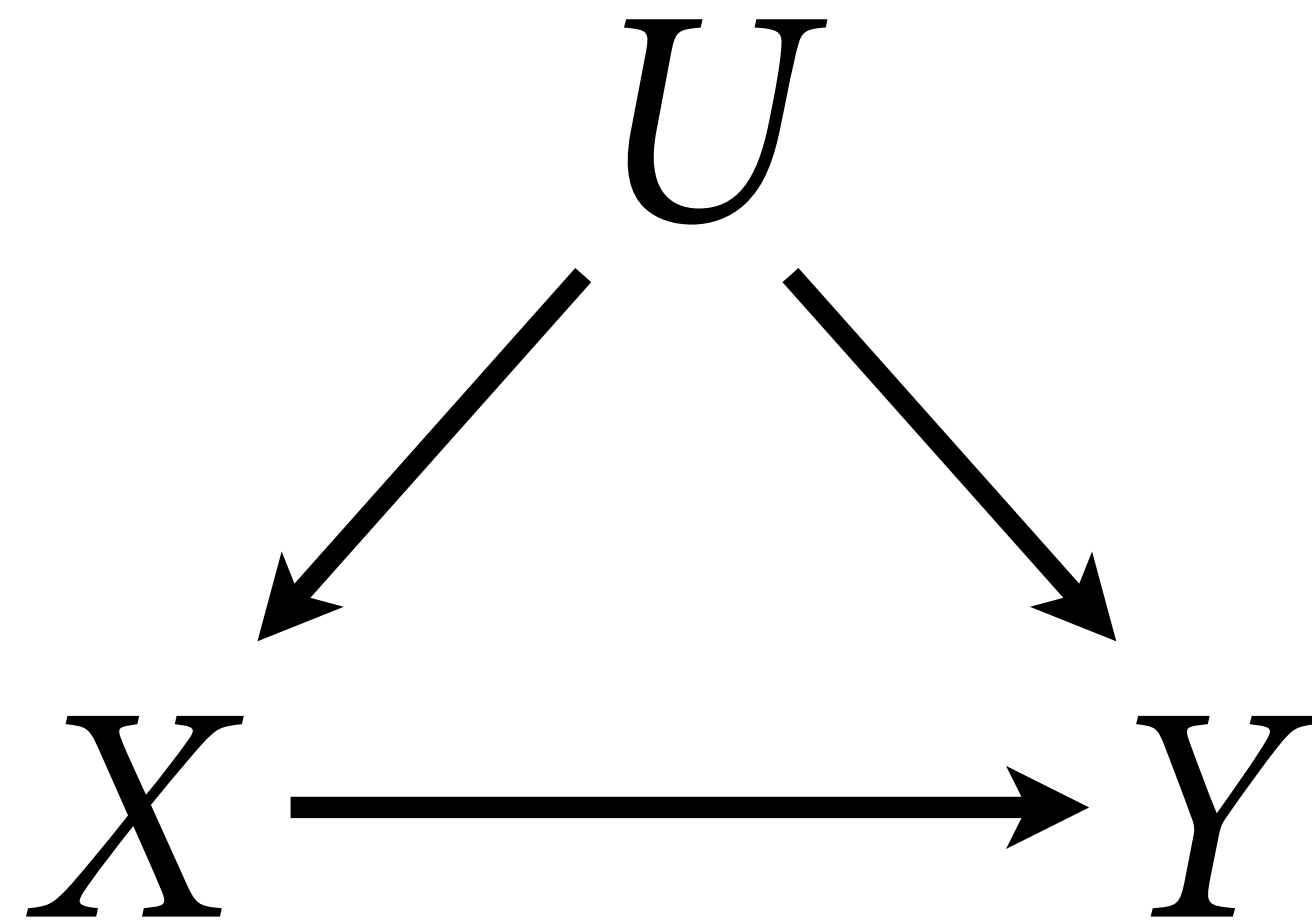
Without randomization



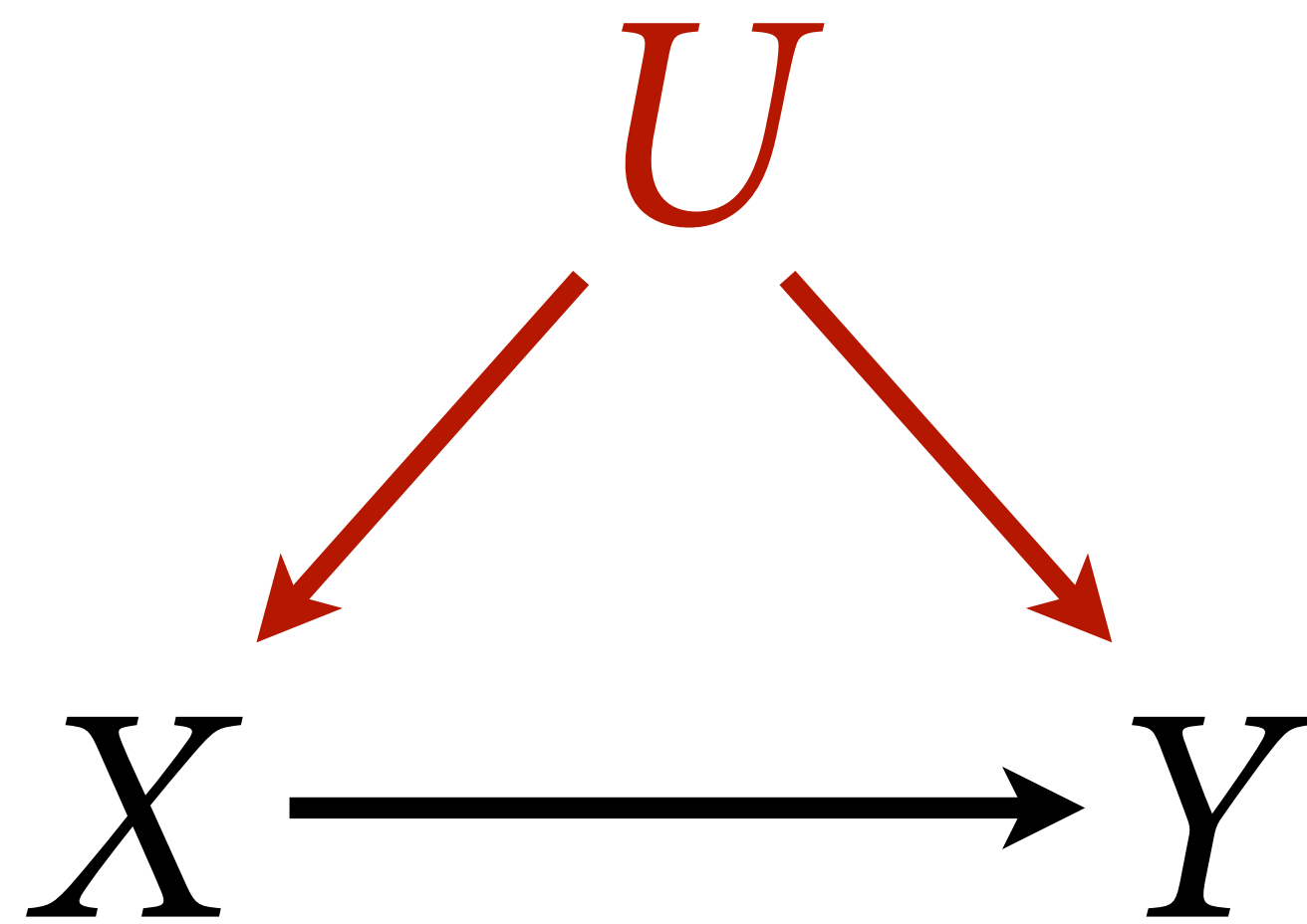
With randomization



Example: Simple Confound



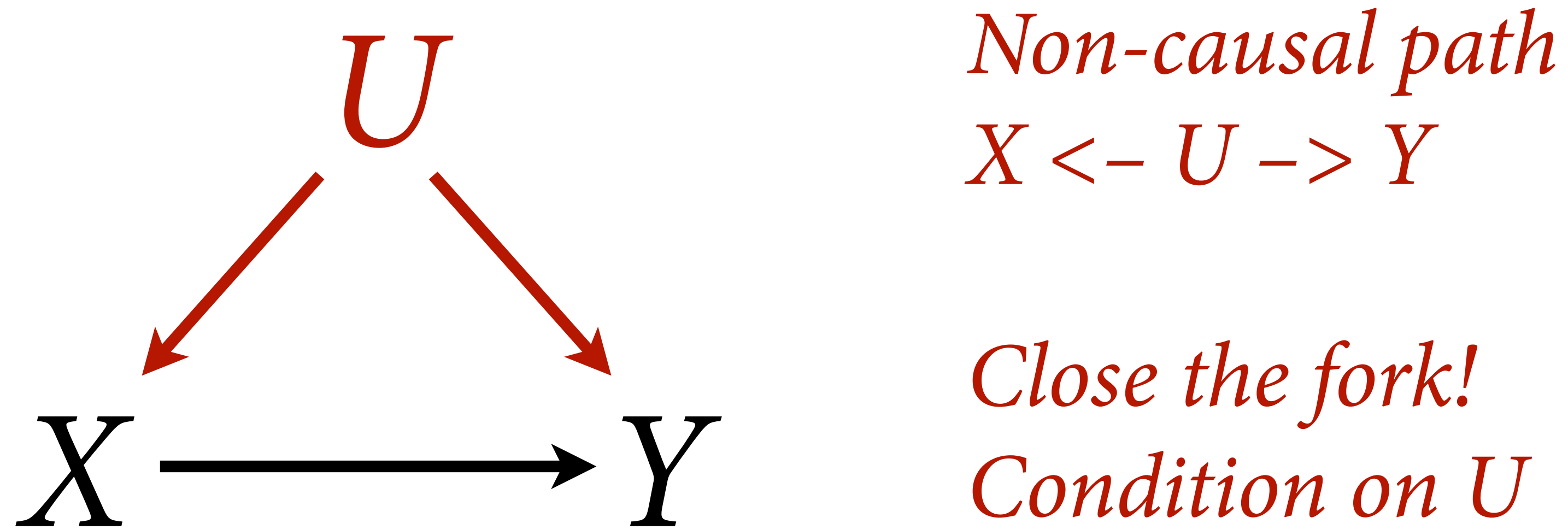
Example: Simple Confound



Non-causal path
 $X \leftarrow U \rightarrow Y$

Close the fork!
Condition on U

Example: Simple Confound



$$P(Y | \text{do}(X)) = \sum_U P(Y | X, U)P(U) = E_U P(Y | X, U)$$

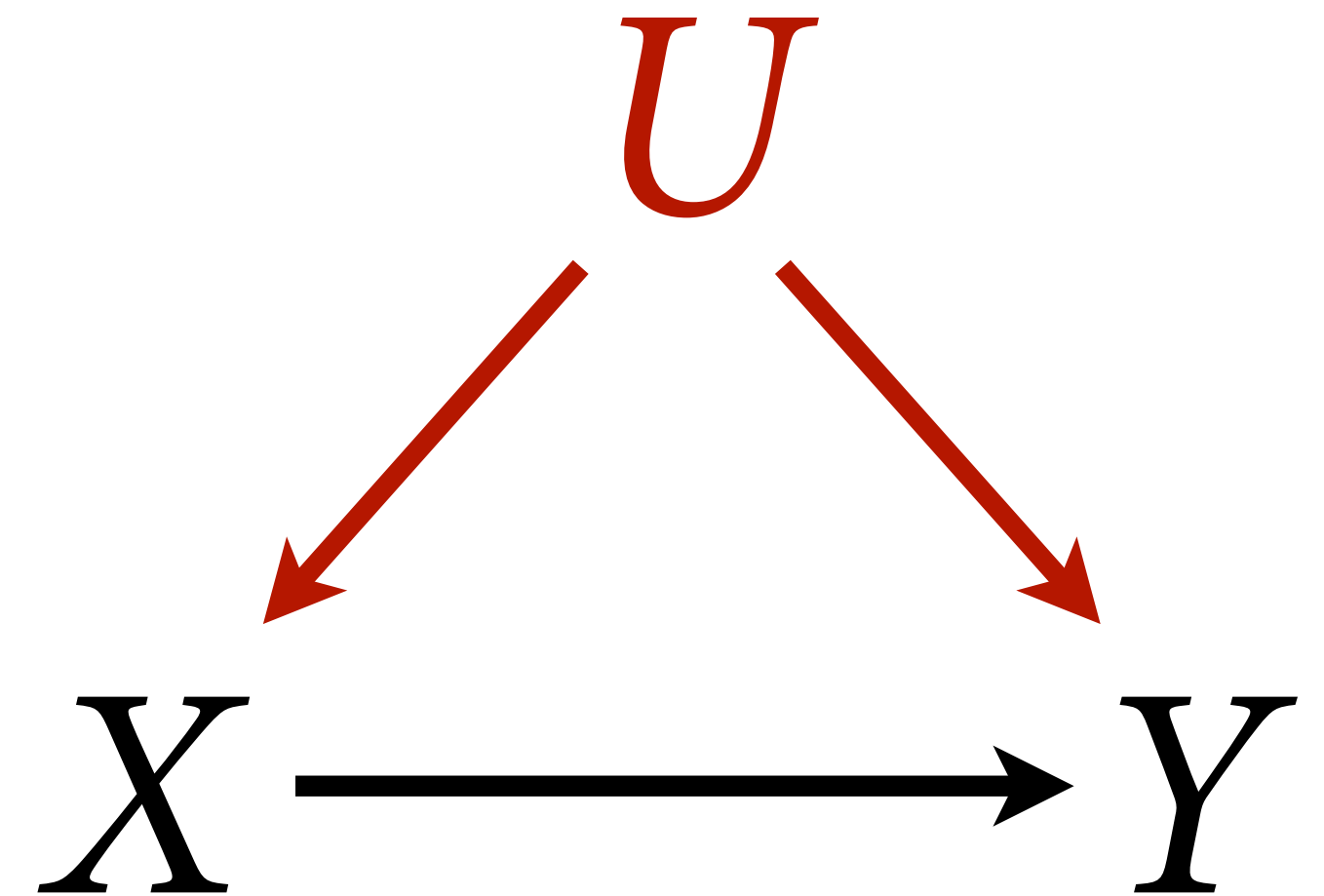
*“The distribution of Y, stratified by X and U,
averaged over the distribution of U.”*

$$P(Y | \text{do}(X)) = \sum_U P(Y | X, U)P(U) = E_U P(Y | X, U)$$

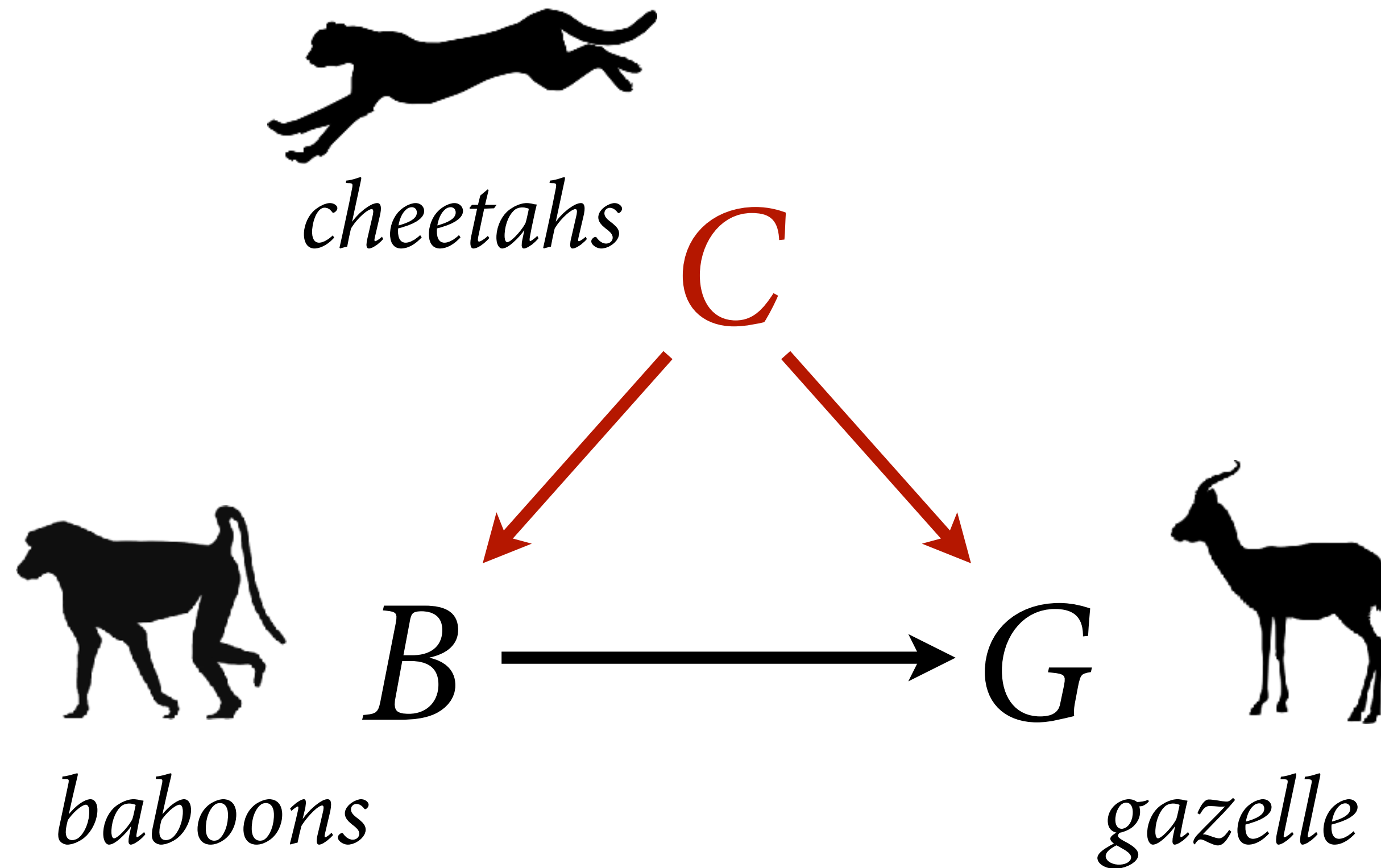
“The distribution of Y , stratified by X and U , averaged over the distribution of U .”

The causal effect of X on Y is **not** (in general) the **coefficient** relating X to Y

It is the distribution of Y when we change X , **averaged** over the distributions of the control variables (here U)



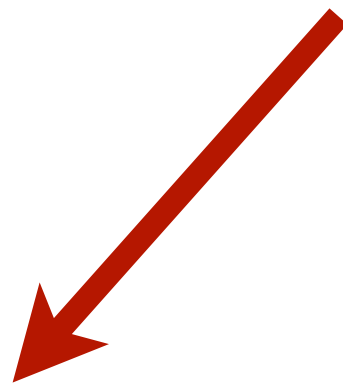
Marginal Effects Example



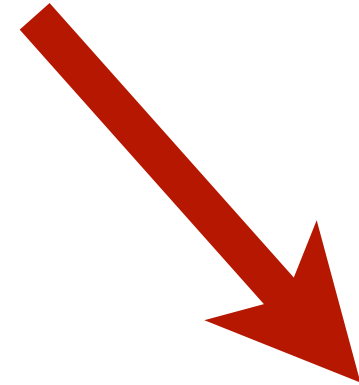
cheetahs present



C

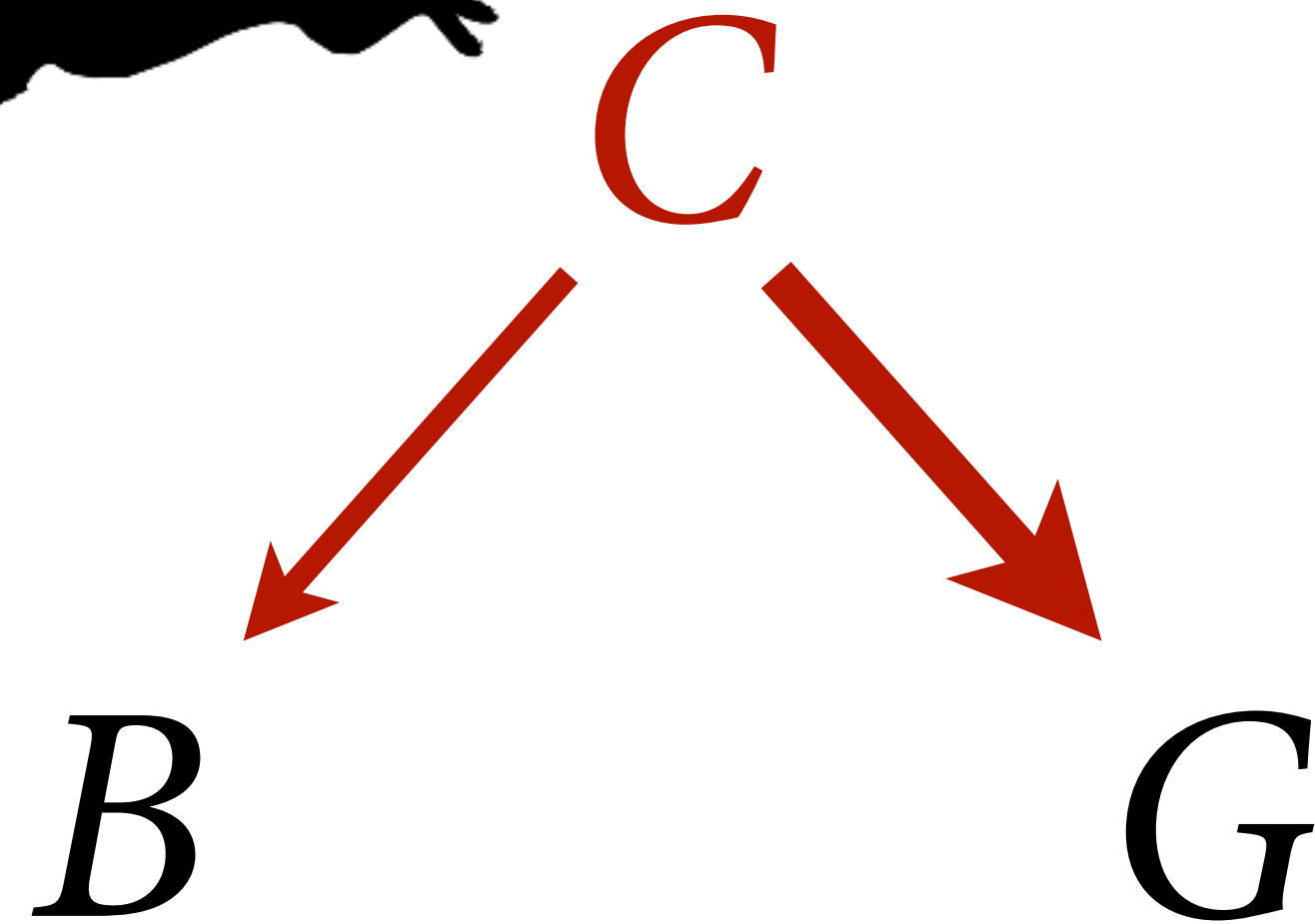


B



G

cheetahs present



cheetahs absent

C



Causal effect of baboons depends upon distribution of cheetahs

do-calculus

For DAGs, rules for finding $P(Y|\text{do}(X))$ known as **do-calculus**

do-calculus says what is possible to say **before** picking functions

Justifies **graphical** analysis

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THE BOOK OF WHY

DO-CALCULUS AT WORK

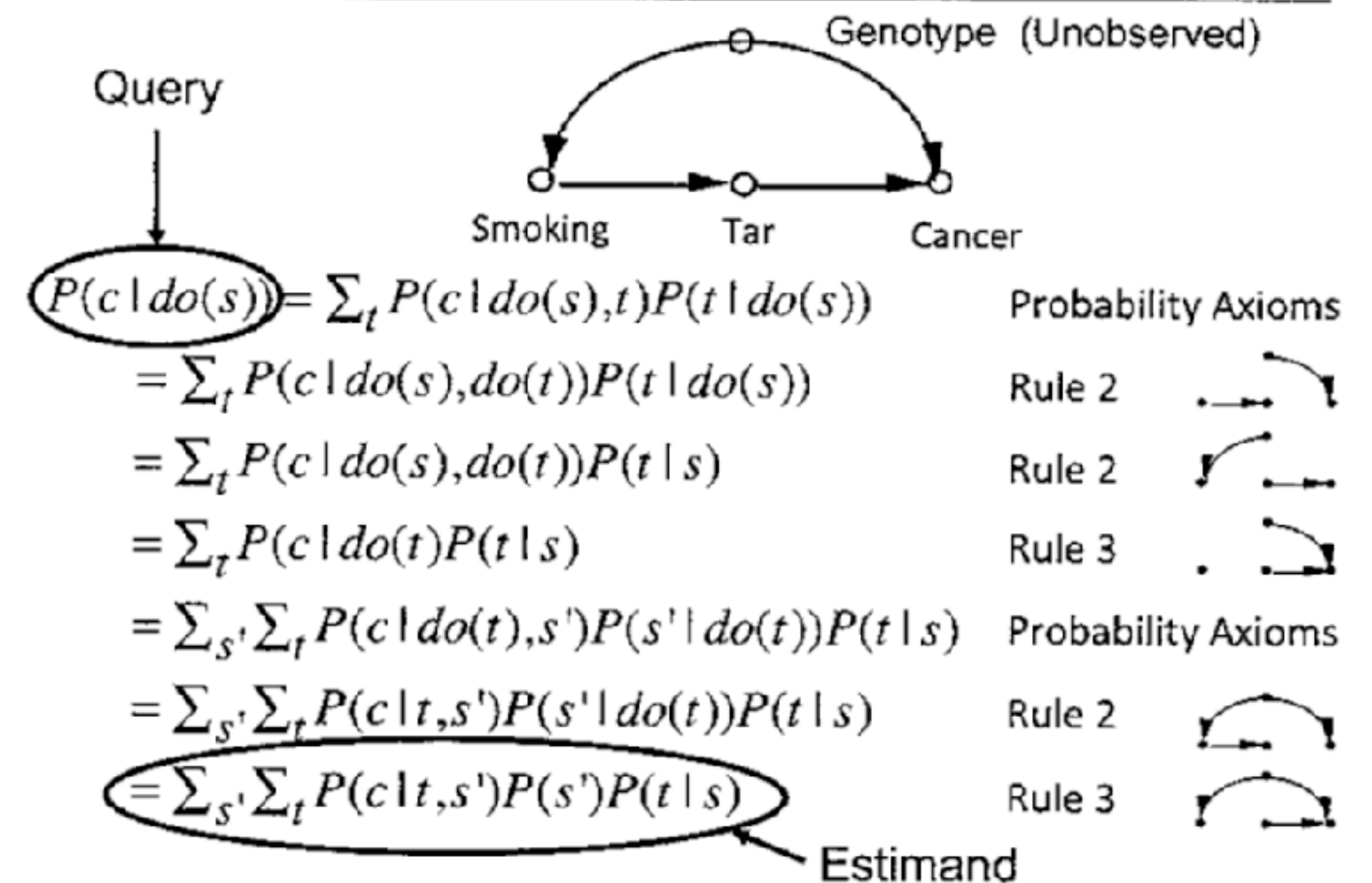


FIGURE 7.4. Derivation of the front-door adjustment formula from the rules of do-calculus.

Do calculus, not too much, mostly graphs

do-calculus

do-calculus is **worst case**:
additional assumptions often
allow stronger inference

do-calculus is **best case**:
if inference possible by do-
calculus, does not depend on
special assumptions

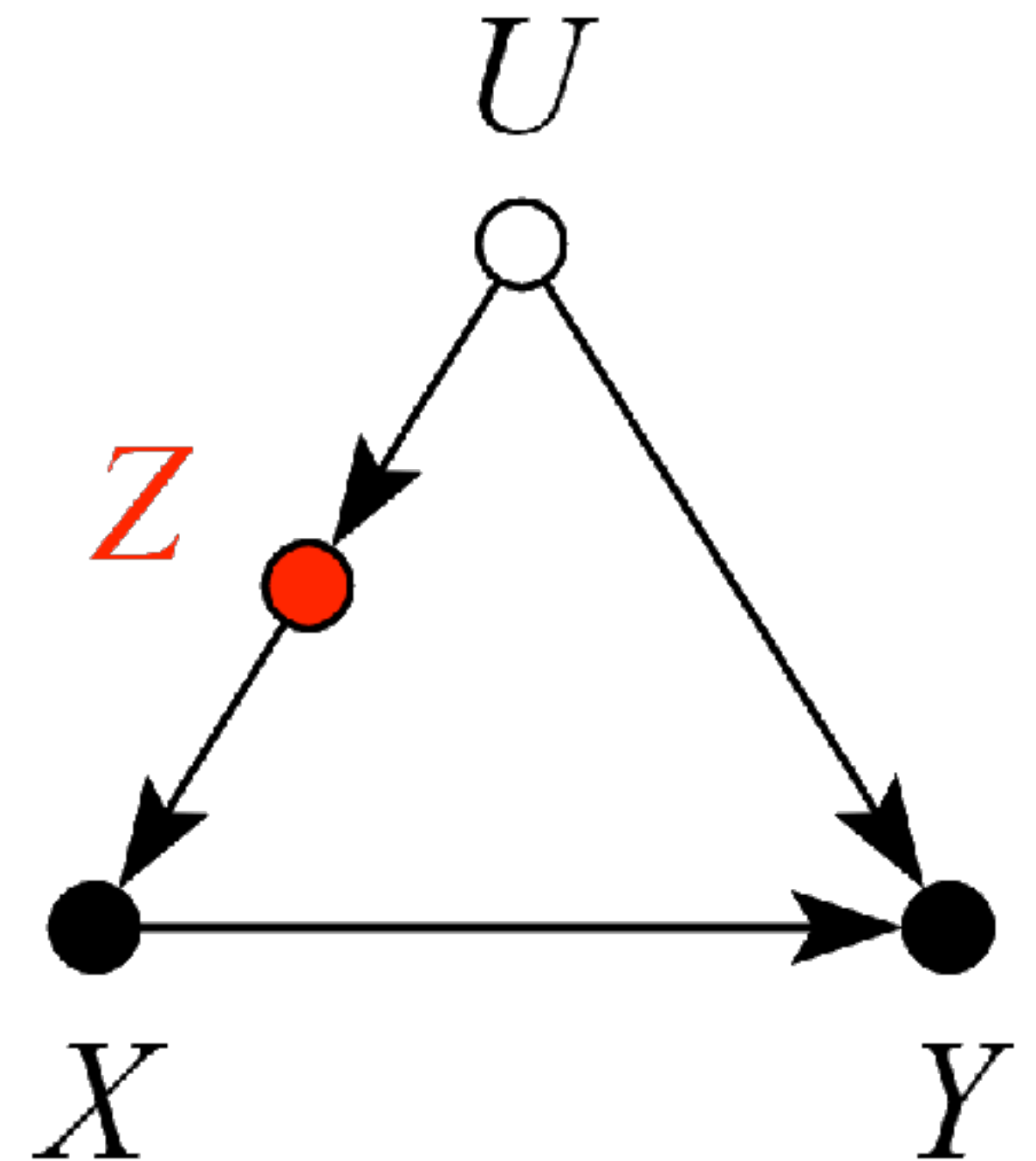


Judea Pearl, father of do-calculus (1966)

Backdoor Criterion

Backdoor Criterion is a shortcut to applying (some) results of do-calculus

Can be performed with your **eyeballs**

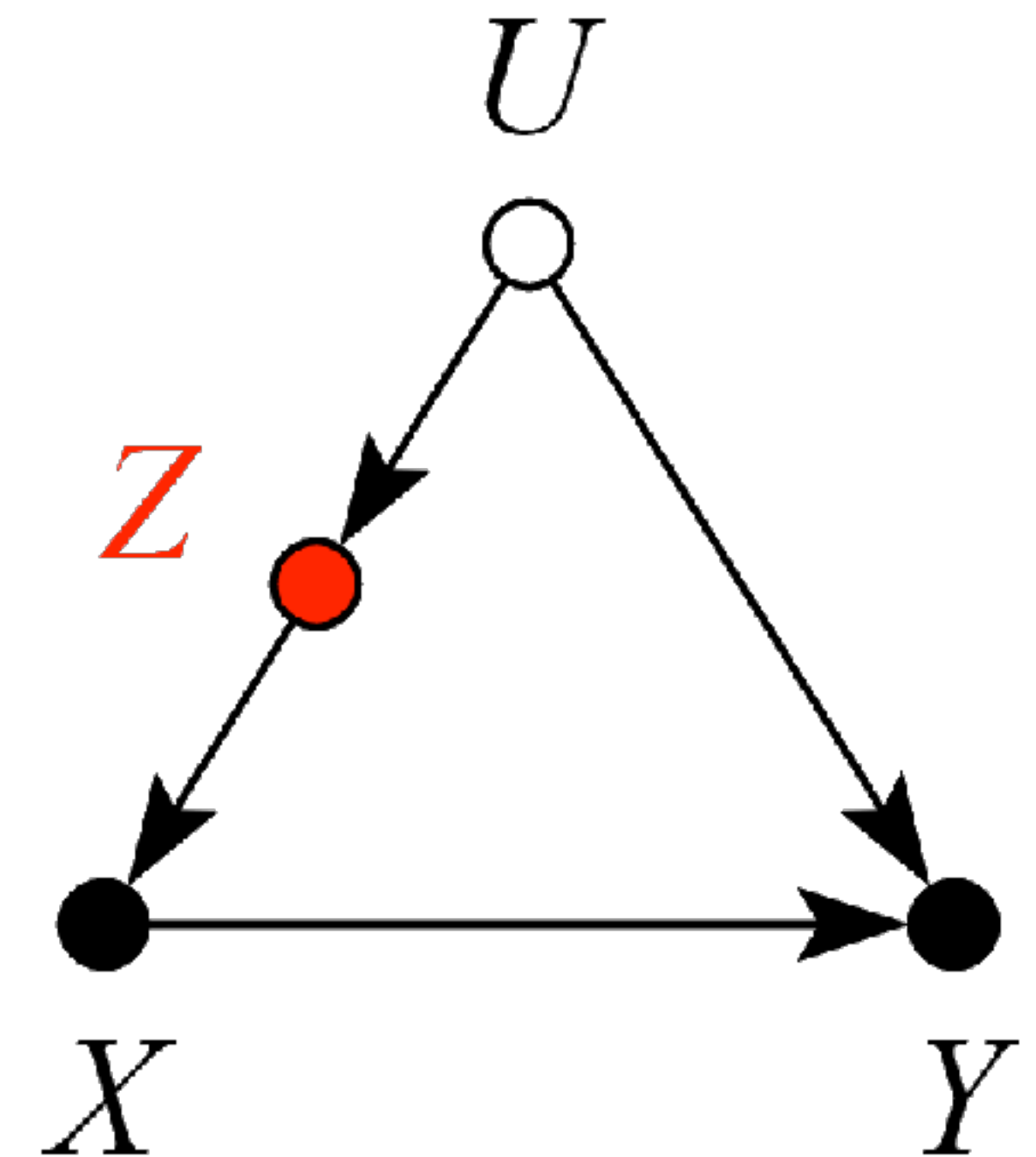


Backdoor Criterion: Rule to find a set of variables to stratify by to yield $P(Y|\text{do}(X))$

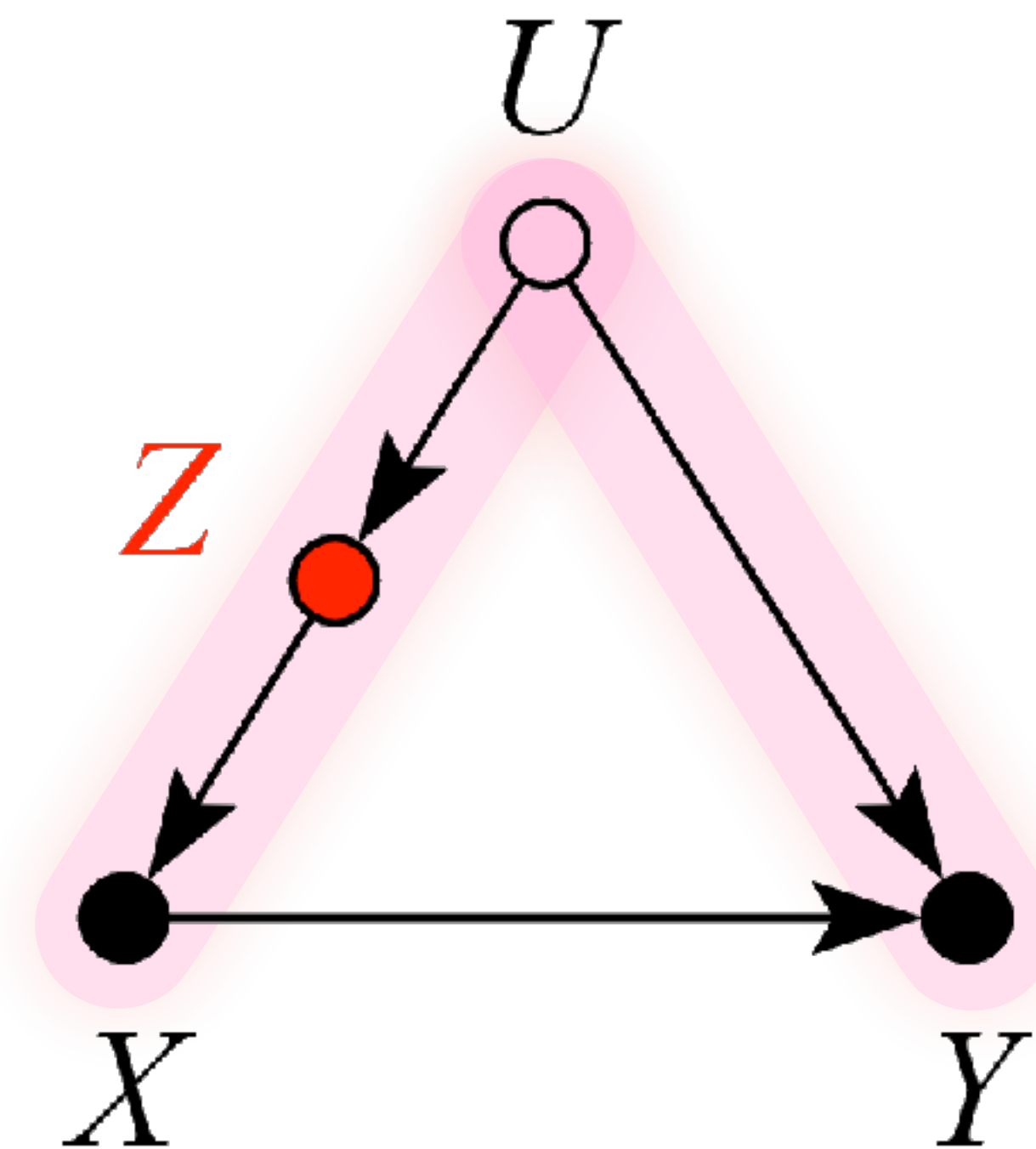
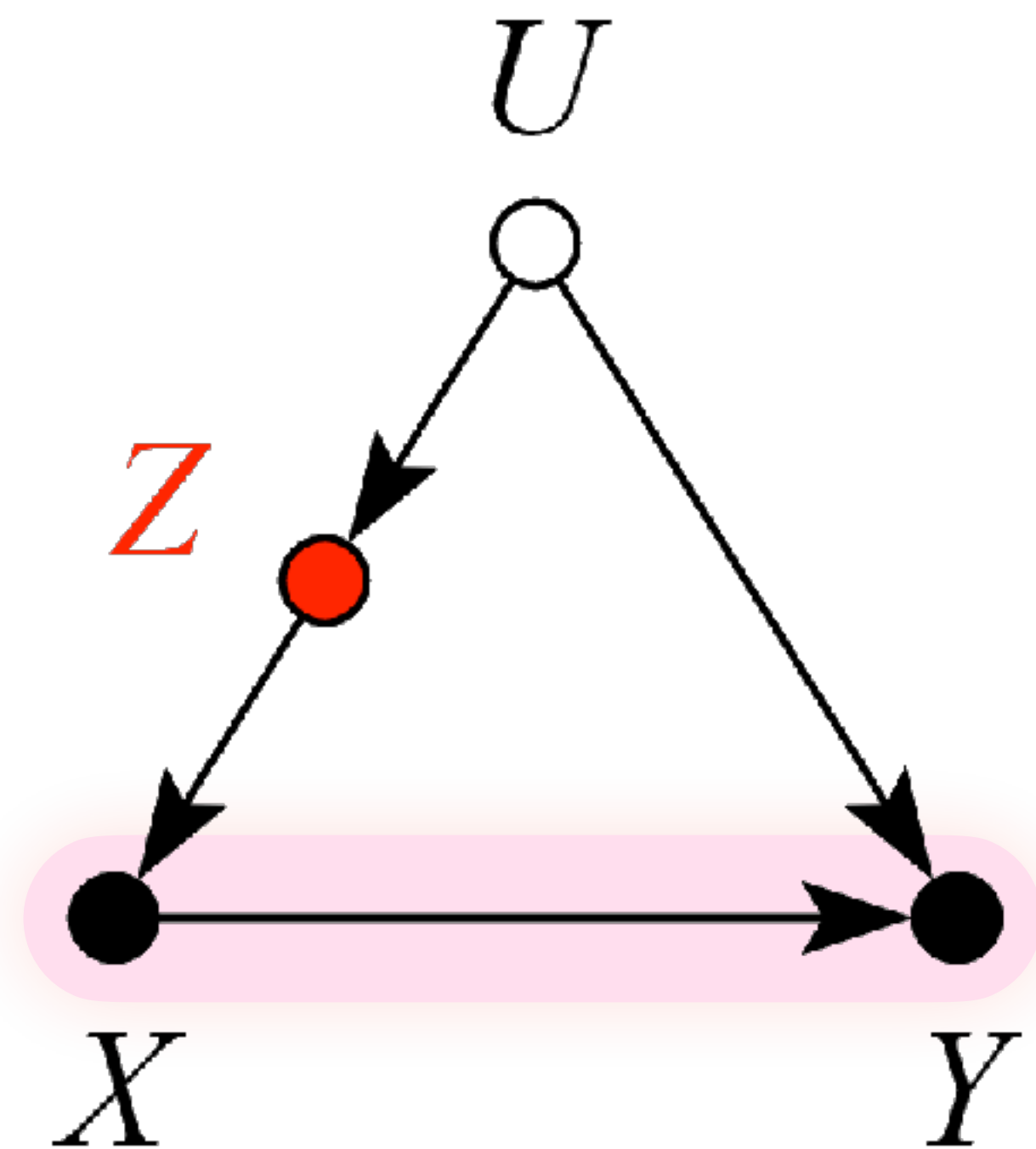
(1) Identify all **paths** connecting the treatment (X) to the outcome (Y)

(2) Paths with arrows **entering** X are backdoor paths (non-causal paths)

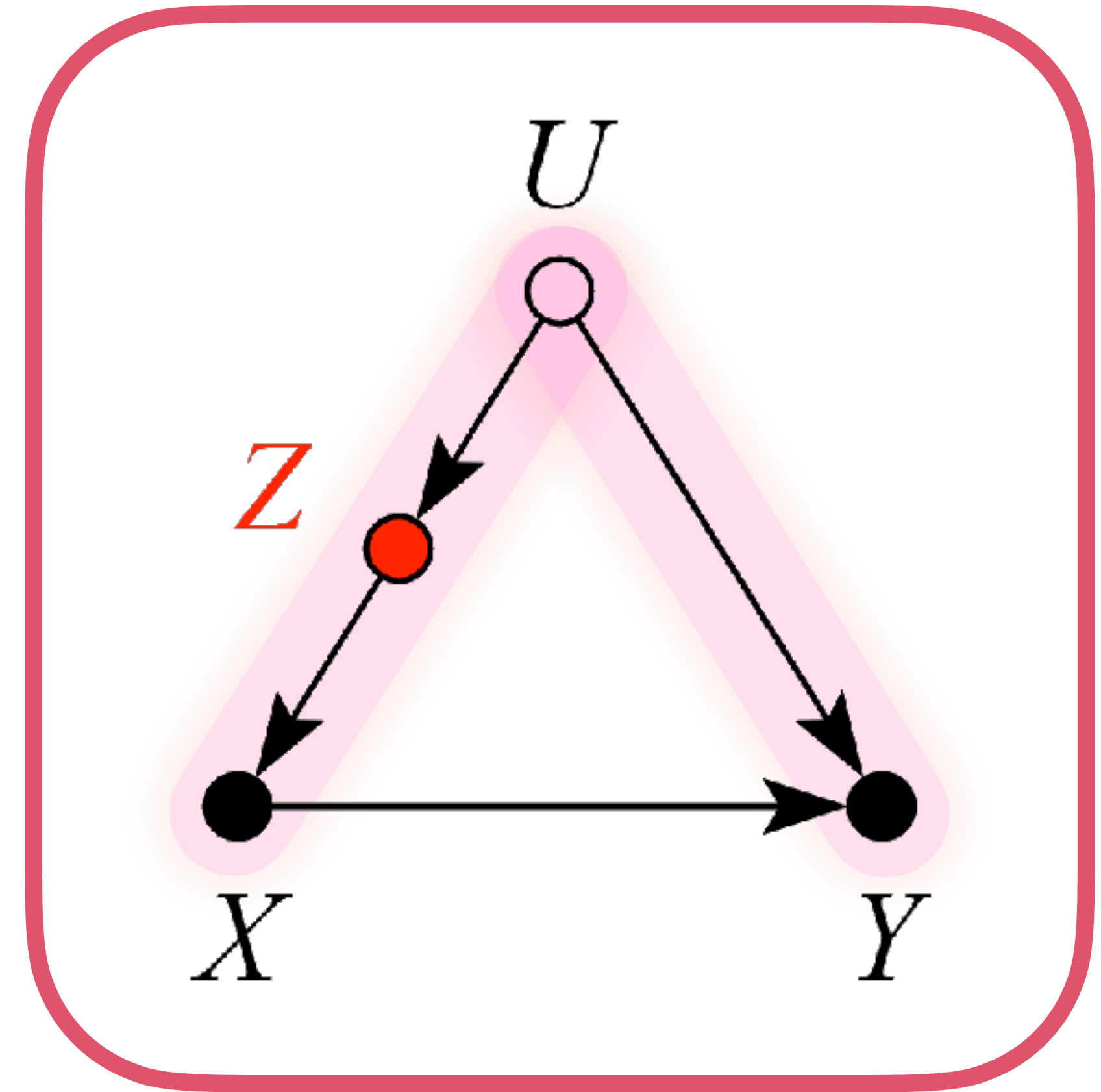
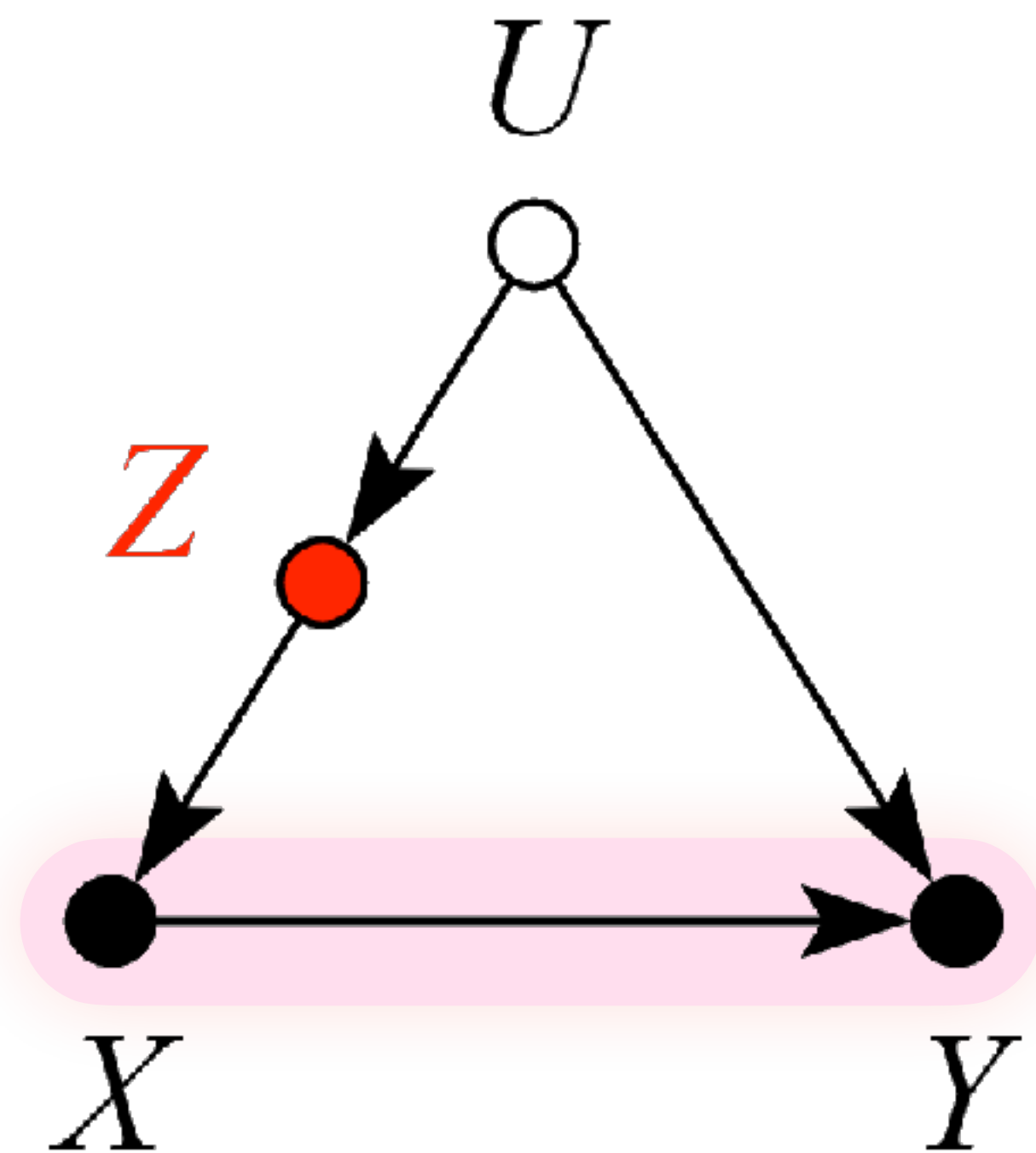
(3) Find **adjustment set** that closes/blocks all backdoor paths



(1) Identify all **paths** connecting the treatment (X) to the outcome (Y)



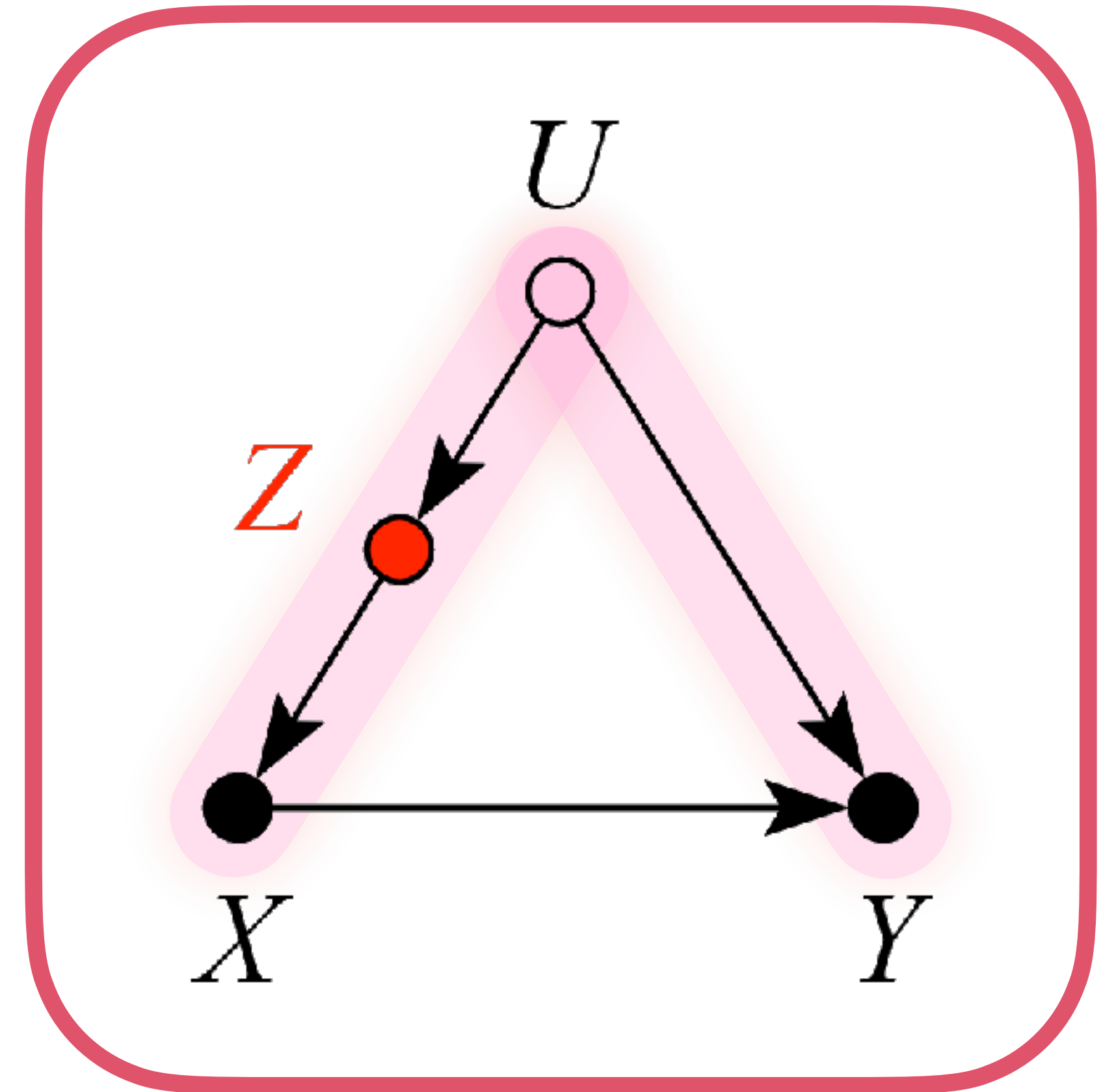
(2) Paths with arrows **entering** X are backdoor paths (confounding paths)



(3) Find a set of control variables that close/block all backdoor paths

Block the pipe: $X \perp\!\!\!\perp U \mid Z$

Z “knows” all of the association between X, Y that is due to U



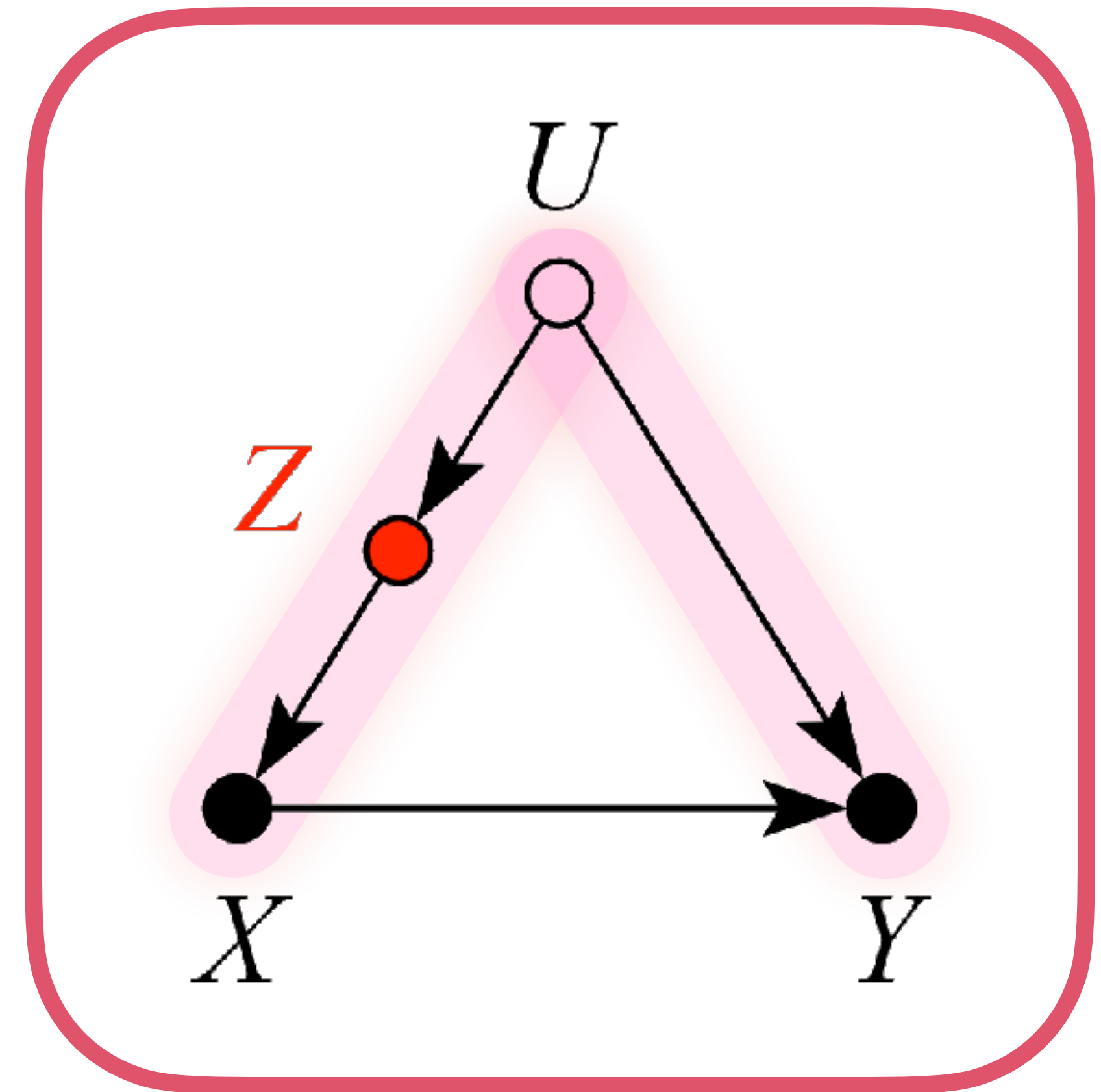
(3) Find a set of control variables that close/block all backdoor paths

Block the pipe: $X \perp\!\!\!\perp U \mid Z$

$$P(Y \mid \text{do}(X)) = \sum_z P(Y \mid X, Z)P(Z = z)$$

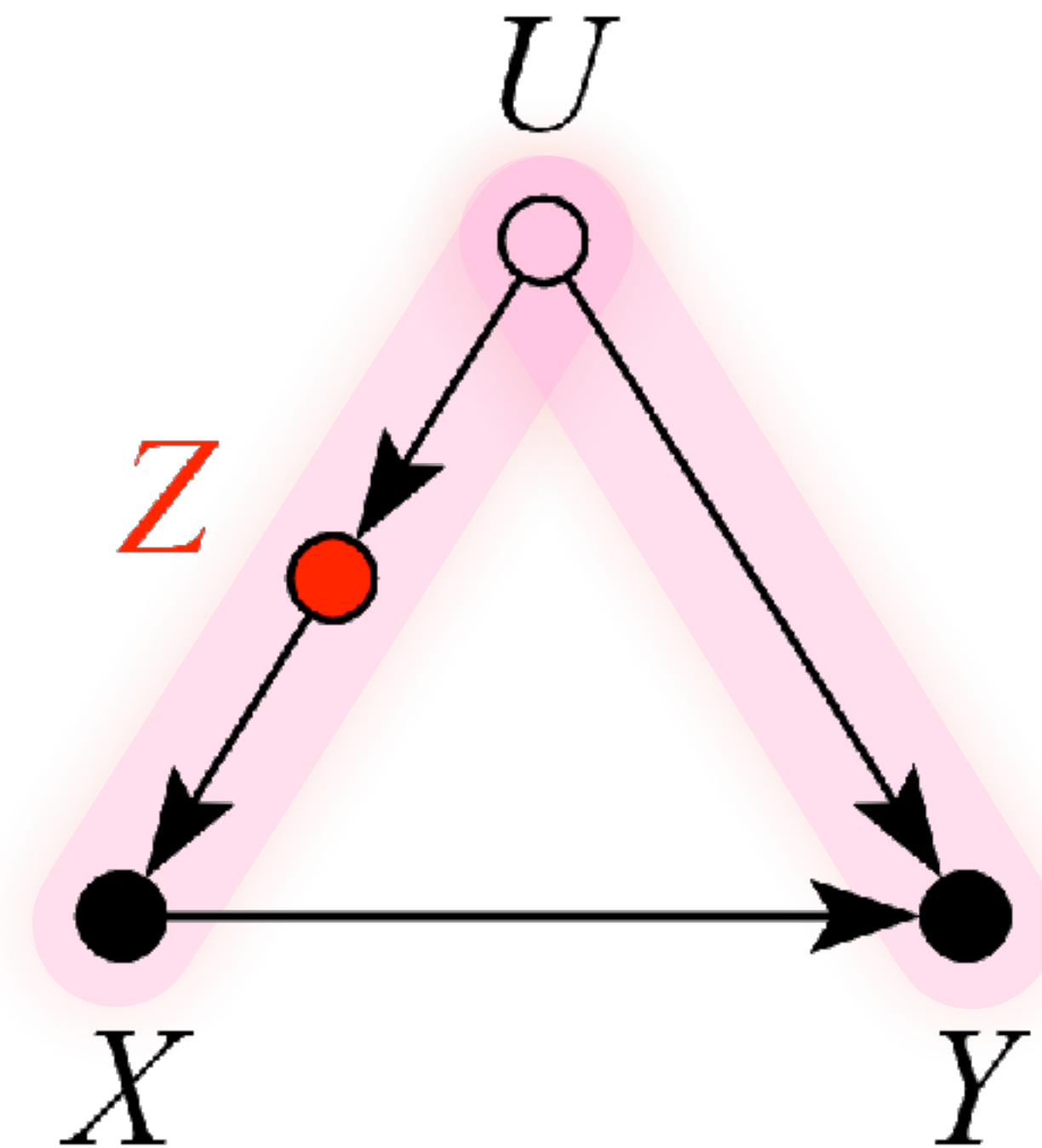
$$Y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta_X X_i + \beta_Z Z_i$$



```
# simulate confounded Y
N <- 200
b_XY <- 0
b_UY <- -1
b_UZ <- -1
b_ZX <- 1

set.seed(10)
U <- rbern(N)
Z <- rnorm(N, b_UZ*U)
X <- rnorm(N, b_ZX*Z)
Y <- rnorm(N, b_XY*X + b_UY*U)
d <- list(Y=Y, X=X, Z=Z)
```



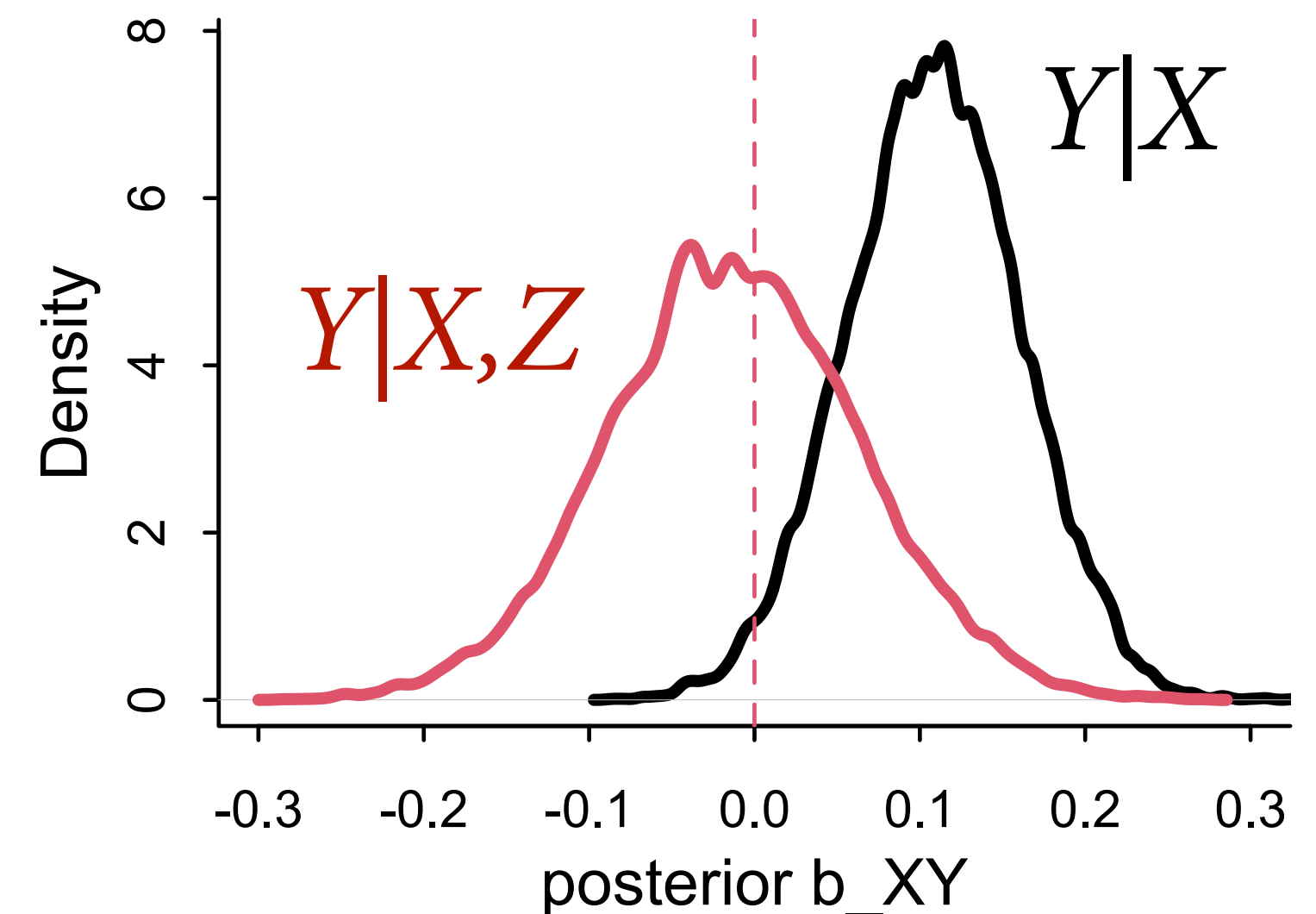
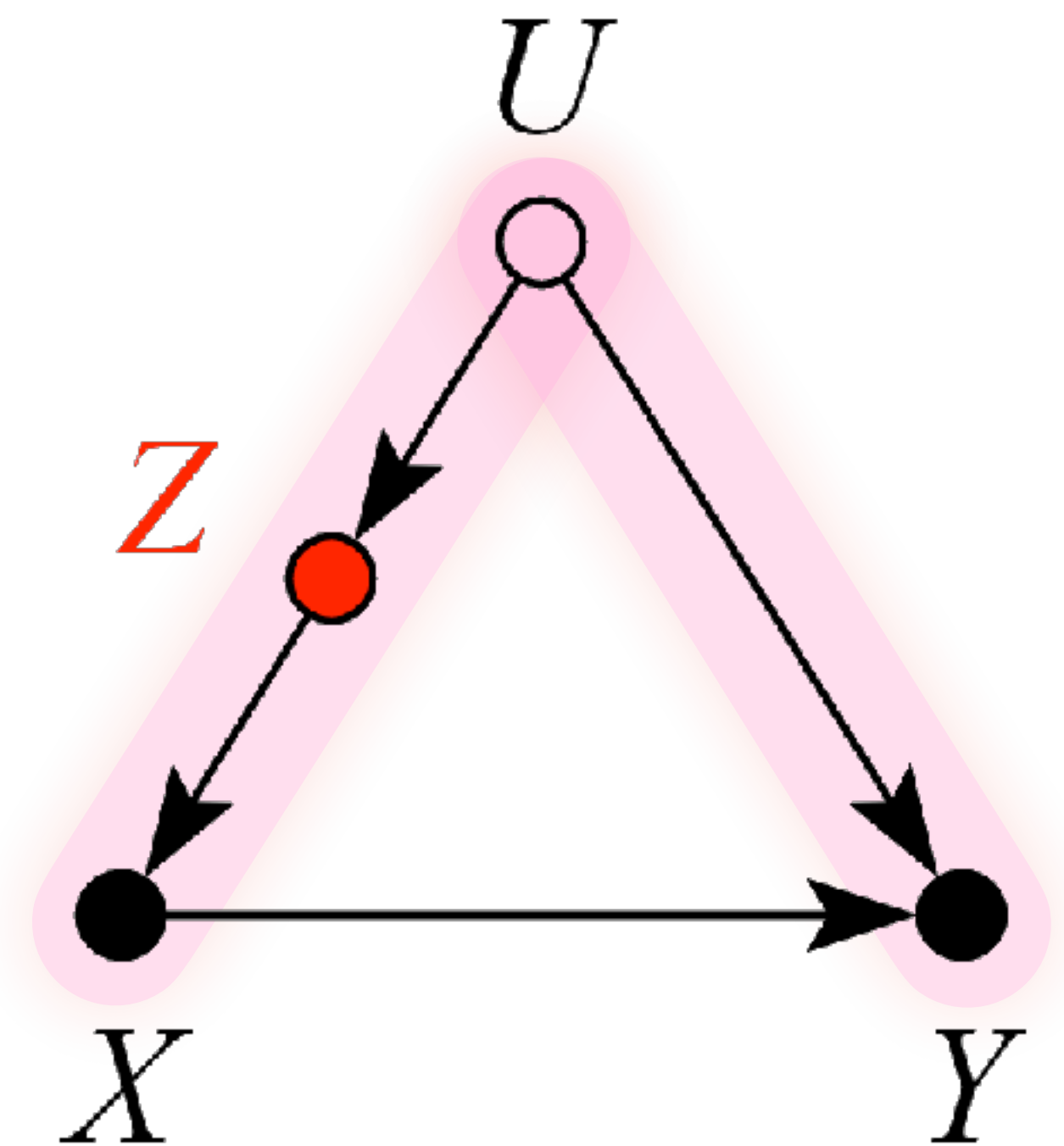
```
# ignore U,Z
```

```
m_YX <- quap(  
  alist(  
    Y ~ dnorm( mu , sigma ),  
    mu <- a + b_XY*X,  
    a ~ dnorm( 0 , 1 ),  
    b_XY ~ dnorm( 0 , 1 ),  
    sigma ~ dexp( 1 )  
  ), data=d )
```

```
# stratify by Z
```

```
m_YXZ <- quap(  
  alist(  
    Y ~ dnorm( mu , sigma ),  
    mu <- a + b_XY*X + b_Z*Z,  
    a ~ dnorm( 0 , 1 ),  
    c(b_XY,b_Z) ~ dnorm( 0 , 1 ),  
    sigma ~ dexp( 1 )  
  ), data=d )
```

```
post <- extract.samples(m_YX)  
post2 <- extract.samples(m_YXZ)  
dens(post$b_XY, lwd=3, col=1, xlab="posterior  
b_XY", xlim=c(-0.3,0.3))  
dens(post2$b_XY, lwd=3, col=2, add=TRUE)
```



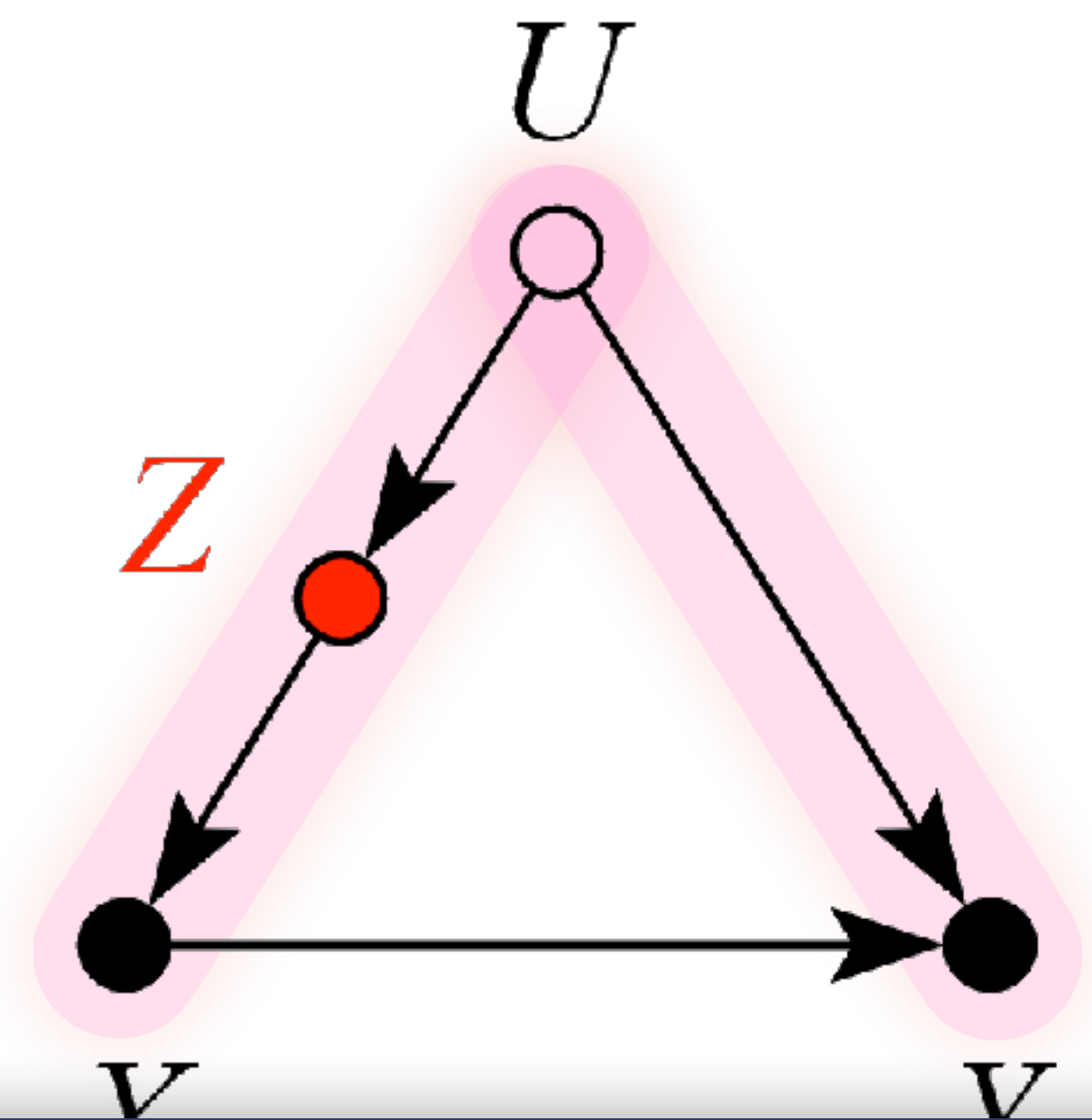
```
# ignore U,Z
```

```
m_YX <- quap(  
  alist(  
    Y ~ dnorm( mu , sigma ),  
    mu <- a + b_XY*X,  
    a ~ dnorm( 0 , 1 ),  
    b_XY ~ dnorm( 0 , 1 ),  
    sigma ~ dexp( 1 )  
  ), data=d )
```

```
# stratify by Z
```

```
m_YXZ <- quap(  
  alist(  
    Y ~ dnorm( mu , sigma ),  
    mu <- a + b_XY*X + b_Z*Z,  
    a ~ dnorm( 0 , 1 ),  
    c(b_XY,b_Z) ~ dnorm( 0 , 1 ),  
    sigma ~ dexp( 1 )  
  ), data=d )
```

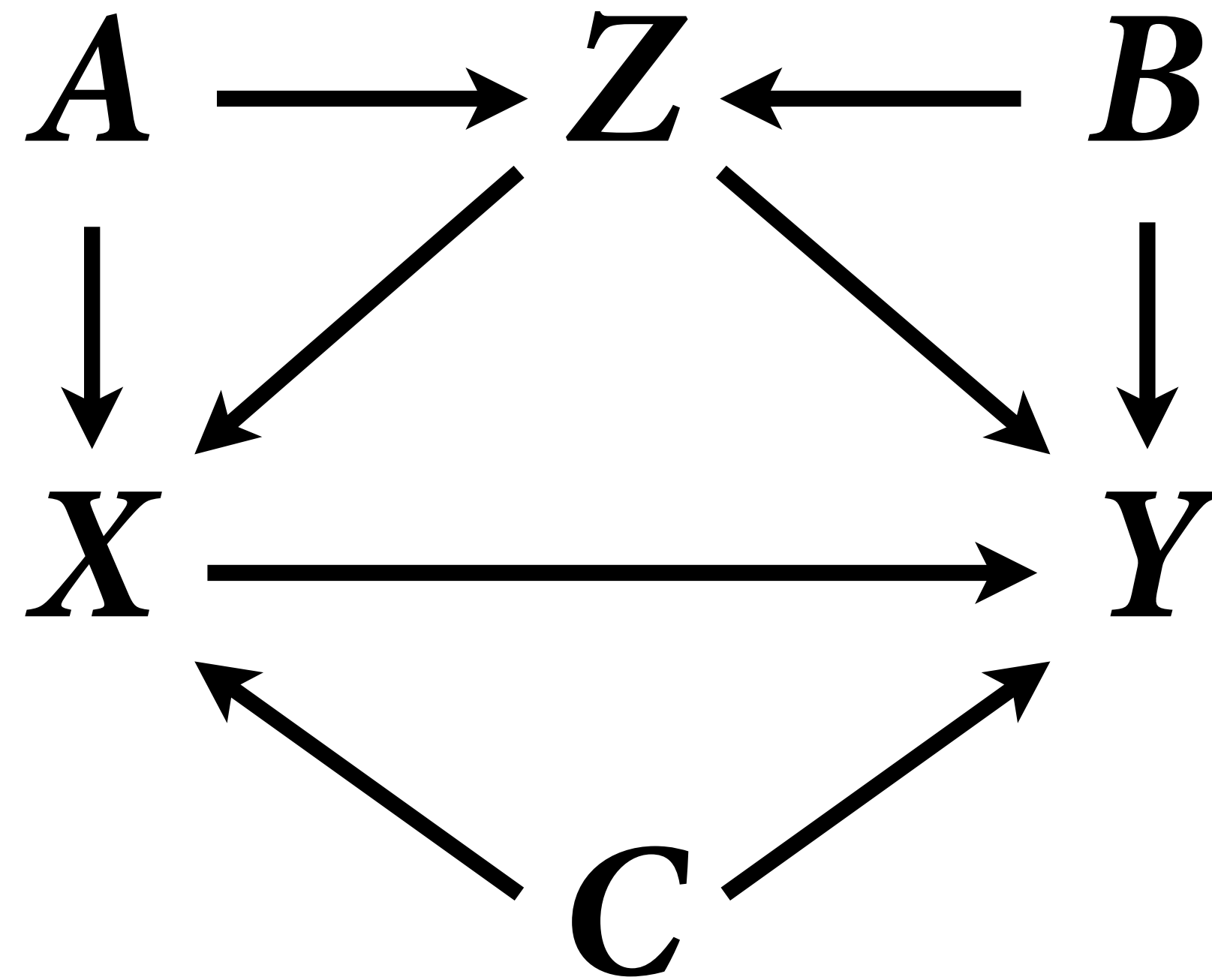
```
post <- extract.samples(m_YX)  
post2 <- extract.samples(m_YXZ)  
dens(post$b_XY, lwd=3, col=1, xlab="posterior  
b_XY", xlim=c(-0.3,0.3))  
dens(post2$b_XY, lwd=3, col=2, add=TRUE)
```



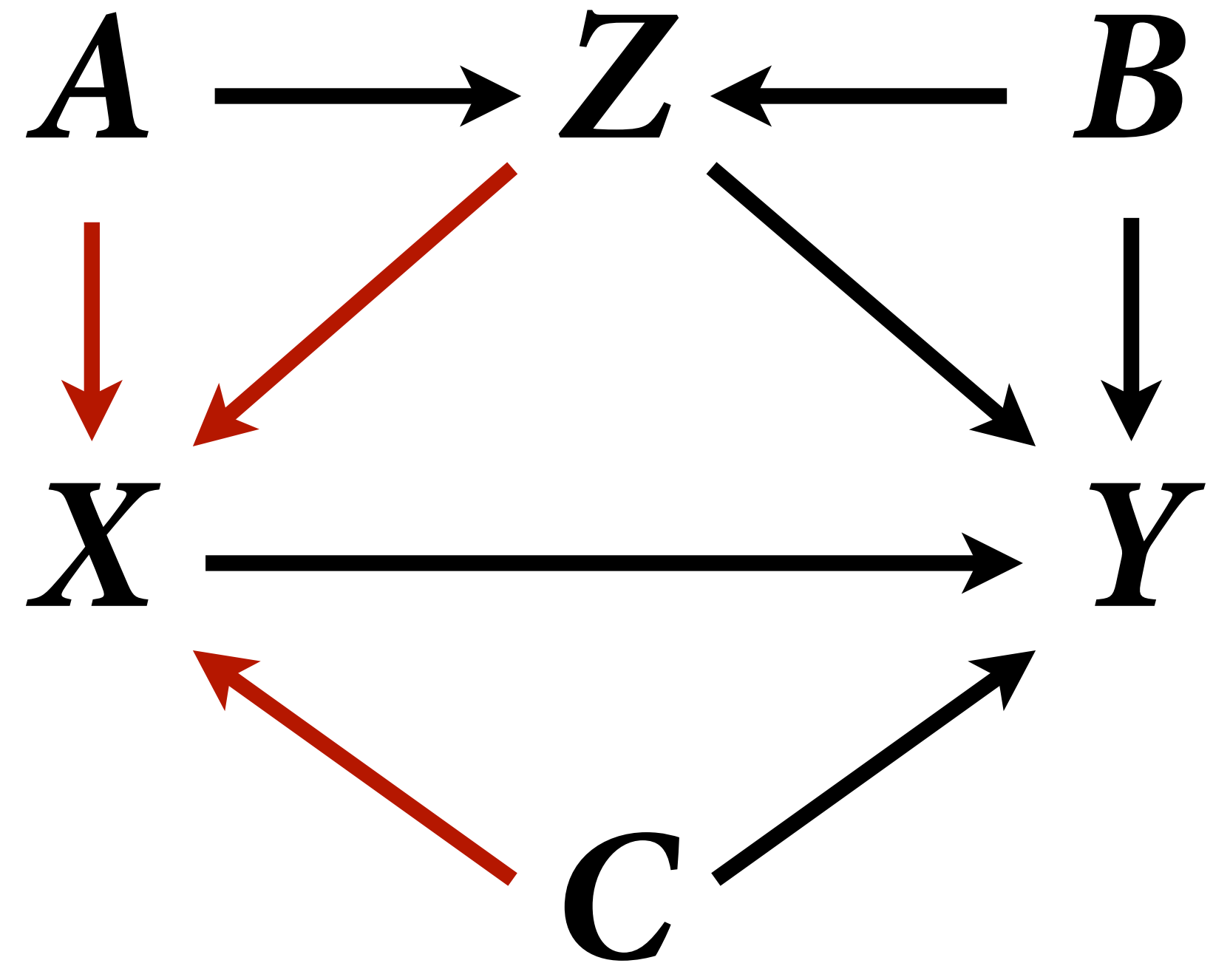
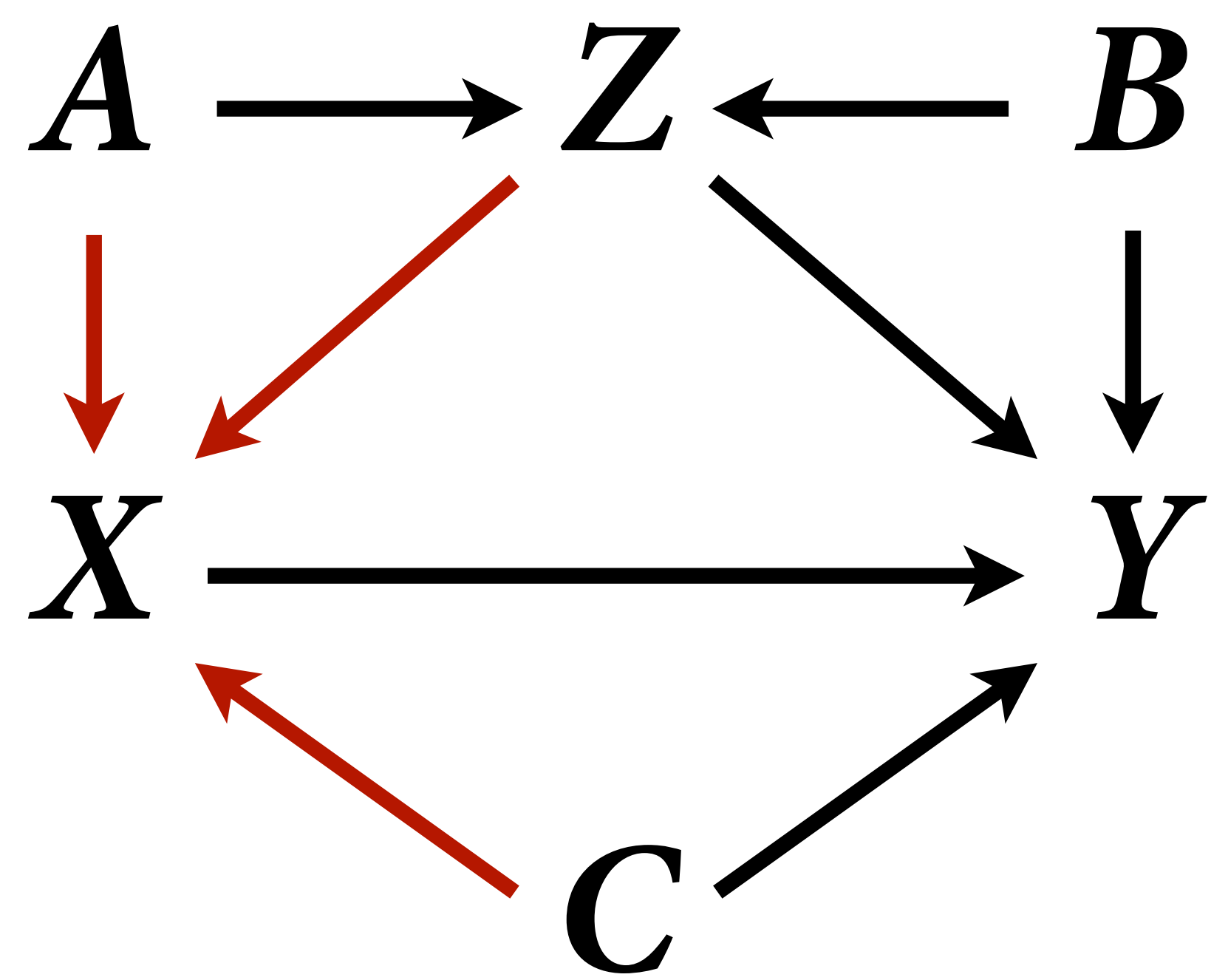
```
> precis(m_YXZ)  
      mean    sd  5.5% 94.5%  
a    -0.32  0.09 -0.47 -0.18  
b_XY  -0.01  0.08 -0.13  0.11  
b_Z    0.24  0.11  0.06  0.42  
sigma  1.18  0.06  1.08  1.27
```

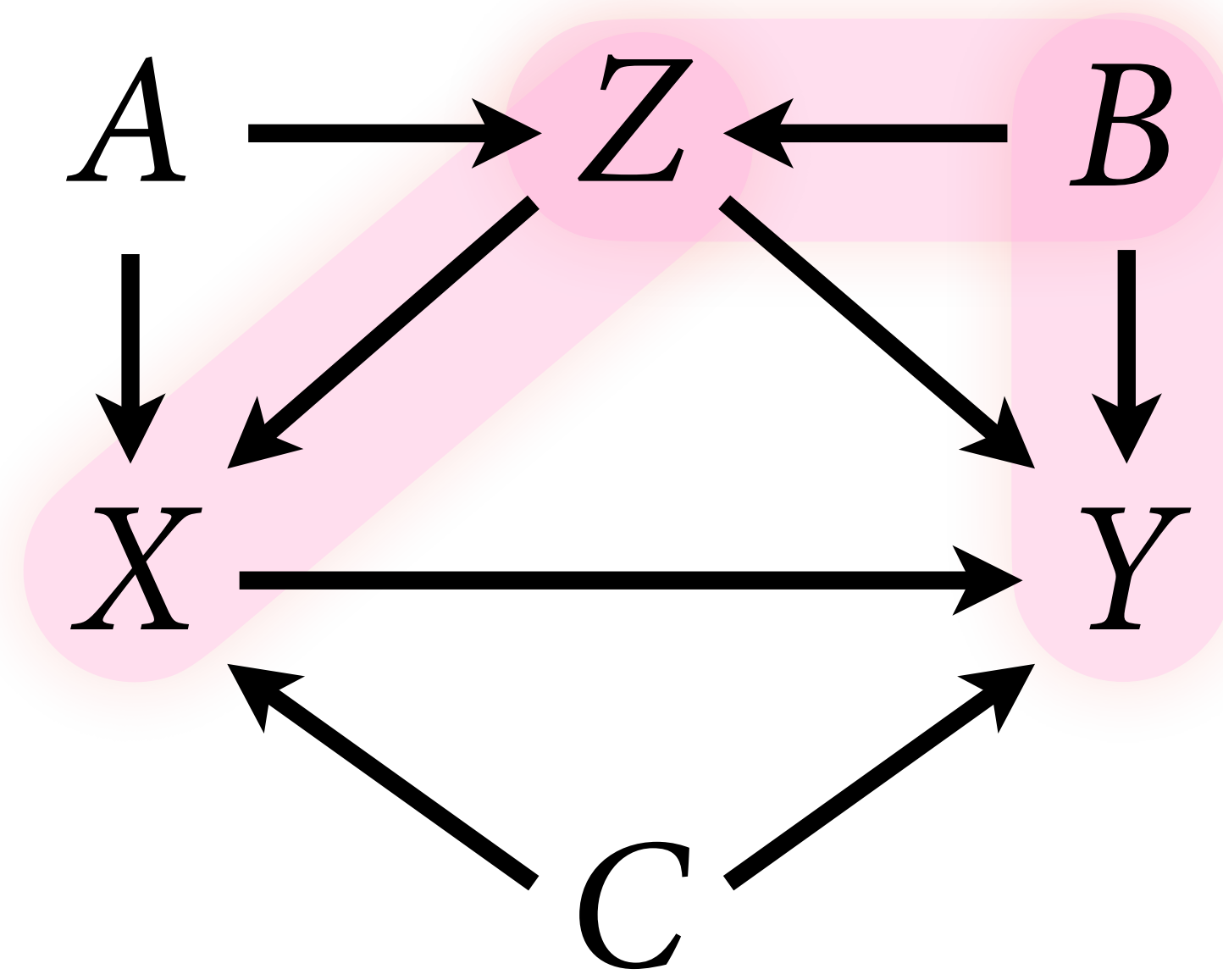
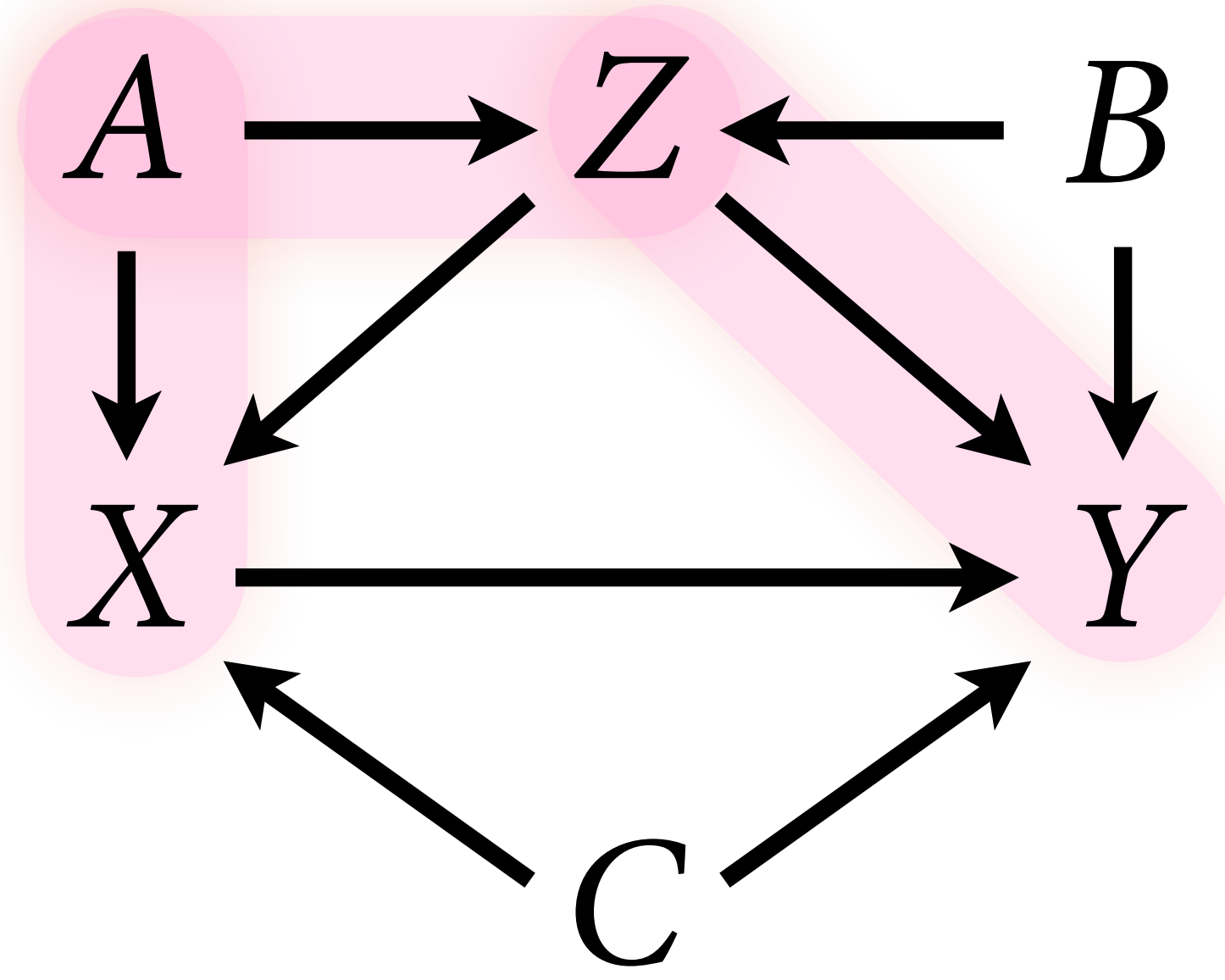
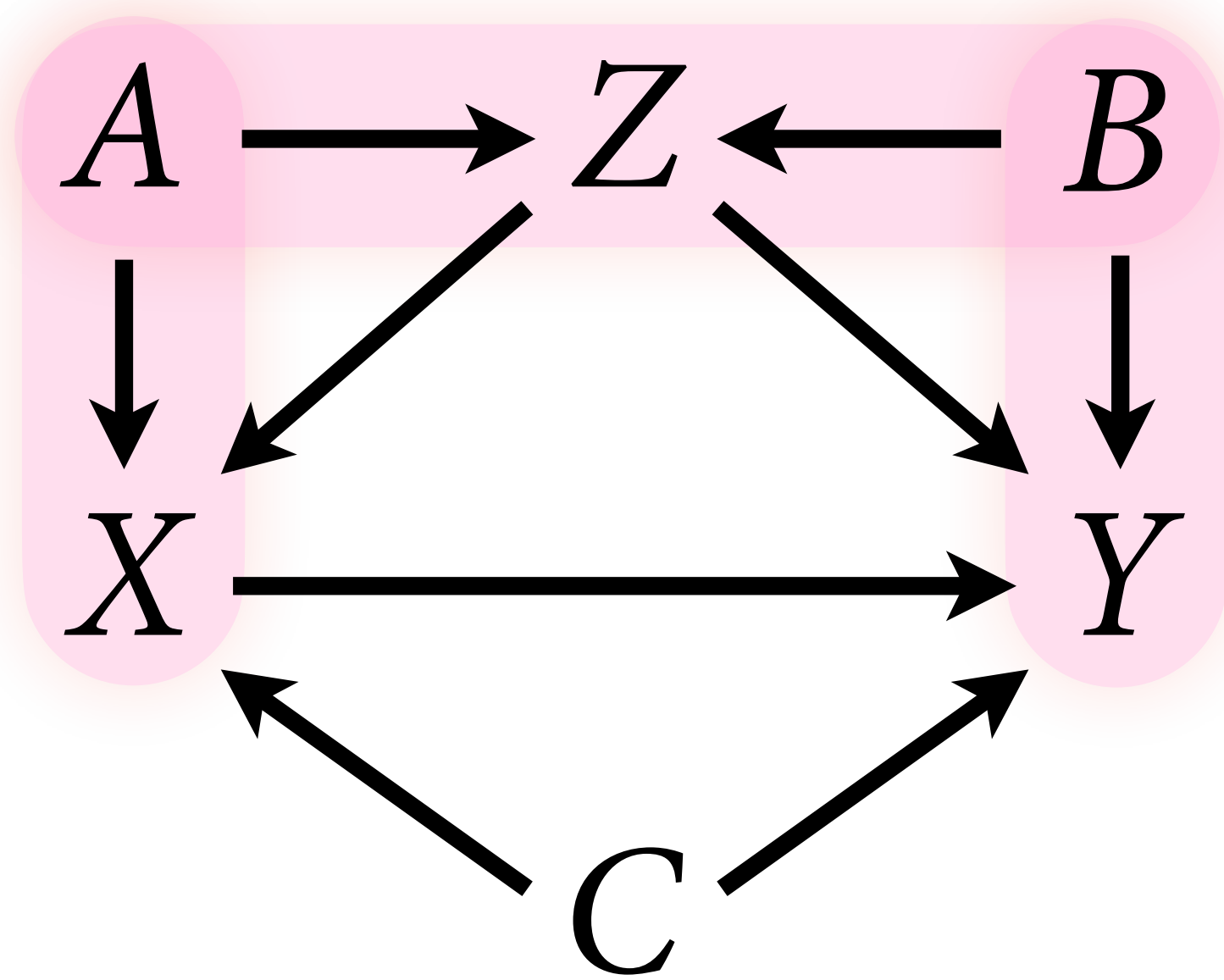
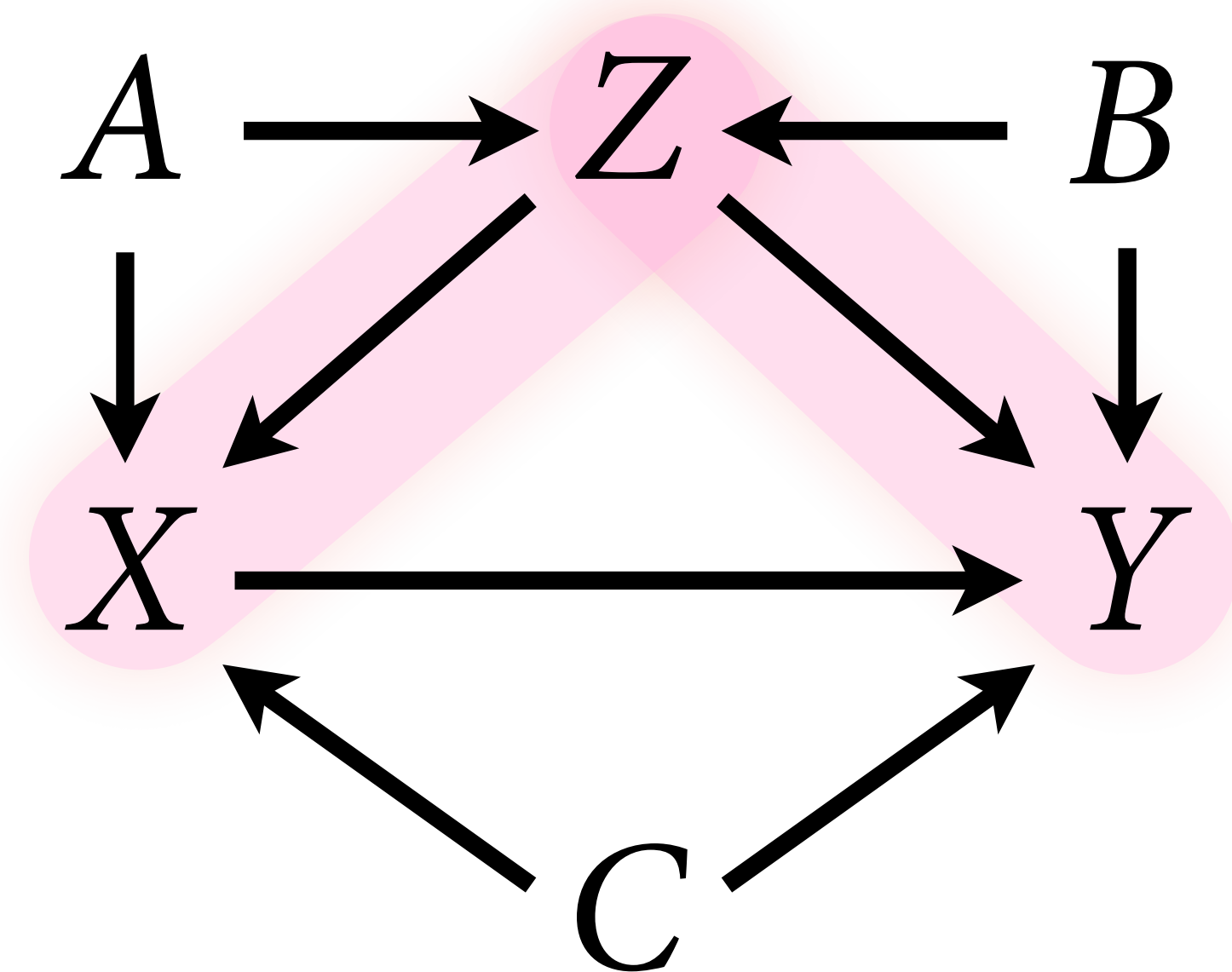
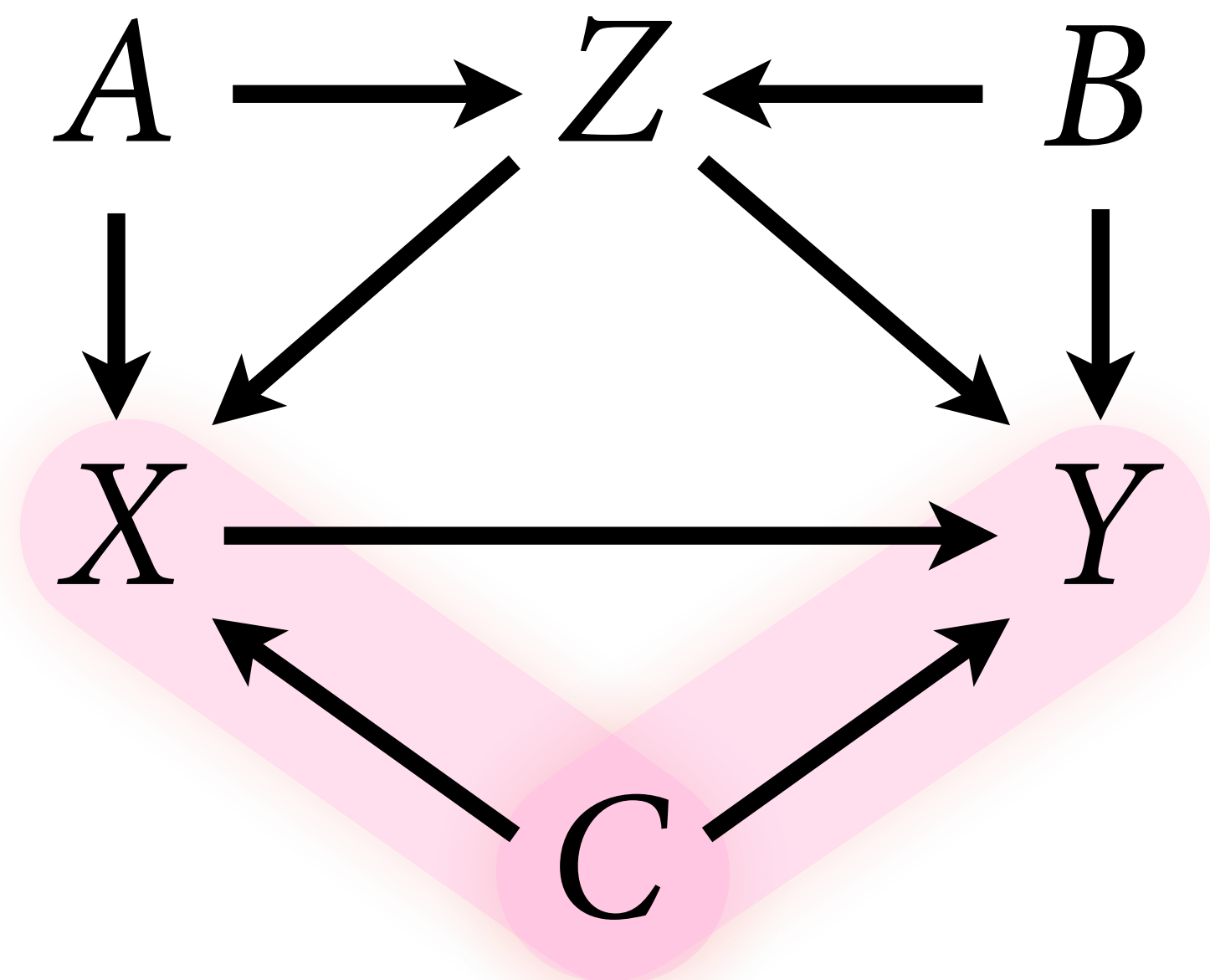
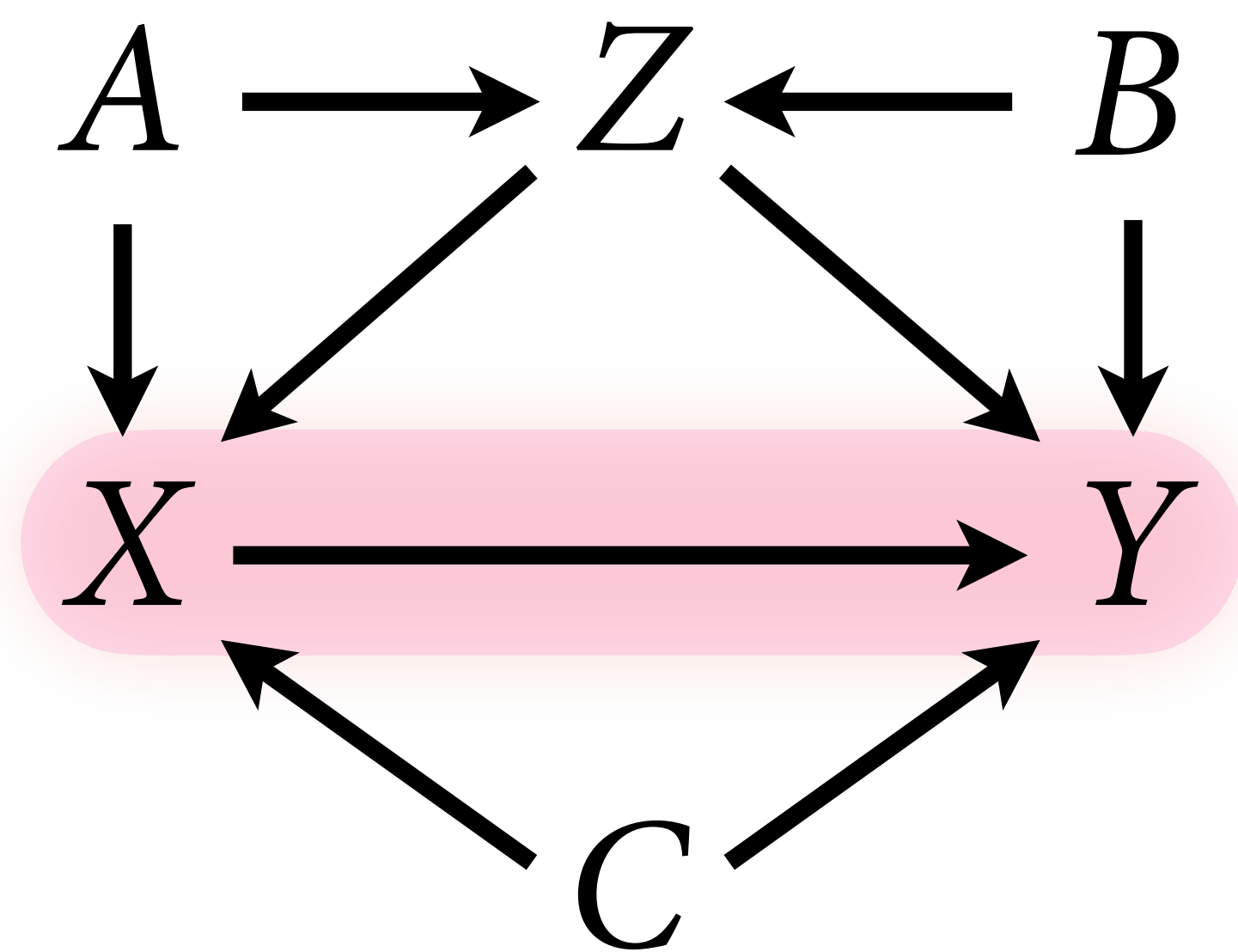
Coefficient on Z means
nothing. “Table 2 Fallacy”

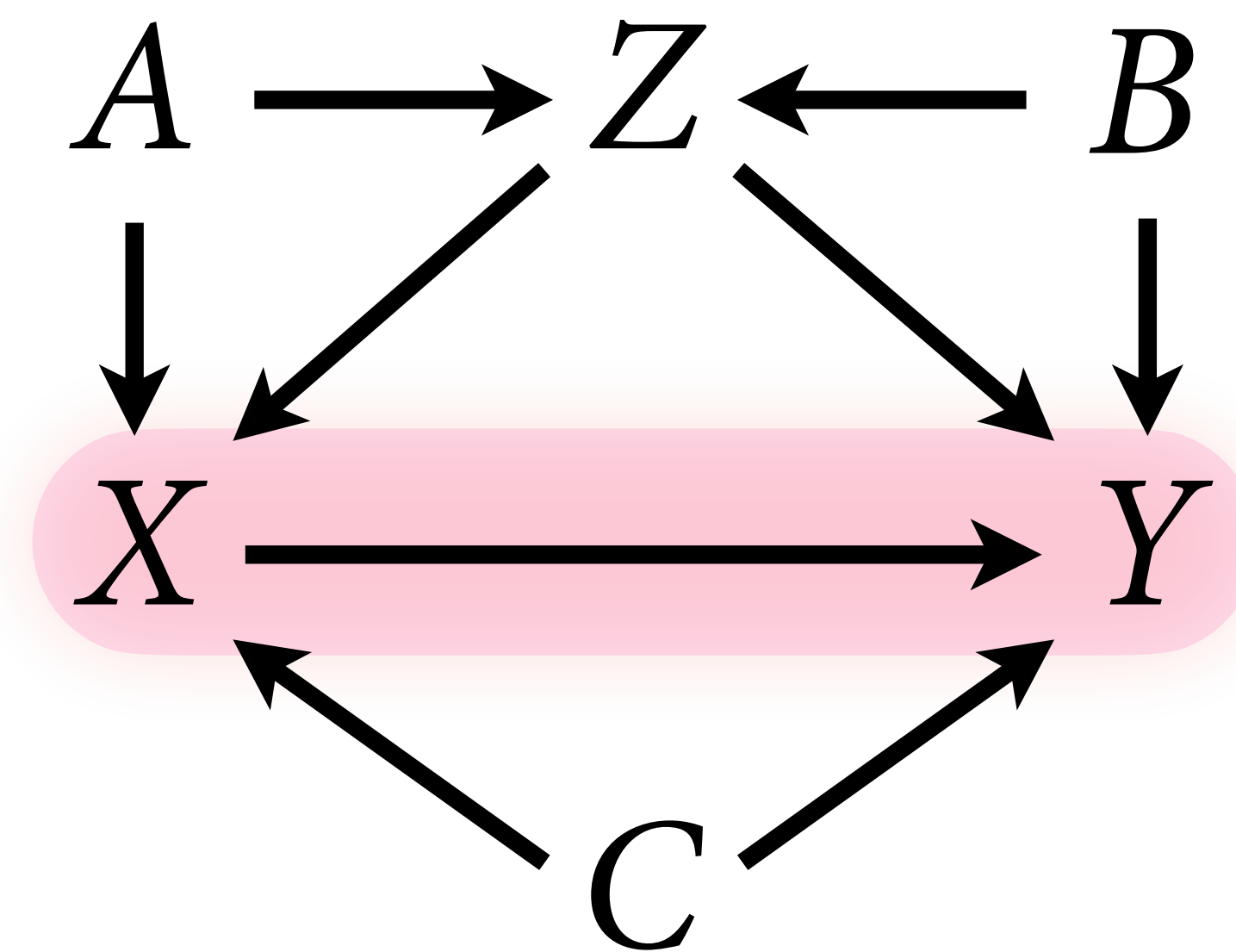
List all the paths connecting **X** and **Y**.
Which need to be closed to estimate
effect of **X** on **Y**?



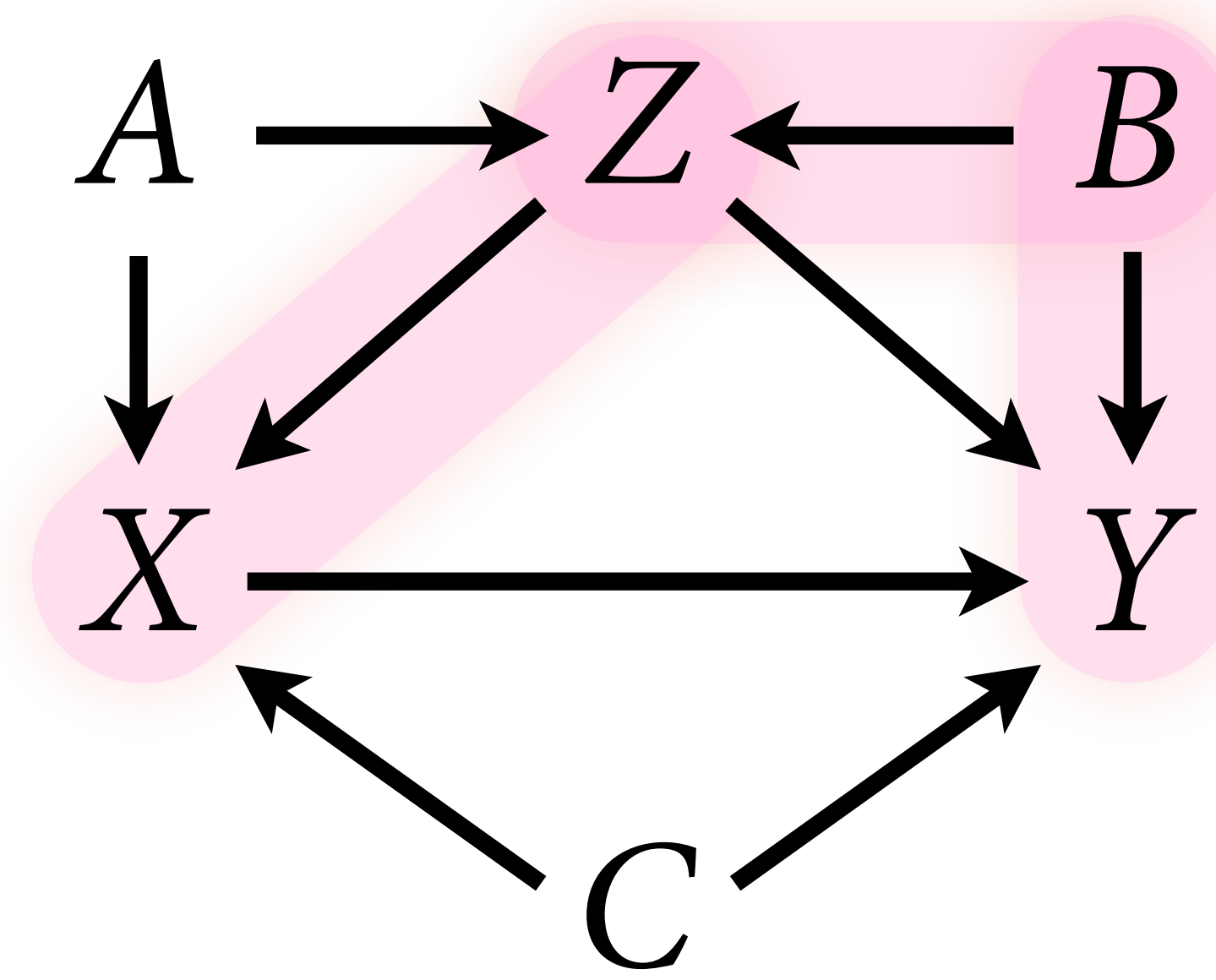
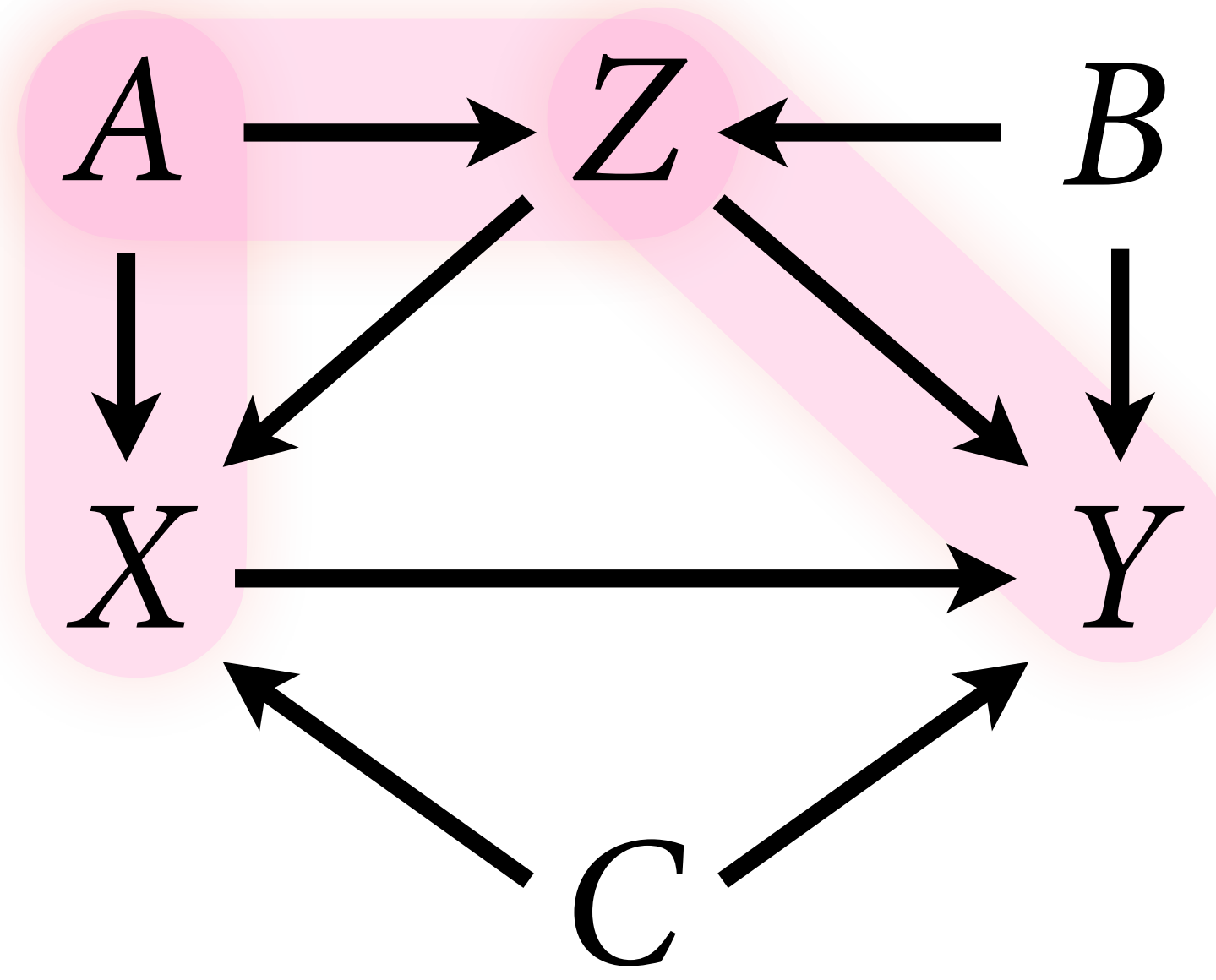
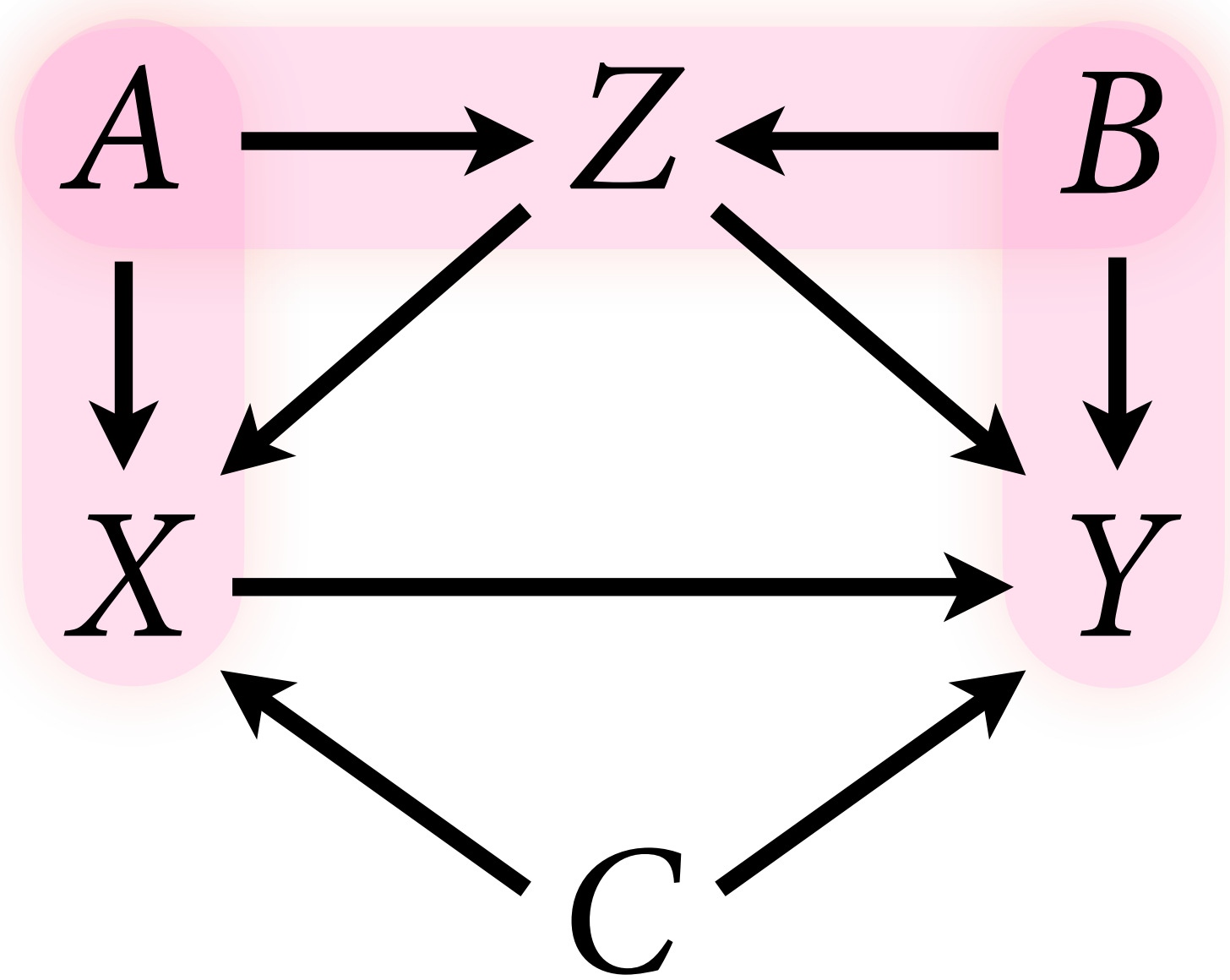
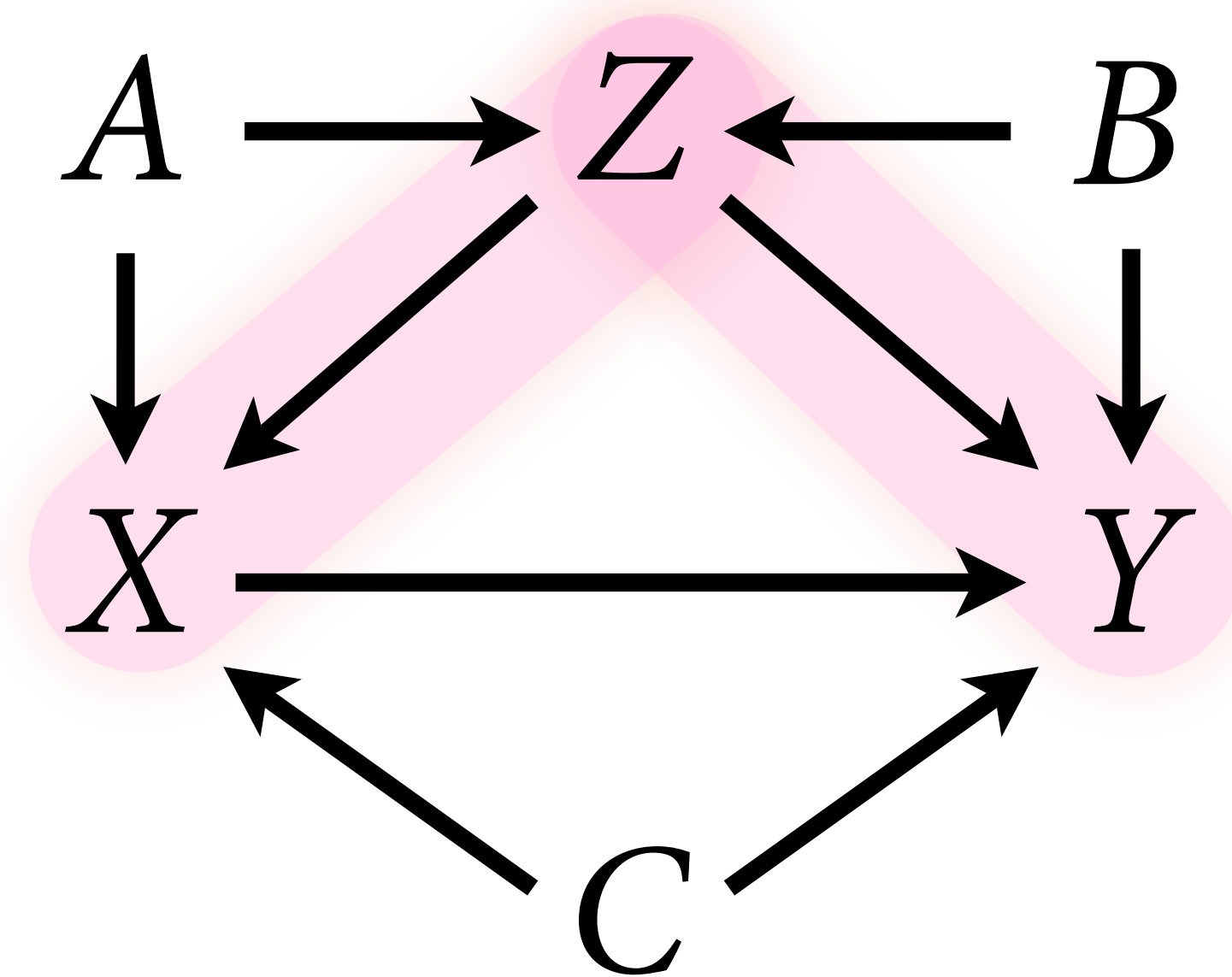
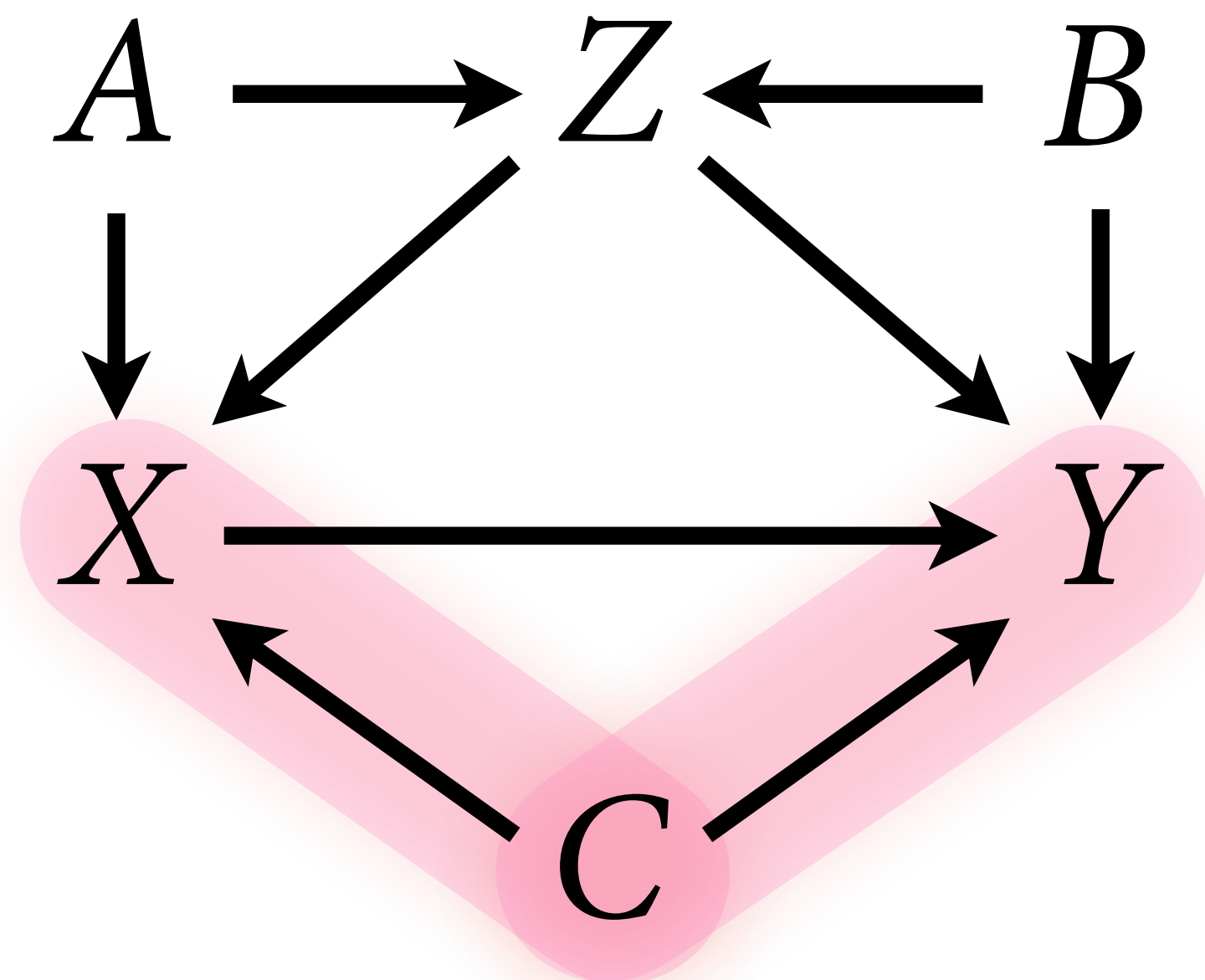
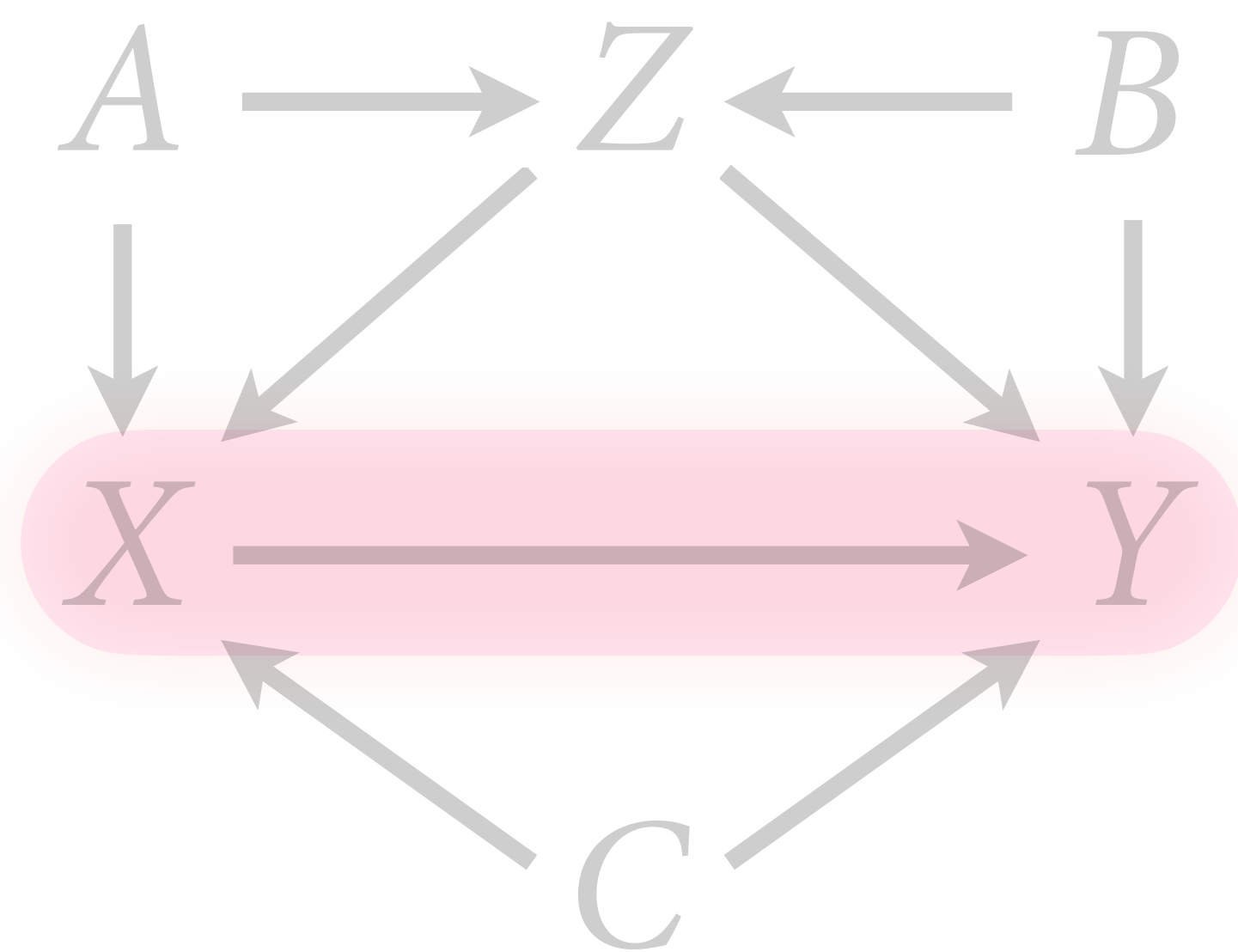
$P(Y|\text{do}(X))$

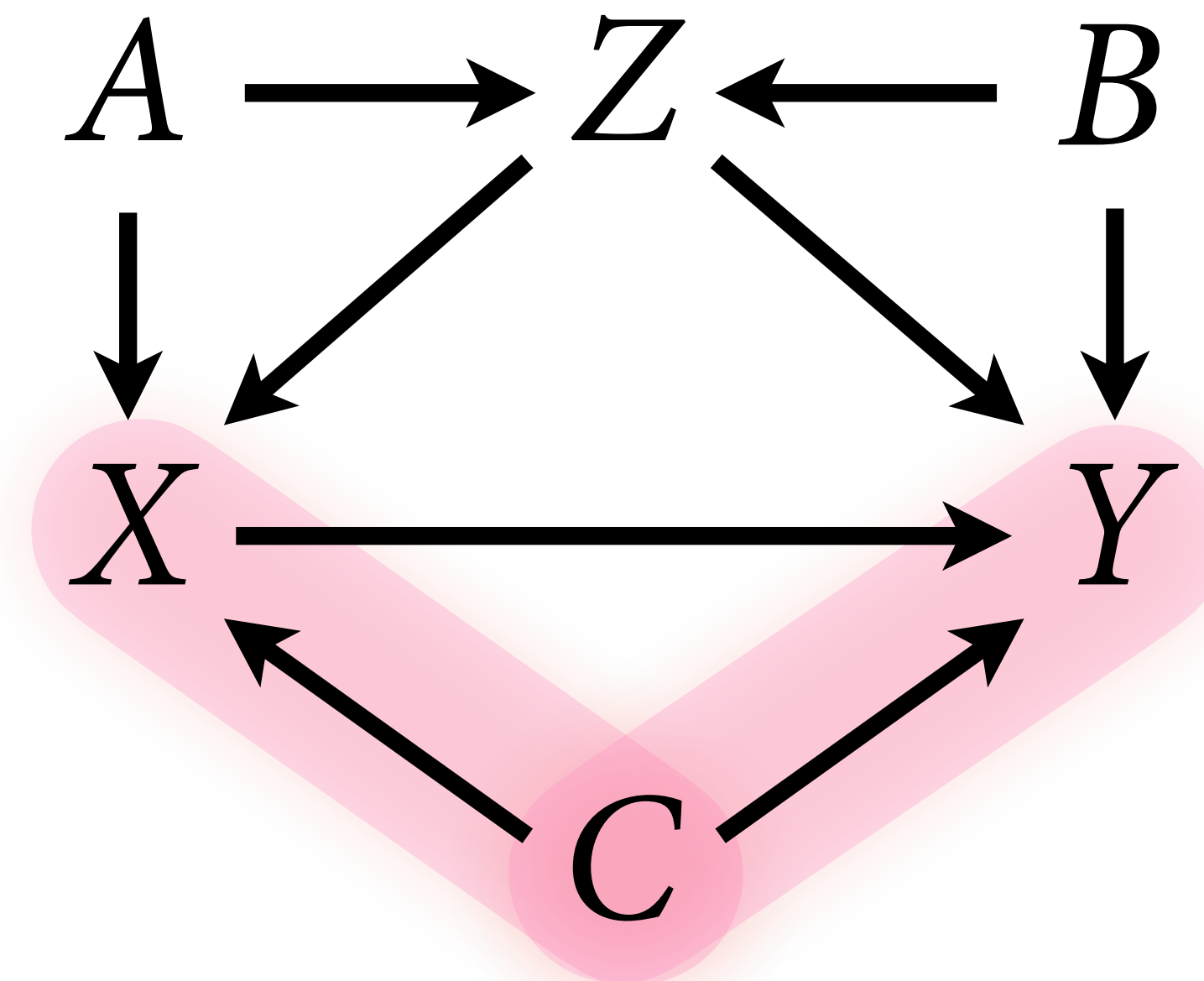




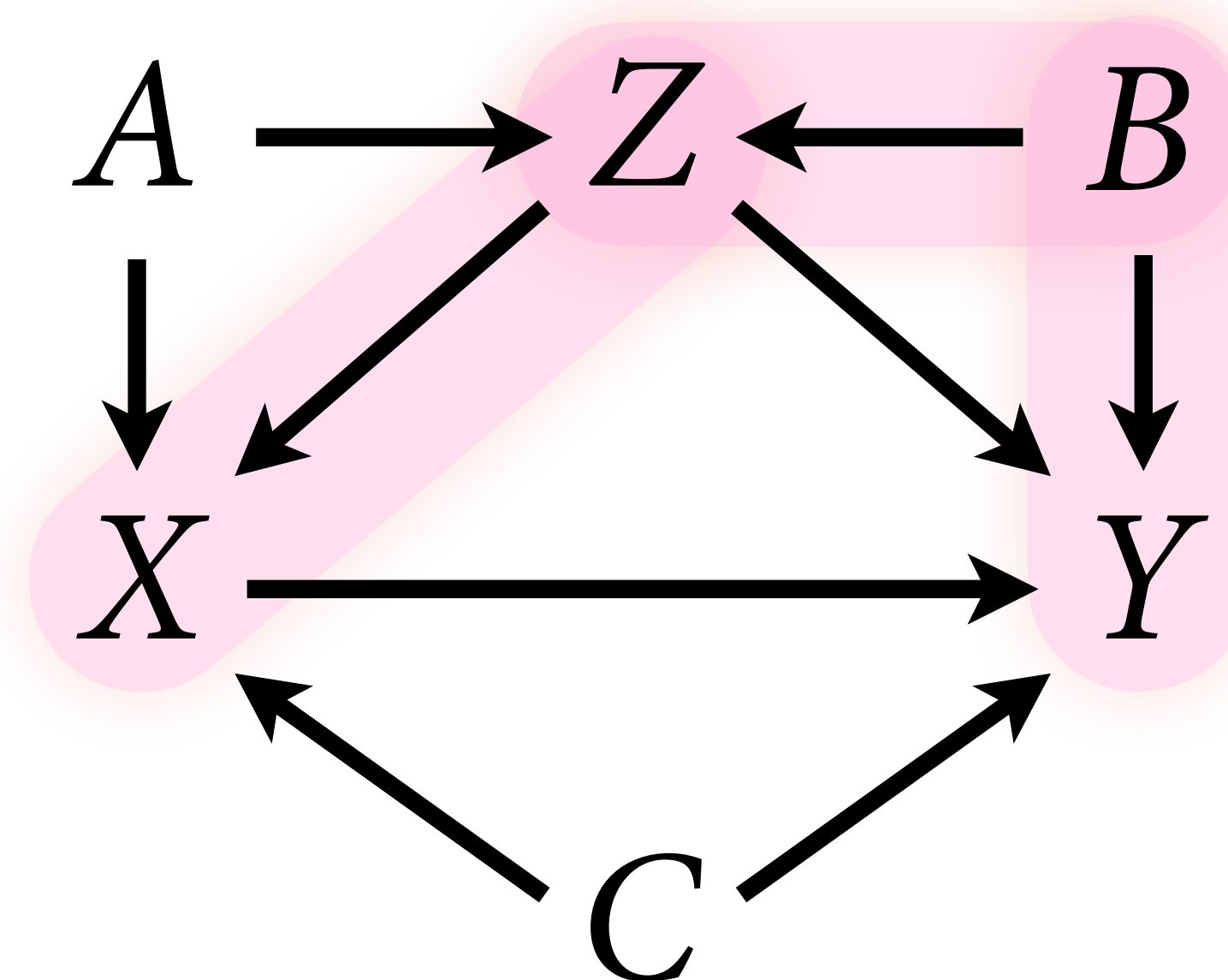
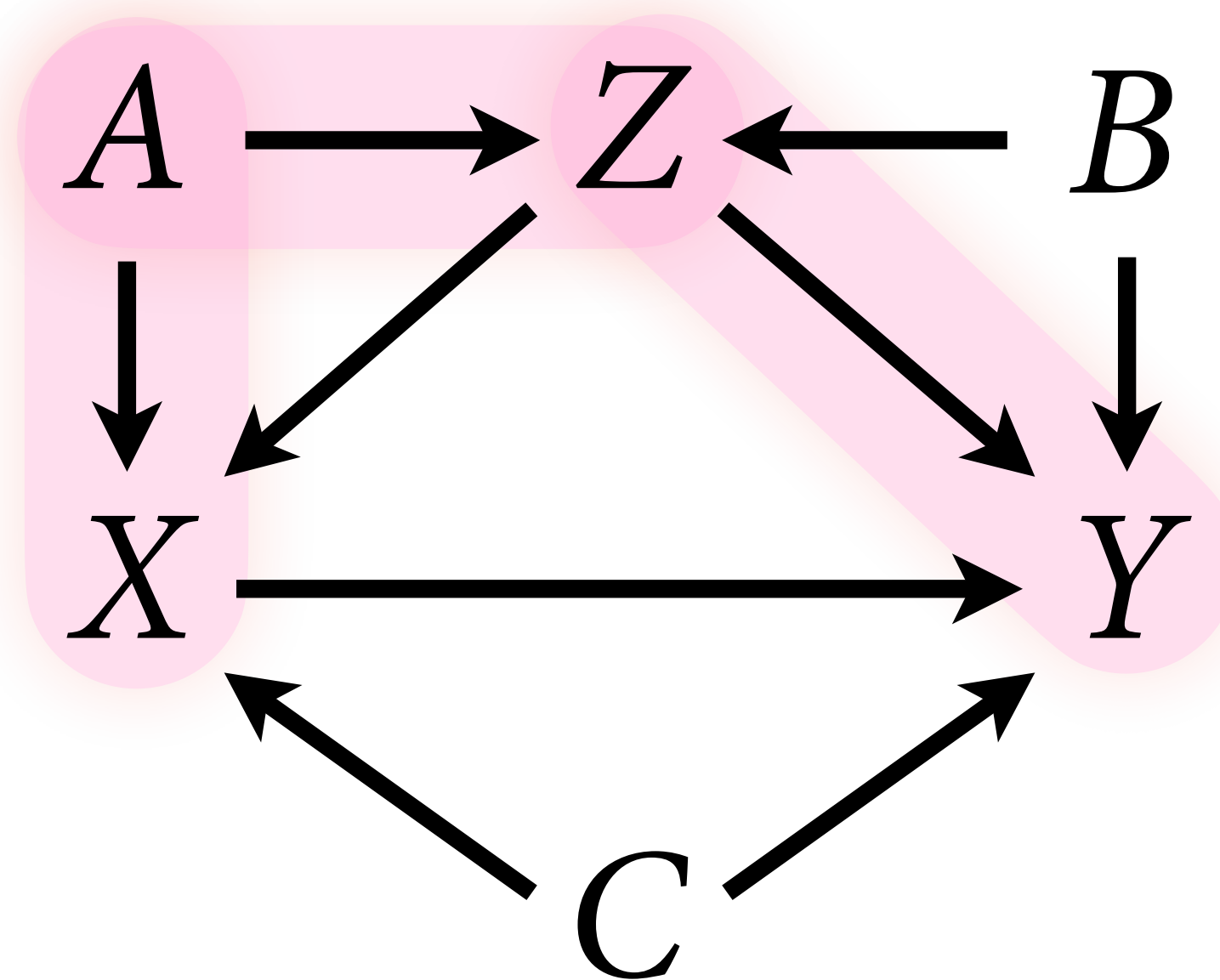
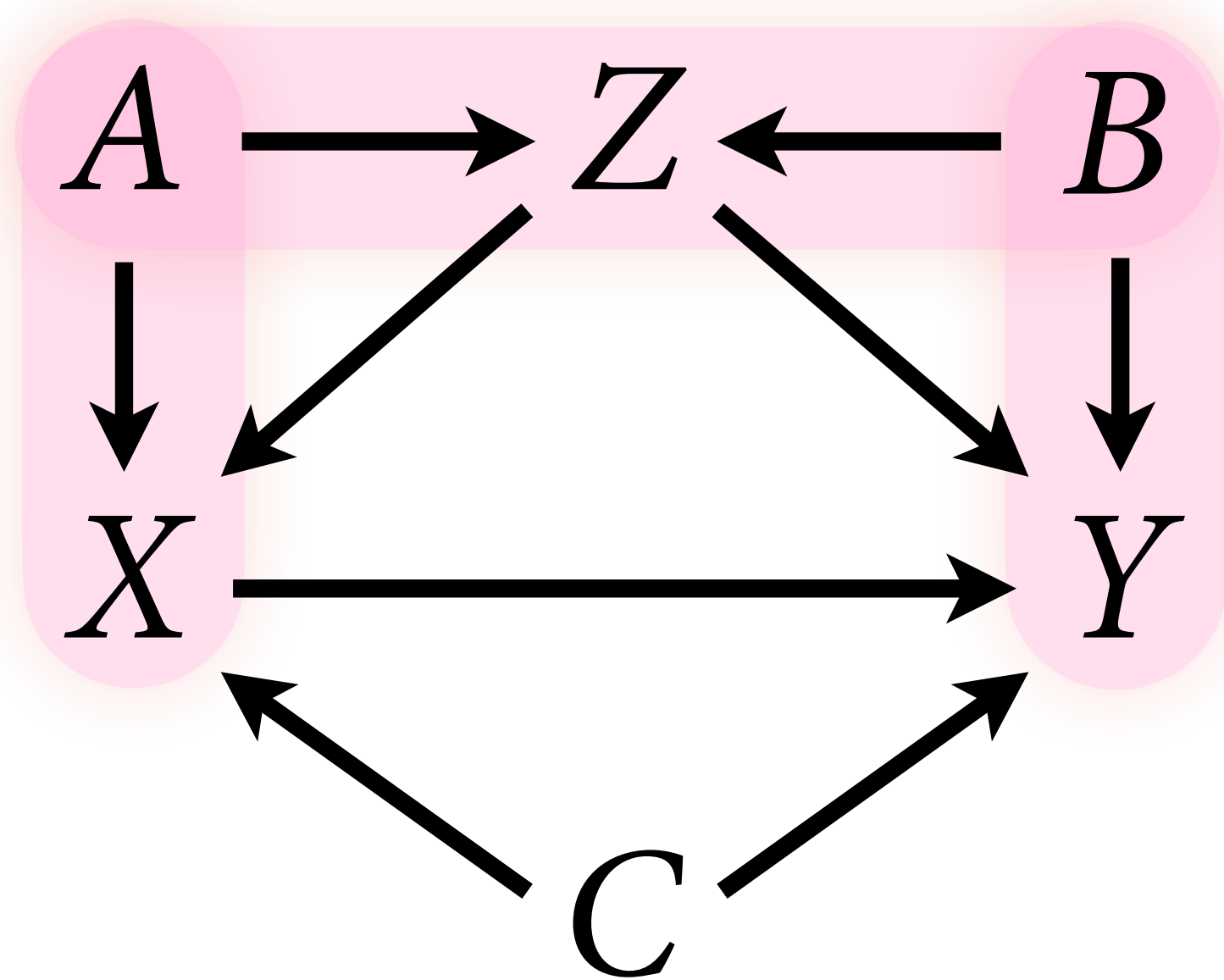
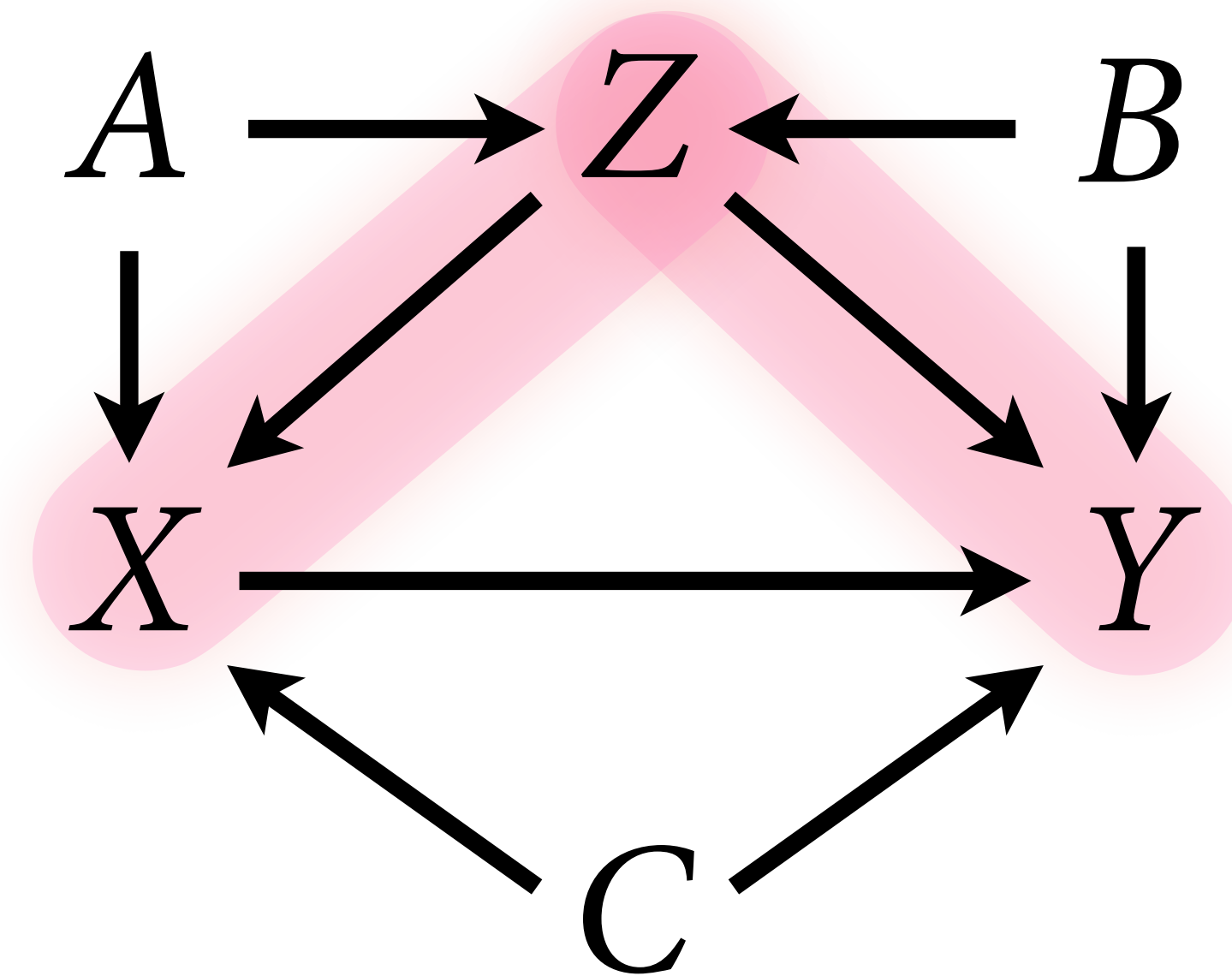
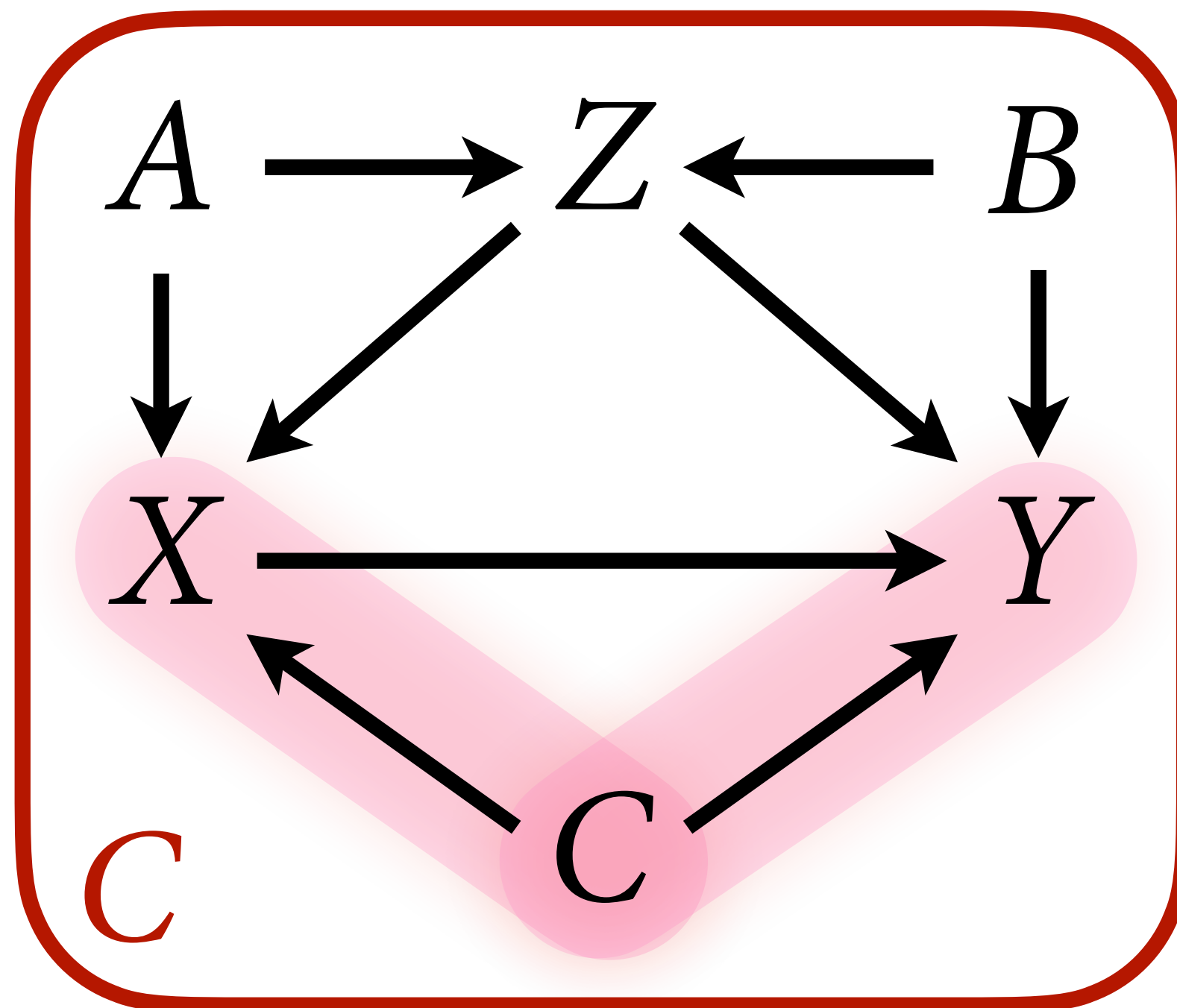
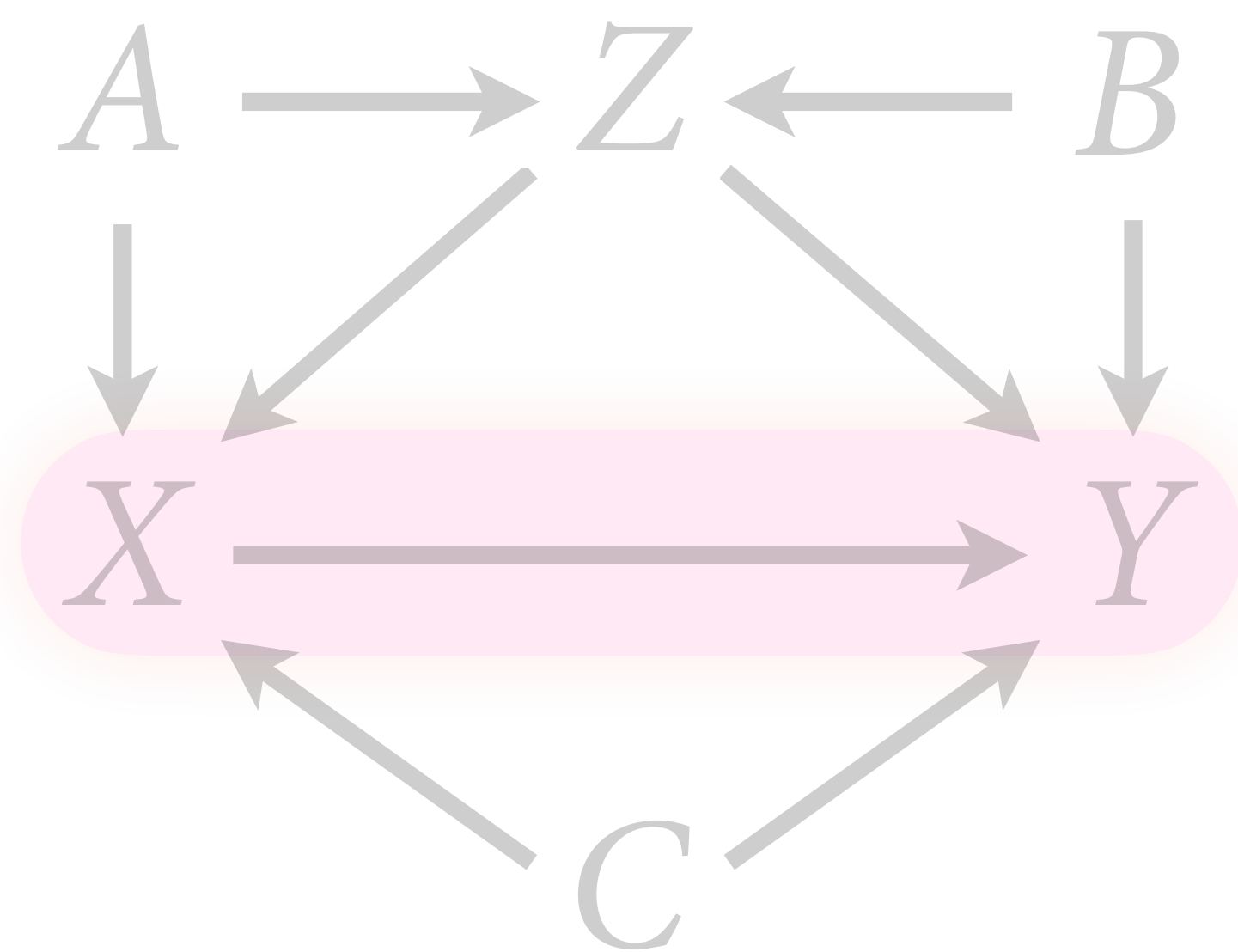


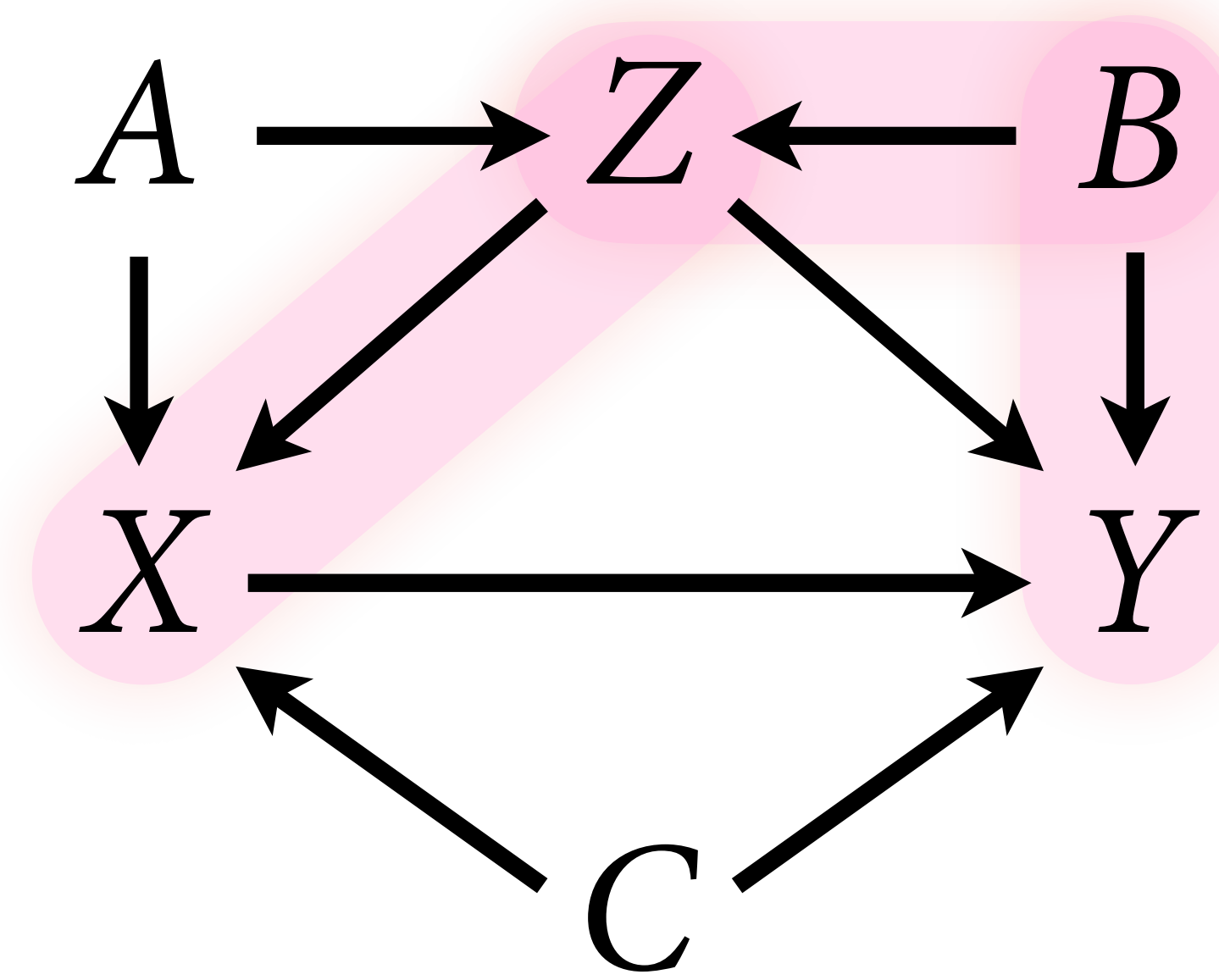
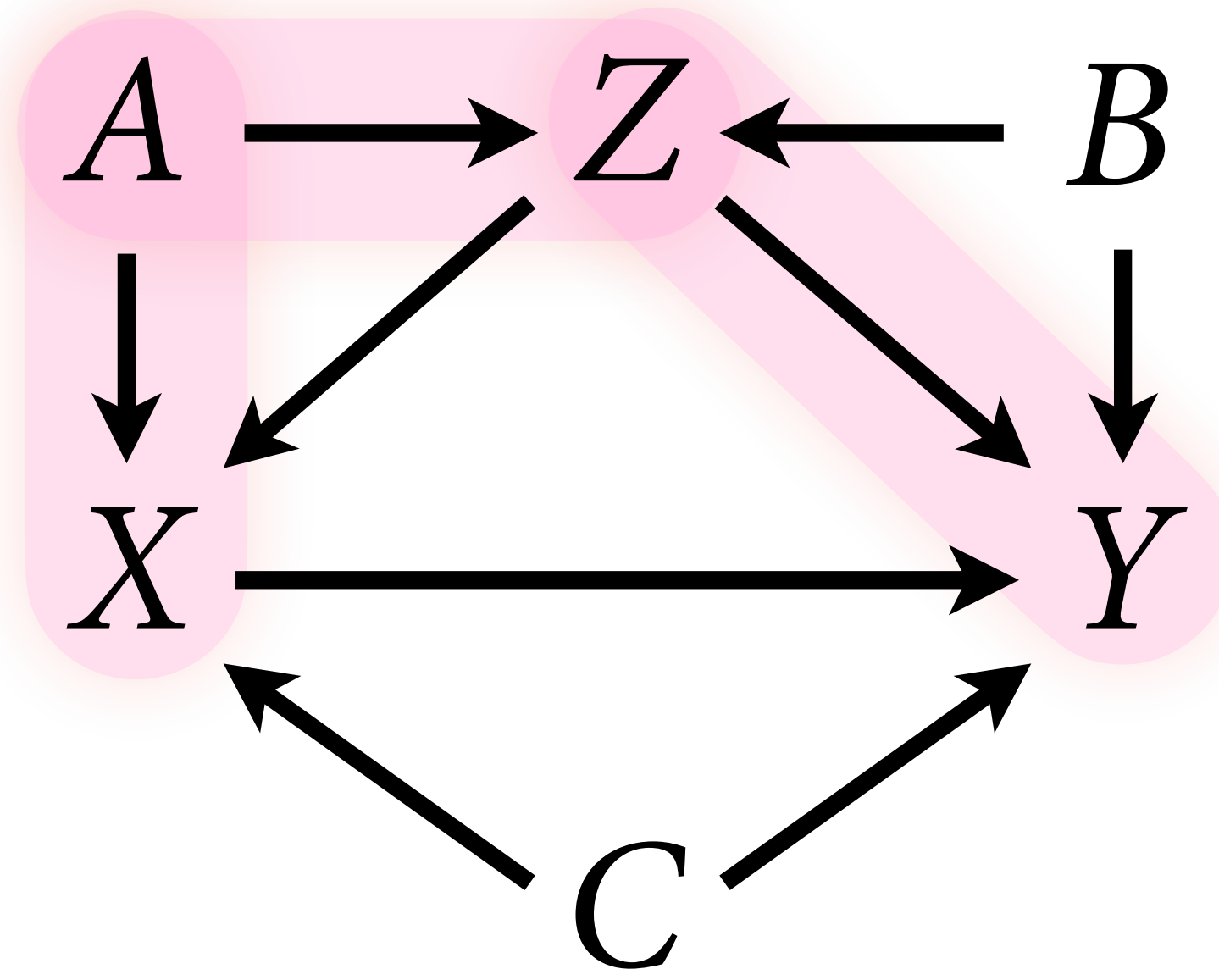
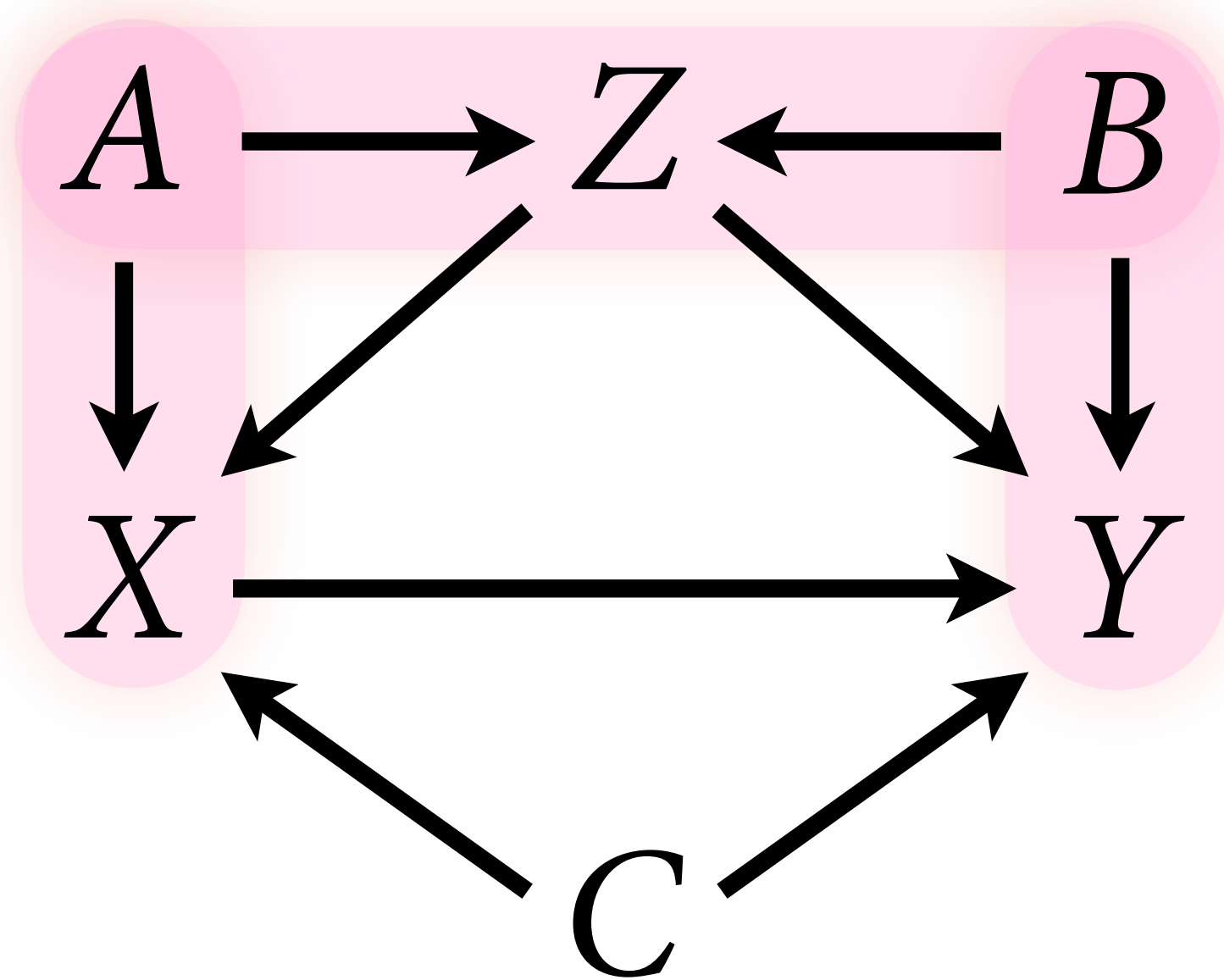
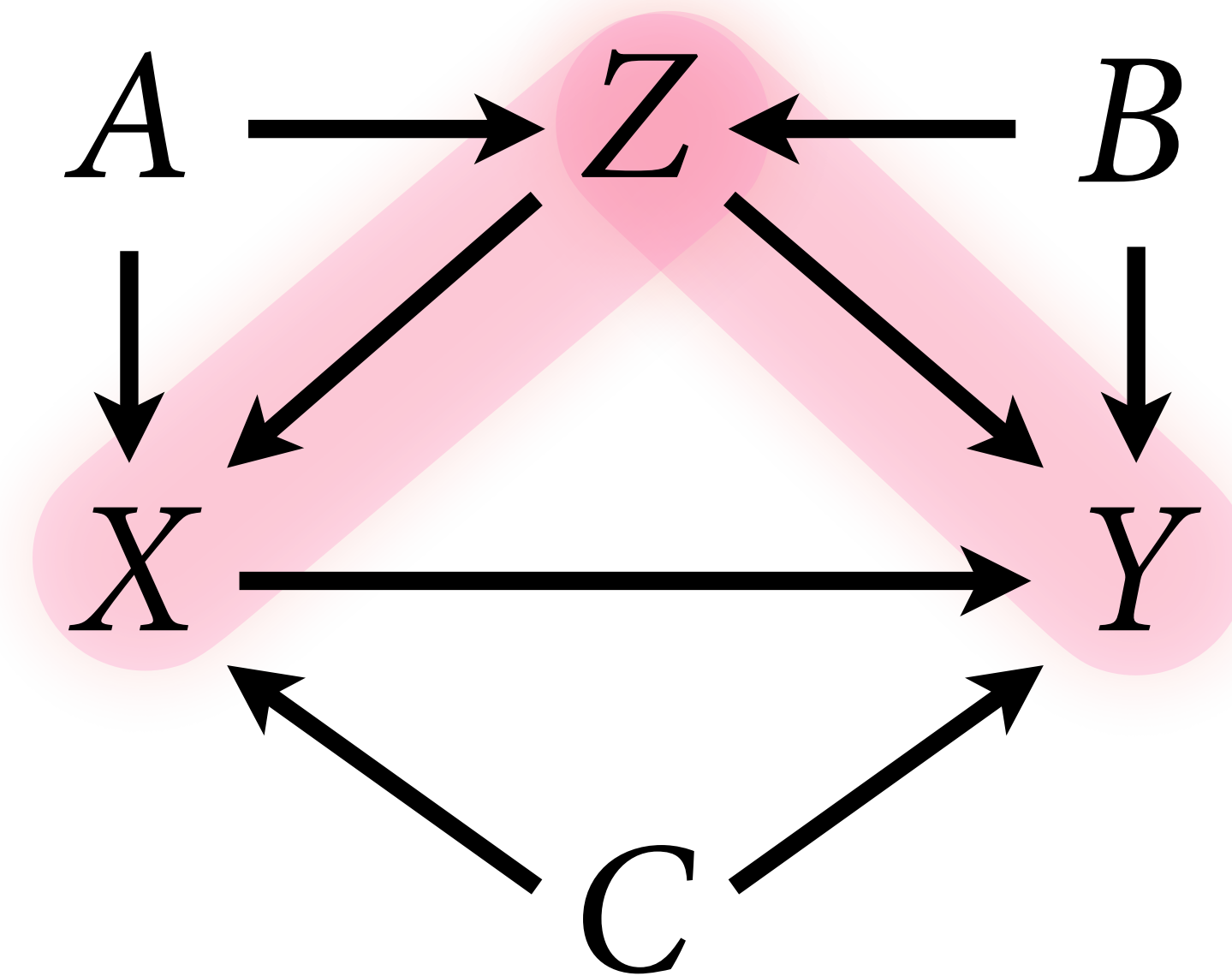
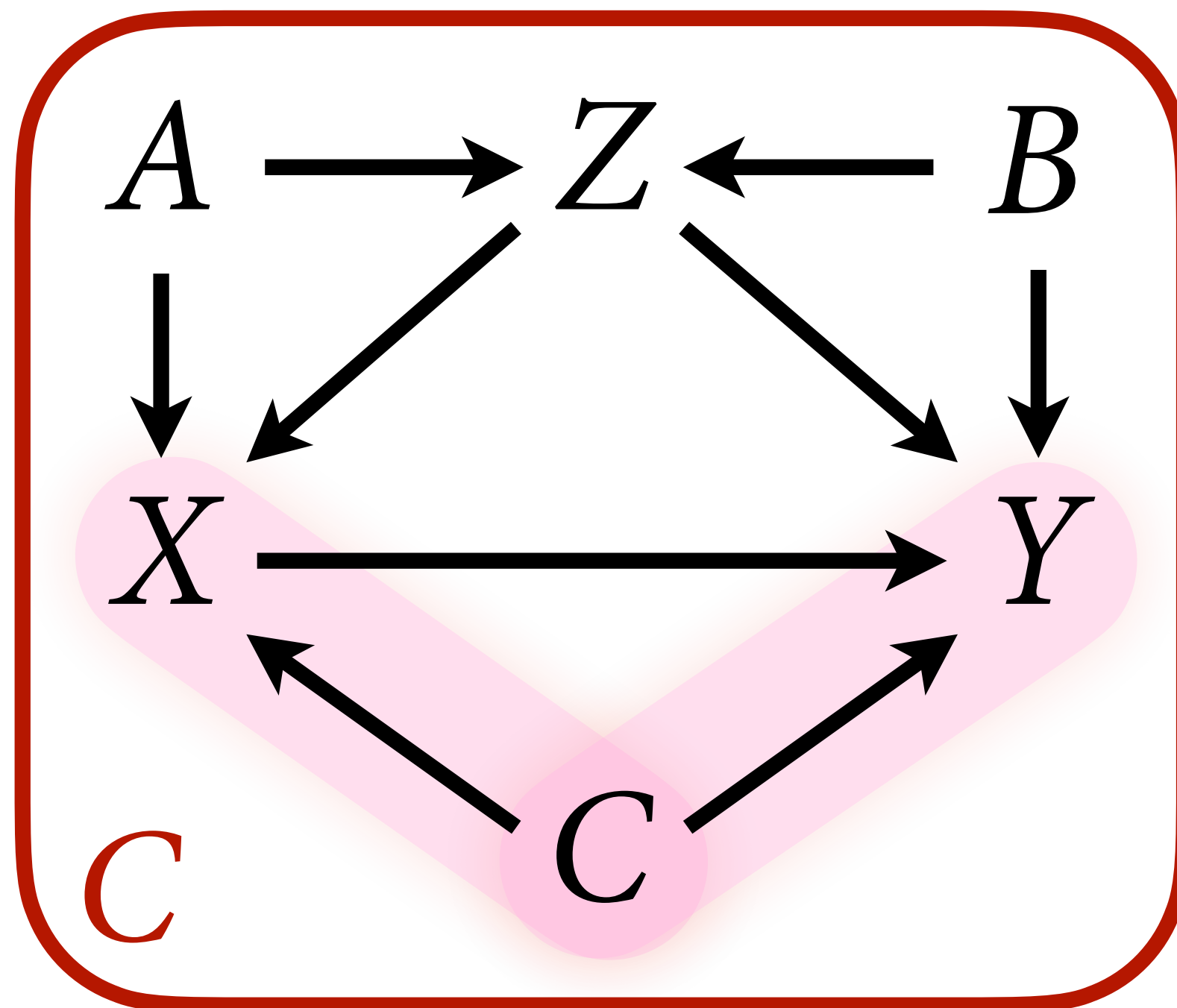
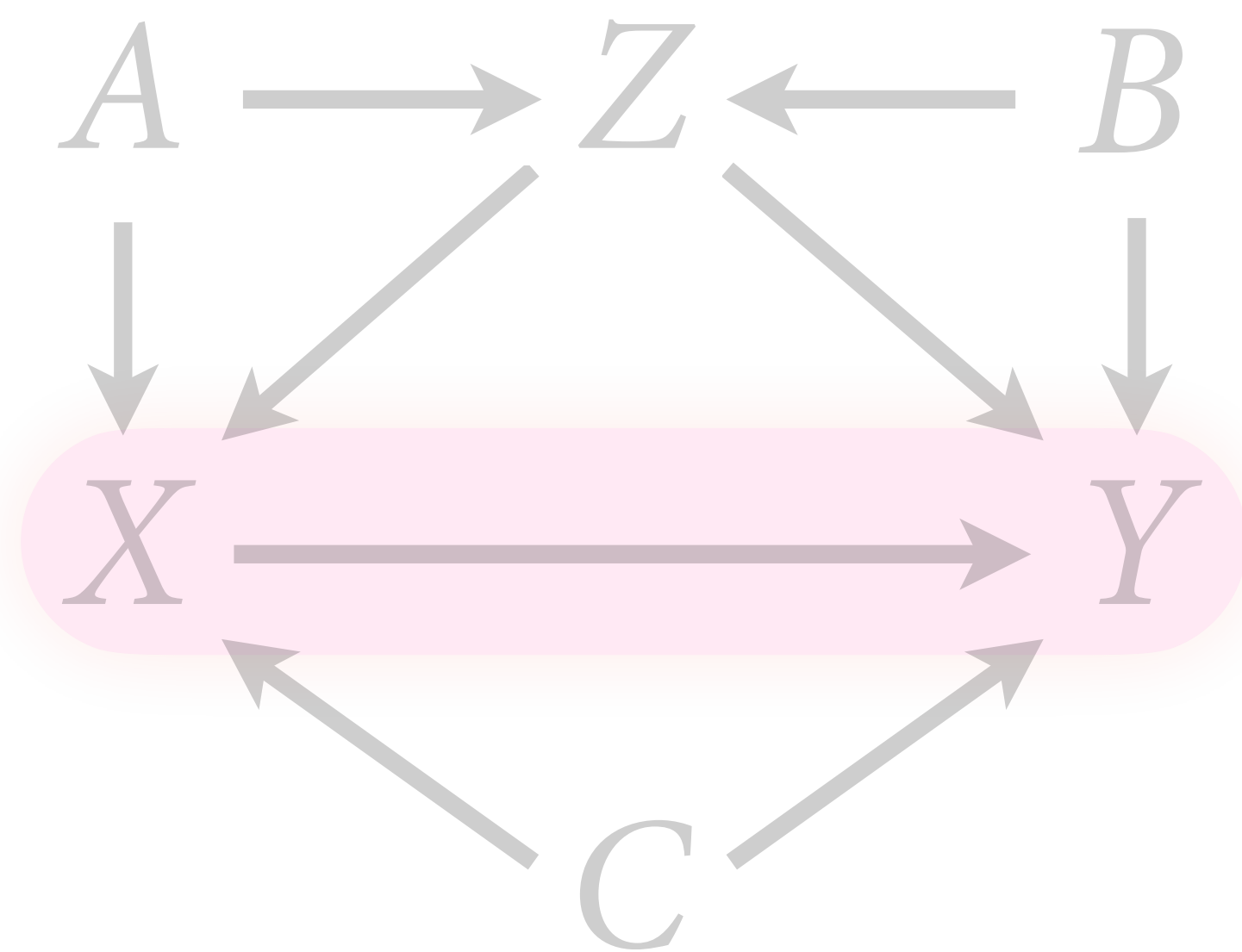
Causal path, open

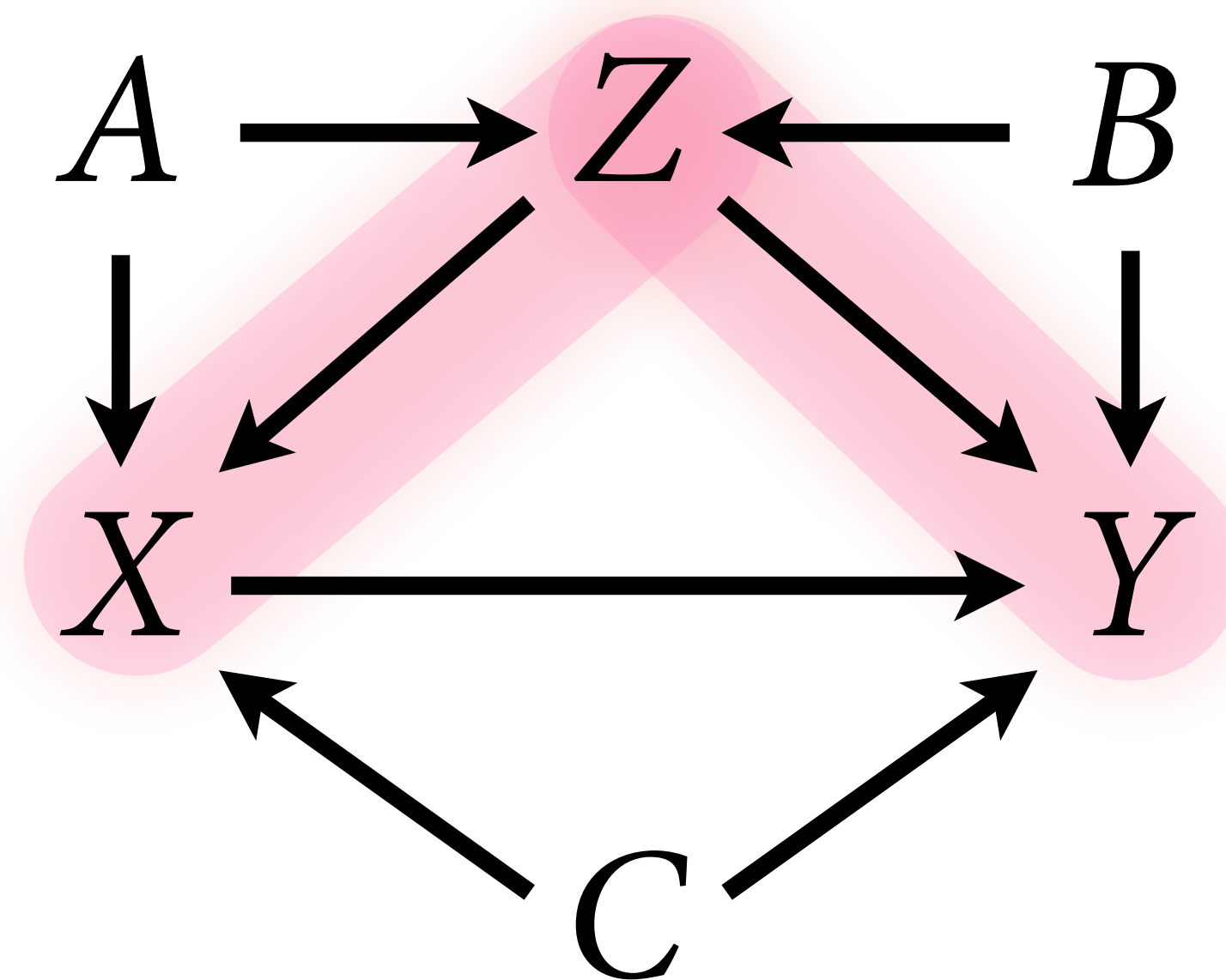




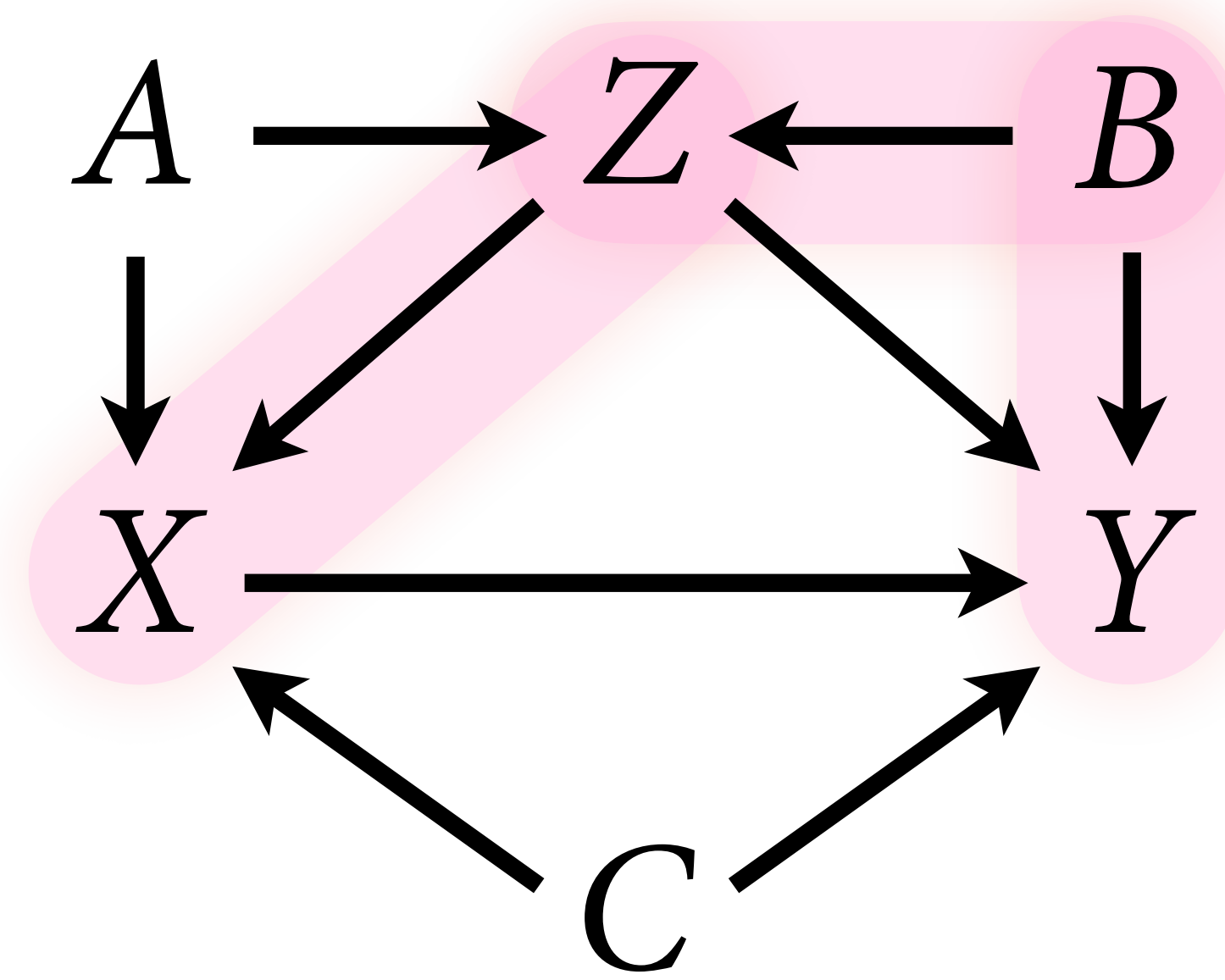
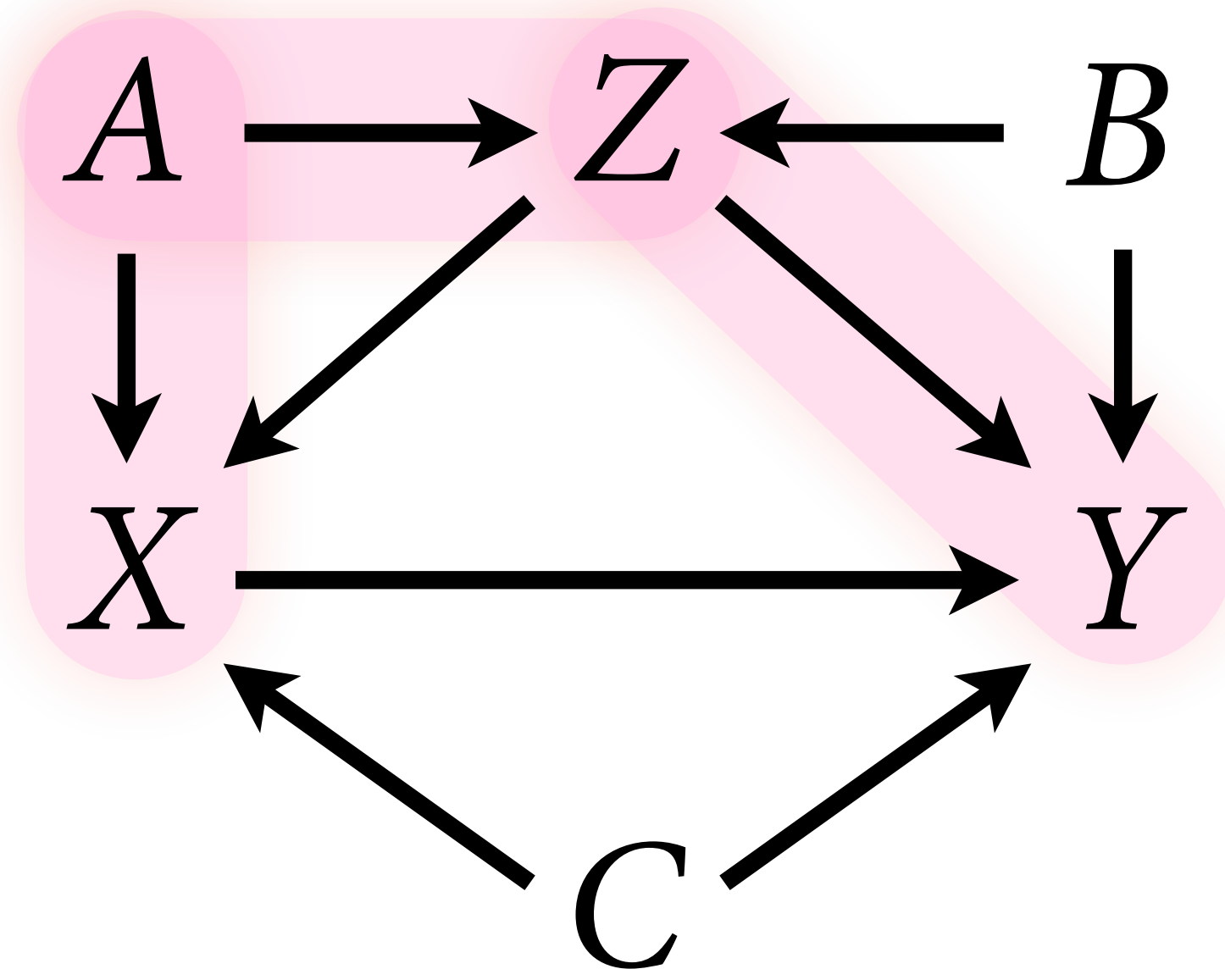
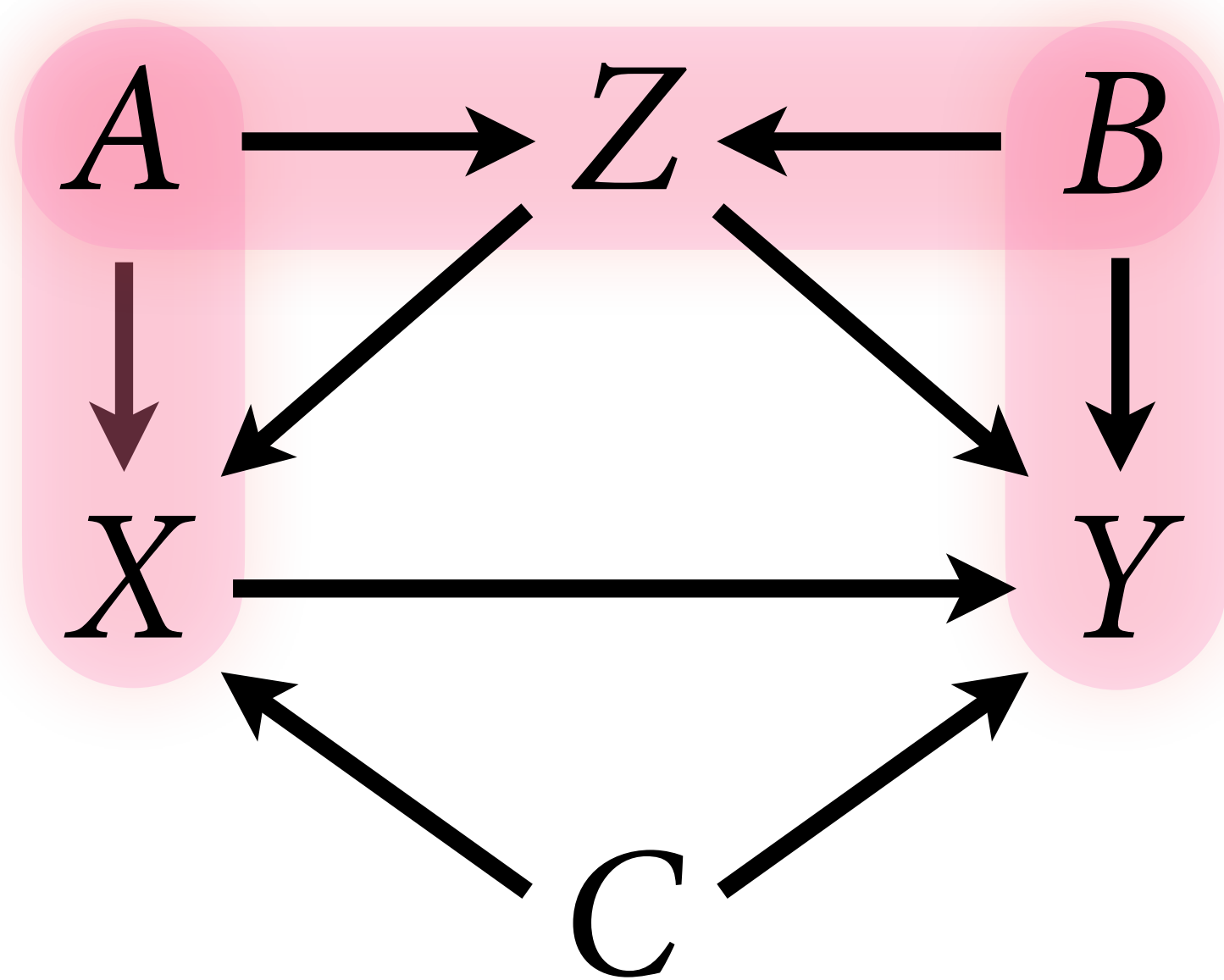
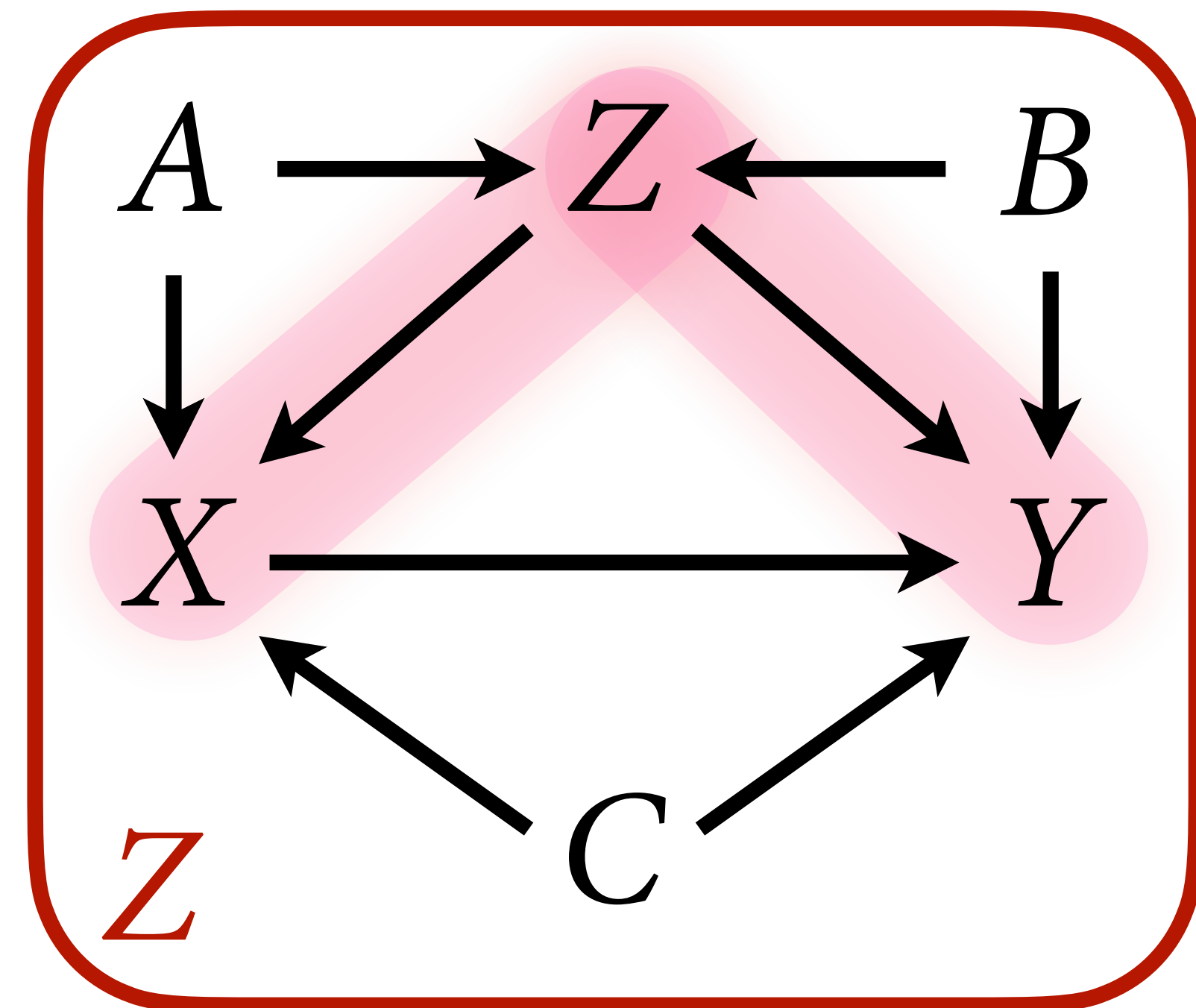
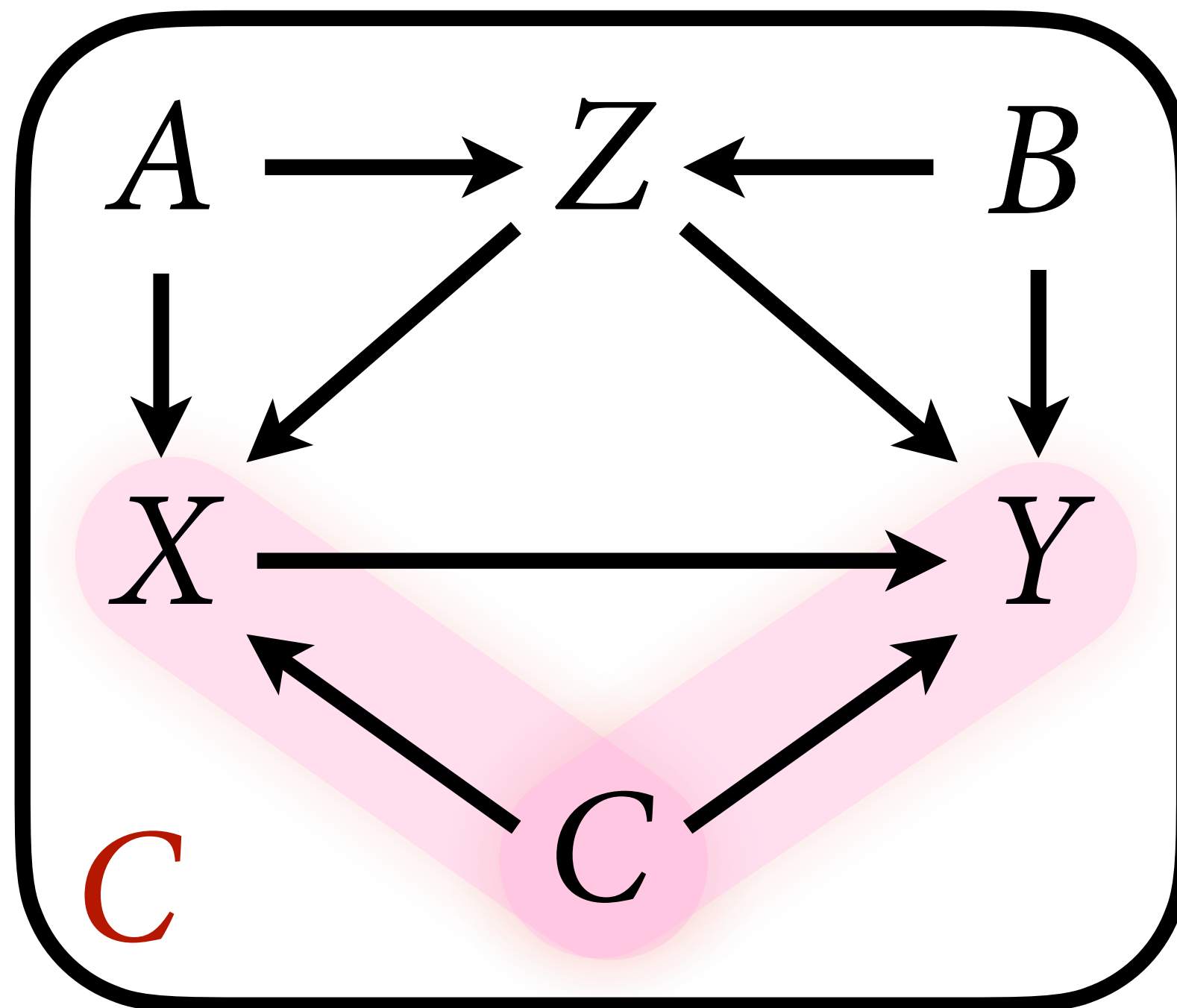
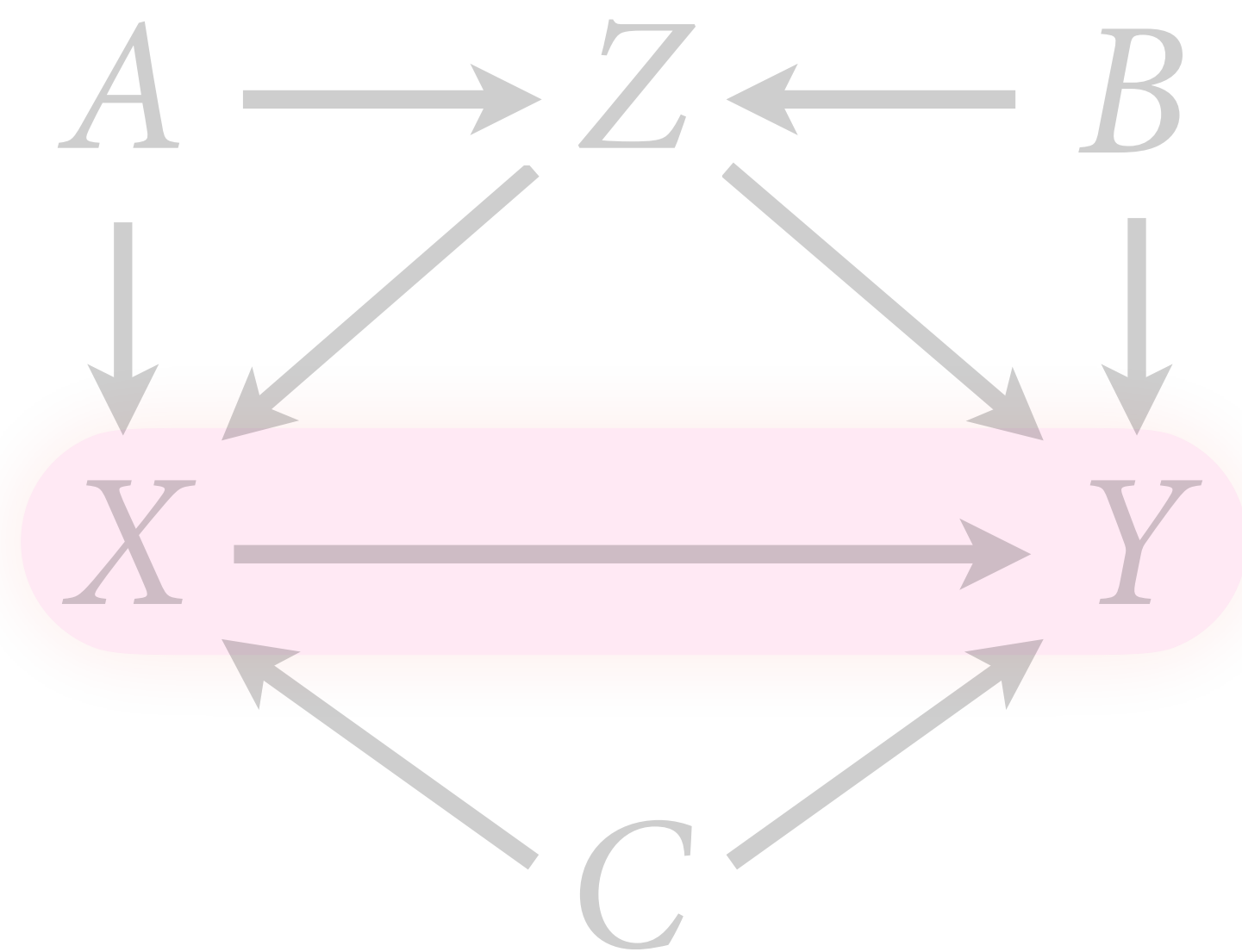
Backdoor path, open
Close with C

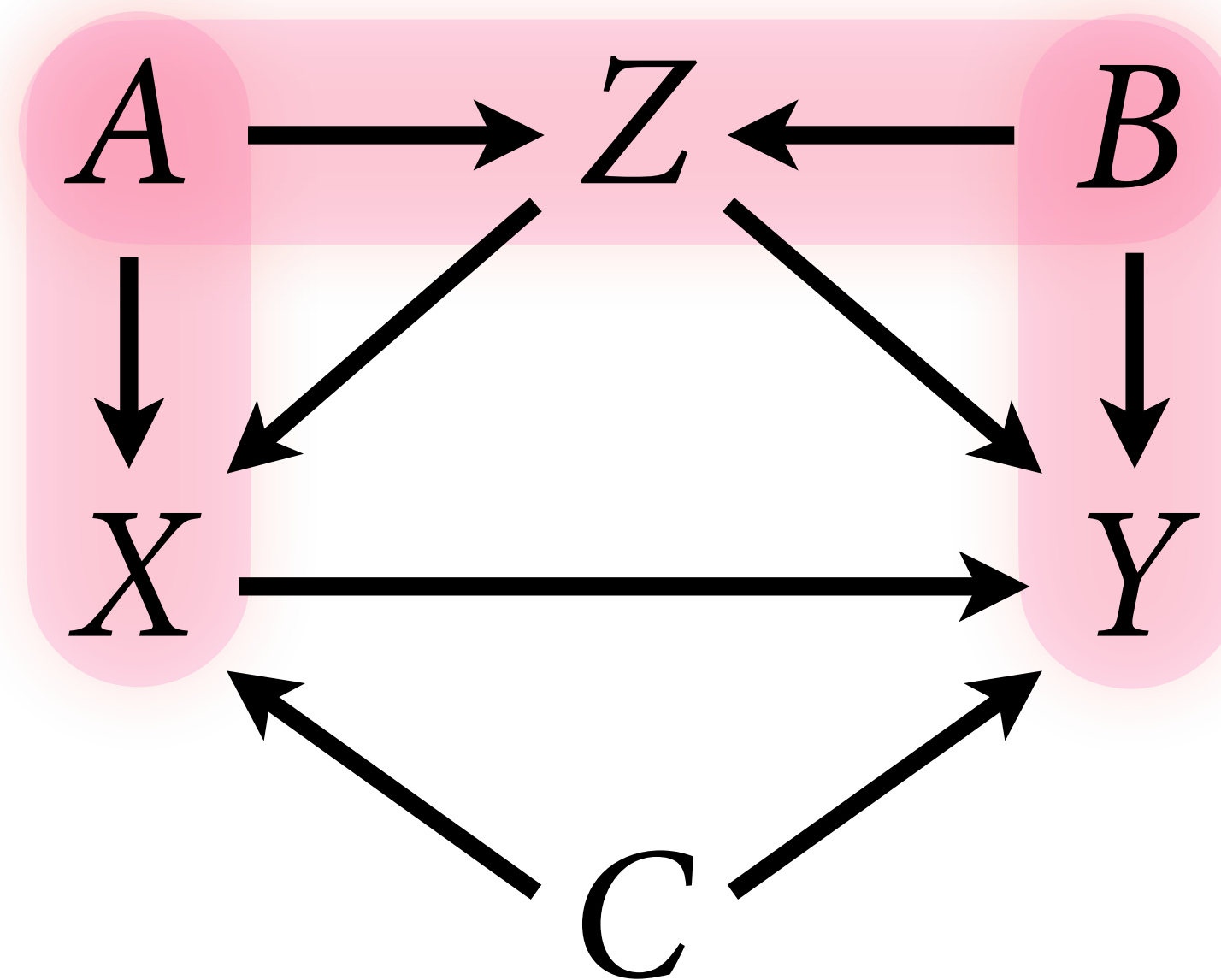




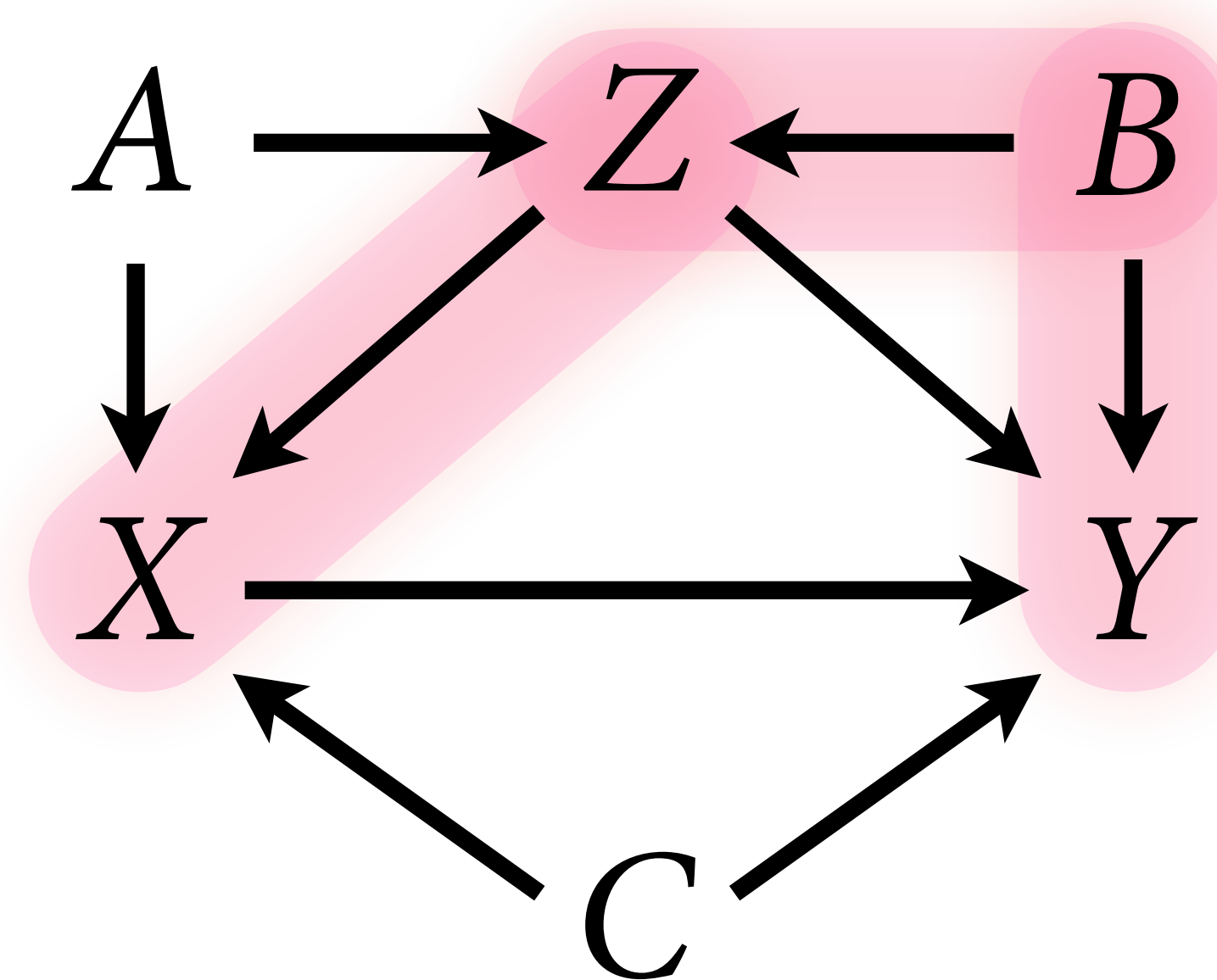
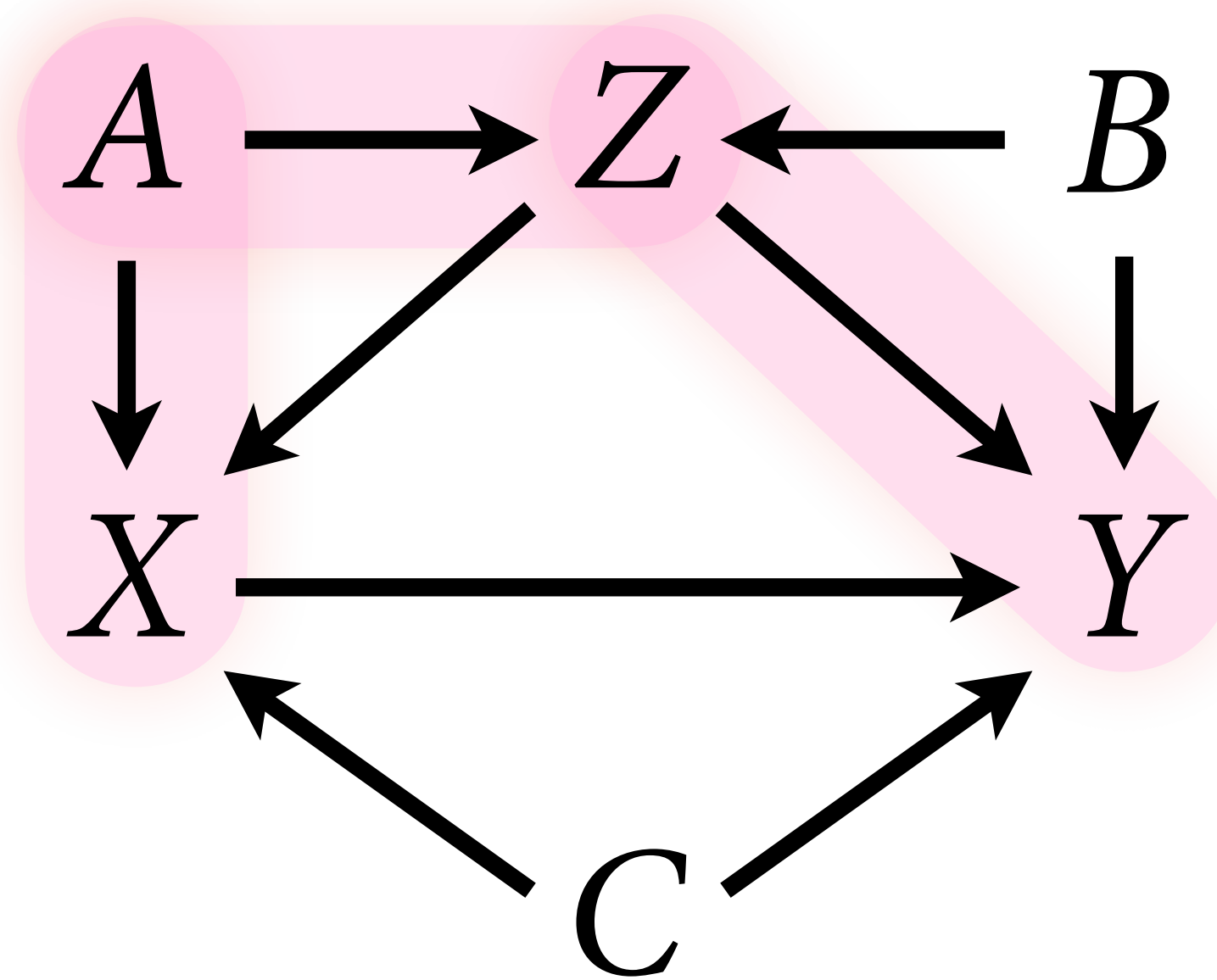
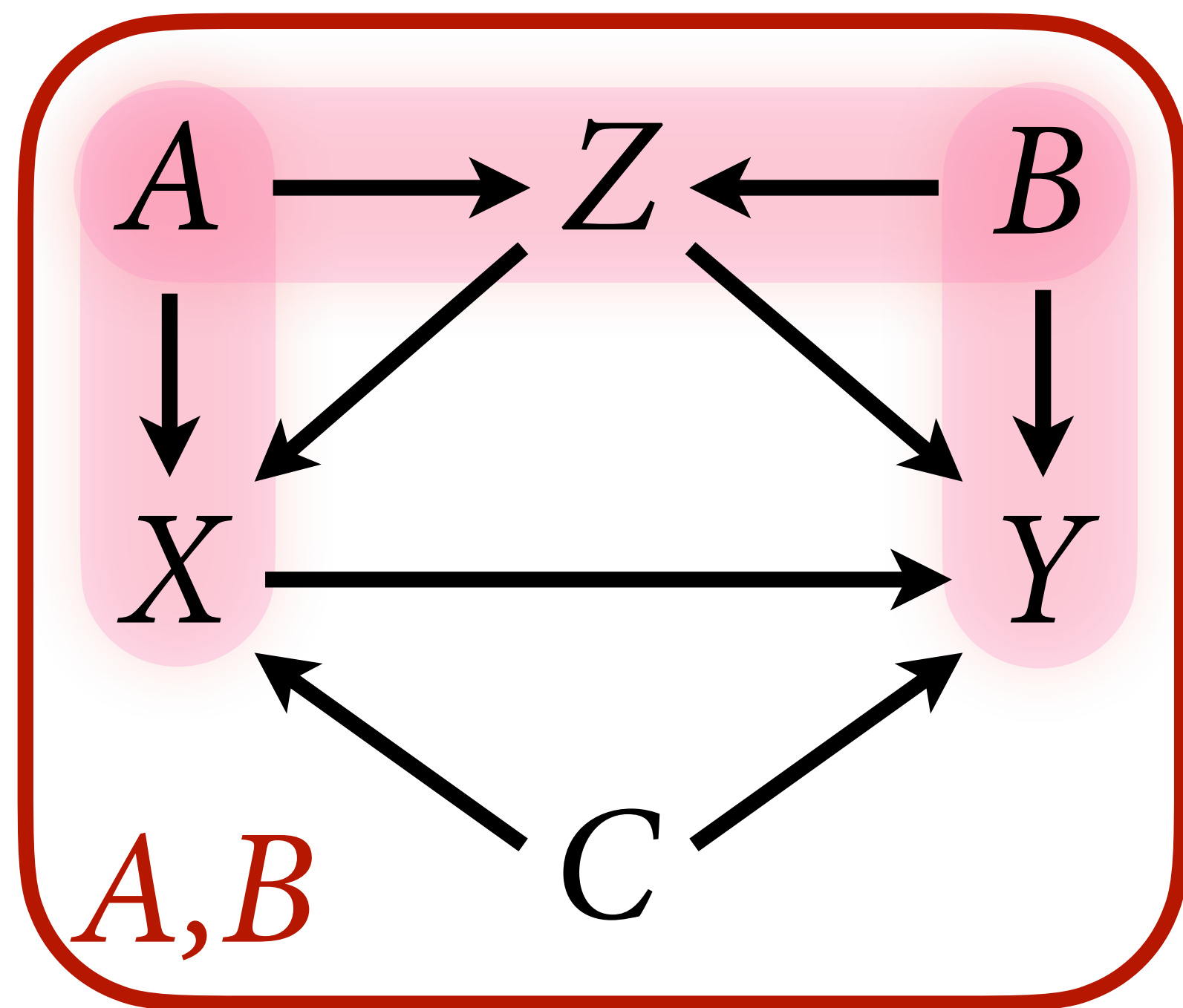
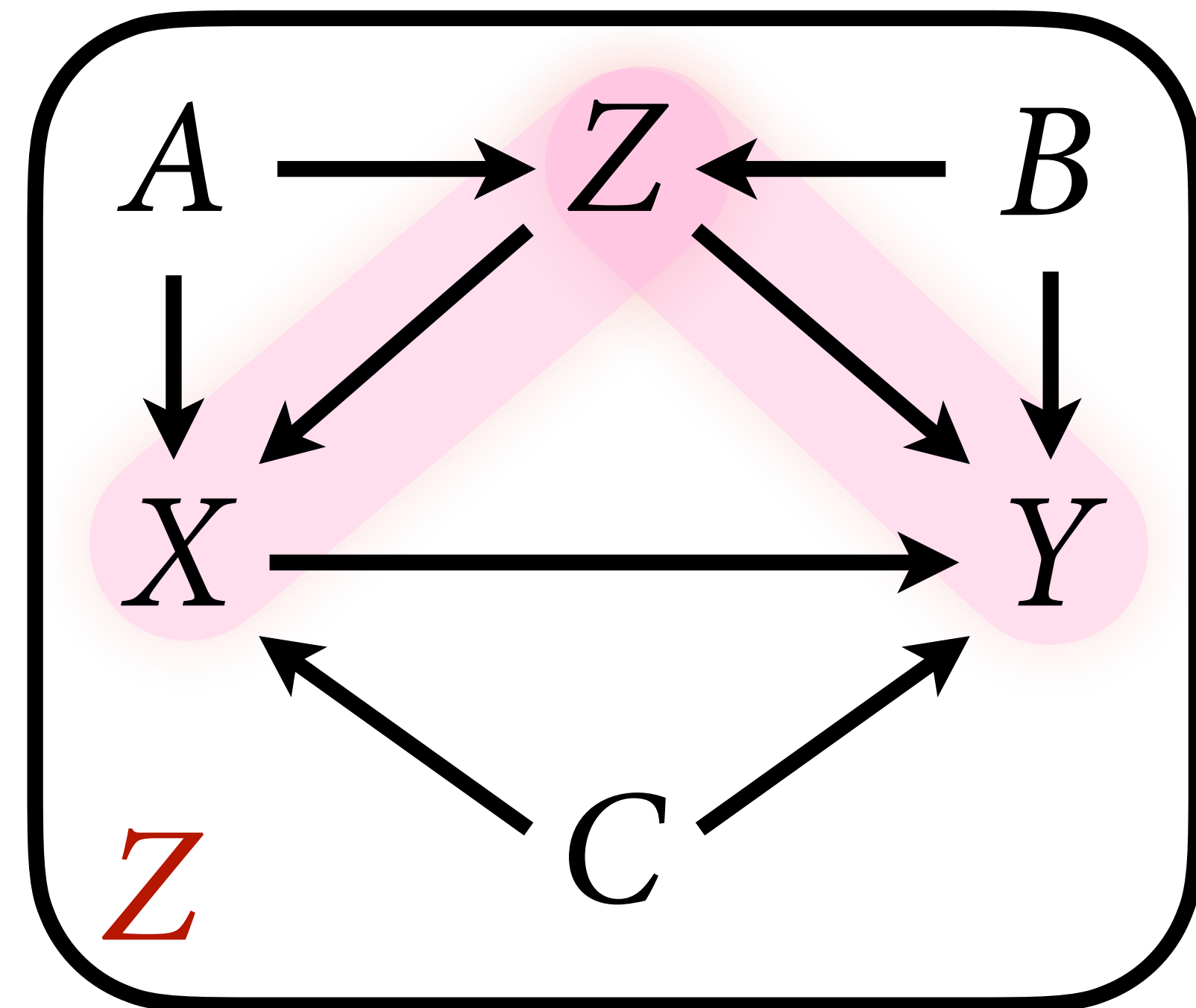
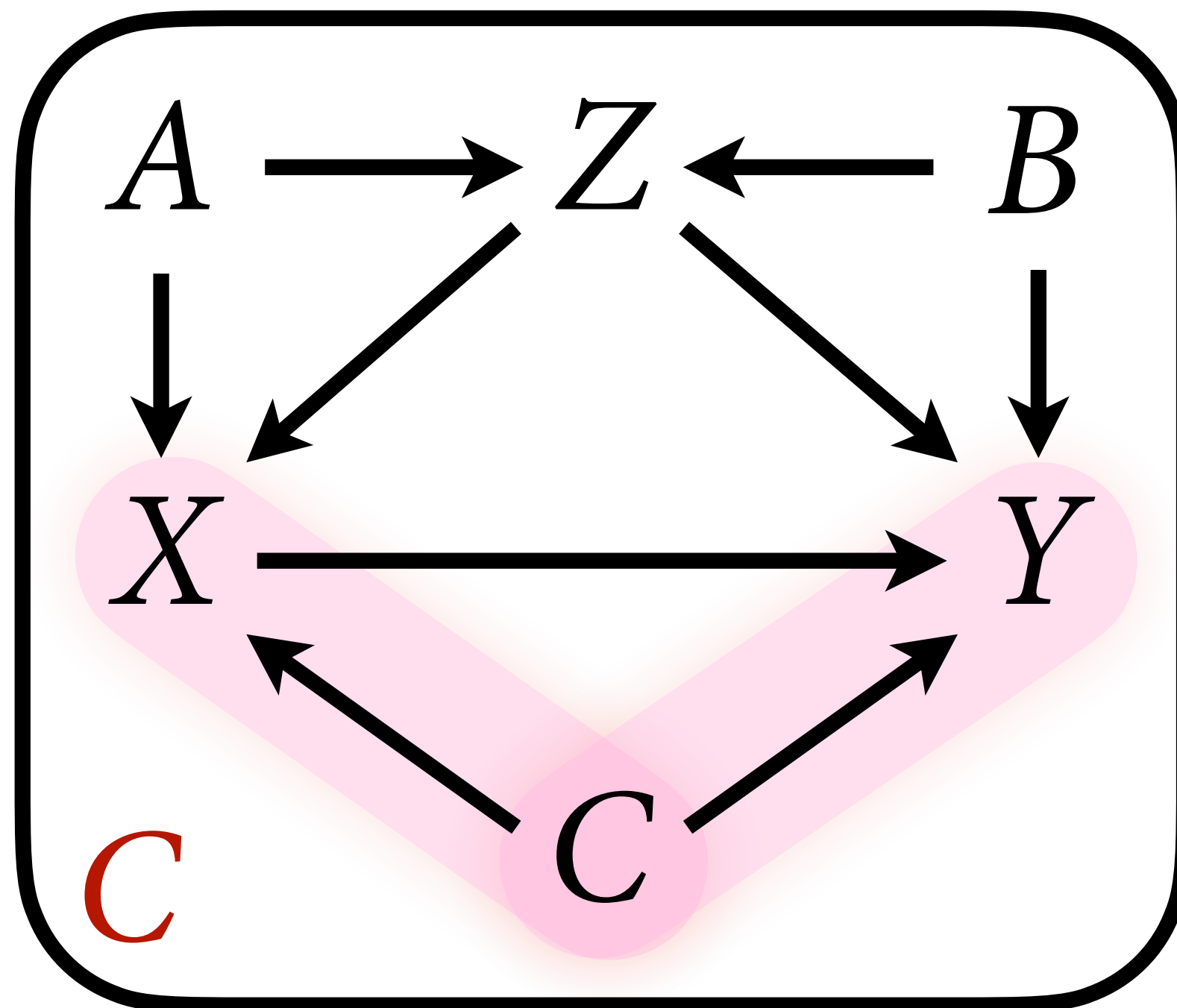
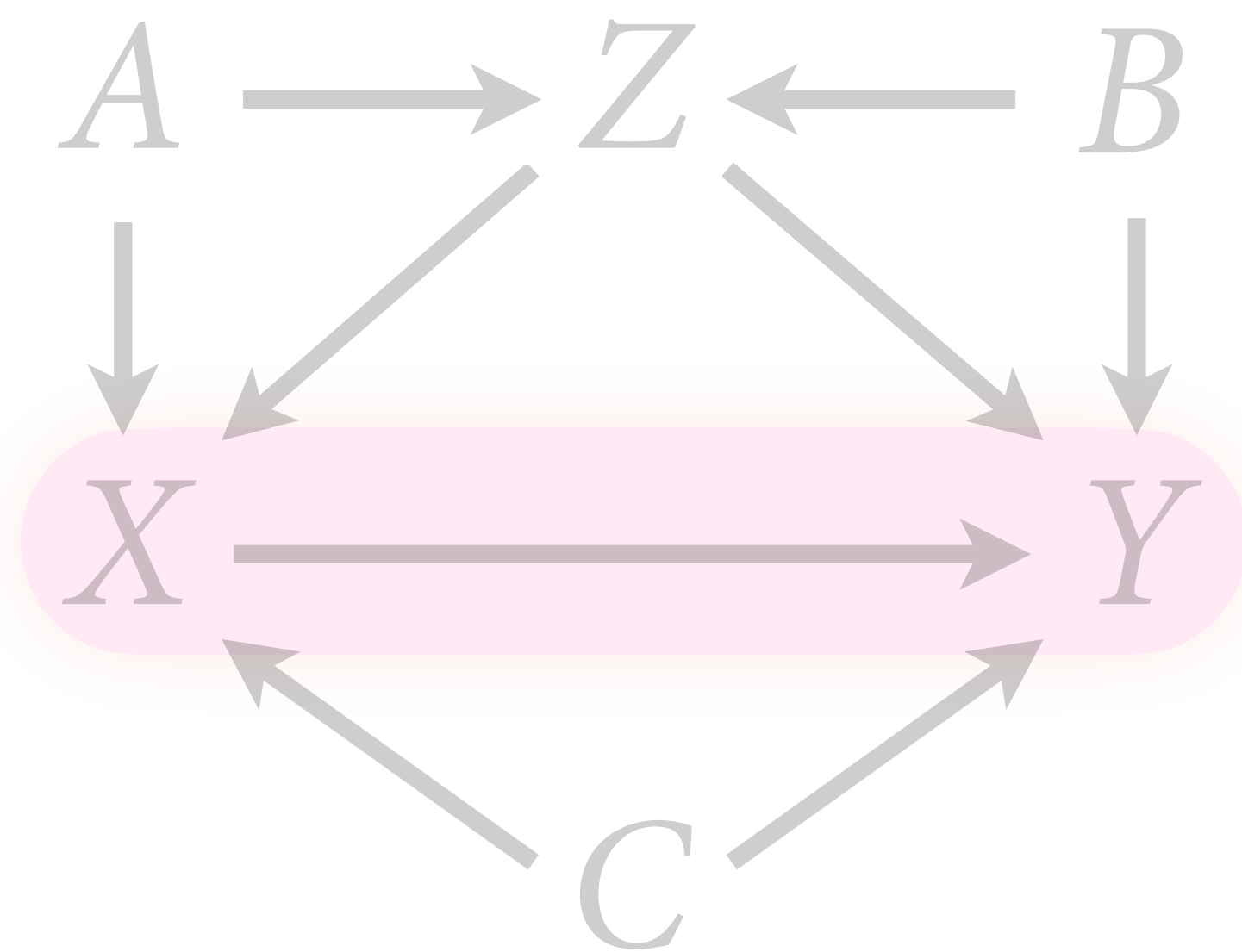


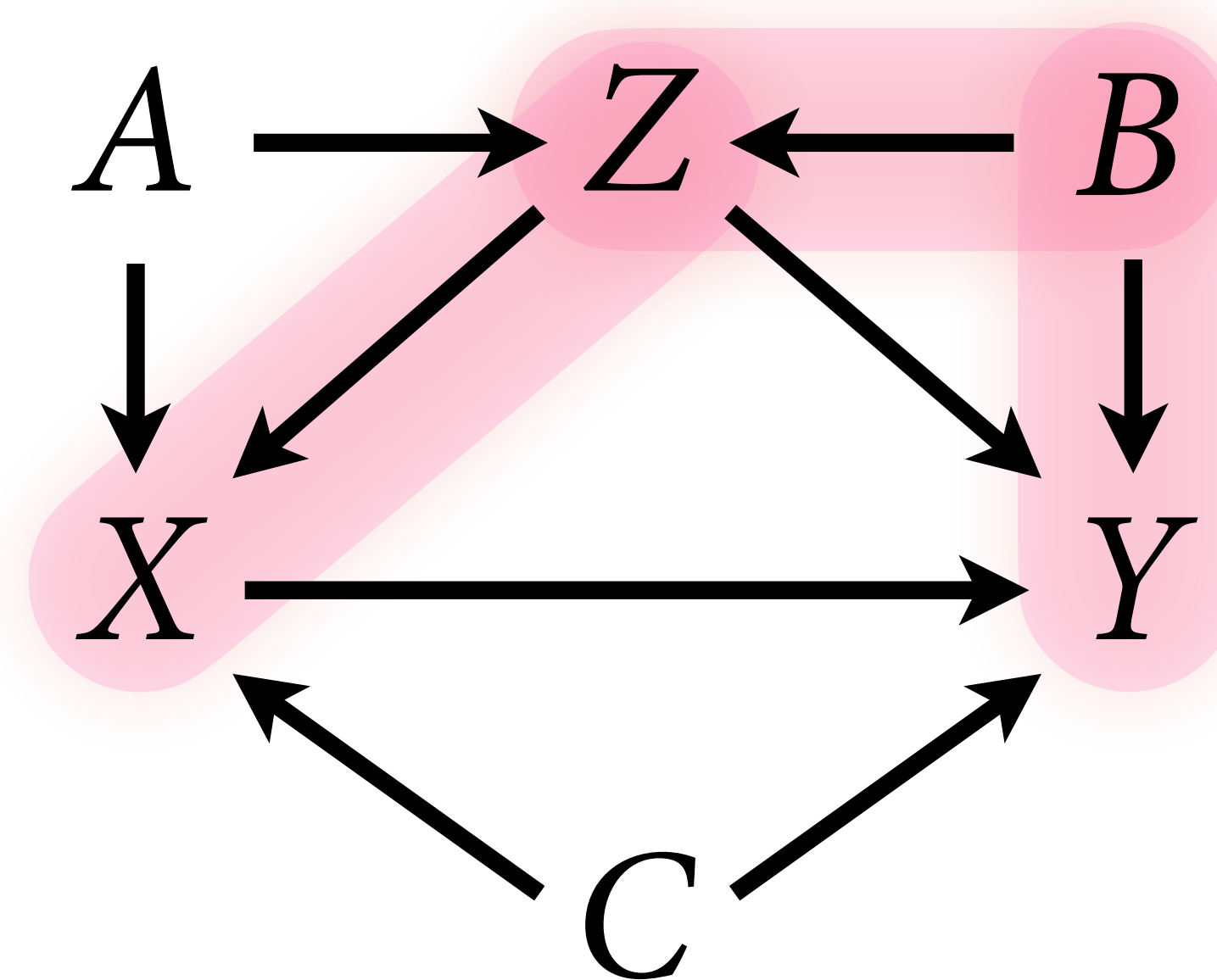
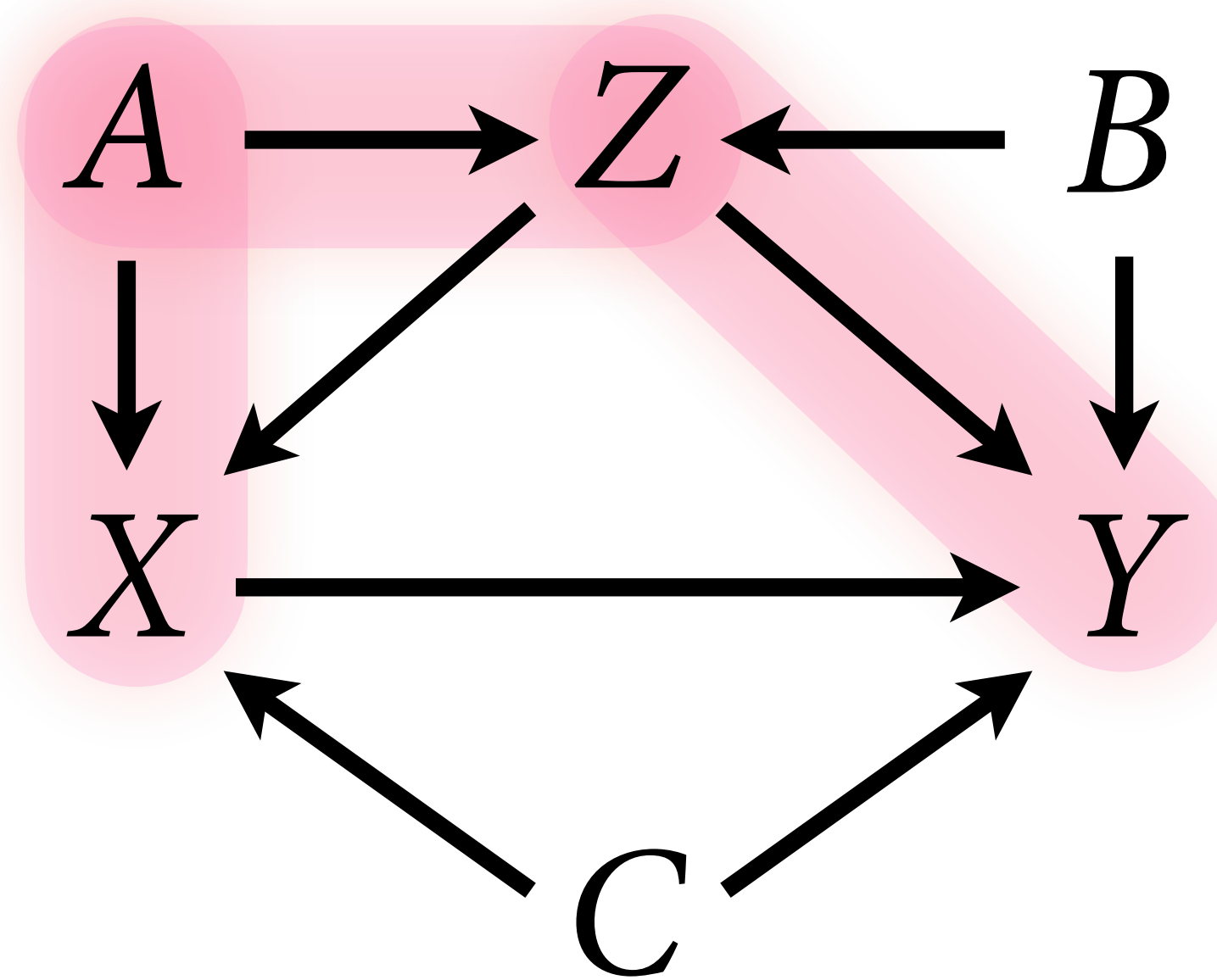
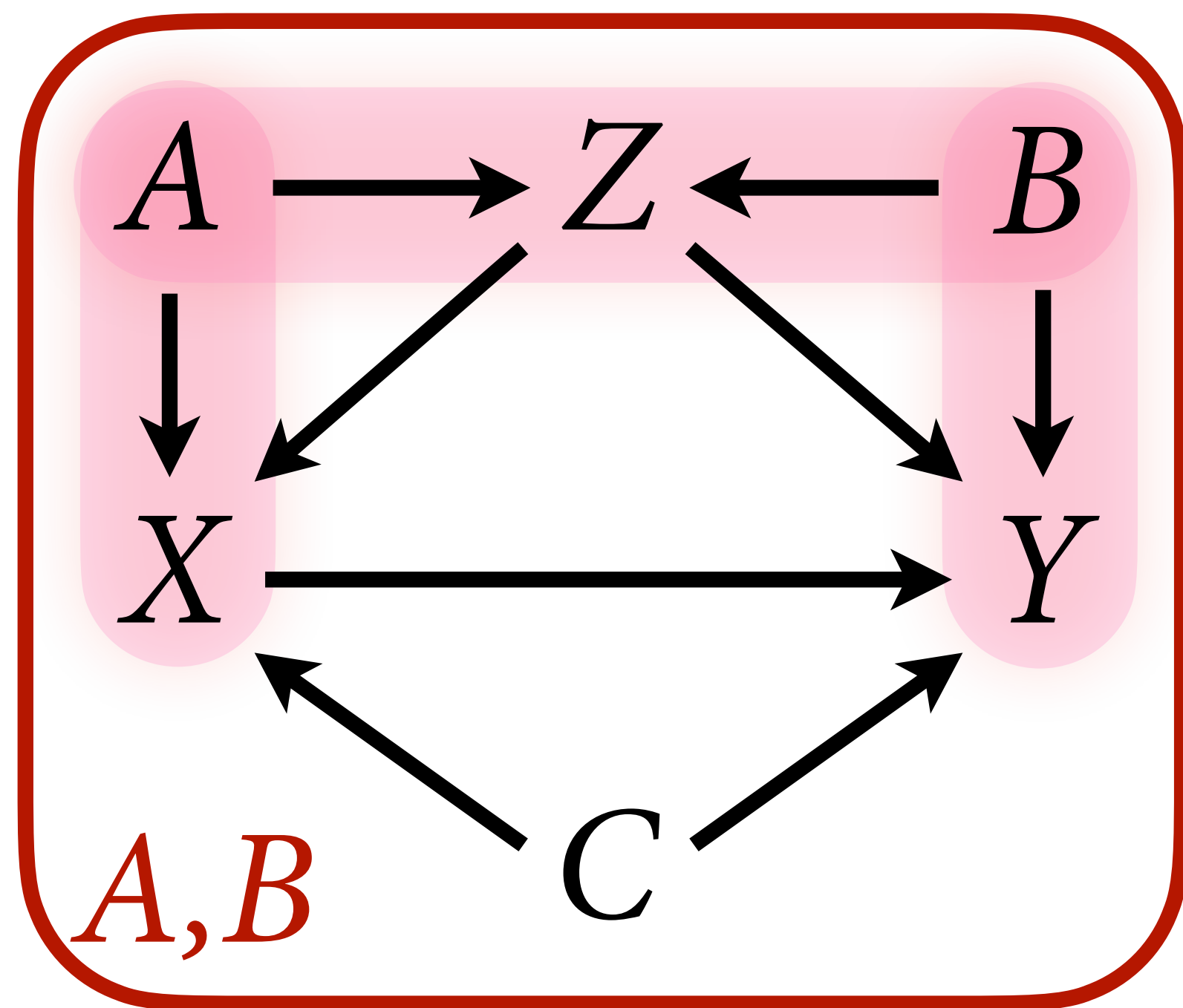
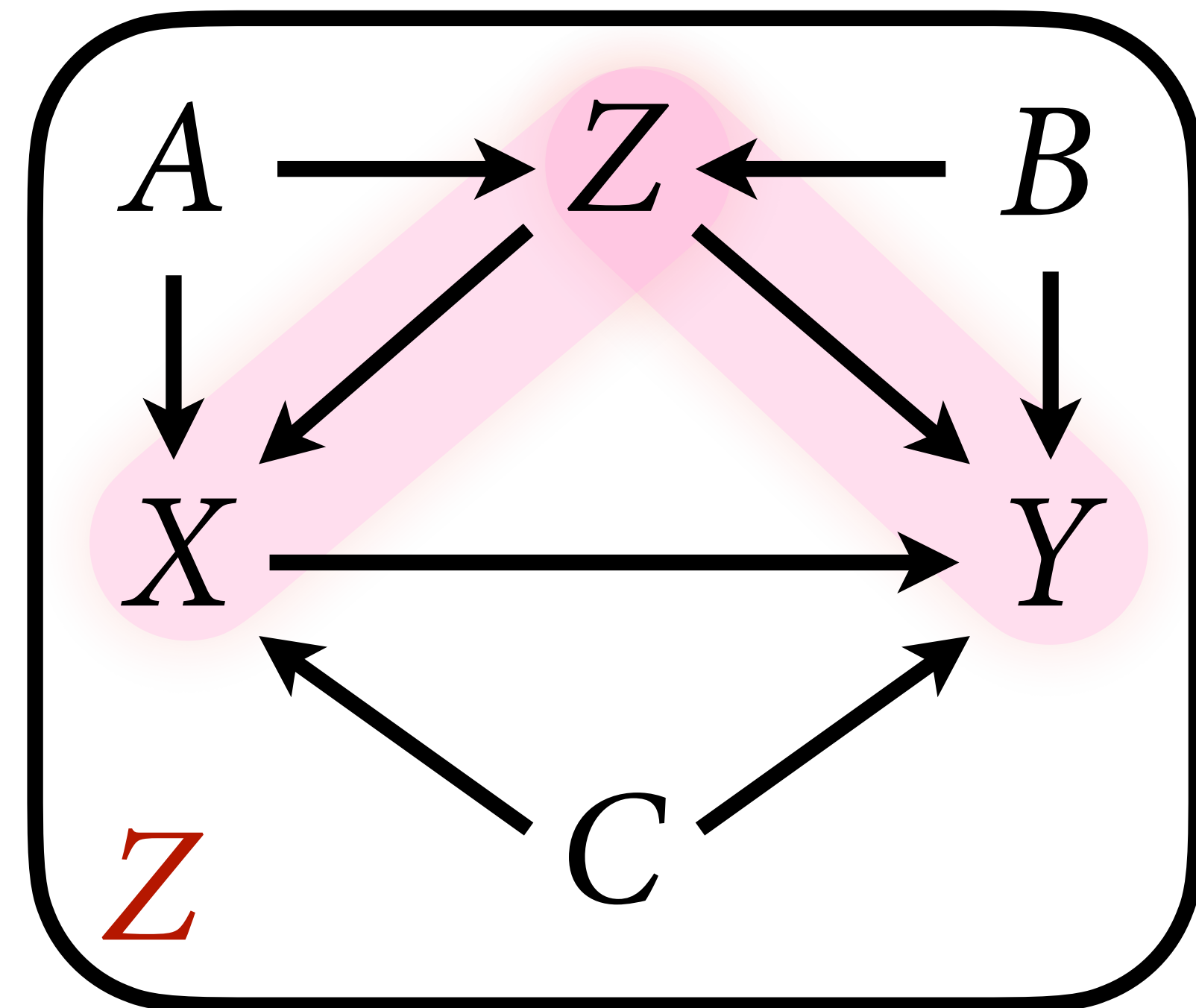
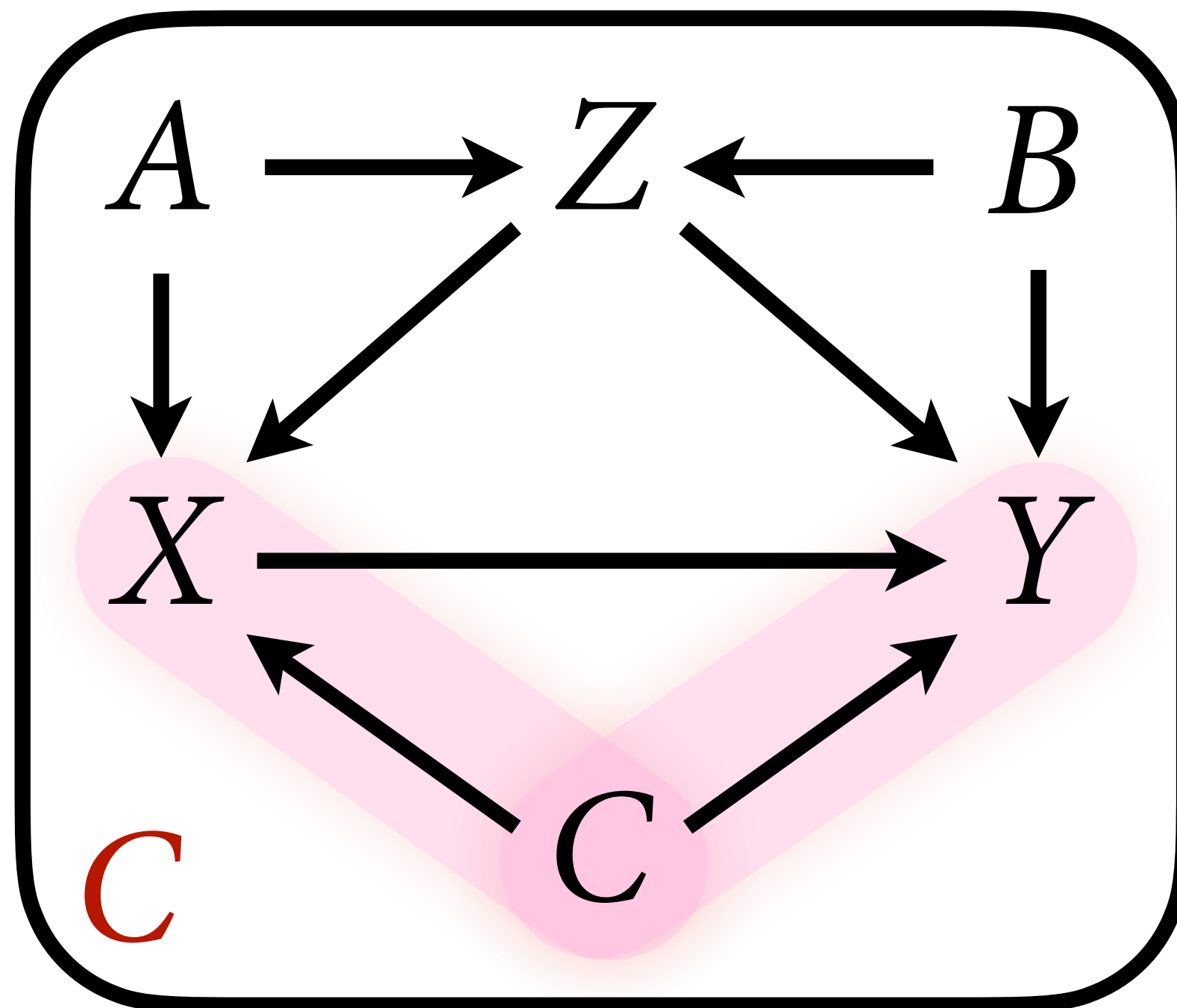
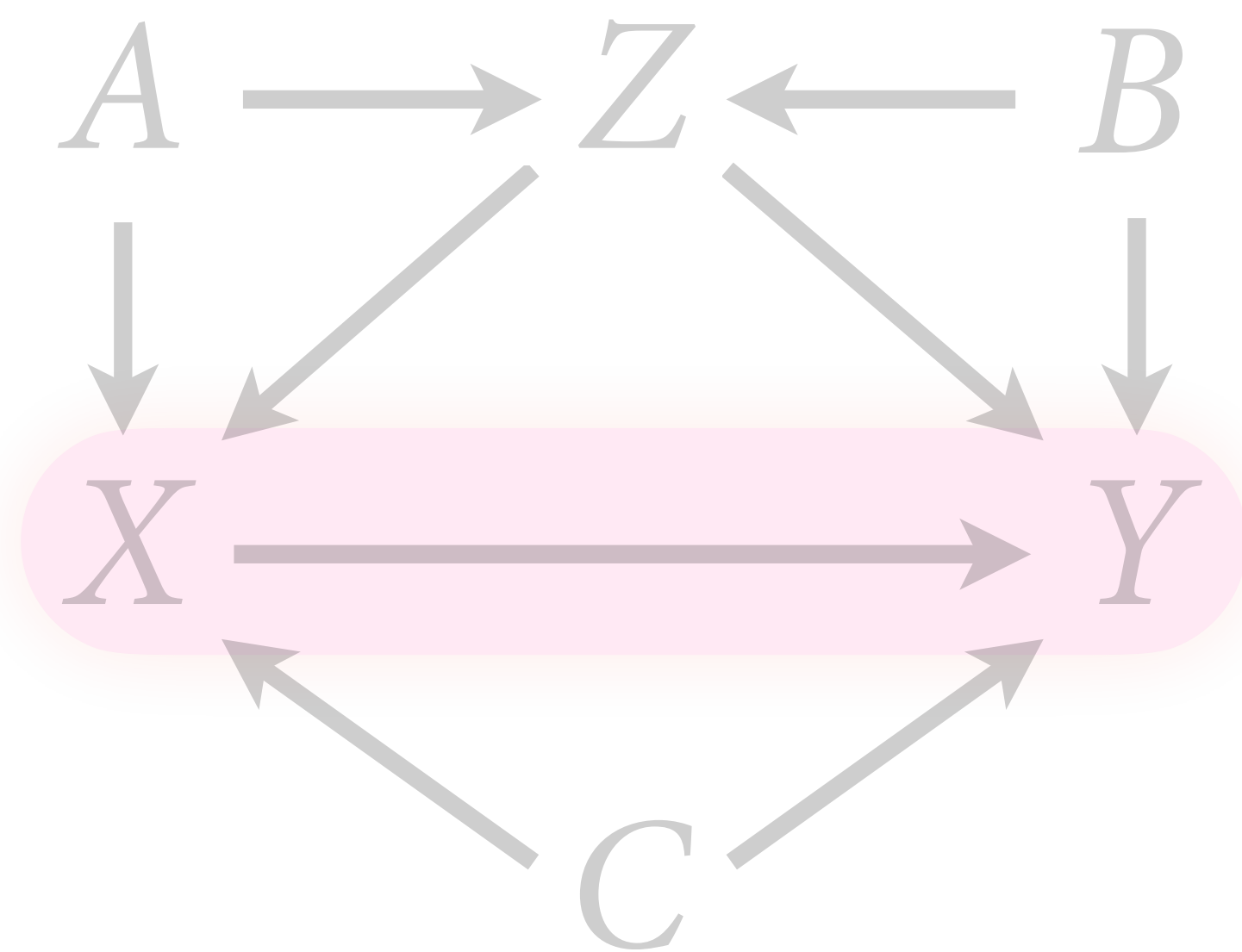
Backdoor path, open
Close with Z

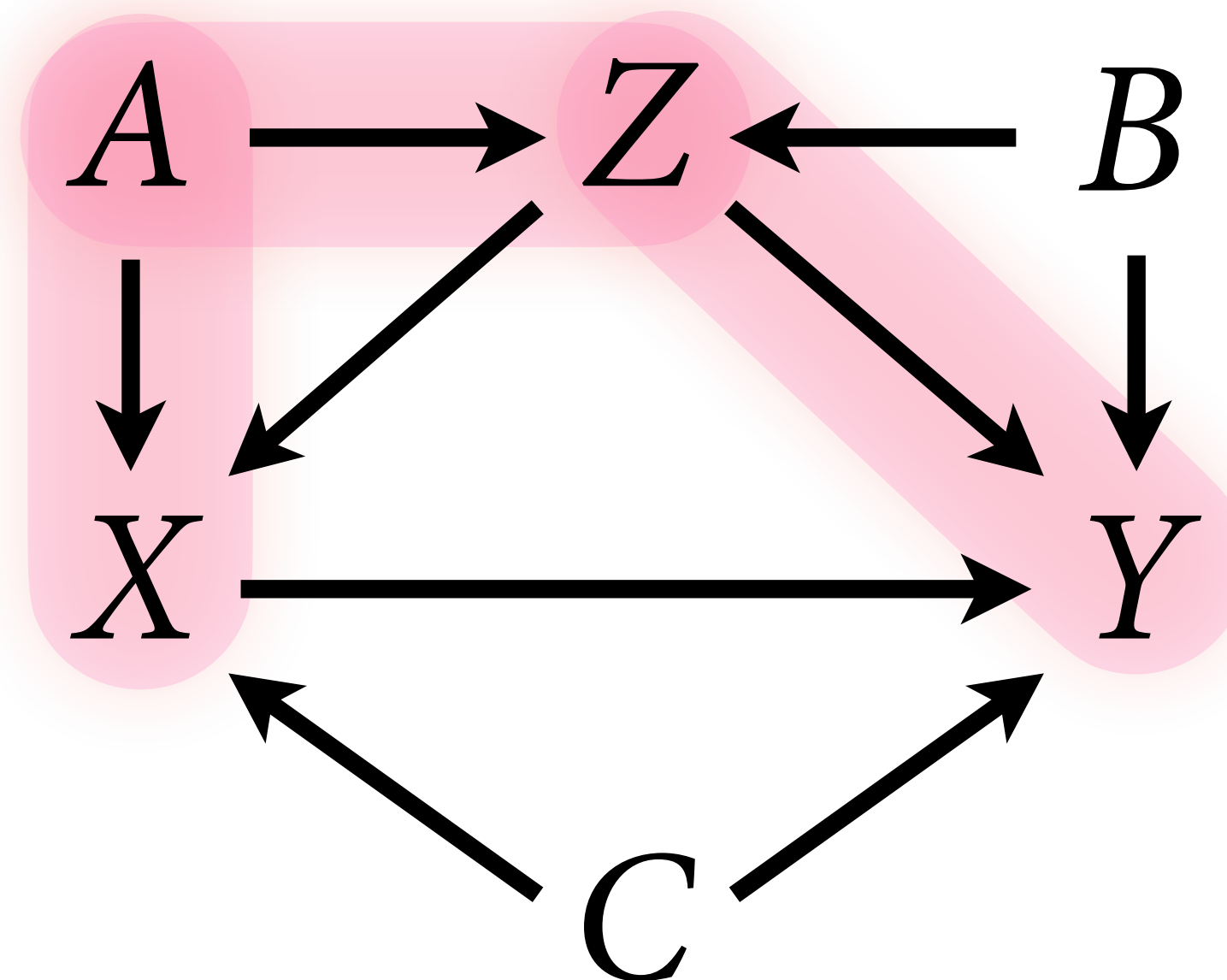




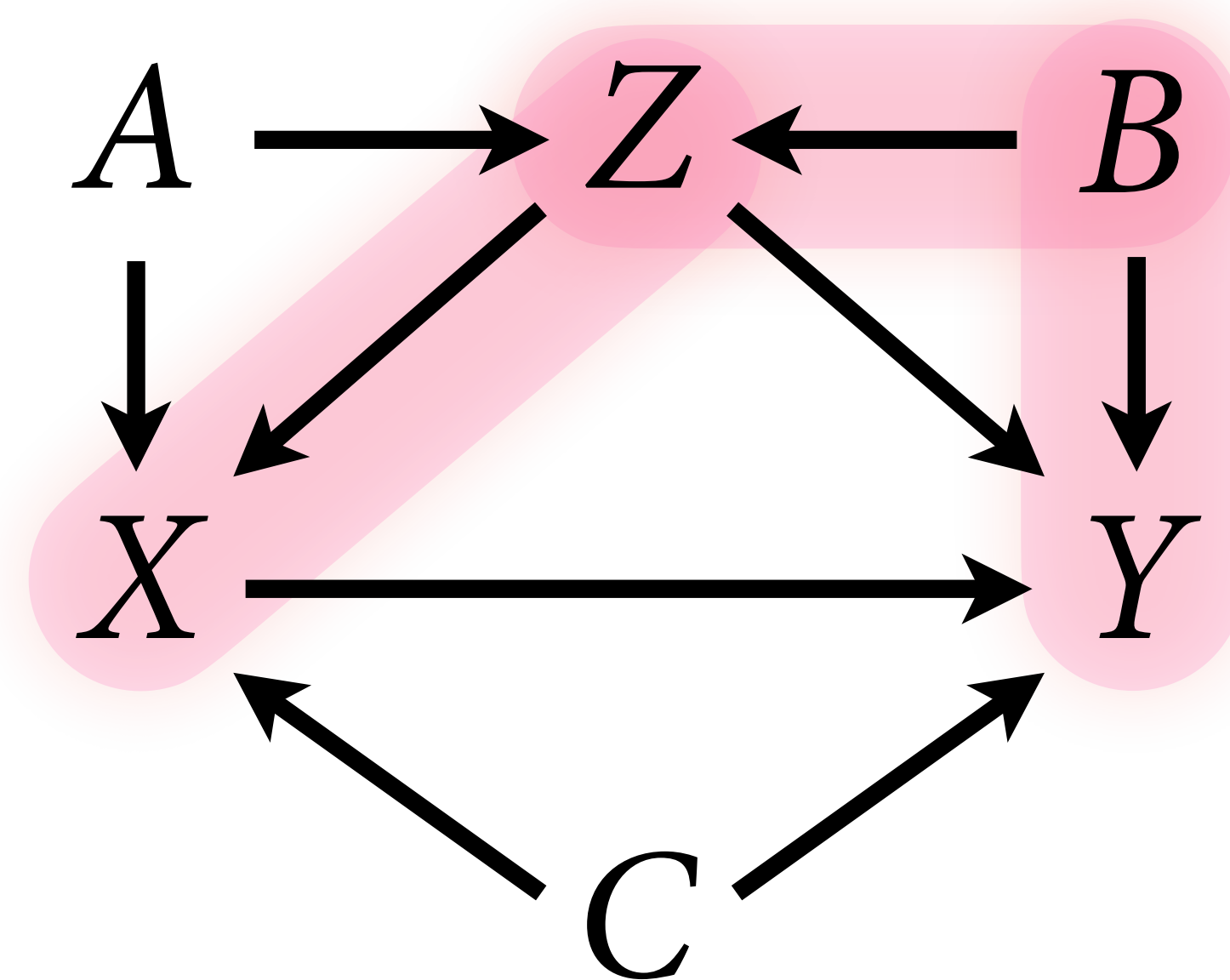
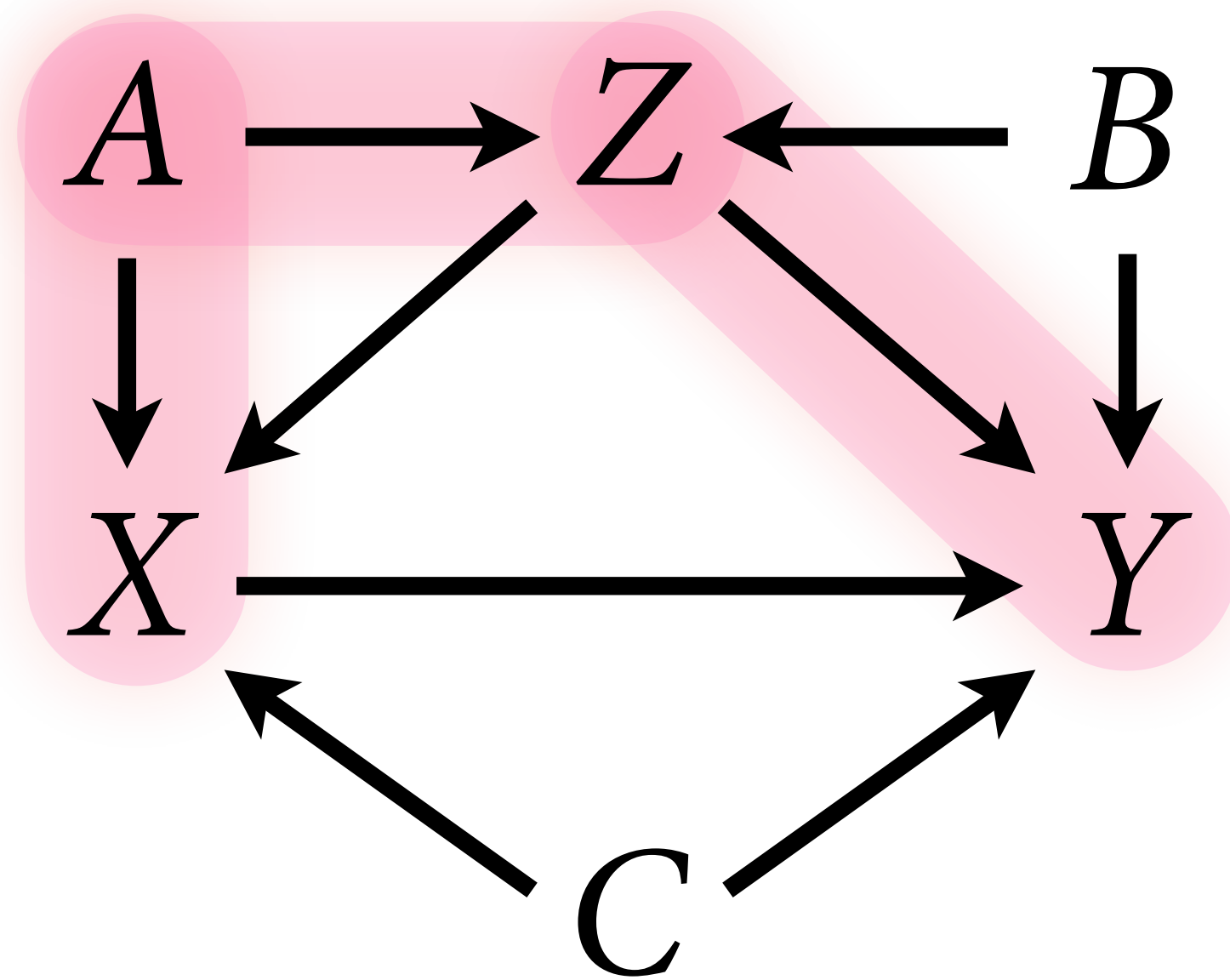
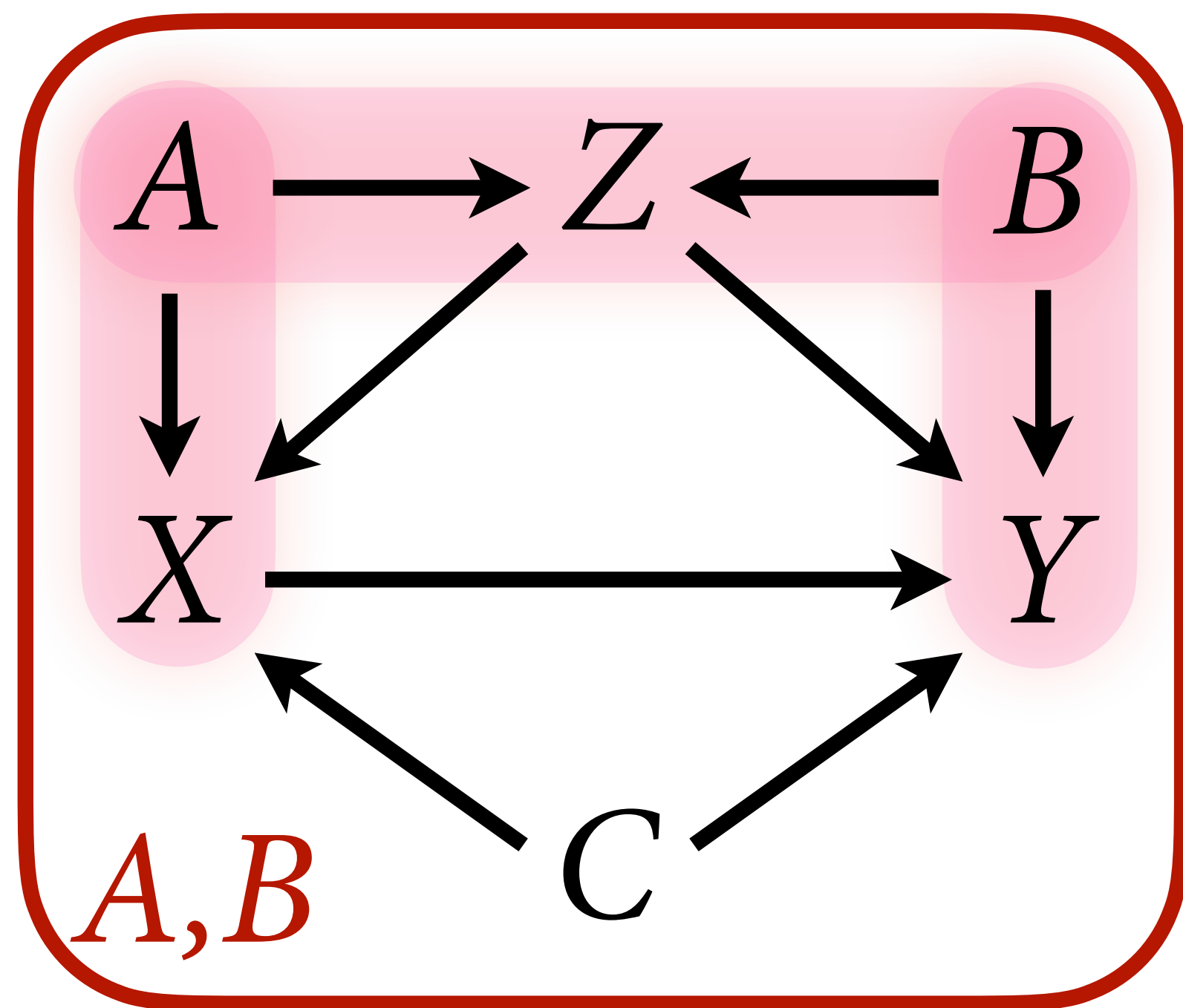
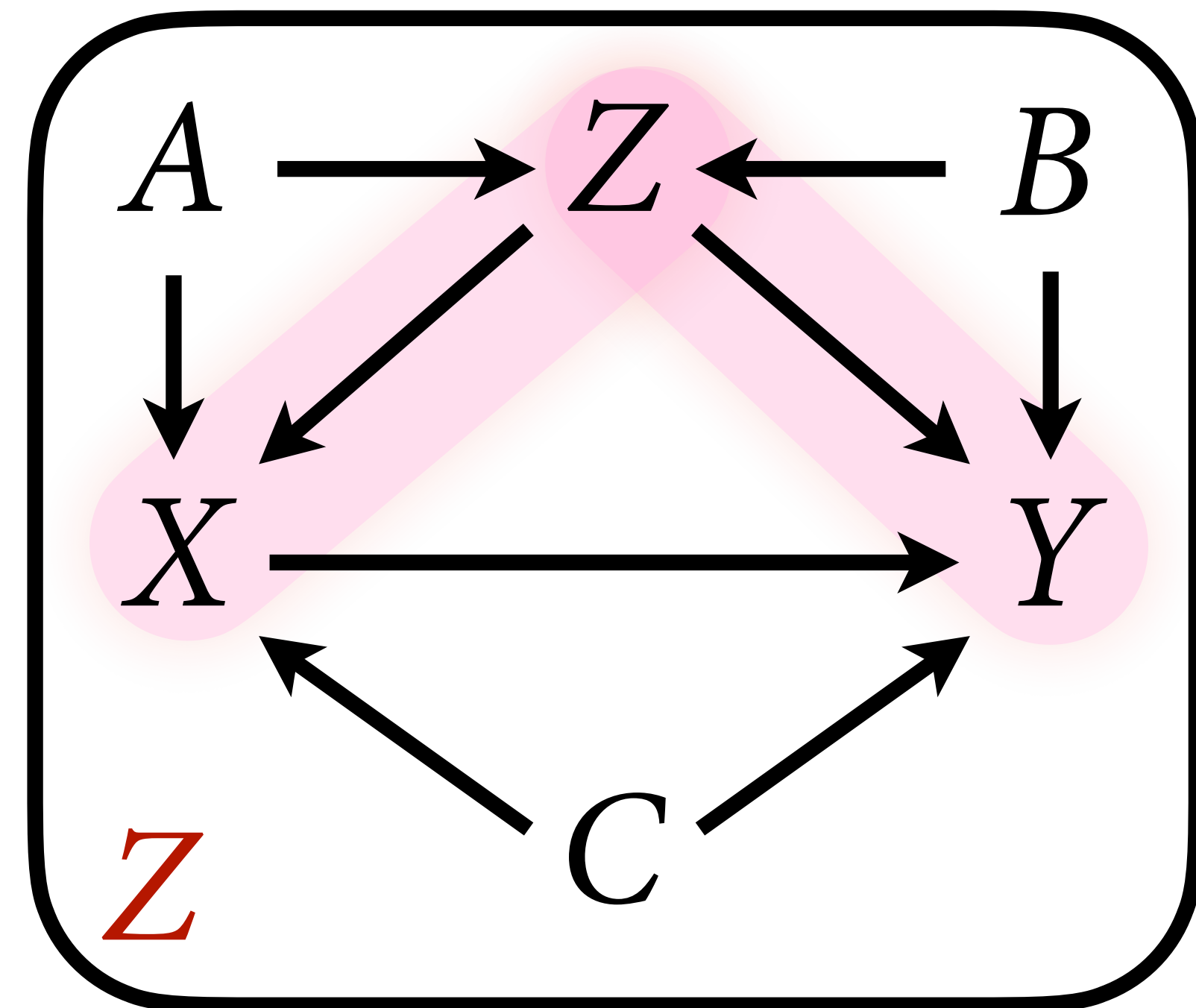
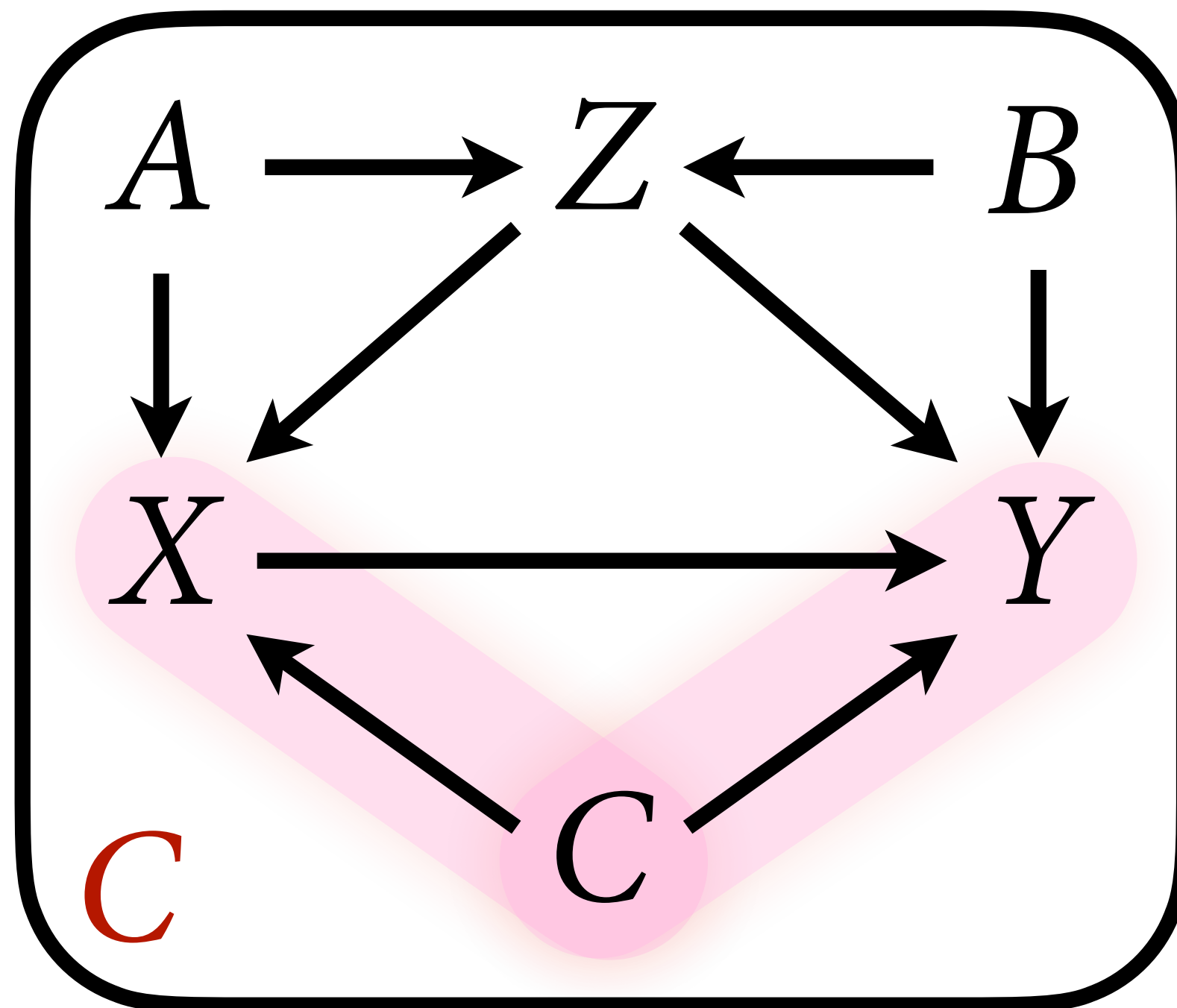
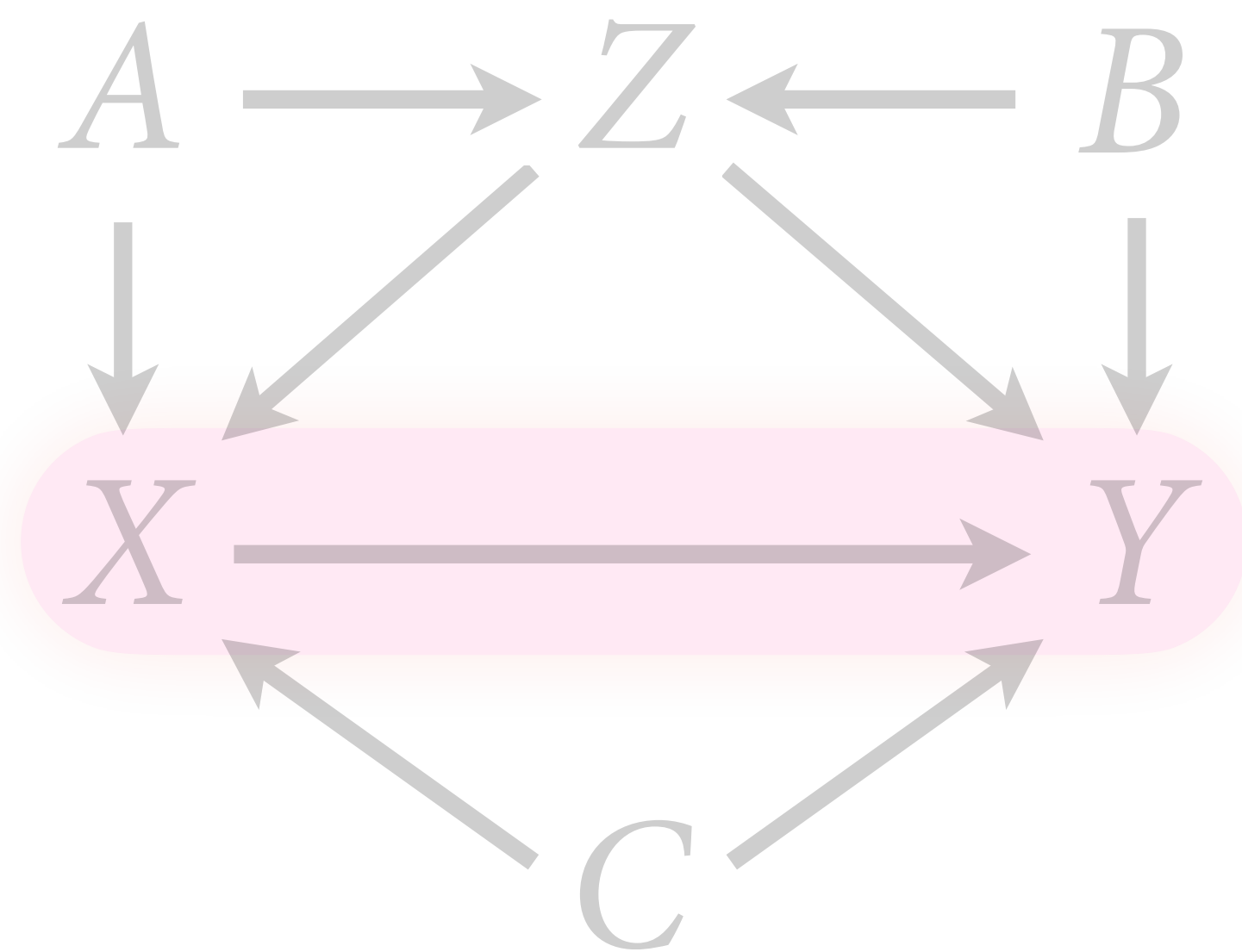
Backdoor path, opened by Z
 A or B to close

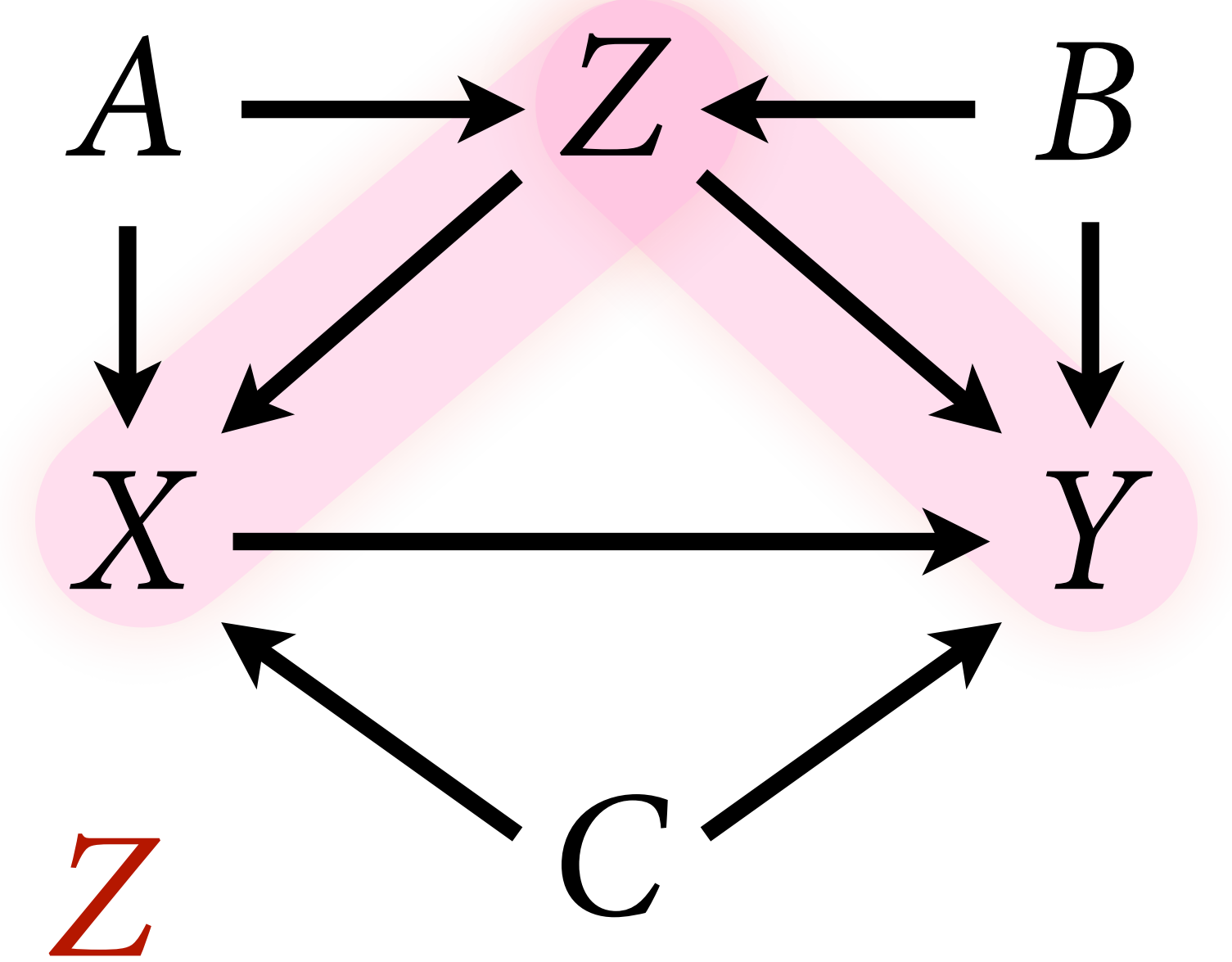
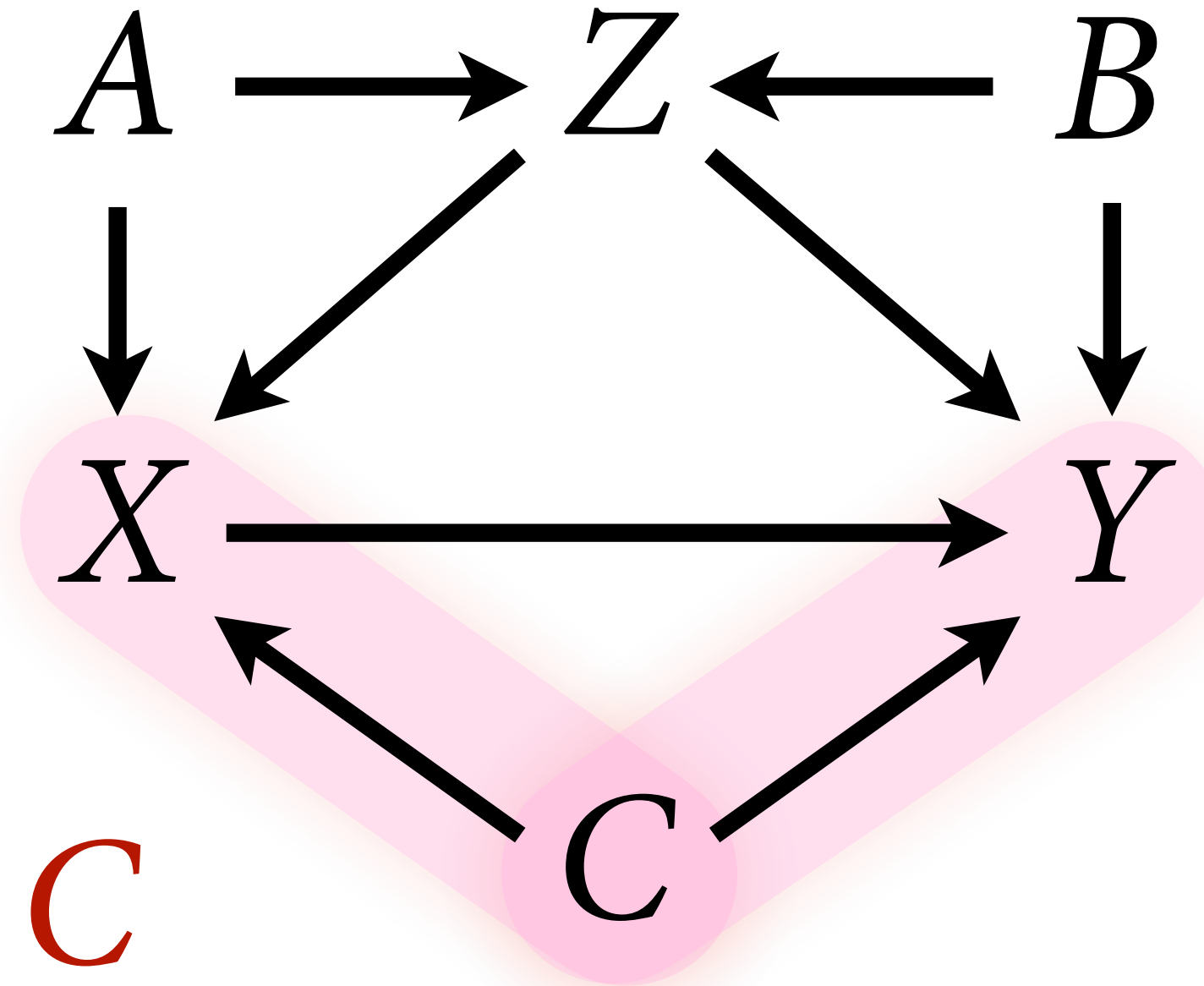
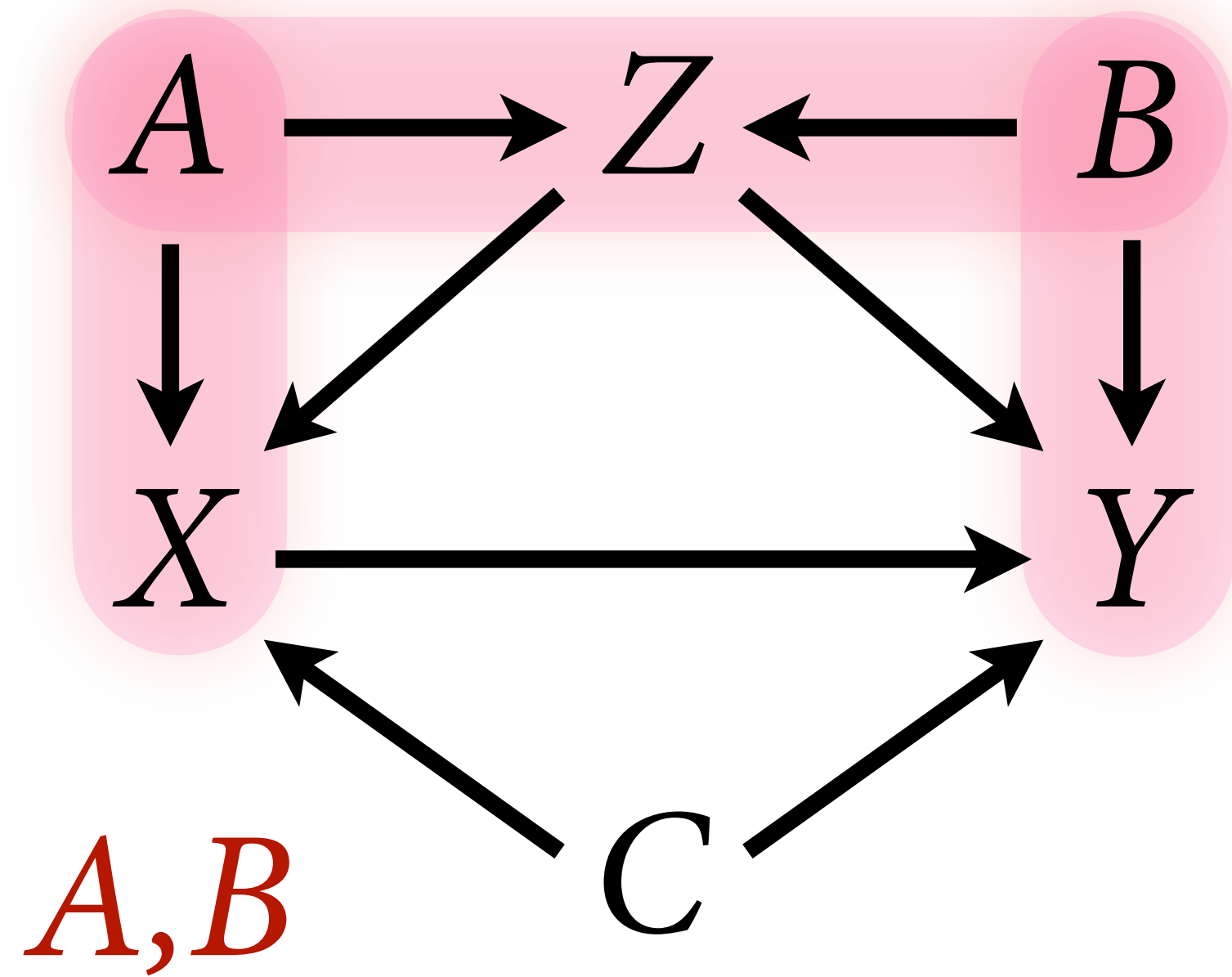




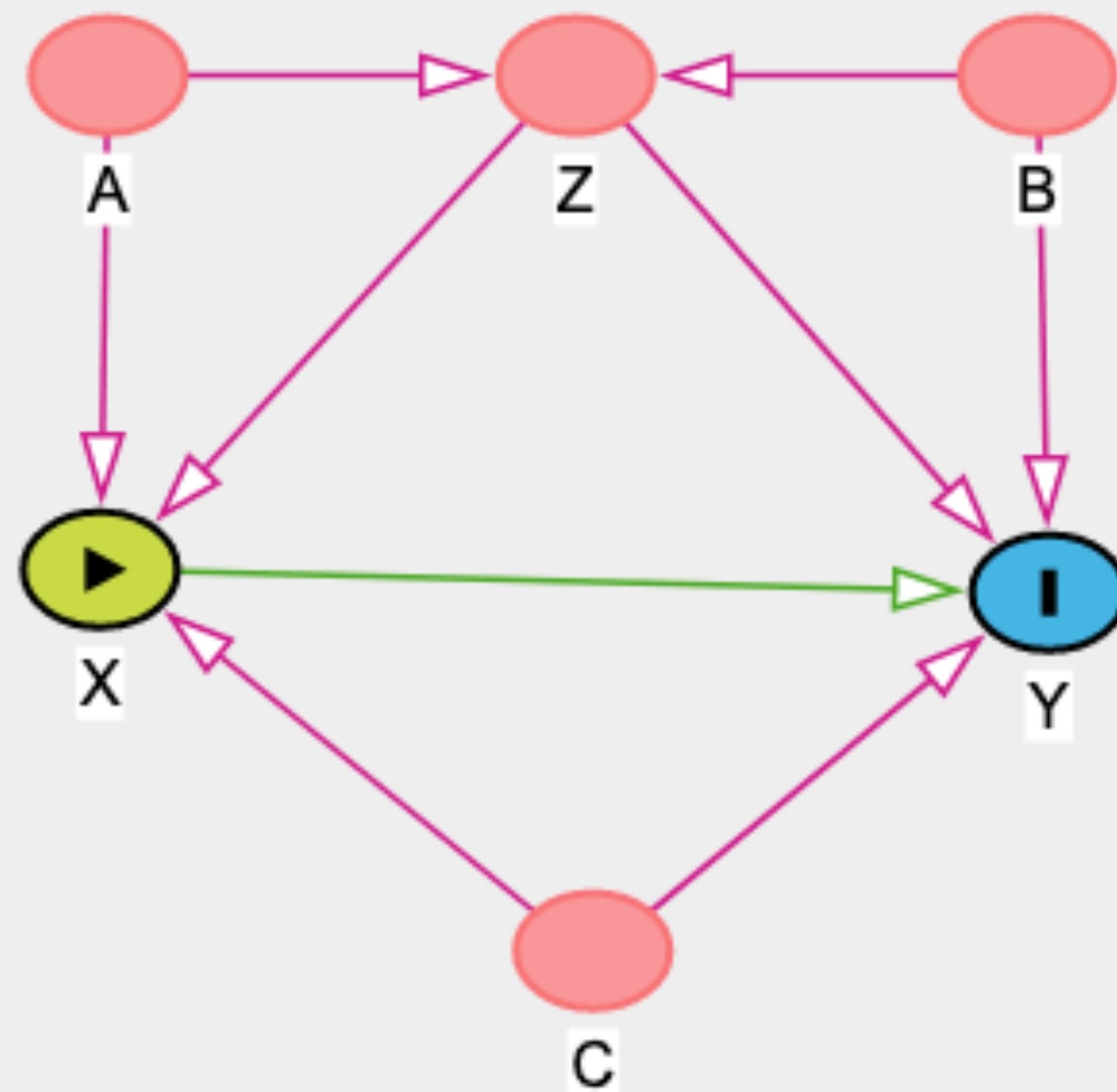


Backdoor path, open
Close with A or Z





Minimum adjustment set:
 C, Z , and either A or B
 (B is better choice)



☑ Causal effect identification

Adjustment (total effect) ▾

Minimal sufficient adjustment sets for estimating the total effect of X on Y:

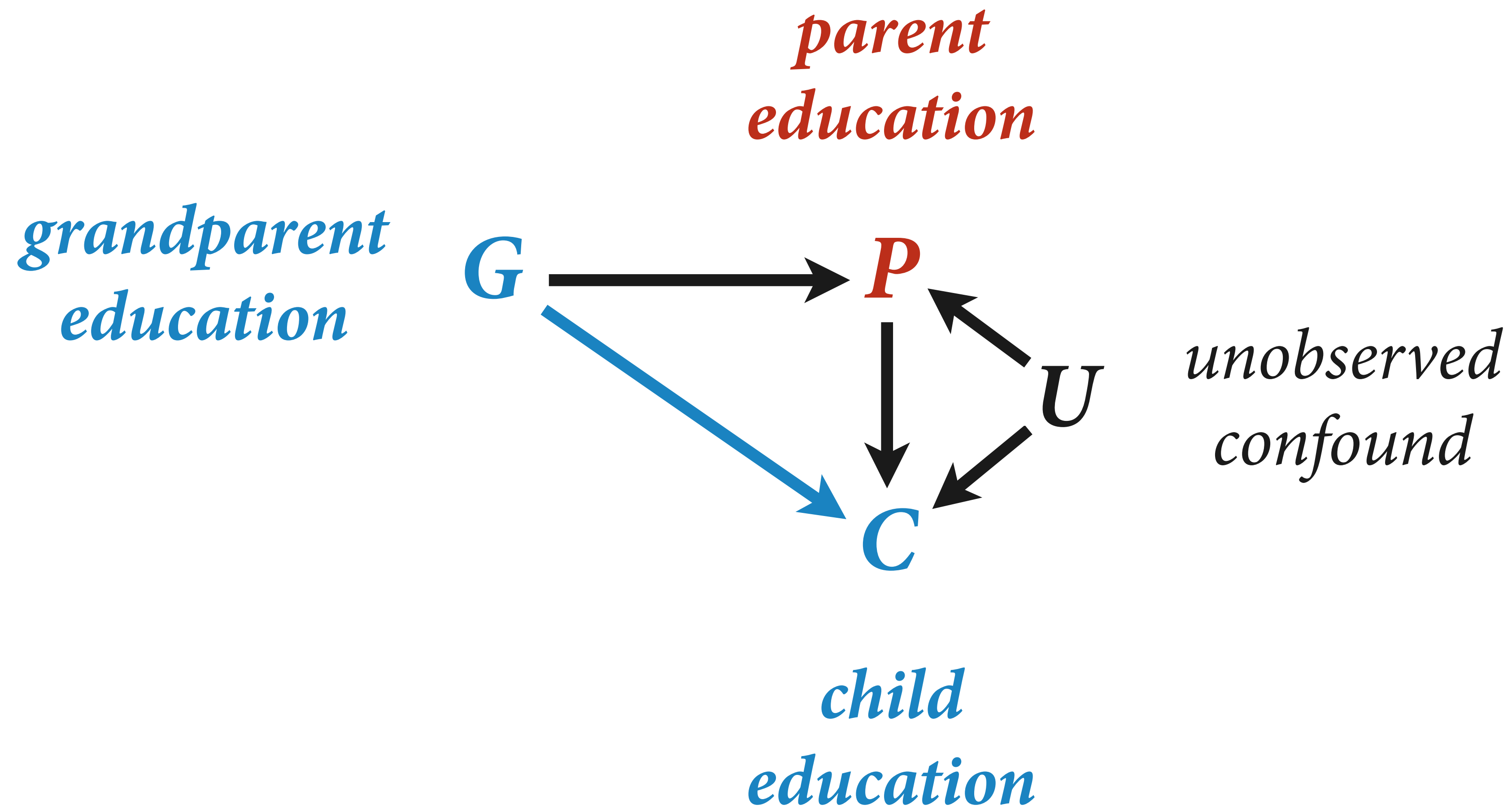
- A, C, Z
- B, C, Z

☑ Testable implications

The model implies the following conditional independences:

- $X \perp B \mid A, Z$
- $Y \perp A \mid B, C, X, Z$
- $A \perp B$
- $A \perp C$
- $B \perp C$
- $Z \perp C$

Export R code



Backdoor Criterion

do-calc more than backdoors & adjustment sets

Full Luxury Bayes: use all variables, but in separate sub-models instead of single regression

do-calc less demanding: finds relevant variables; saves us having to make some assumptions; not always a regression

this is
YOUR BRAIN



this is
REGRESSION



this is
YOUR BRAIN
on
REGRESSION



any questions?

Good & Bad Controls

“Control” variable: Variable introduced to an analysis so that a causal estimate is possible

Common **wrong** heuristics for choosing control variables

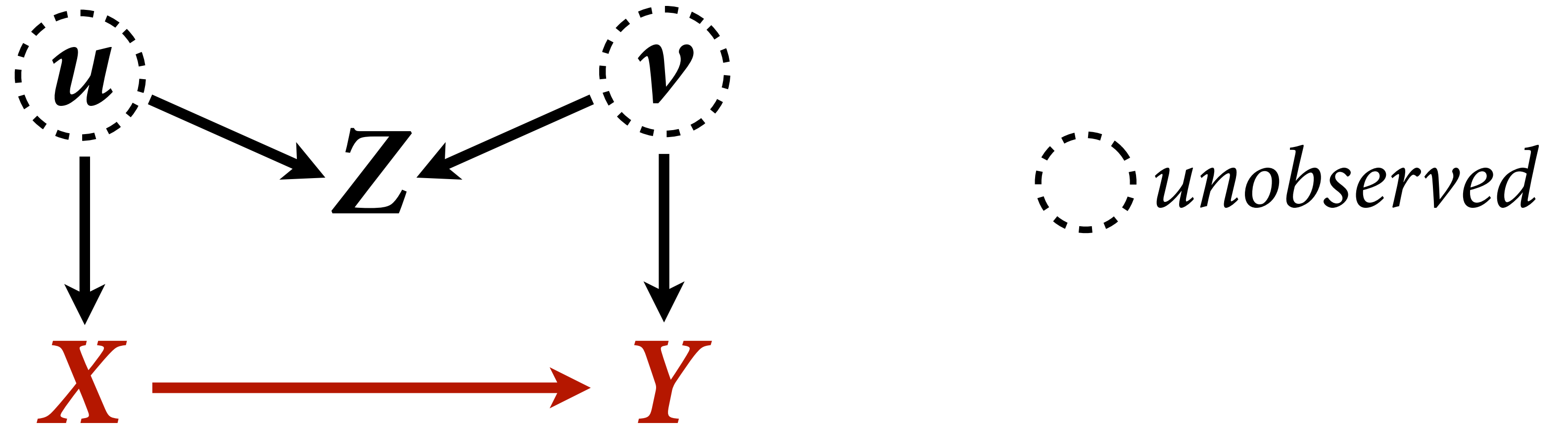
Anything in the spreadsheet **YOLO!**

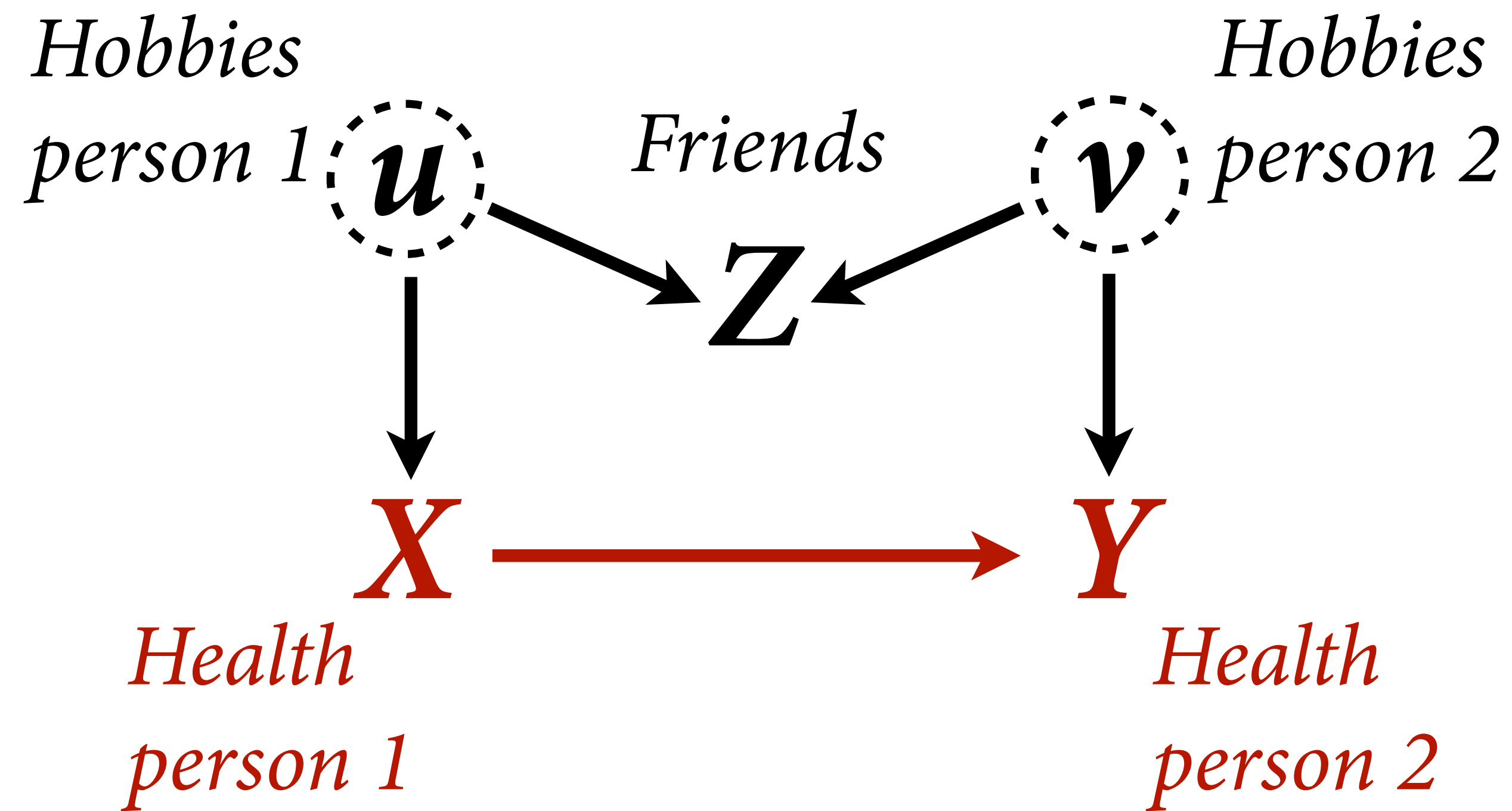
Any variables not highly **collinear**

Any **pre-treatment** measurement (baseline)

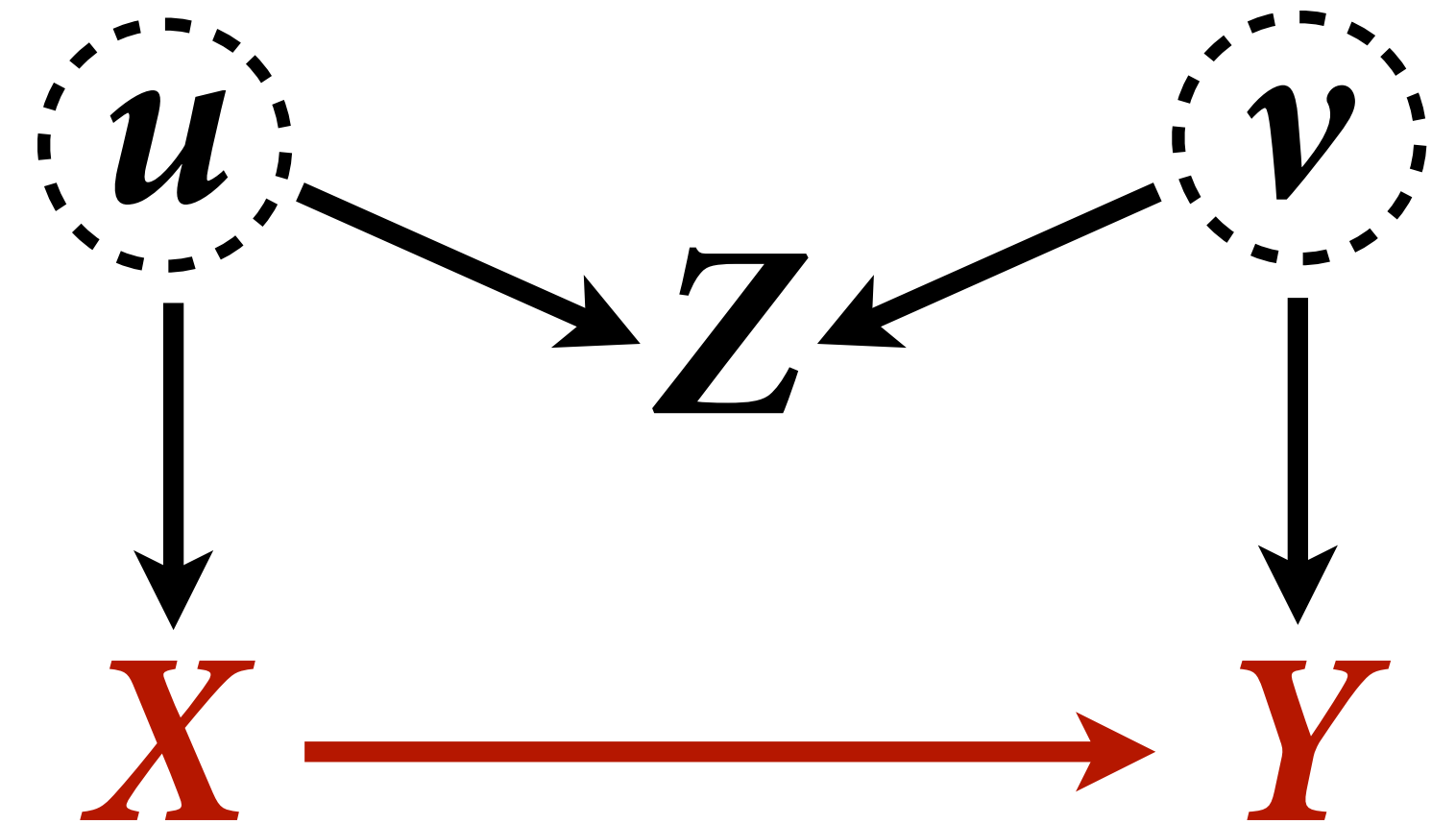


$$\mathbf{X} \longrightarrow \mathbf{Y}$$



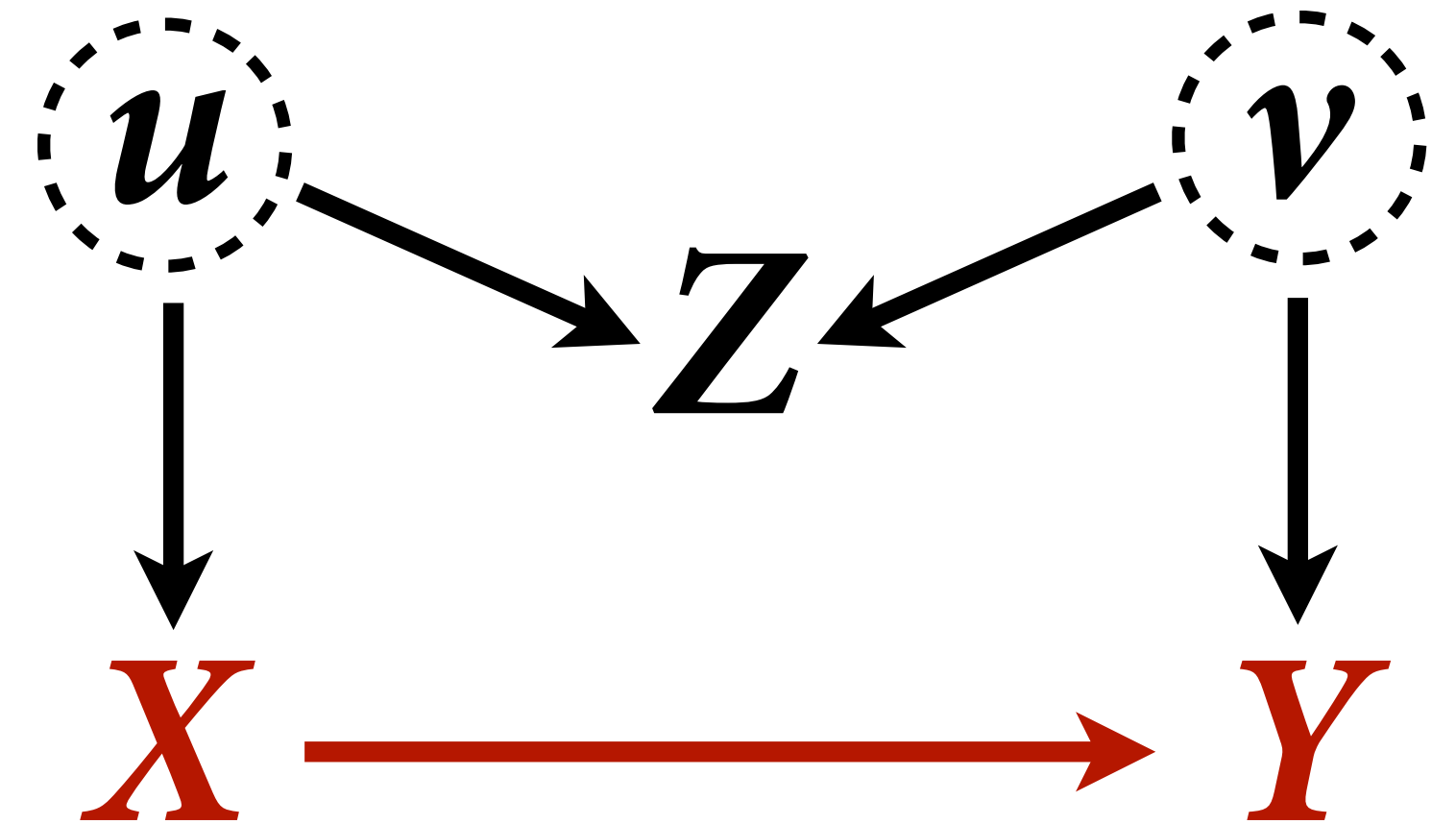


(1) List the paths



(1) List the paths

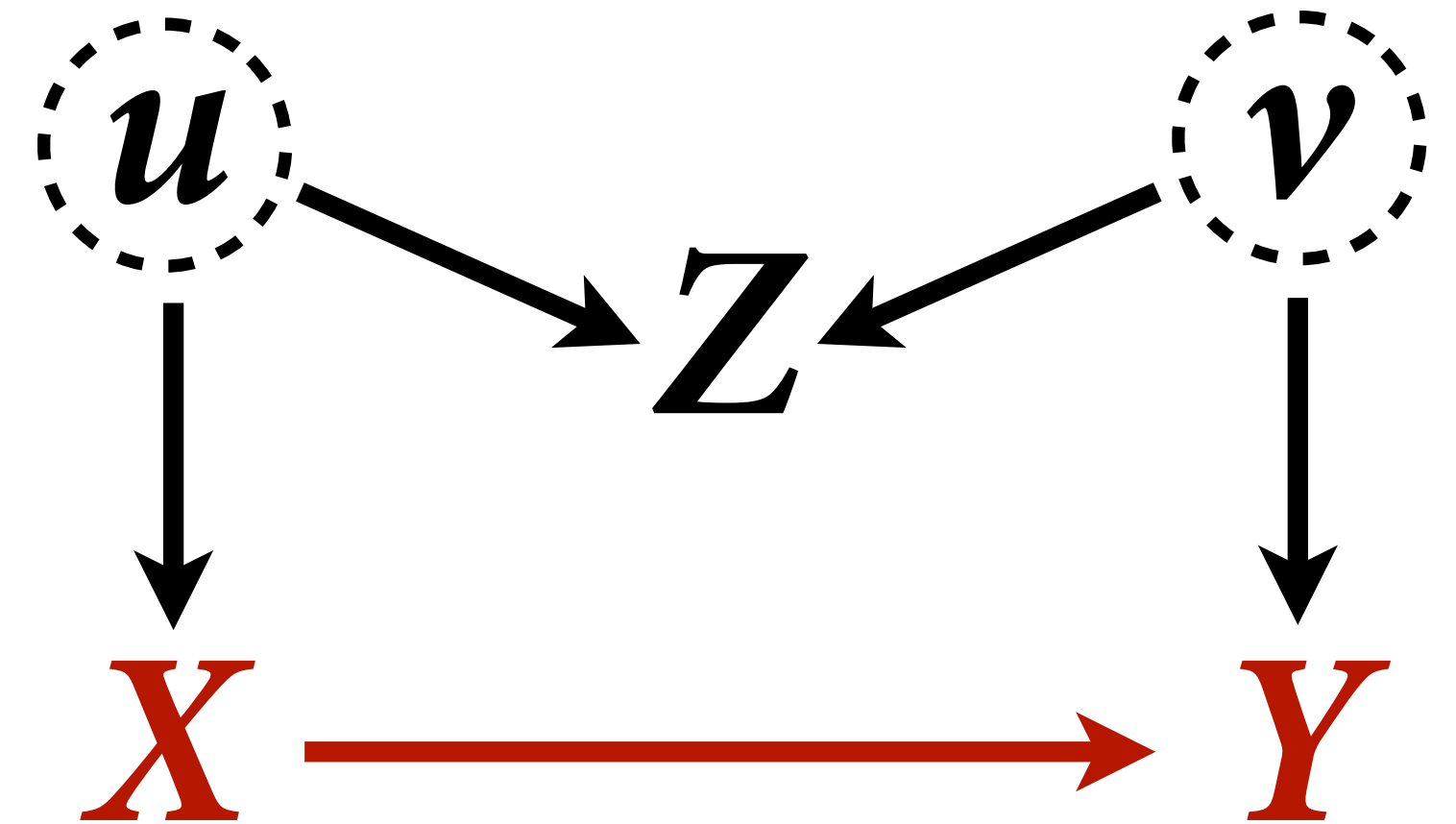
$X \rightarrow Y$



(1) List the paths

$$X \rightarrow Y$$

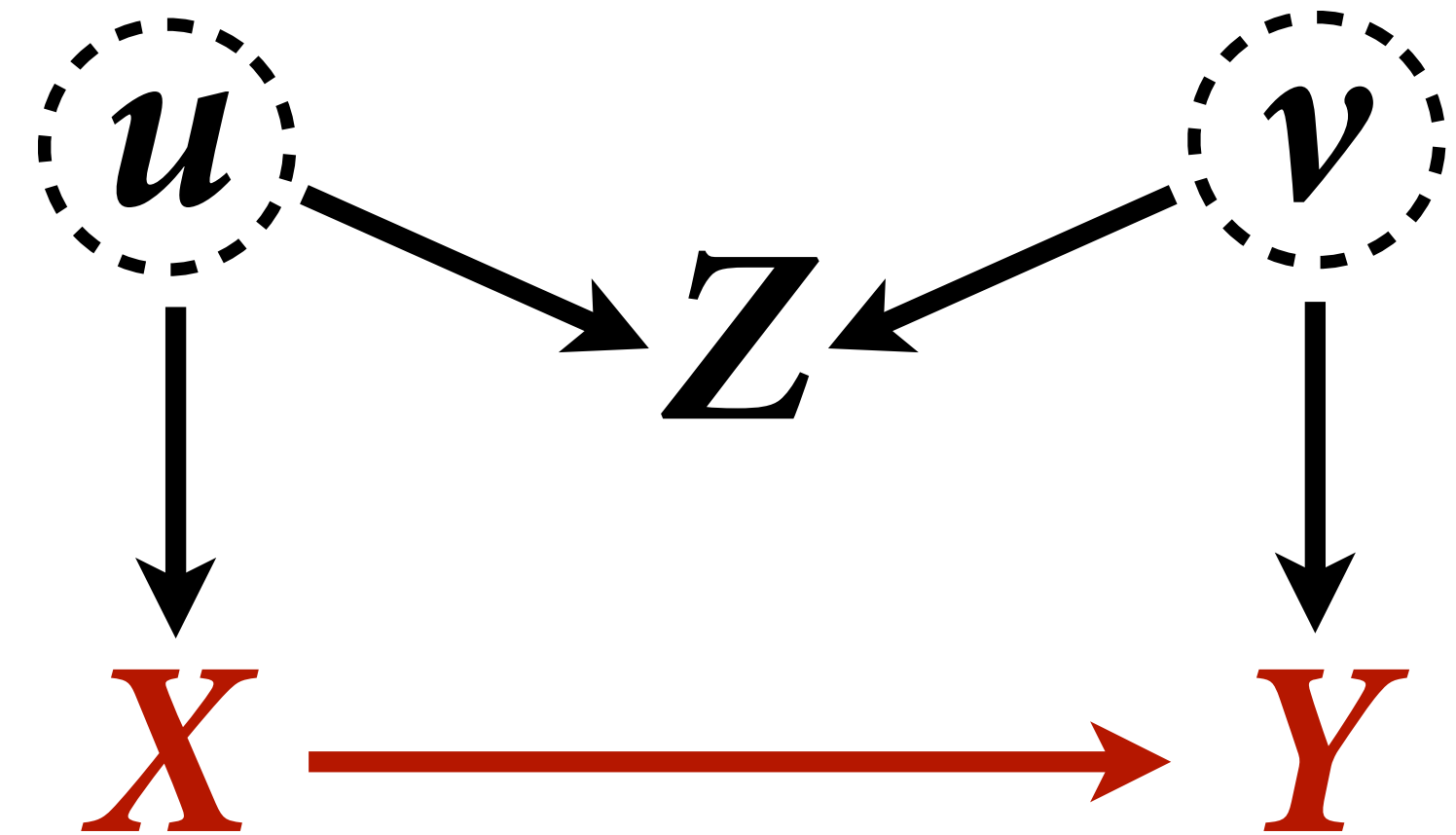
$$X \leftarrow u \rightarrow Z \leftarrow v \rightarrow Y$$



(1) List the paths (2) Find backdoors

$X \rightarrow Y$
frontdoor & open

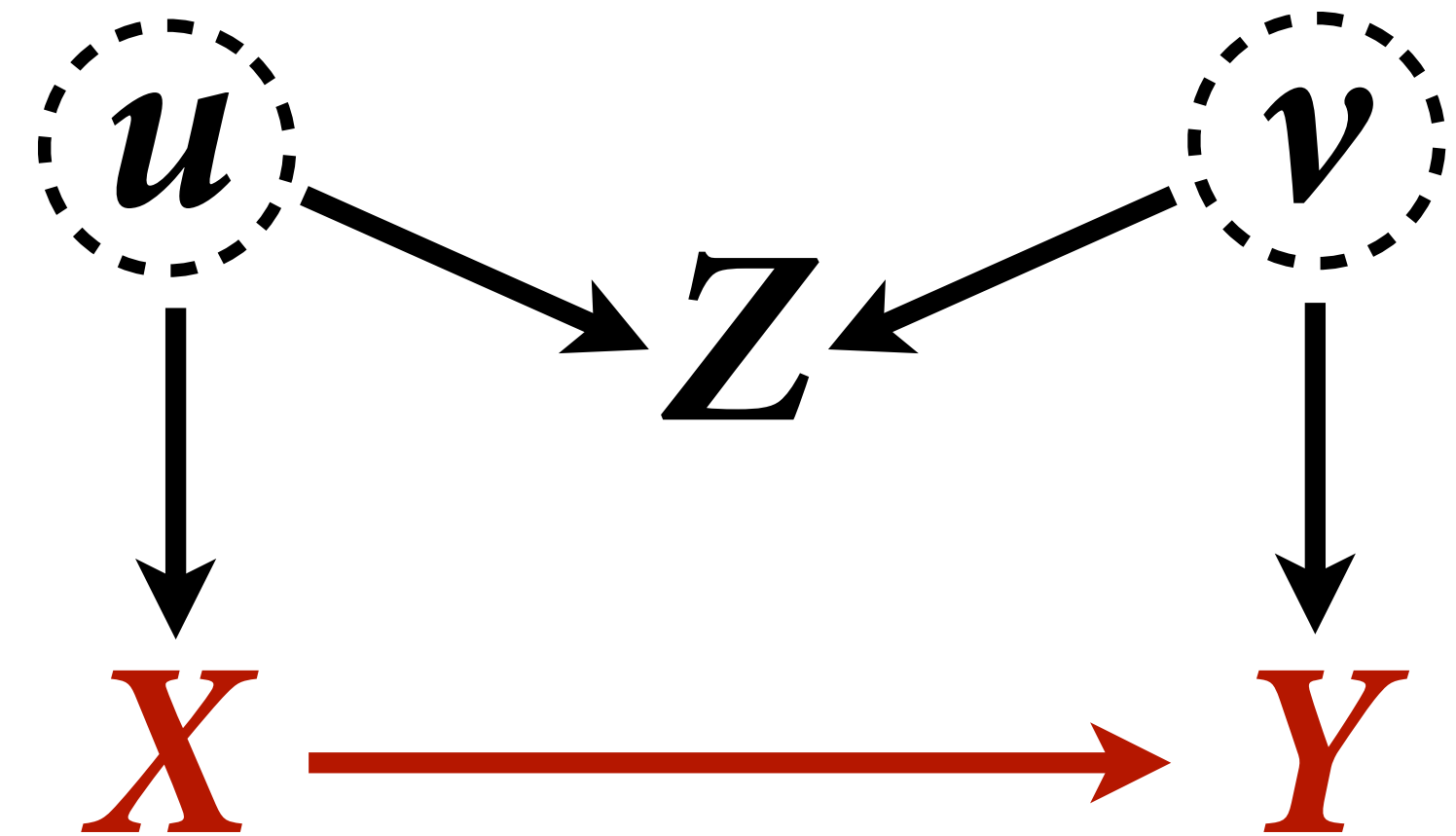
$X \leftarrow u \rightarrow Z \leftarrow v \rightarrow Y$
backdoor & closed



(1) List the paths (2) Find backdoors

$X \rightarrow Y$
frontdoor & open

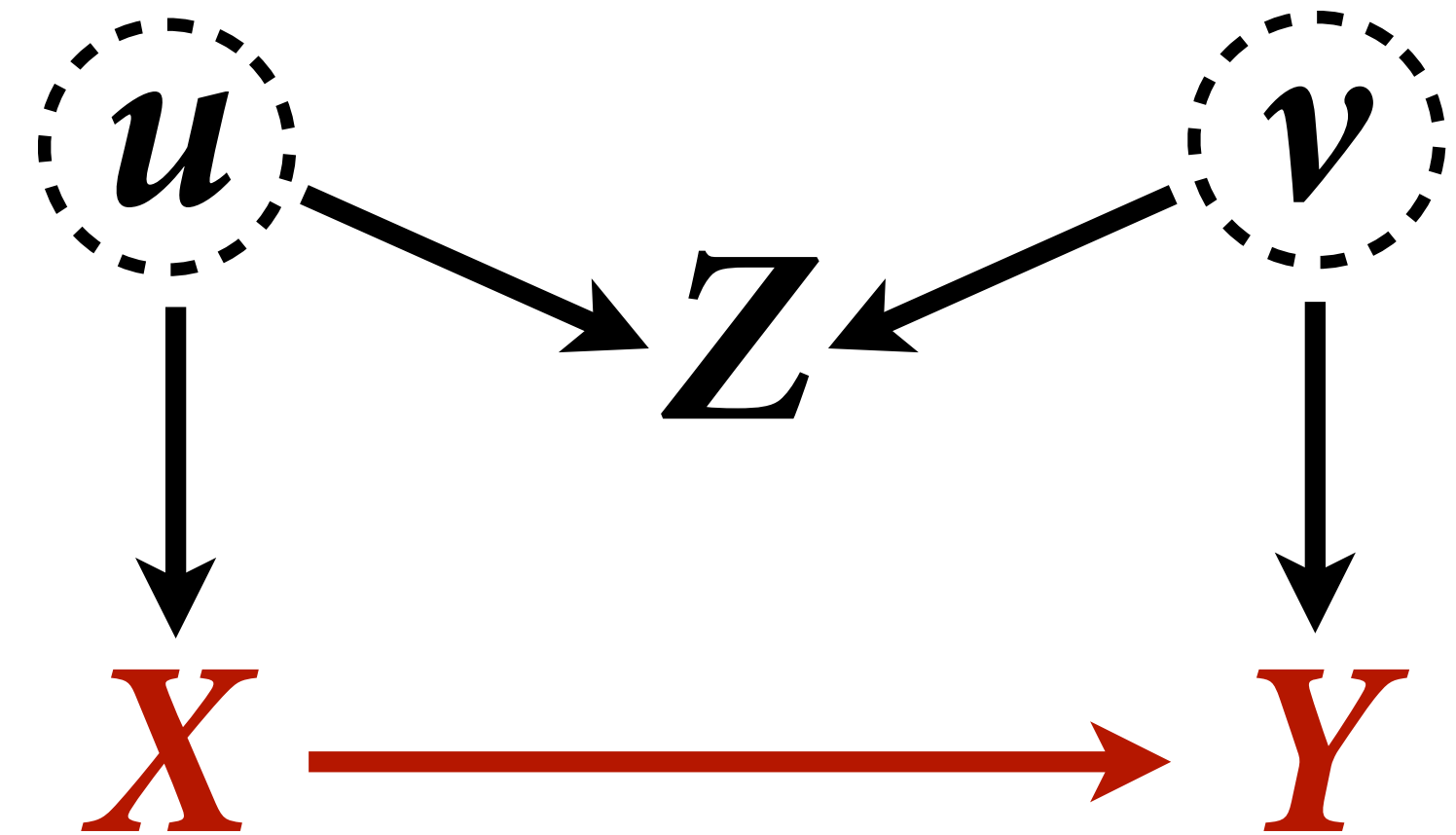
$X \leftarrow u \rightarrow \textcircled{Z} \leftarrow v \rightarrow Y$
backdoor & closed



(1) List the paths (2) Find backdoors (3) Close backdoors

$X \rightarrow Y$
frontdoor & open

$X \leftarrow u \rightarrow Z \leftarrow v \rightarrow Y$
backdoor & closed

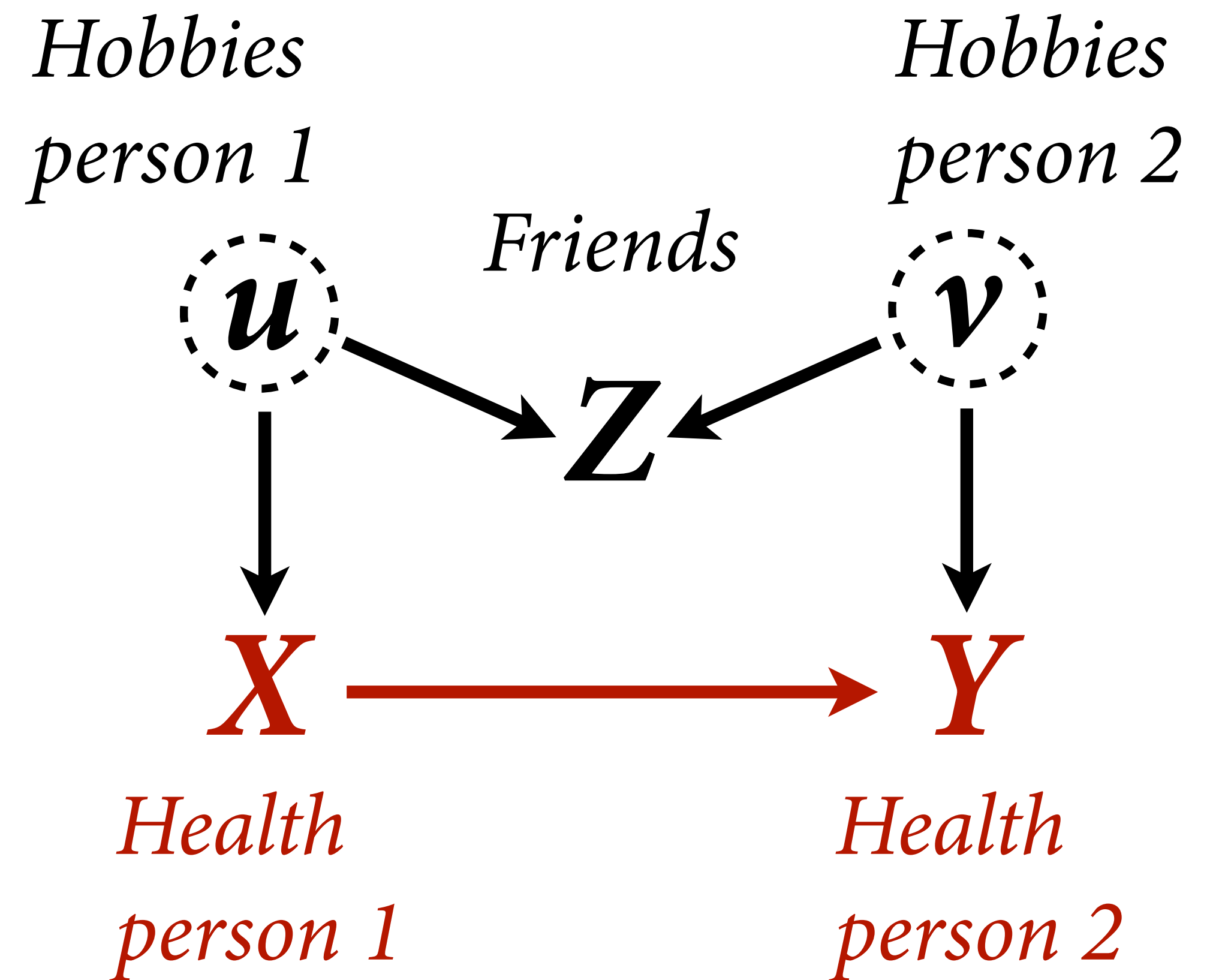


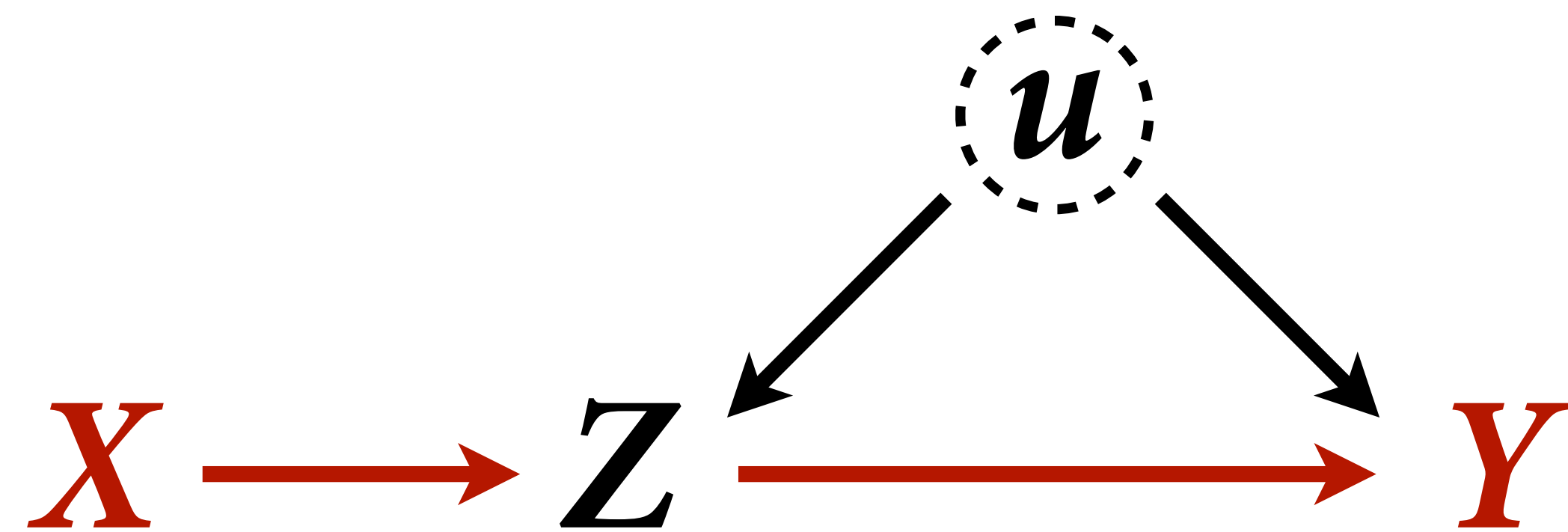
What happens if you stratify by Z ?

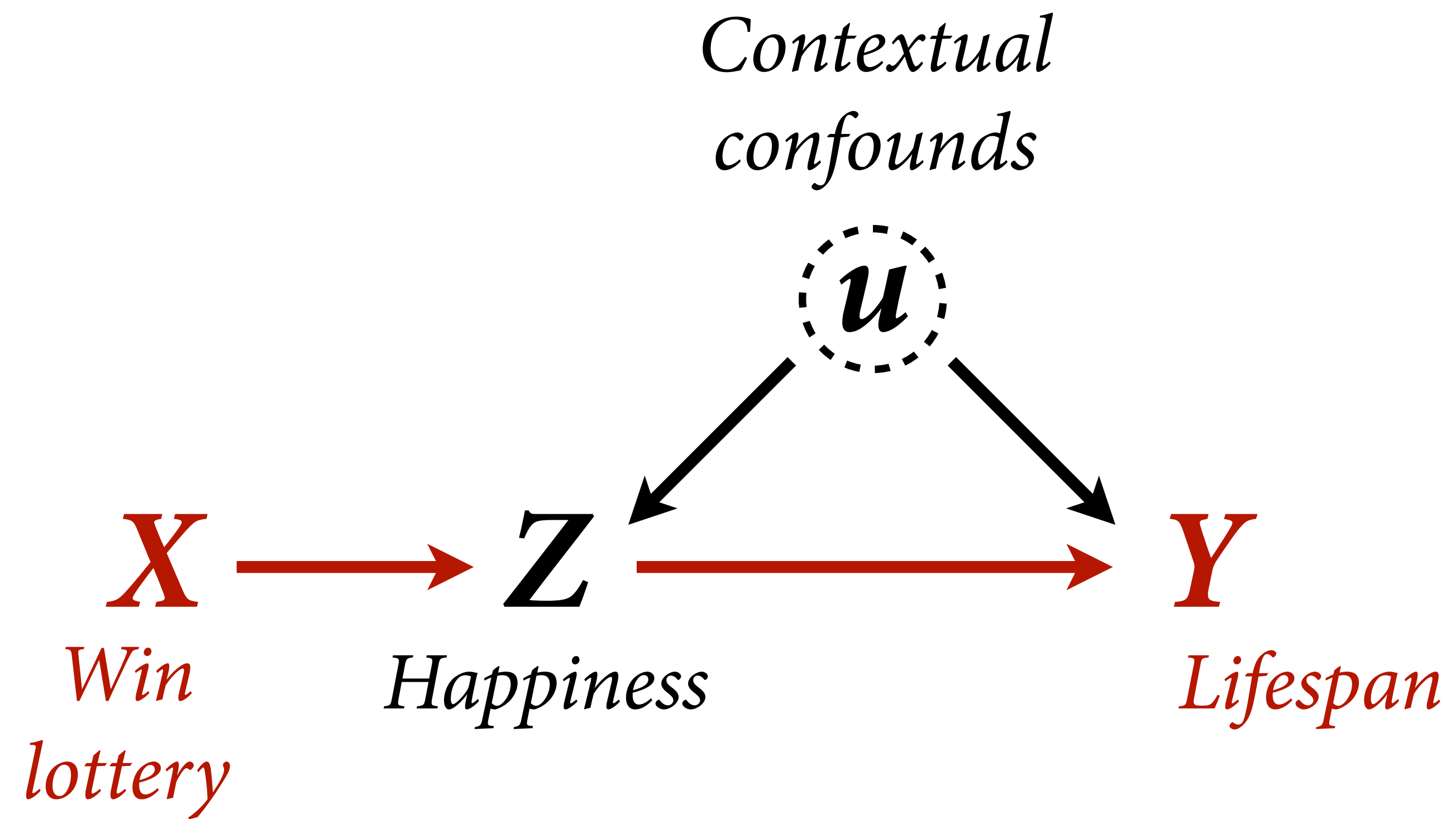
Opens the backdoor path

Z could be a **pre-treatment** variable

Not safe to always control pre-treatment measurements



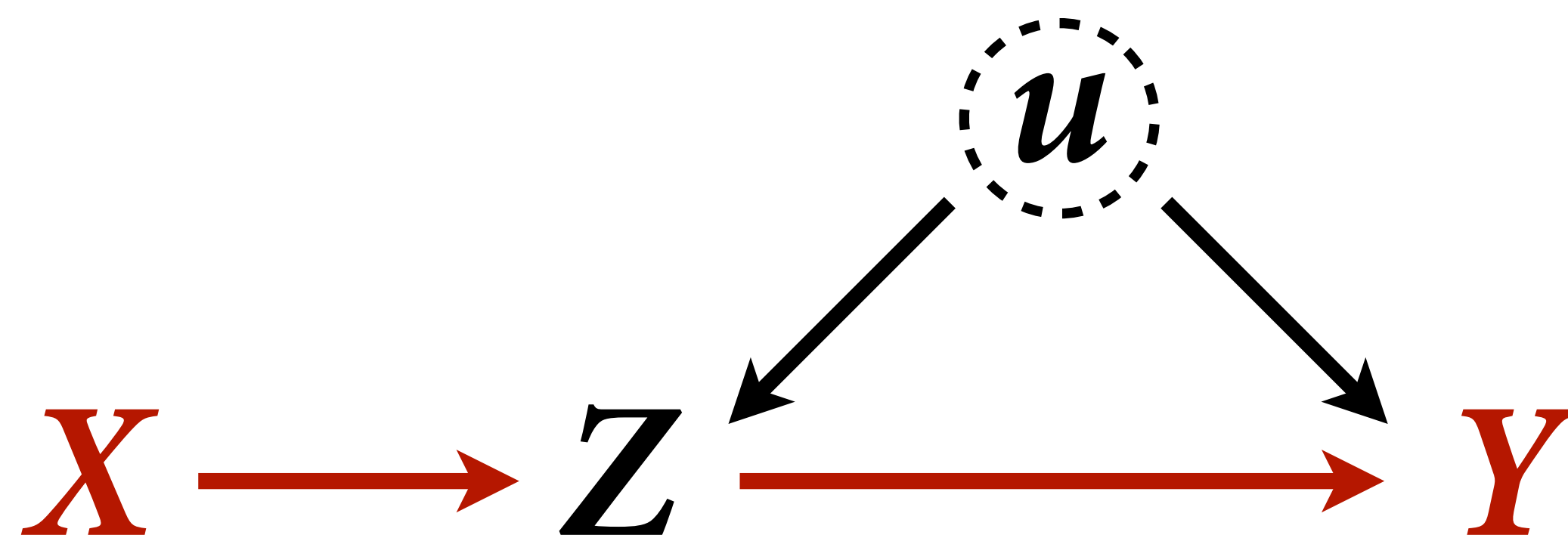




$$X \rightarrow Z \rightarrow Y$$

$$X \rightarrow Z \leftarrow u \rightarrow Y$$

No backdoor, no need
to control for Z



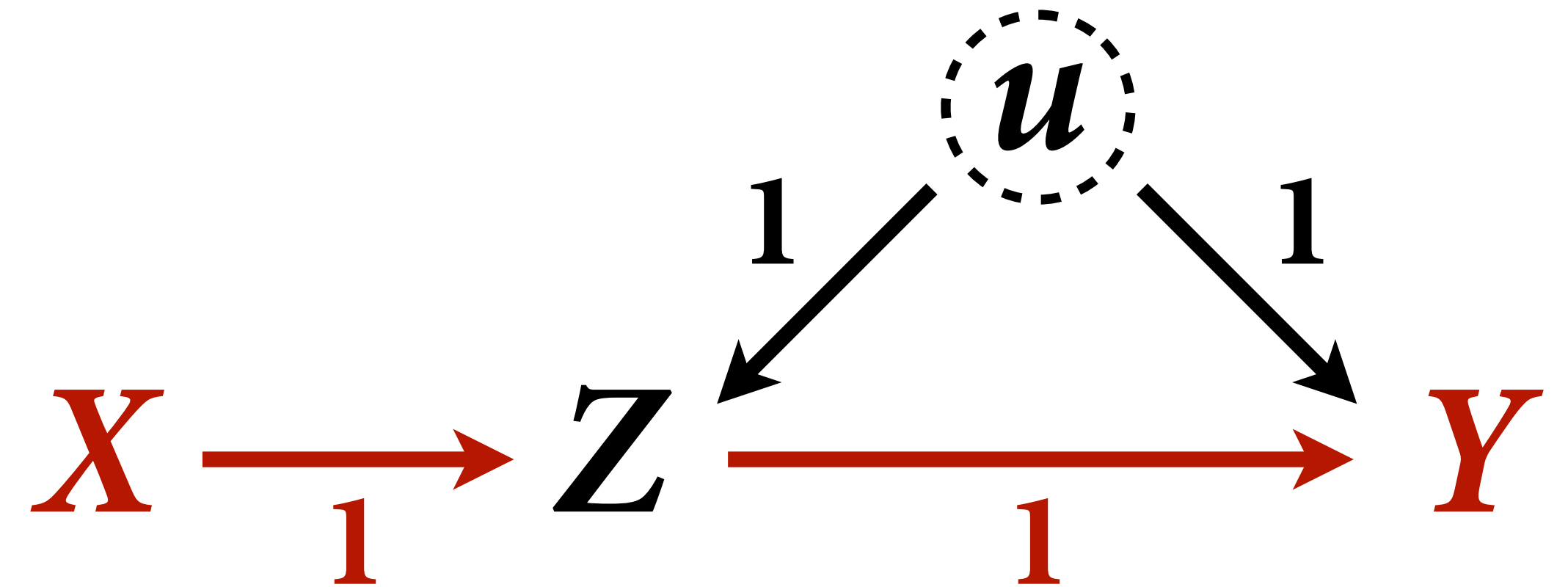
```

f <- function(n=100,bXZ=1,bZY=1) {
  X <- rnorm(n)
  u <- rnorm(n)
  Z <- rnorm(n, bXZ*X + u)
  Y <- rnorm(n, bZY*Z + u )
  bX <- coef( lm(Y ~ X) )['X']
  bXZ <- coef( lm(Y ~ X + Z) )['X']
  return( c(bX,bXZ) )
}

sim <- mcreplicate( 1e4 , f() , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )

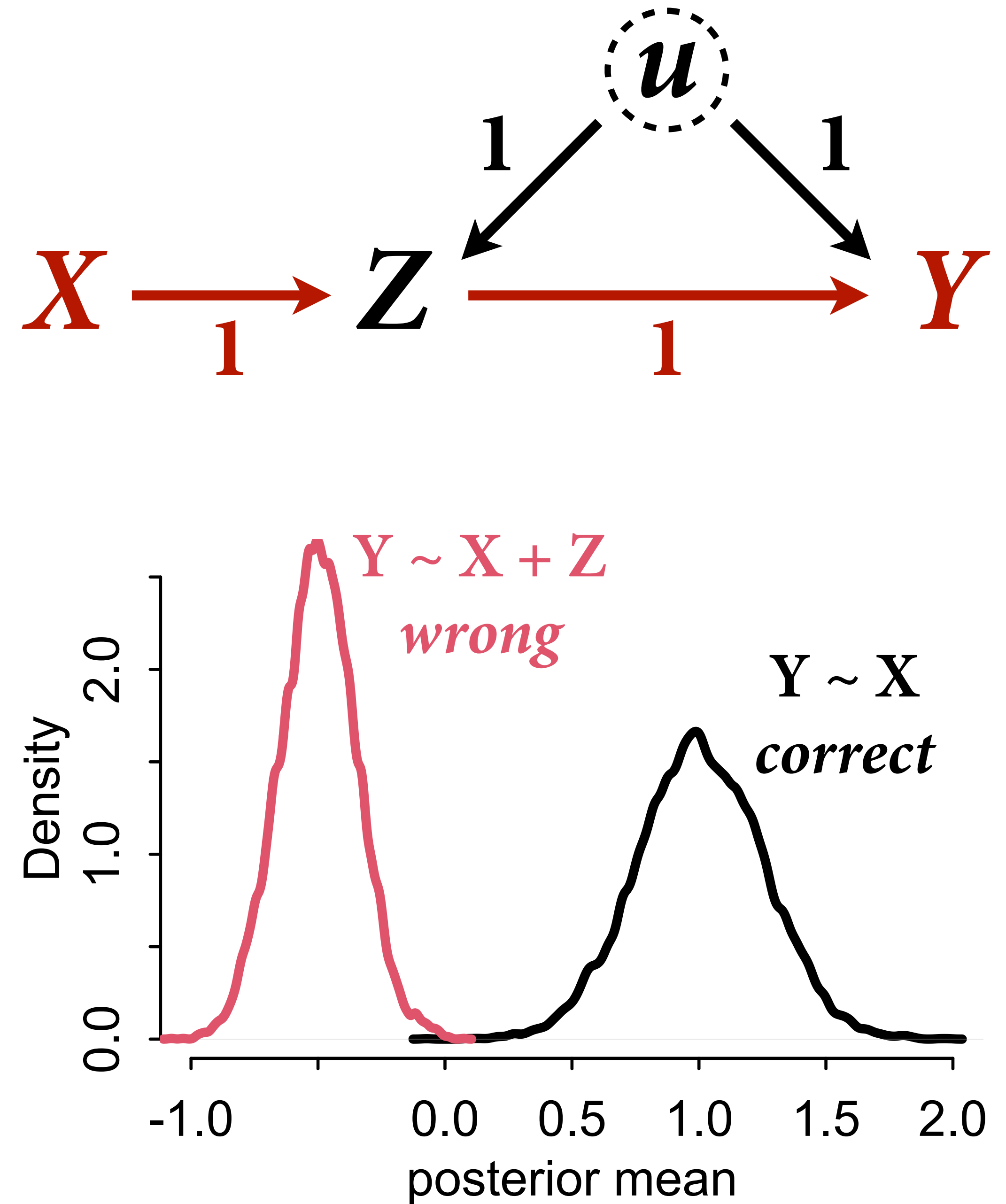
```



```
f <- function(n=100,bXZ=1,bZY=1) {
  X <- rnorm(n)
  u <- rnorm(n)
  Z <- rnorm(n, bXZ*X + u)
  Y <- rnorm(n, bZY*Z + u )
  bX <- coef( lm(Y ~ X) )['X']
  bXZ <- coef( lm(Y ~ X + Z) )['X']
  return( c(bX,bXZ) )
}

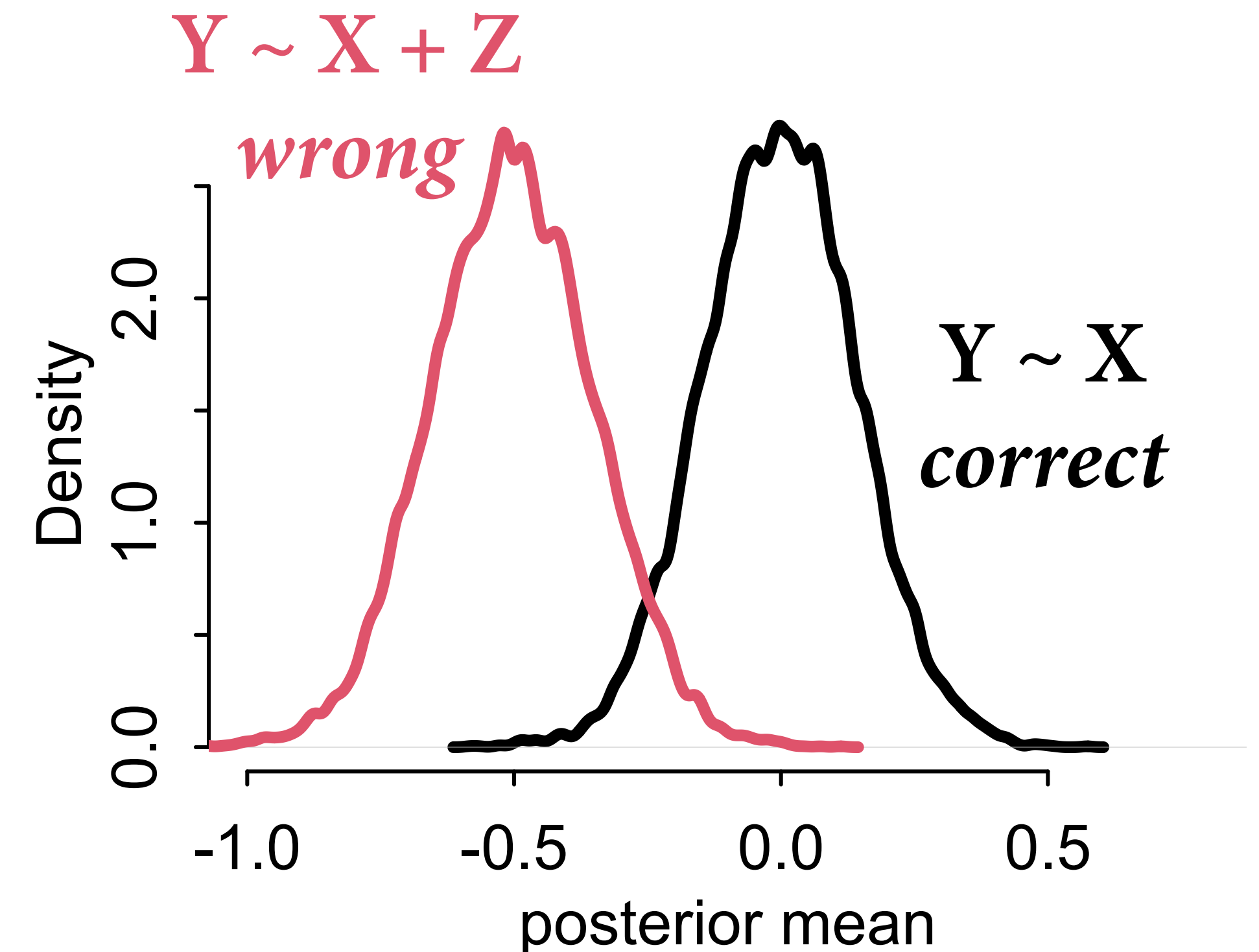
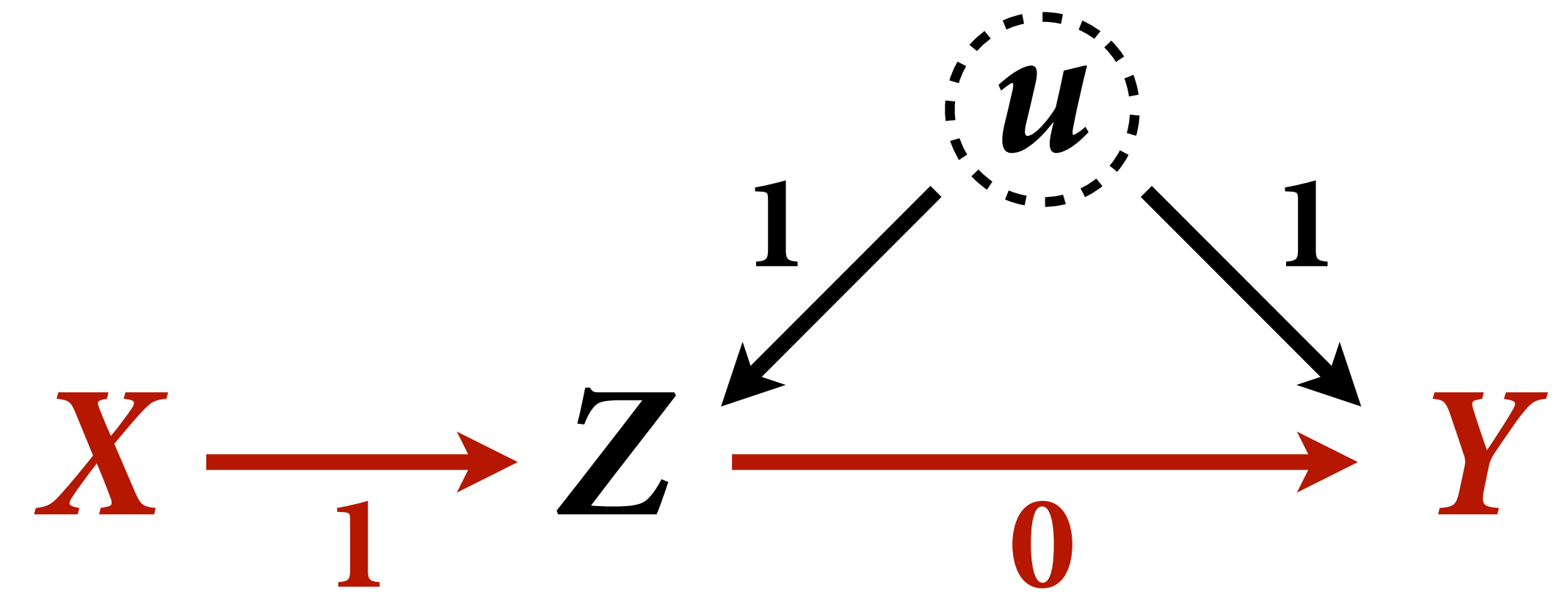
sim <- mcreplicate( 1e4 , f() , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



Change bZY to zero

```
f <- function(n=100,bXZ=1,bZY=1) {  
  X <- rnorm(n)  
  u <- rnorm(n)  
  Z <- rnorm(n, bXZ*X + u)  
  Y <- rnorm(n, bZY*Z + u )  
  bX <- coef( lm(Y ~ X) )['X']  
  bXZ <- coef( lm(Y ~ X + Z) )['X']  
  return( c(bX,bXZ) )  
}  
  
sim <- mcreplicate( 1e4 , f(bZY=0) , mc.cores=8 )  
  
dens( sim[1,] , lwd=3 , xlab="posterior mean" )  
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



$$X \rightarrow Z \rightarrow Y$$

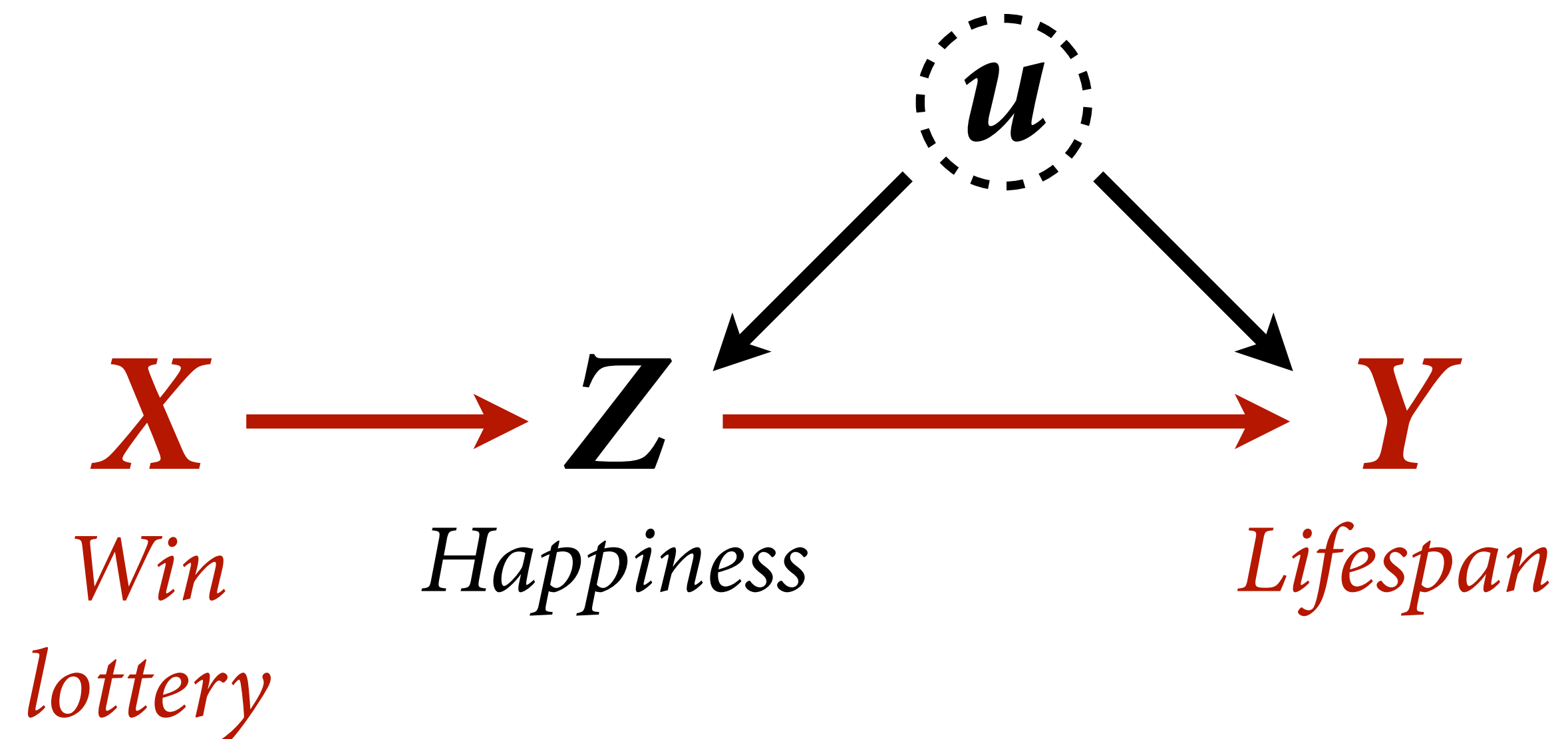
$$X \rightarrow Z \leftarrow u \rightarrow Y$$

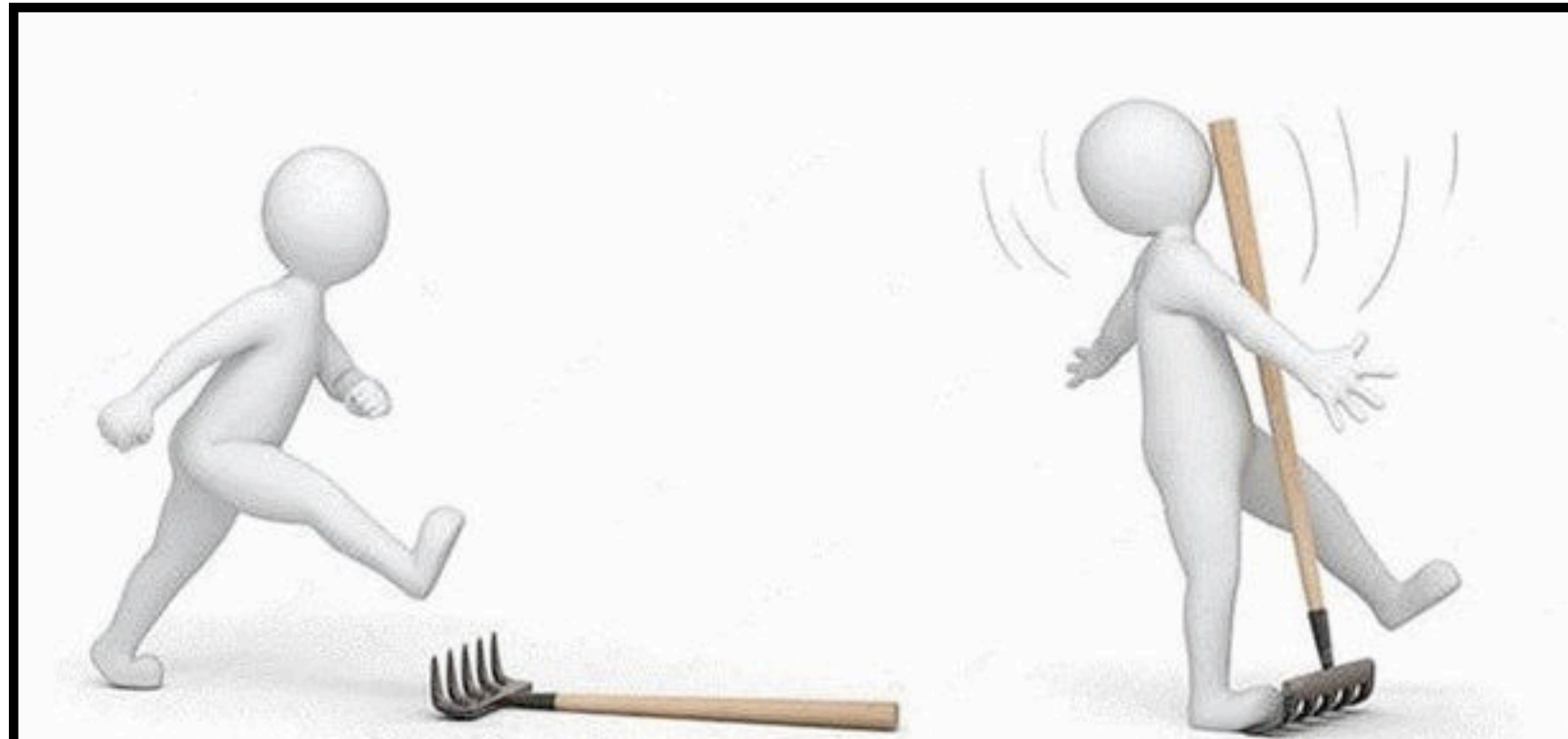
No backdoor, no need
to control for Z

Controlling for Z biases
treatment estimate X

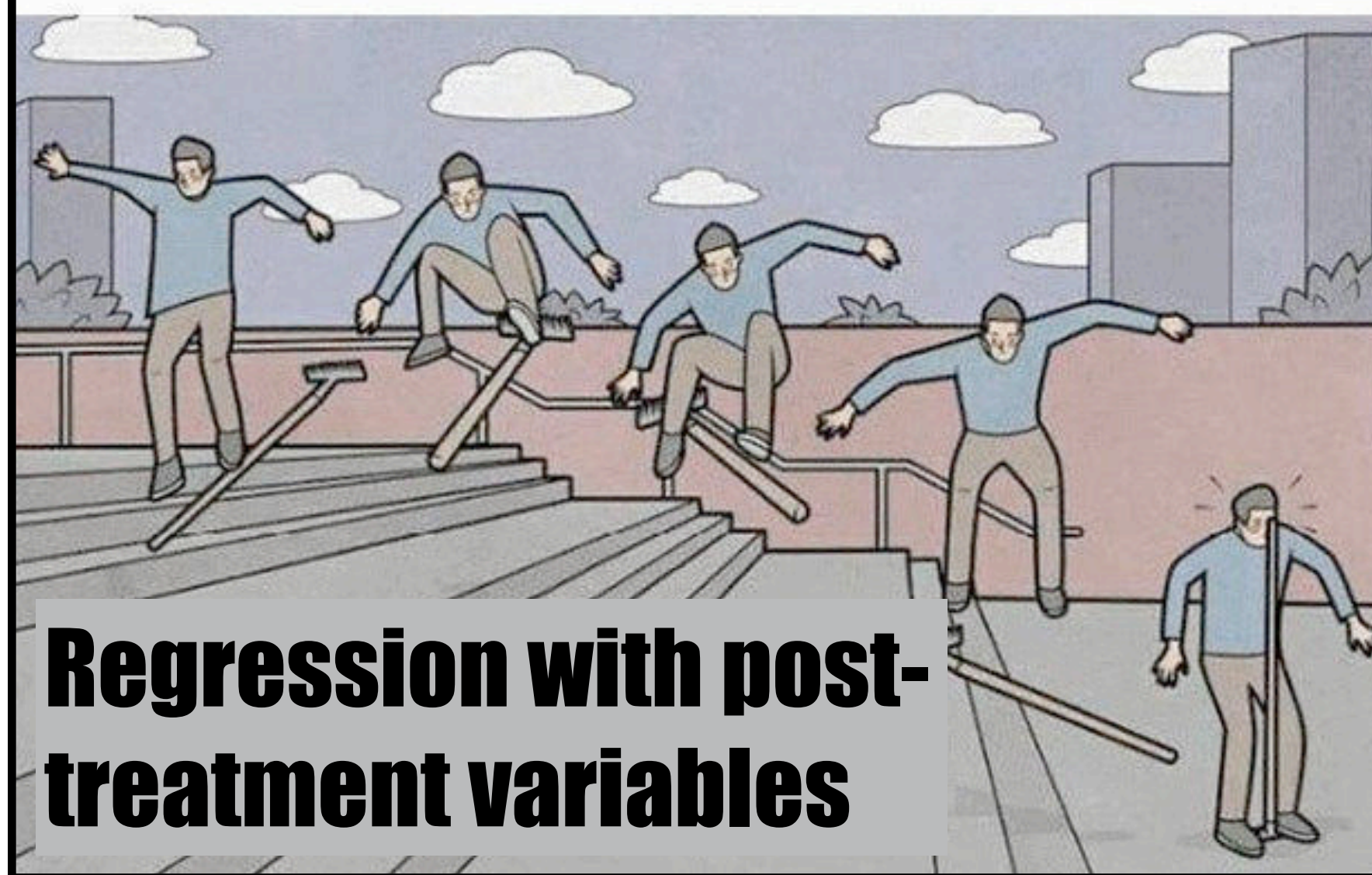
Controlling for Z opens biasing
path through u

Can estimate effect of X ; Cannot
estimate mediation effect Z

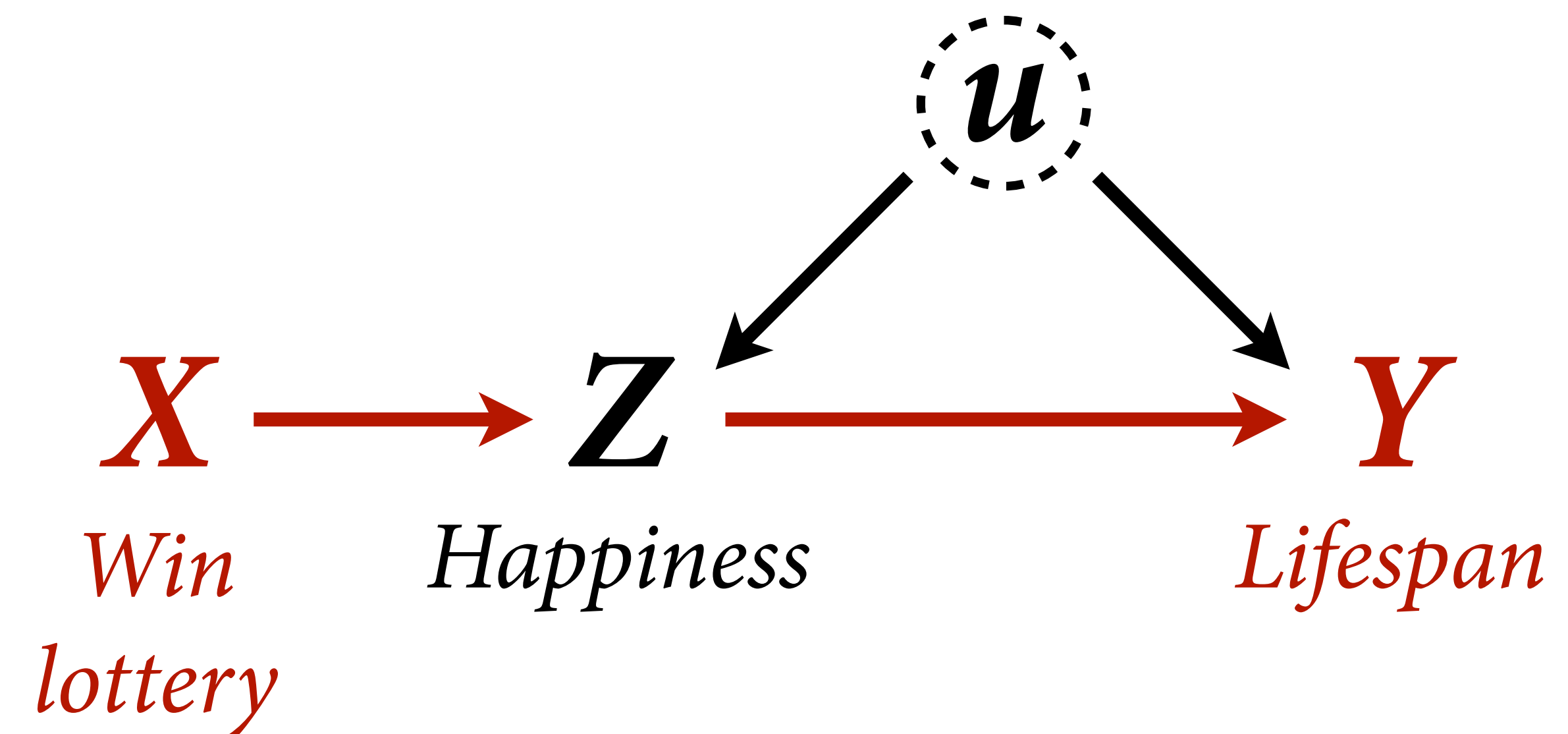




Regression with confounds

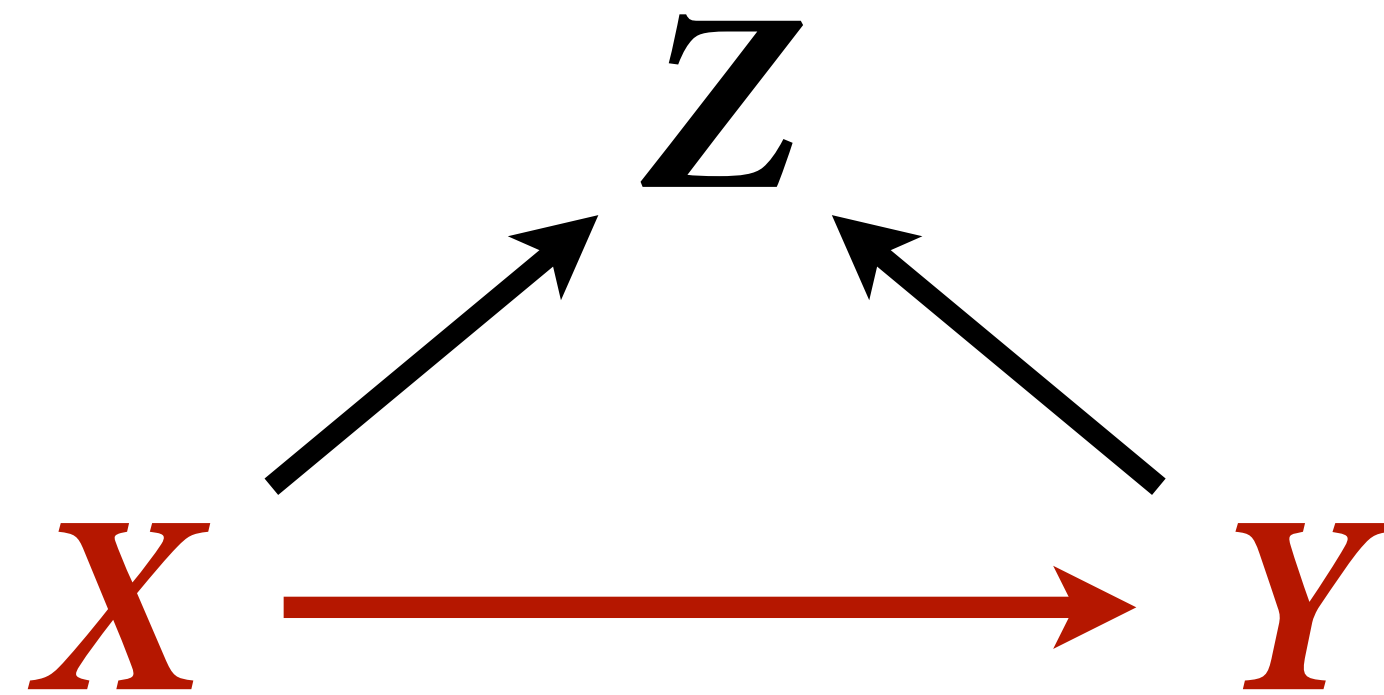


Regression with post-treatment variables

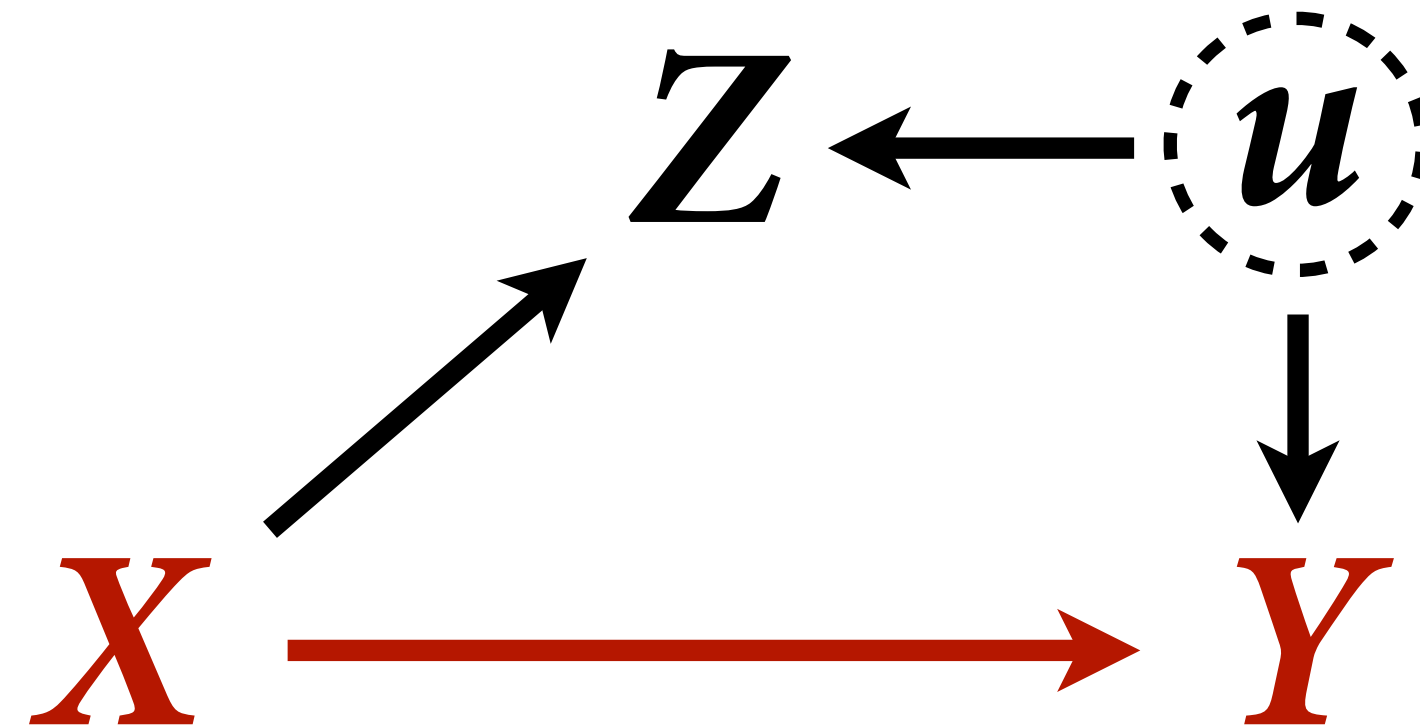


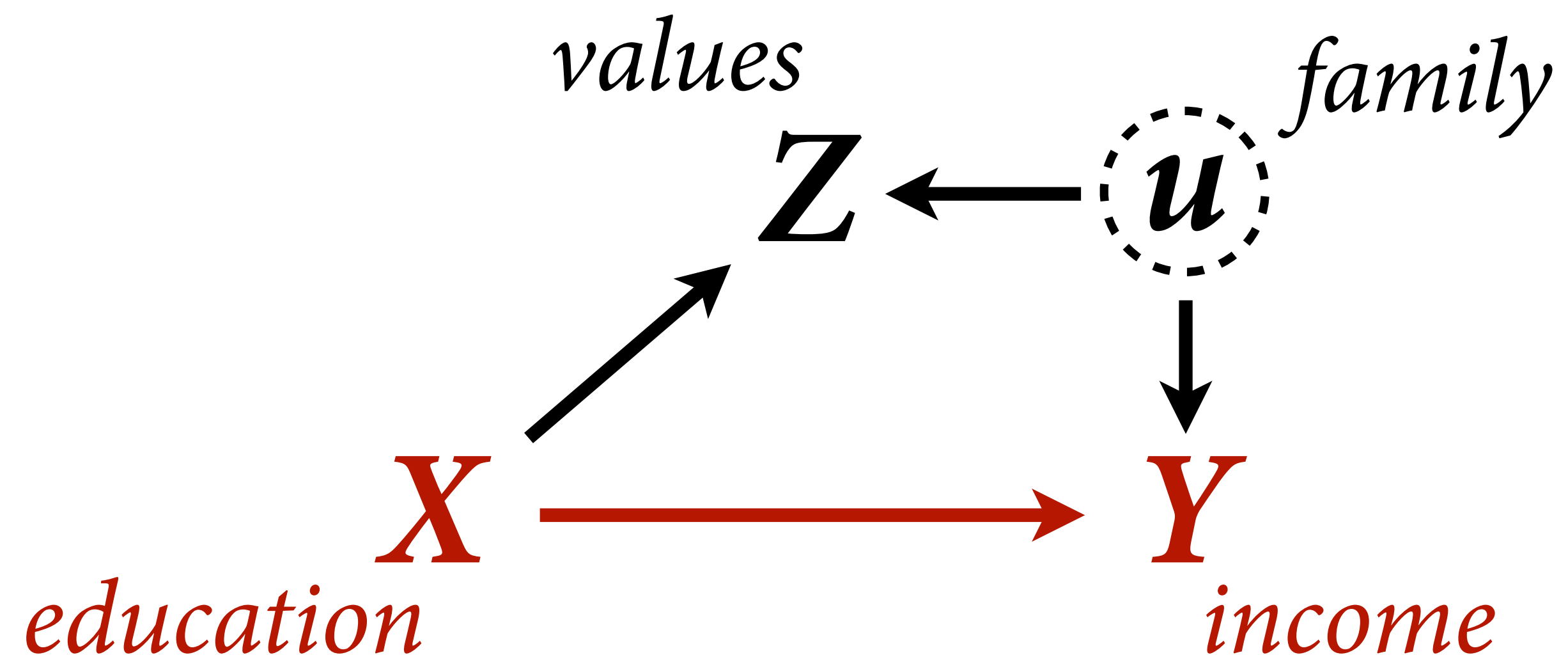
PAUSE

Do not touch the collider!

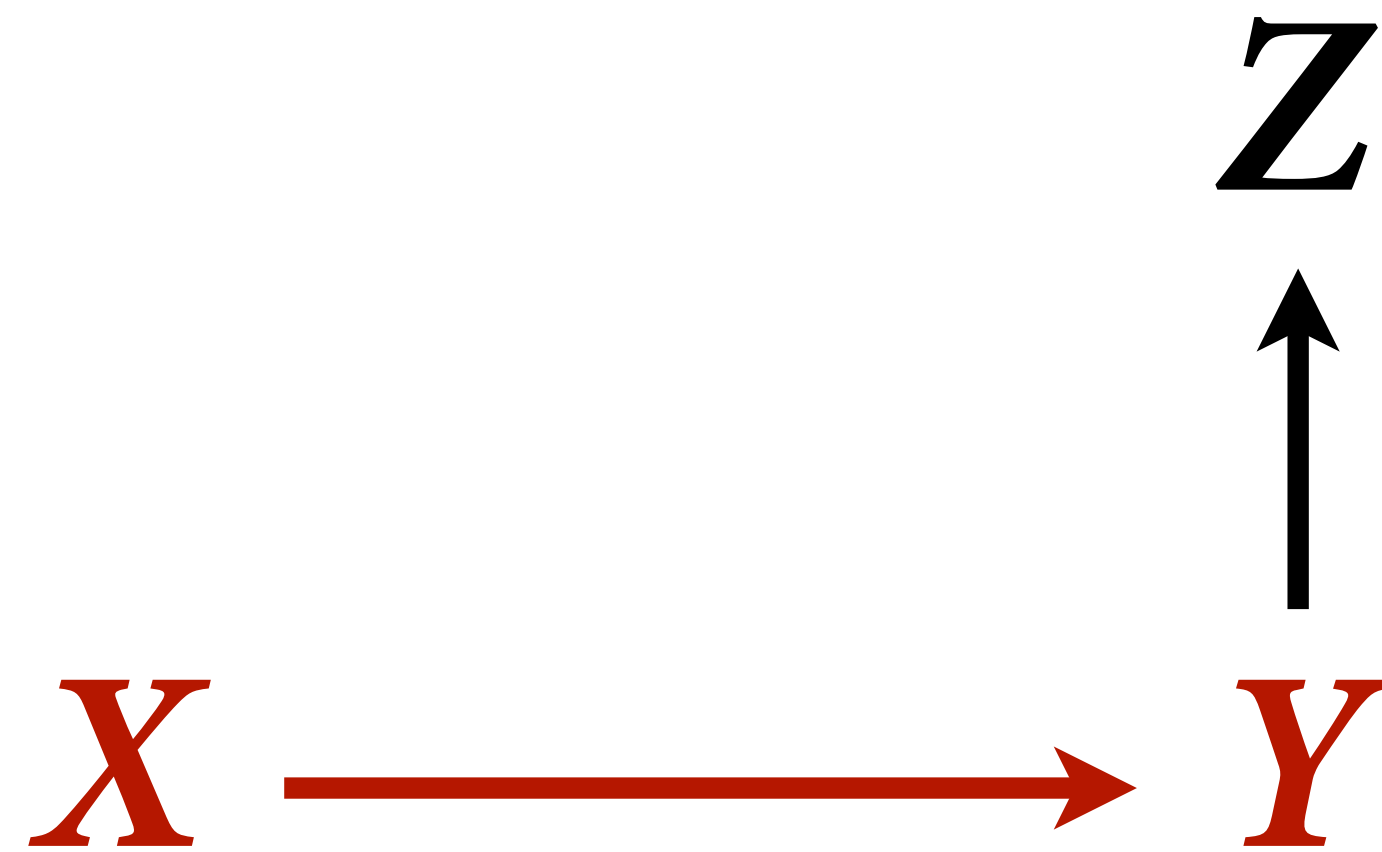


Colliders not always so obvious

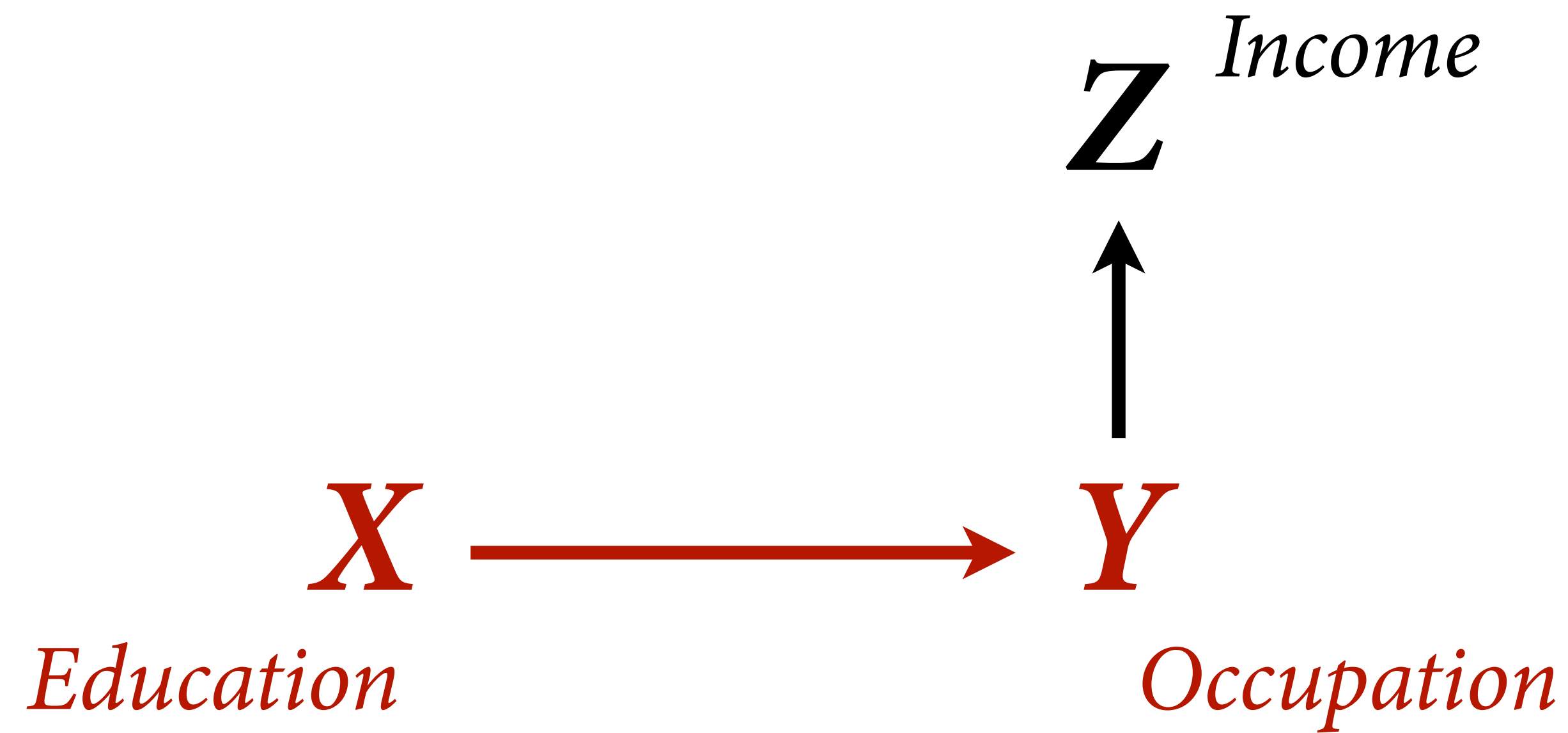




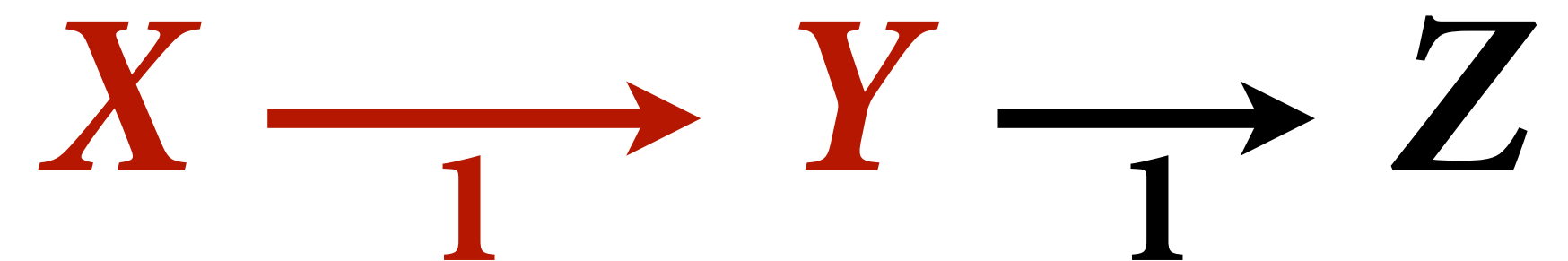
Case-control bias (selection on outcome)



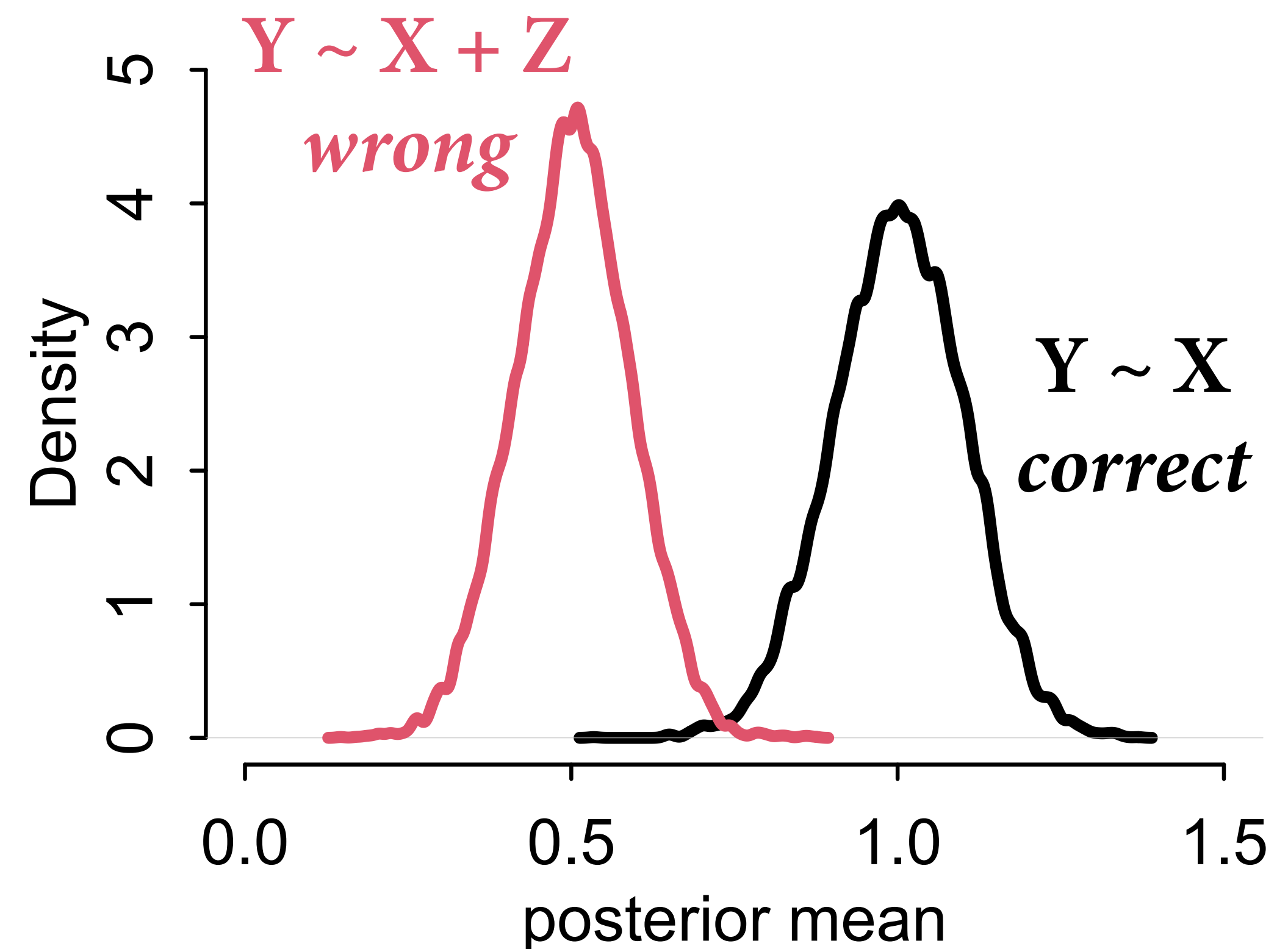
Case-control bias (selection on outcome)



Case-control bias (selection on outcome)



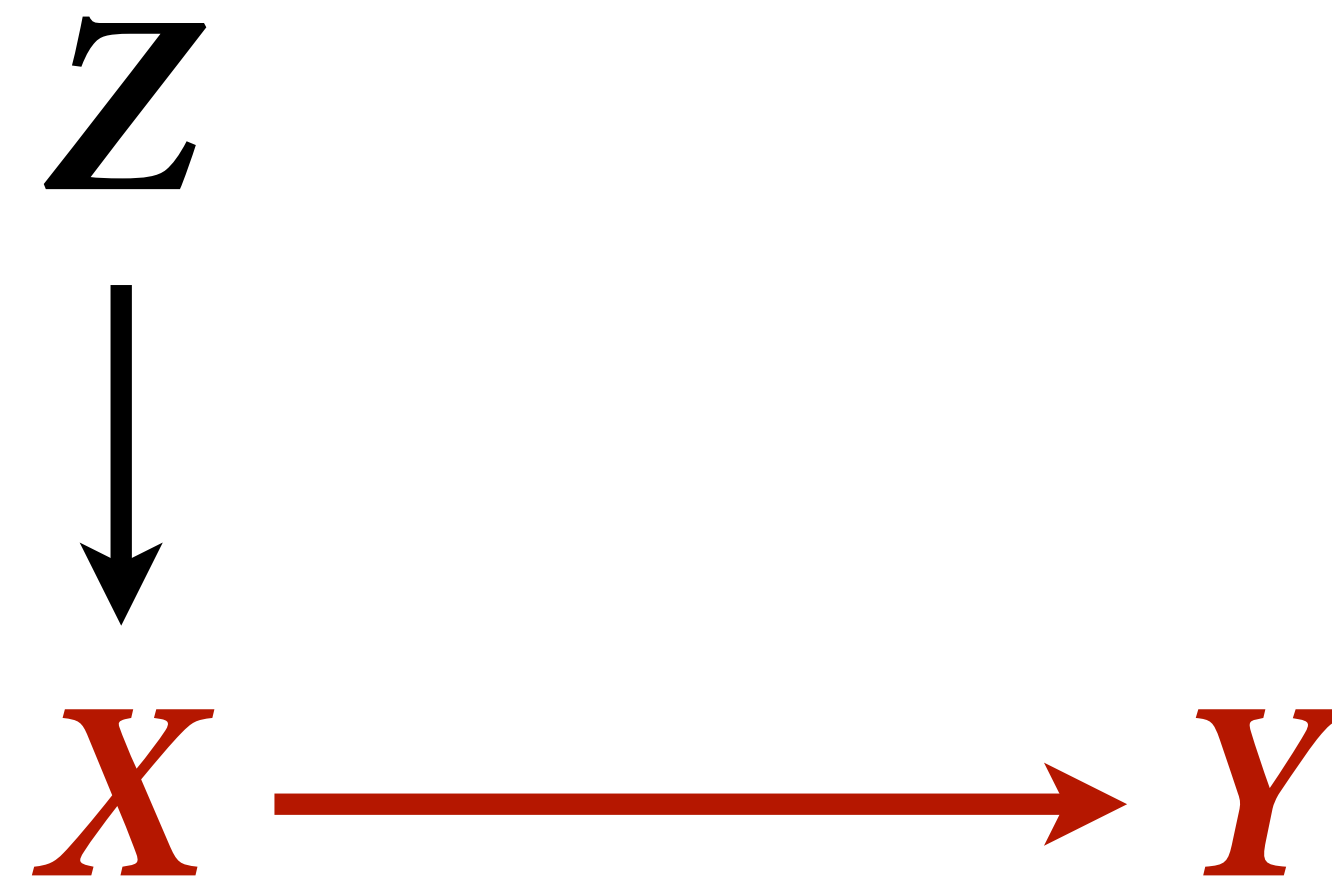
```
f <- function(n=100,bXY=1,bYZ=1) {  
  X <- rnorm(n)  
  Y <- rnorm(n, bXY*X )  
  Z <- rnorm(n, bYZ*Y )  
  bX <- coef( lm(Y ~ X) )['X']  
  bXZ <- coef( lm(Y ~ X + Z) )['X']  
  return( c(bX,bXZ) )  
}  
  
sim <- mcreplicate( 1e4 , f() , mc.cores=8 )  
  
dens( sim[1,] , lwd=3 , xlab="posterior mean" )  
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



“Precision parasite”

No backdoors

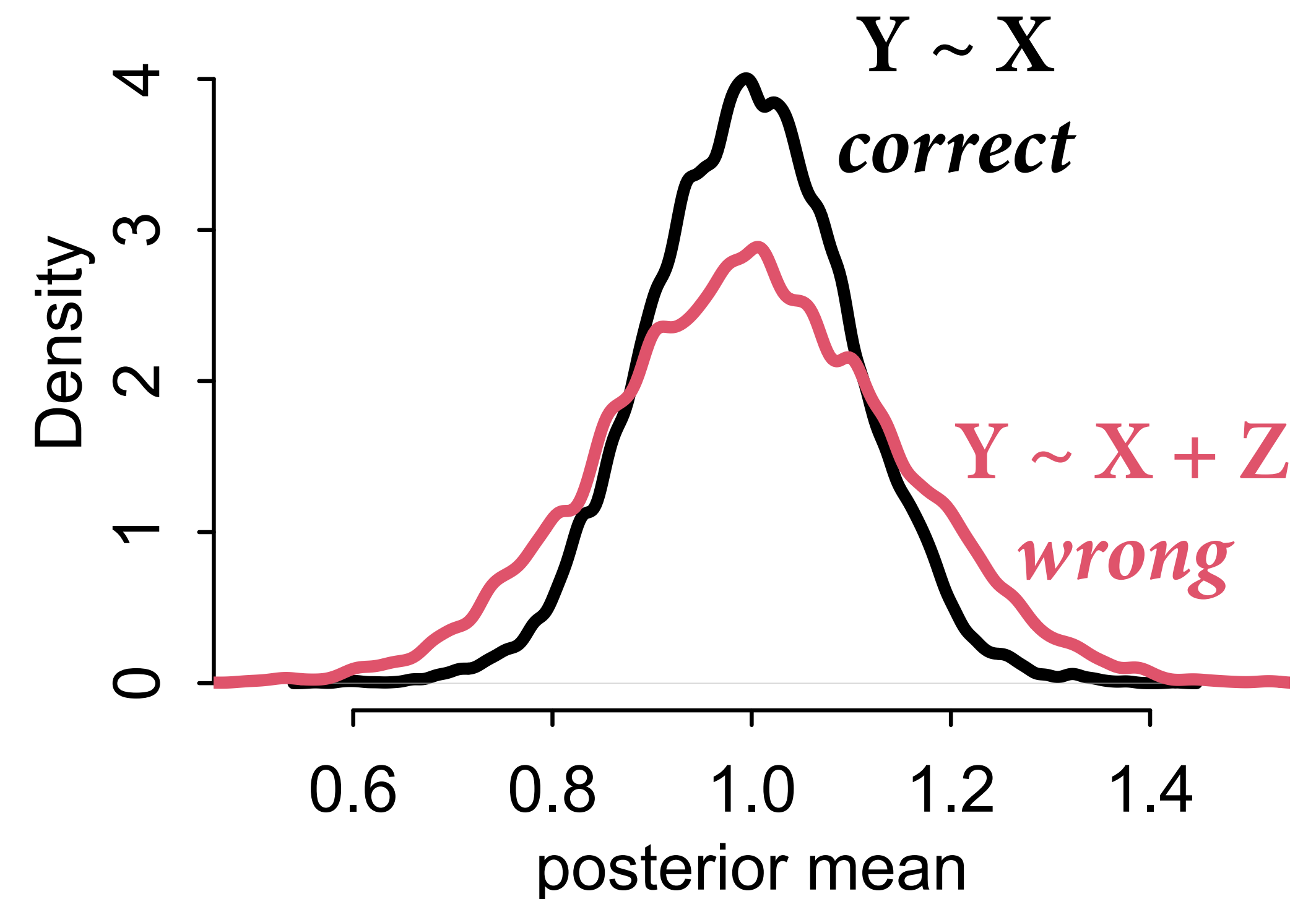
But still not good to
condition on Z



“Precision parasite”



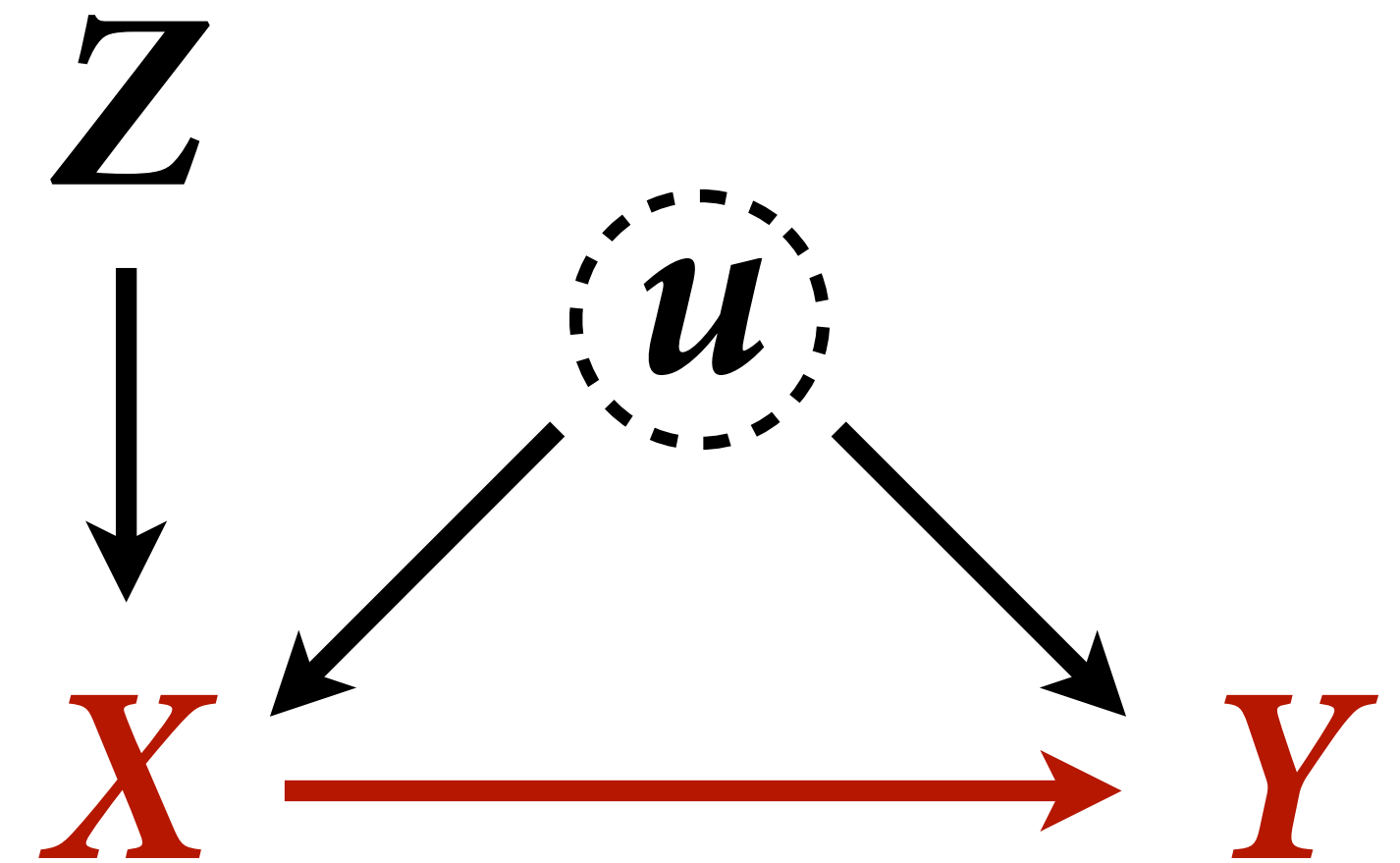
```
f <- function(n=100,bZX=1,bXY=1) {  
  Z <- rnorm(n)  
  X <- rnorm(n, bZX*Z )  
  Y <- rnorm(n, bXY*X )  
  bX <- coef( lm(Y ~ X) )['X']  
  bXZ <- coef( lm(Y ~ X + Z) )['X']  
  return( c(bX,bXZ) )  
}  
  
sim <- mcreplicate( 1e4 , f(n=50) , mc.cores=8 )  
  
dens( sim[1,] , lwd=3 , xlab="posterior mean" )  
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```



“Bias amplification”

X and Y confounded by u

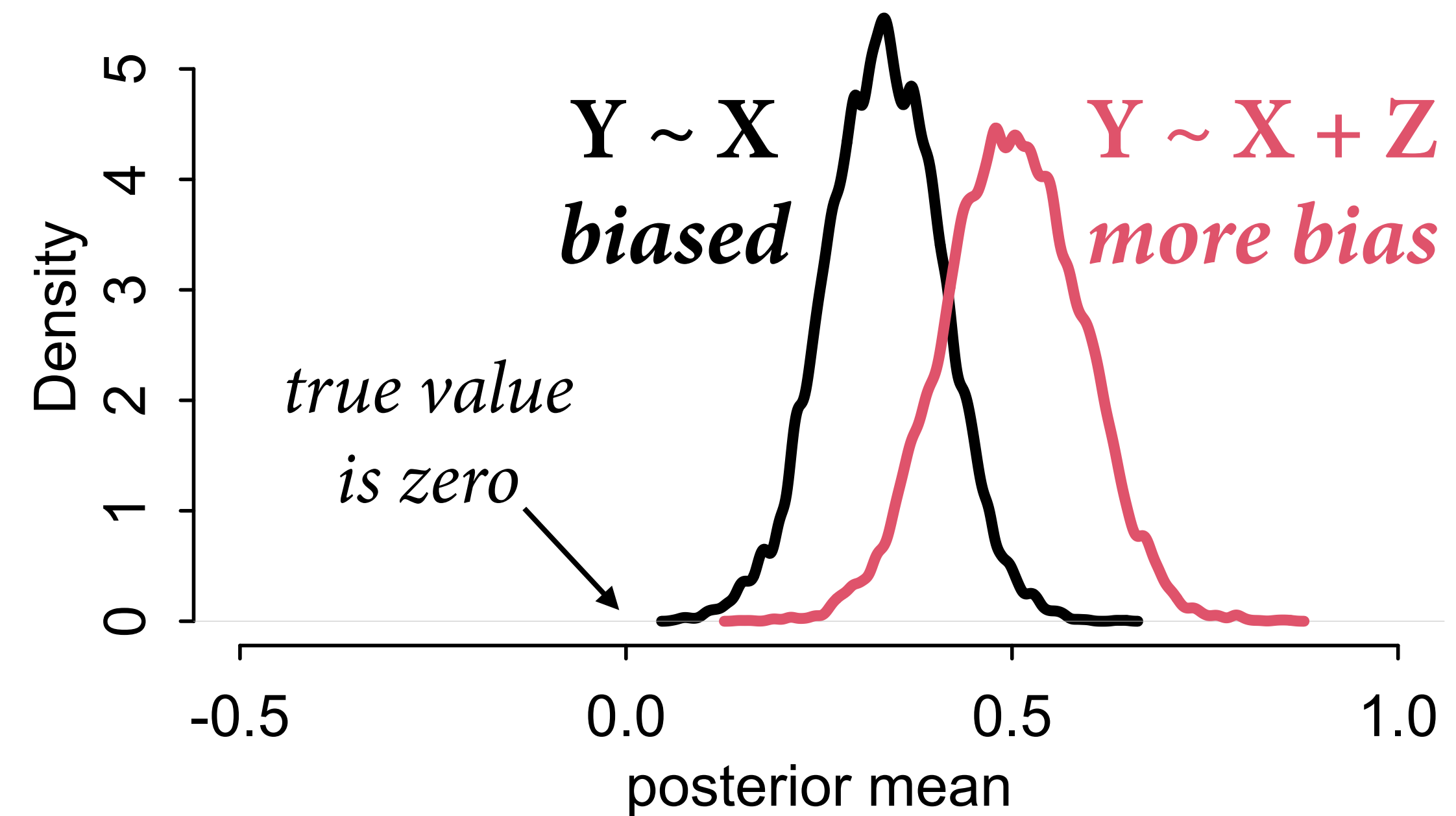
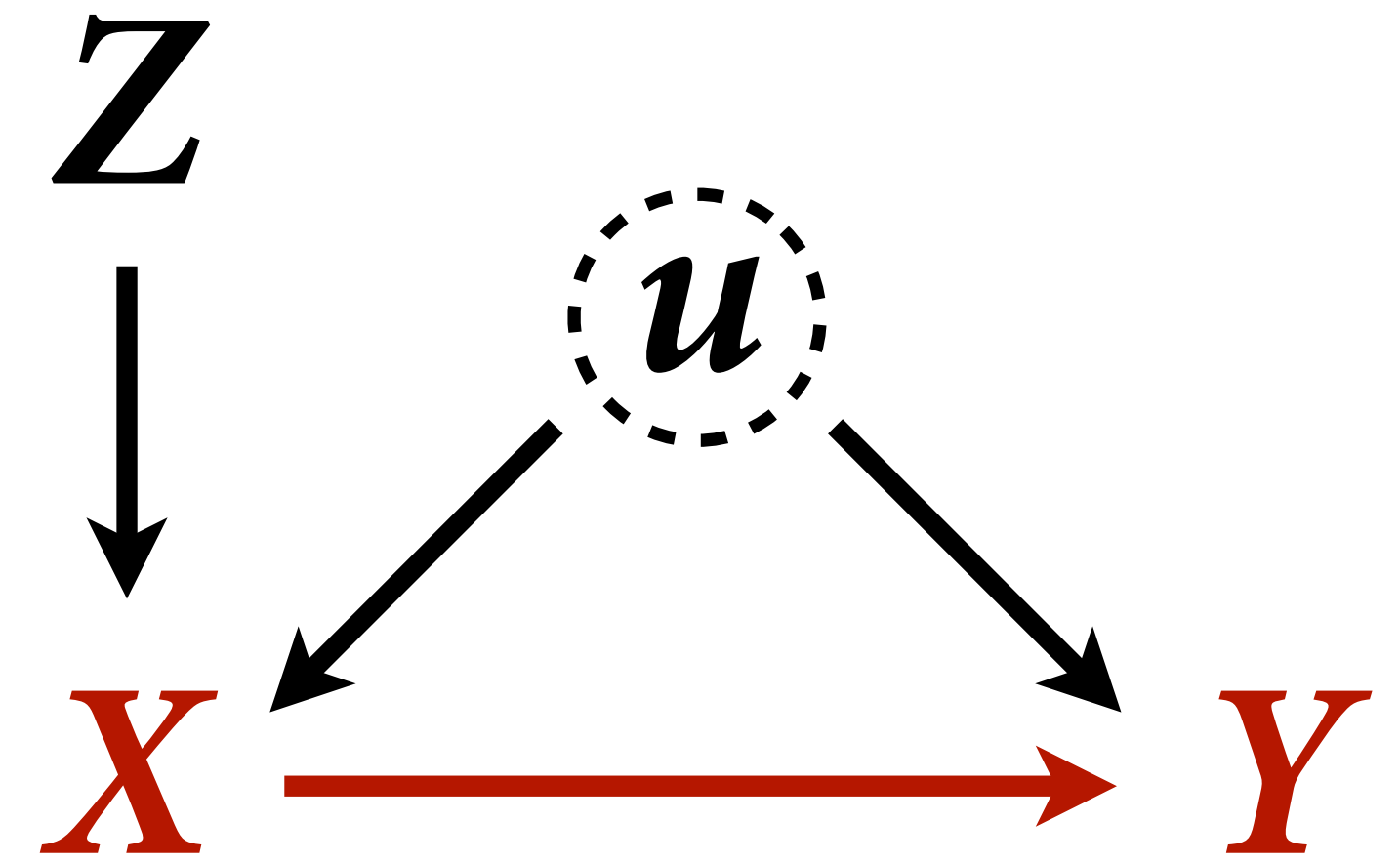
Something **truly awful** happens
when we add Z



```
f <- function(n=100,bZX=1,bXY=1) {
  Z <- rnorm(n)
  u <- rnorm(n)
  X <- rnorm(n, bZX*Z + u )
  Y <- rnorm(n, bXY*X + u )
  bX <- coef( lm(Y ~ X) )['X']
  bXZ <- coef( lm(Y ~ X + Z) )['X']
  return( c(bX,bXZ) )
}

sim <- mcreplicate( 1e4 , f(bXY=0) , mc.cores=8 )

dens( sim[1,] , lwd=3 , xlab="posterior mean" )
dens( sim[2,] , lwd=3 , col=2 , add=TRUE )
```

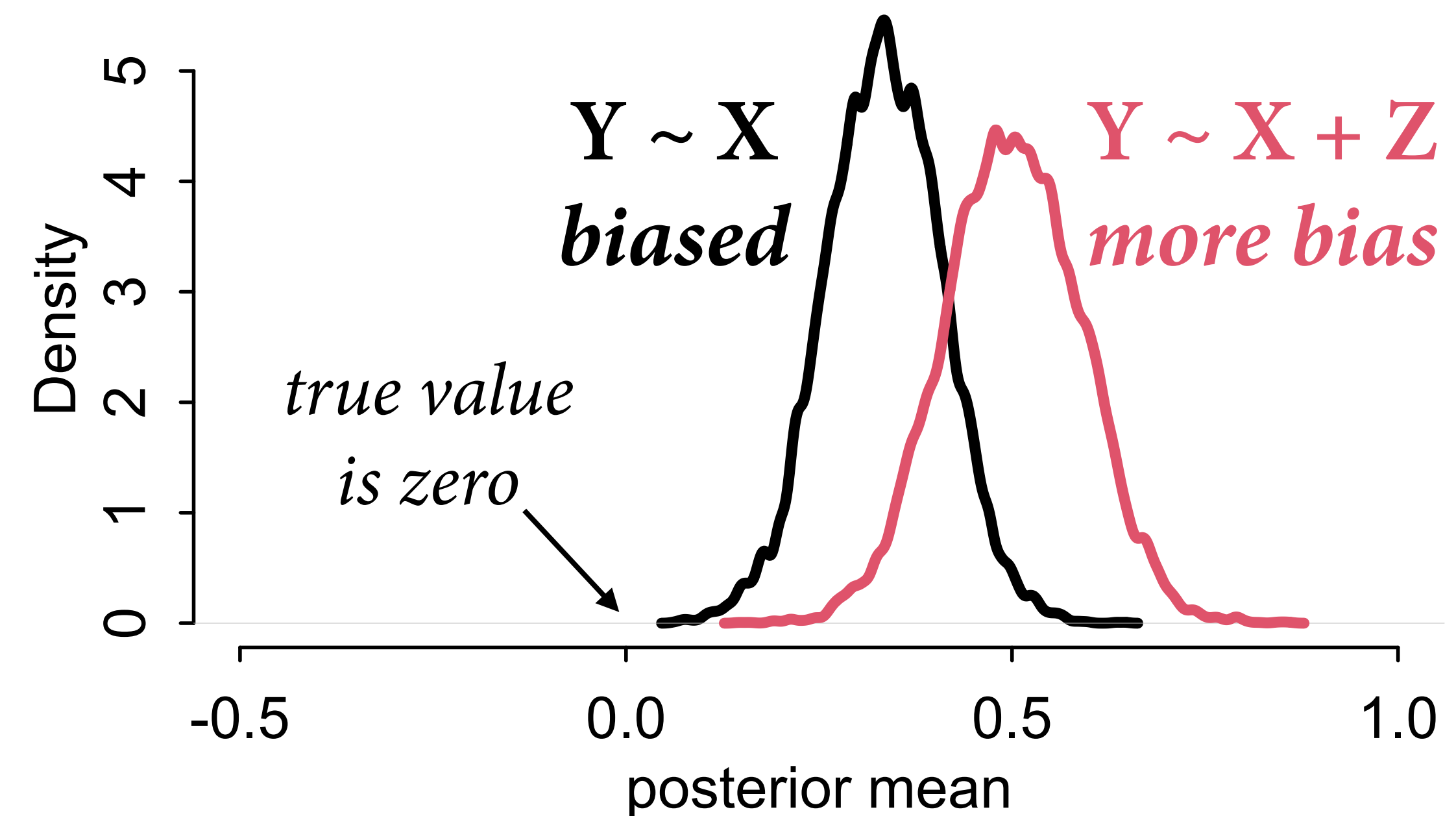
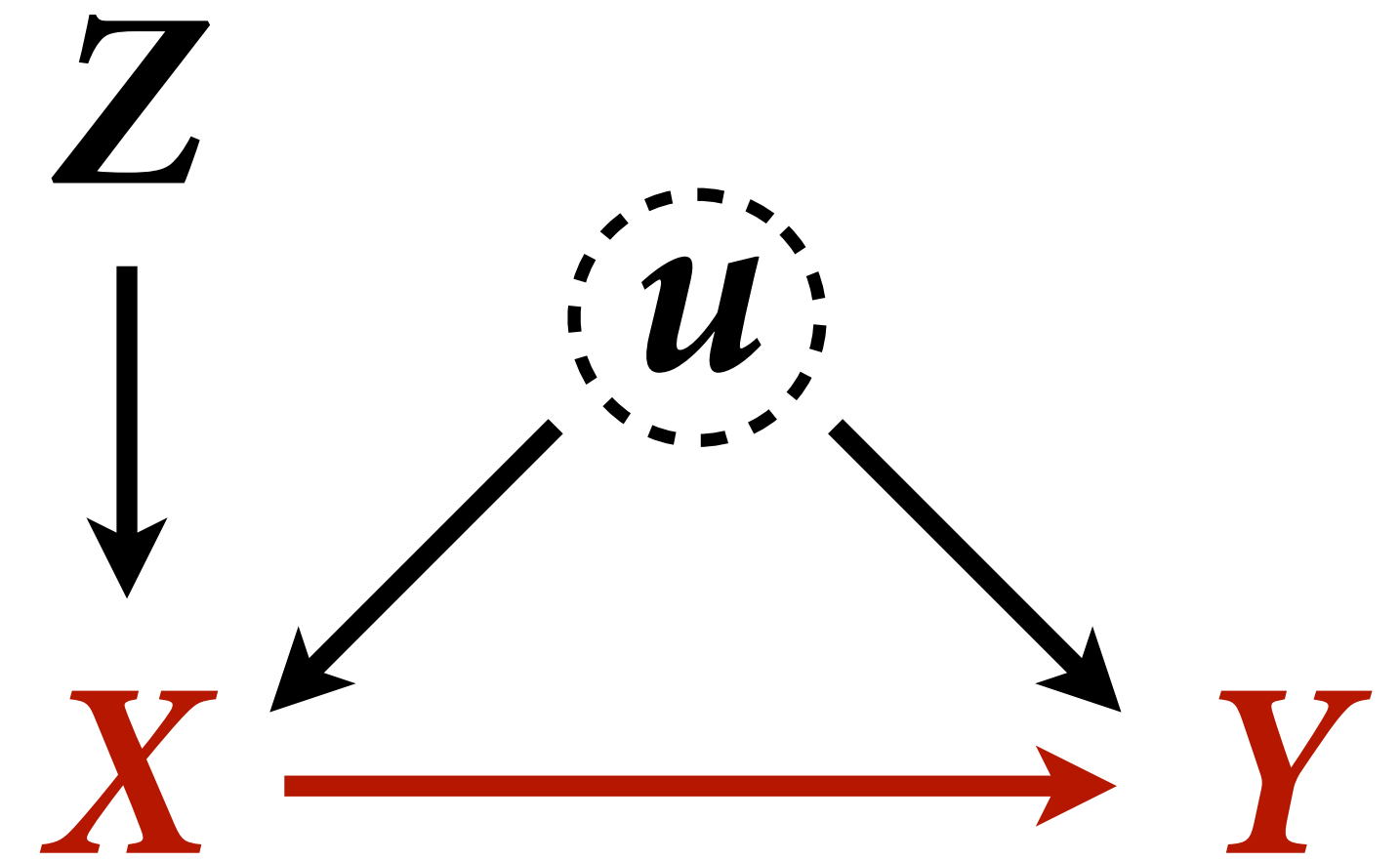


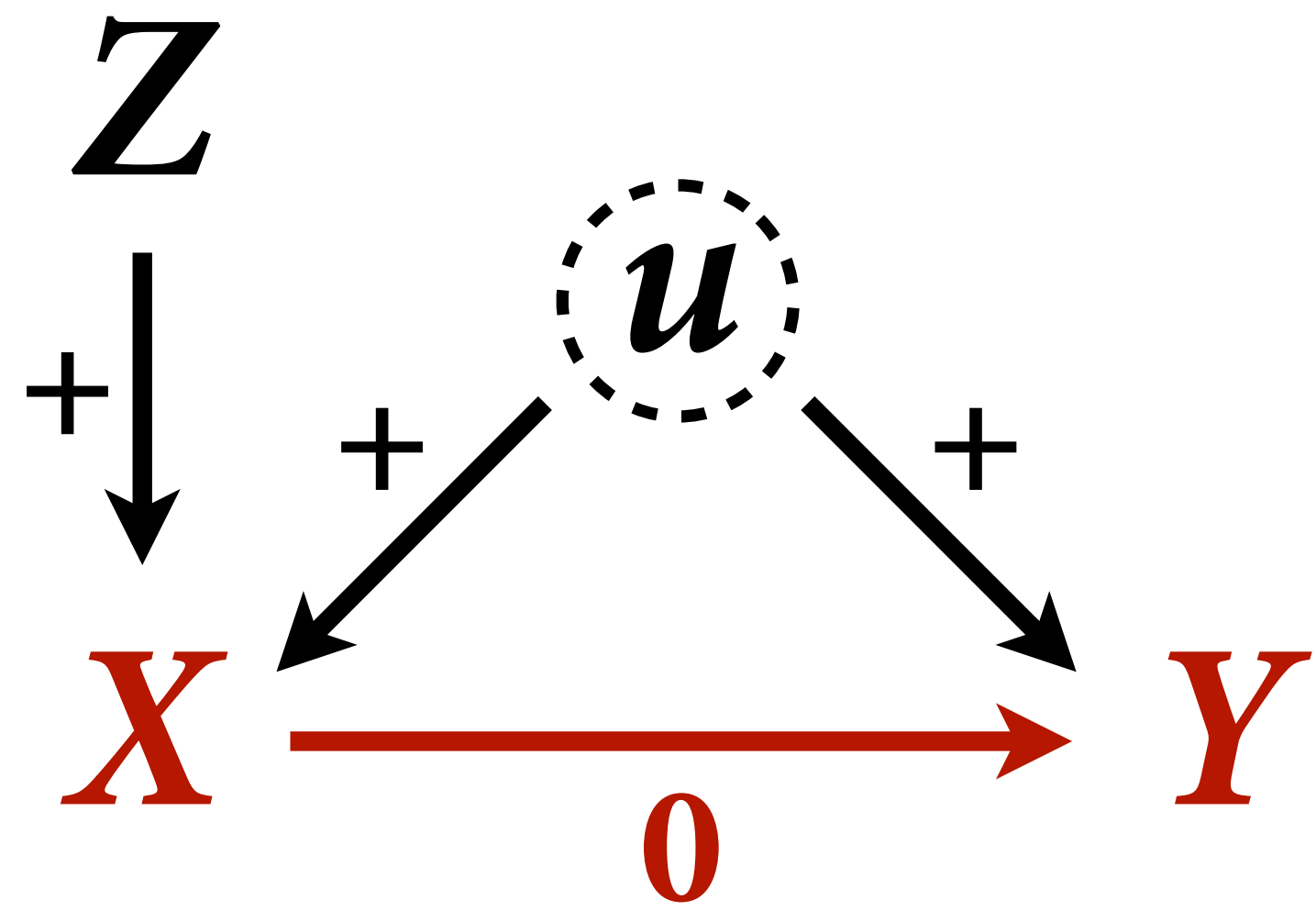
WHY?

Covariation X & Y requires variation in their causes

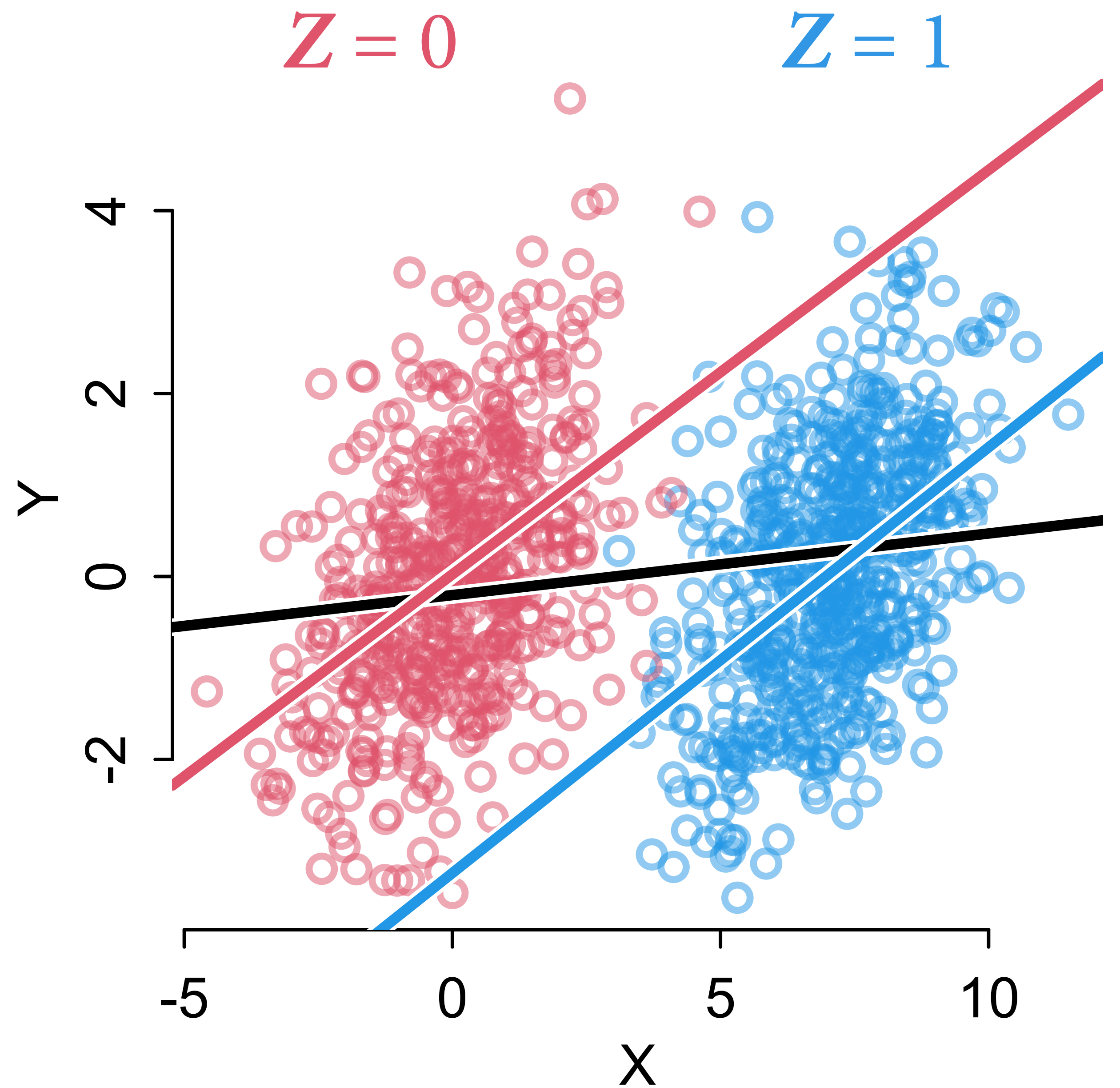
Within each level of Z , less variation in X

Confound u relatively more important within each Z





```
n <- 1000
Z <- rbern(n)
u <- rnorm(n)
X <- rnorm(n, 7*Z + u )
Y <- rnorm(n, 0*X + u )
```

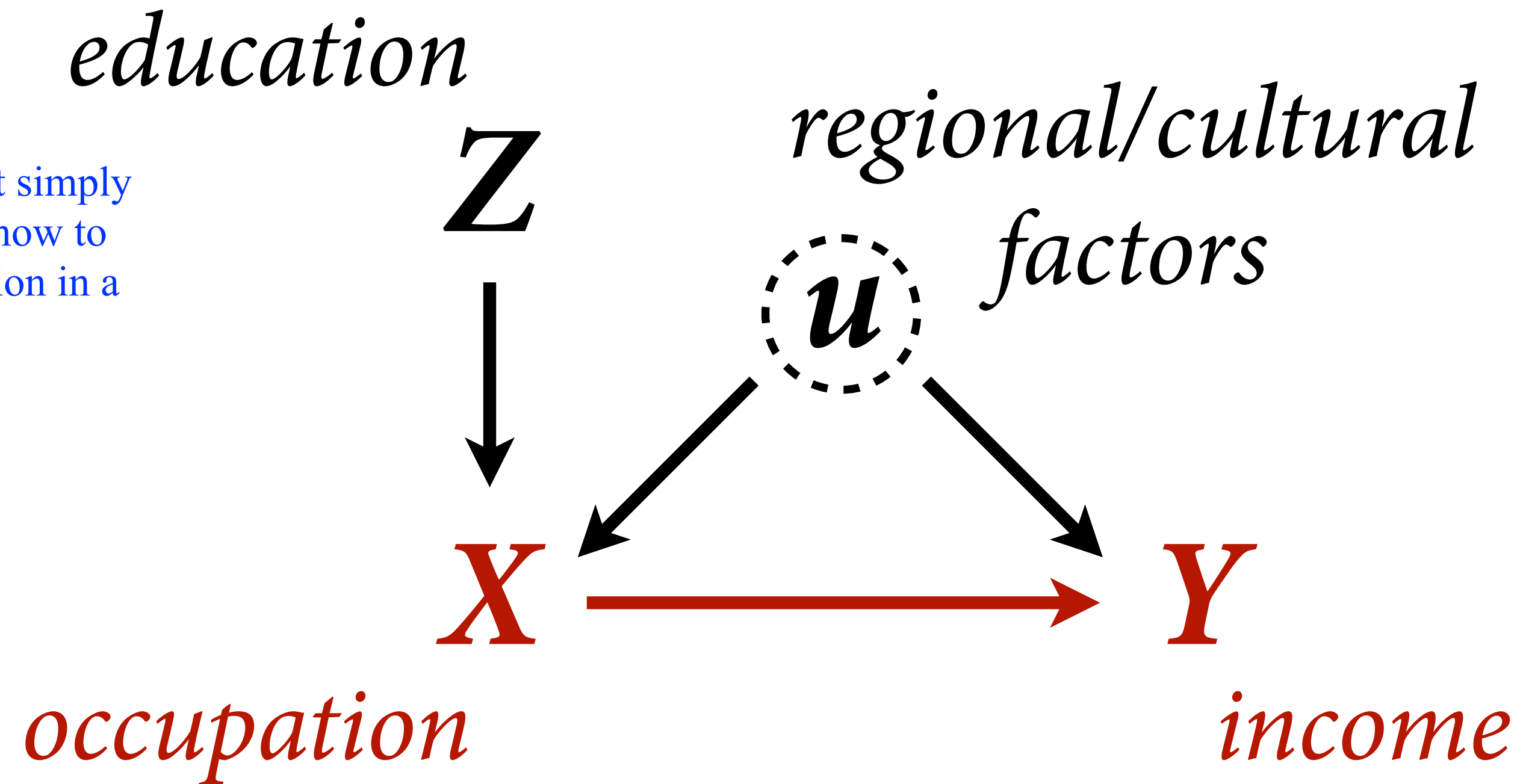


We may want to use Z as an instrumental variable, but we have to do this with care.

In technical terms, Z could be an instrumental variable if it satisfies these criteria:

1. Independent of U ($Z \perp\!\!\!\perp U$)
2. Not independent of X ($X \not\perp\!\!\!\perp Z$)
3. Z cannot influence Y except through X

As noted in the previous slides, we can't simply include Z in the regression. We'll learn how to specify an instrumental variable regression in a few weeks.



Good & Bad Controls

“Control” variable: Variable introduced to an analysis so that a causal estimate is possible

Heuristics fail — adding control variables can be worse than omitting

Make assumptions explicit

**MODEL
ALL THE
THINGS**



Table 2 Fallacy

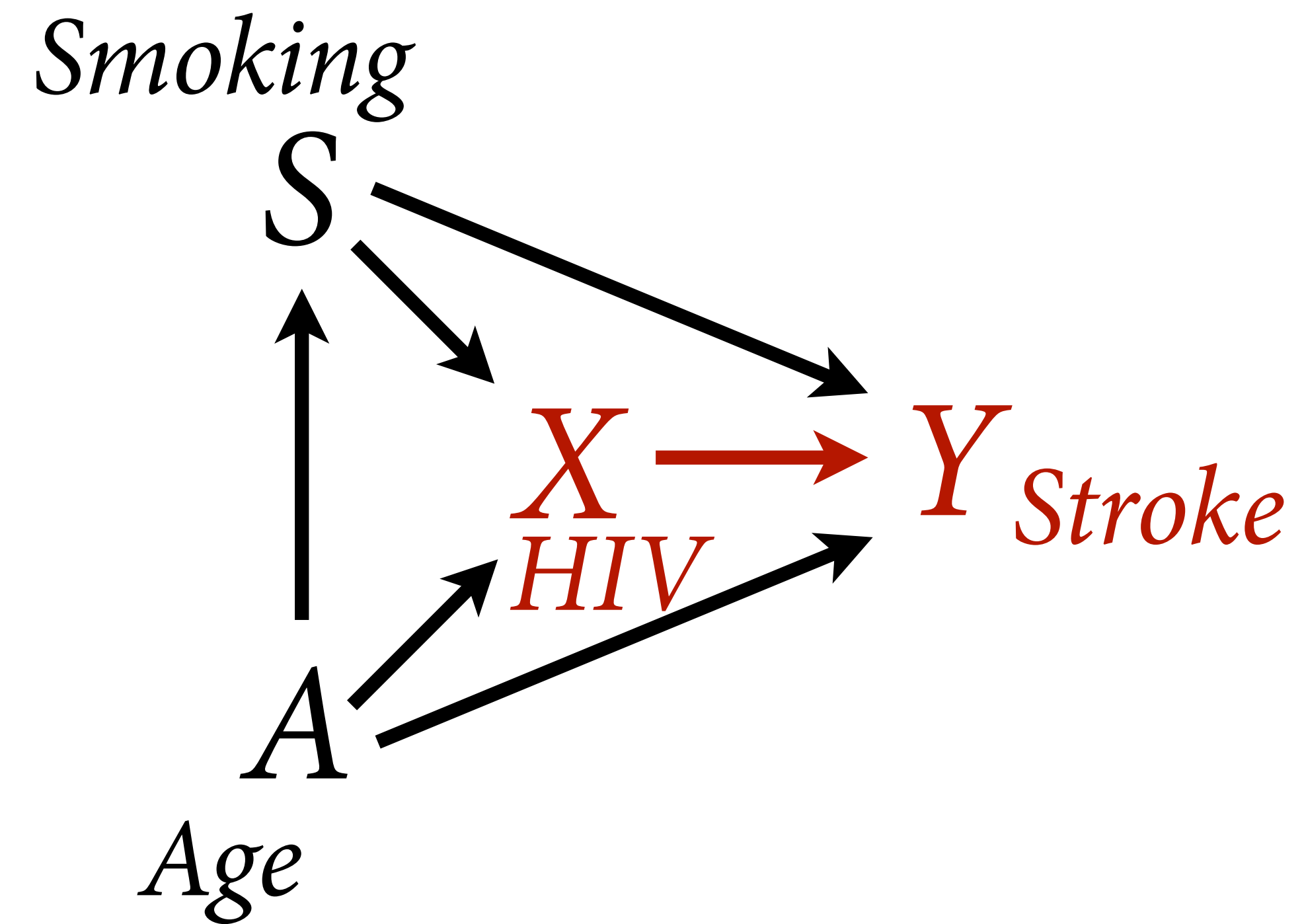
Not all coefficients are causal effects

Statistical model designed to identify $X \rightarrow Y$ will not also identify effects of control variables

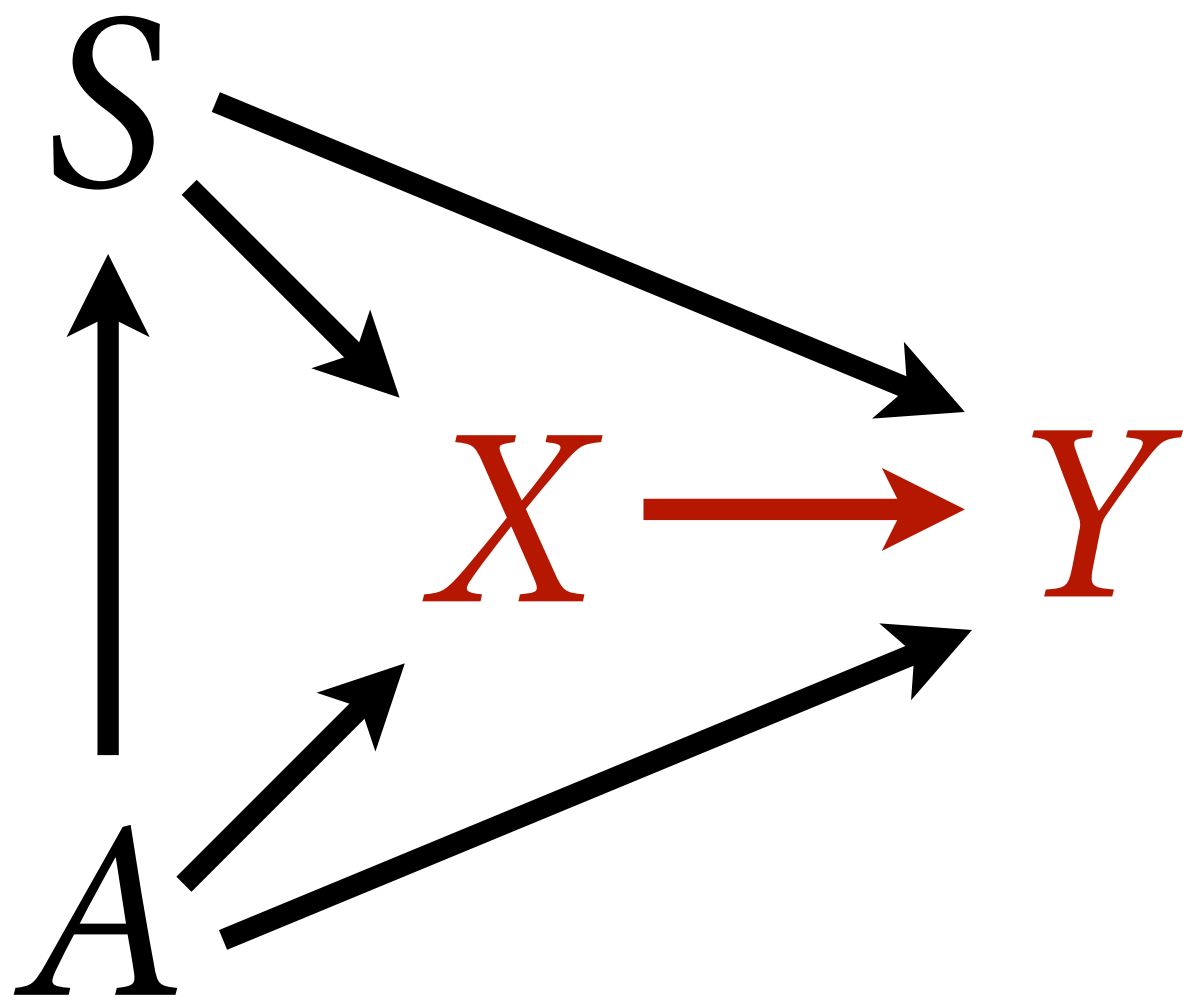
Table 2 is dangerous

TABLE 2—ESTIMATED PROBIT MODELS
FOR THE USE OF A SCREEN

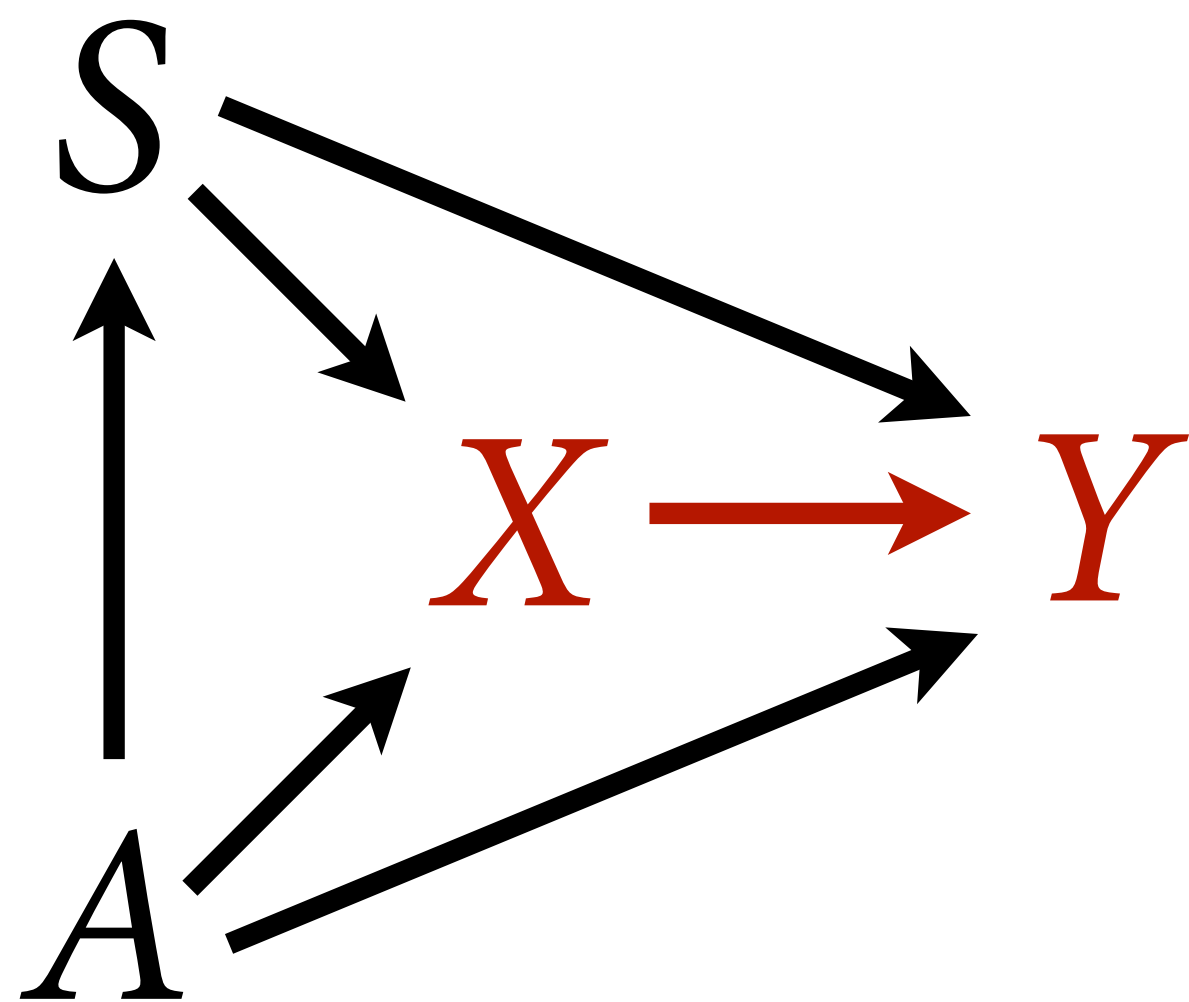
	Preliminaries blind		Finals blind
	(1)	(2)	(3)
(Proportion female) _{<i>t</i>−1}	2.744 (3.265) [0.006]	3.120 (3.271) [0.004]	0.490 (1.163) [0.011]
(Proportion of orchestra personnel with <6 years tenure) _{<i>t</i>−1}	−26.46 (7.314) [−0.058]	−28.13 (8.459) [−0.039]	−9.467 (2.787) [−0.207]
“Big Five” orchestra		0.367 (0.452) [0.001]	
pseudo <i>R</i> ²	0.178	0.193	0.050
Number of observations	294	294	434



Use Backdoor Criterion

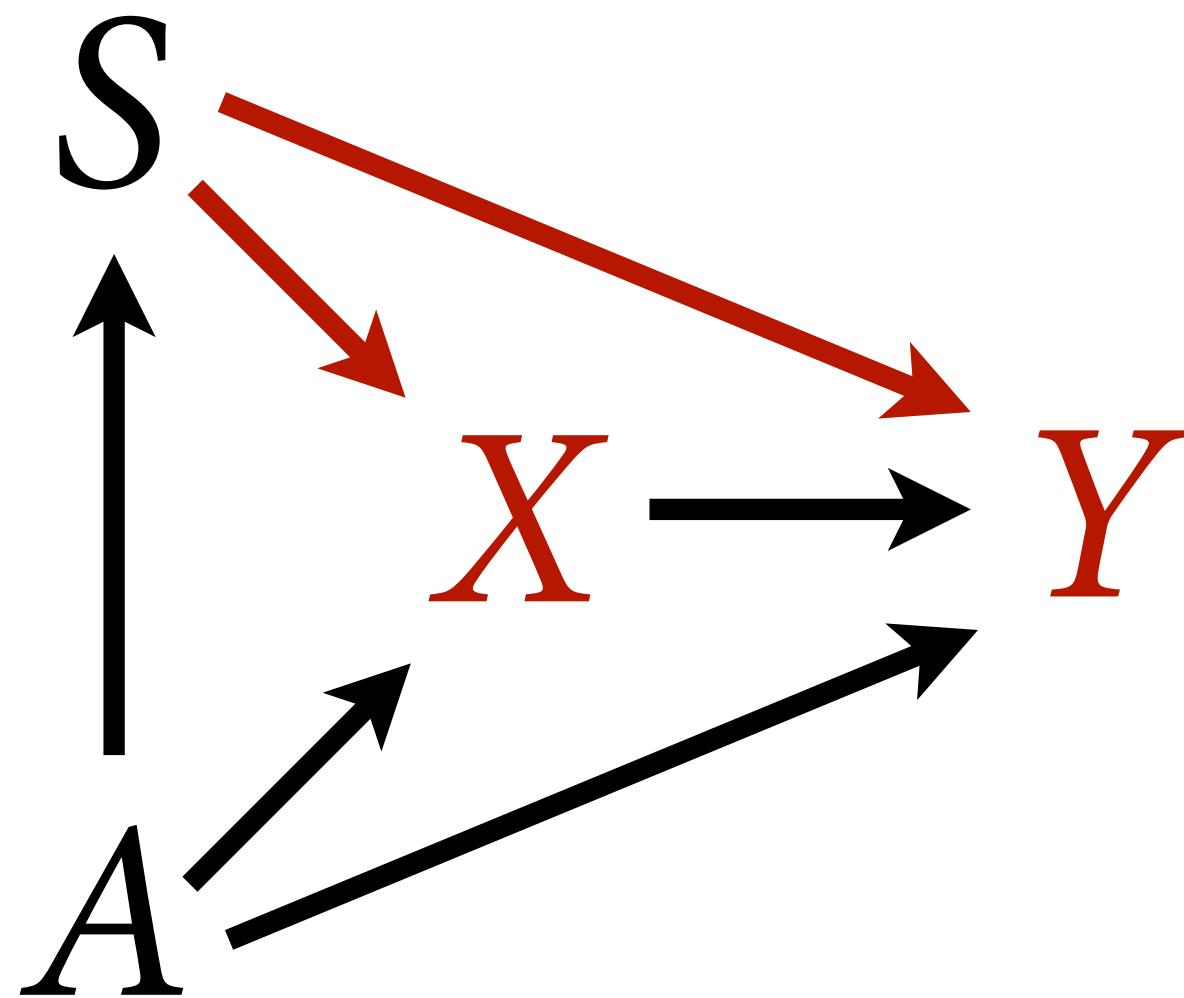


Use Backdoor Criterion

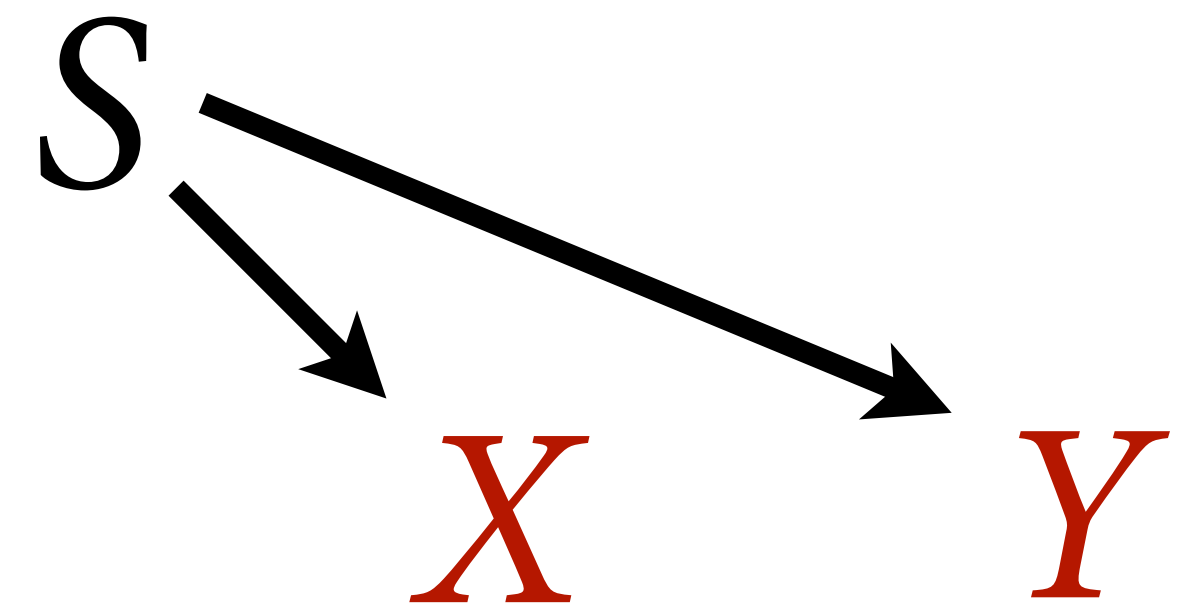


$$X \longrightarrow Y$$

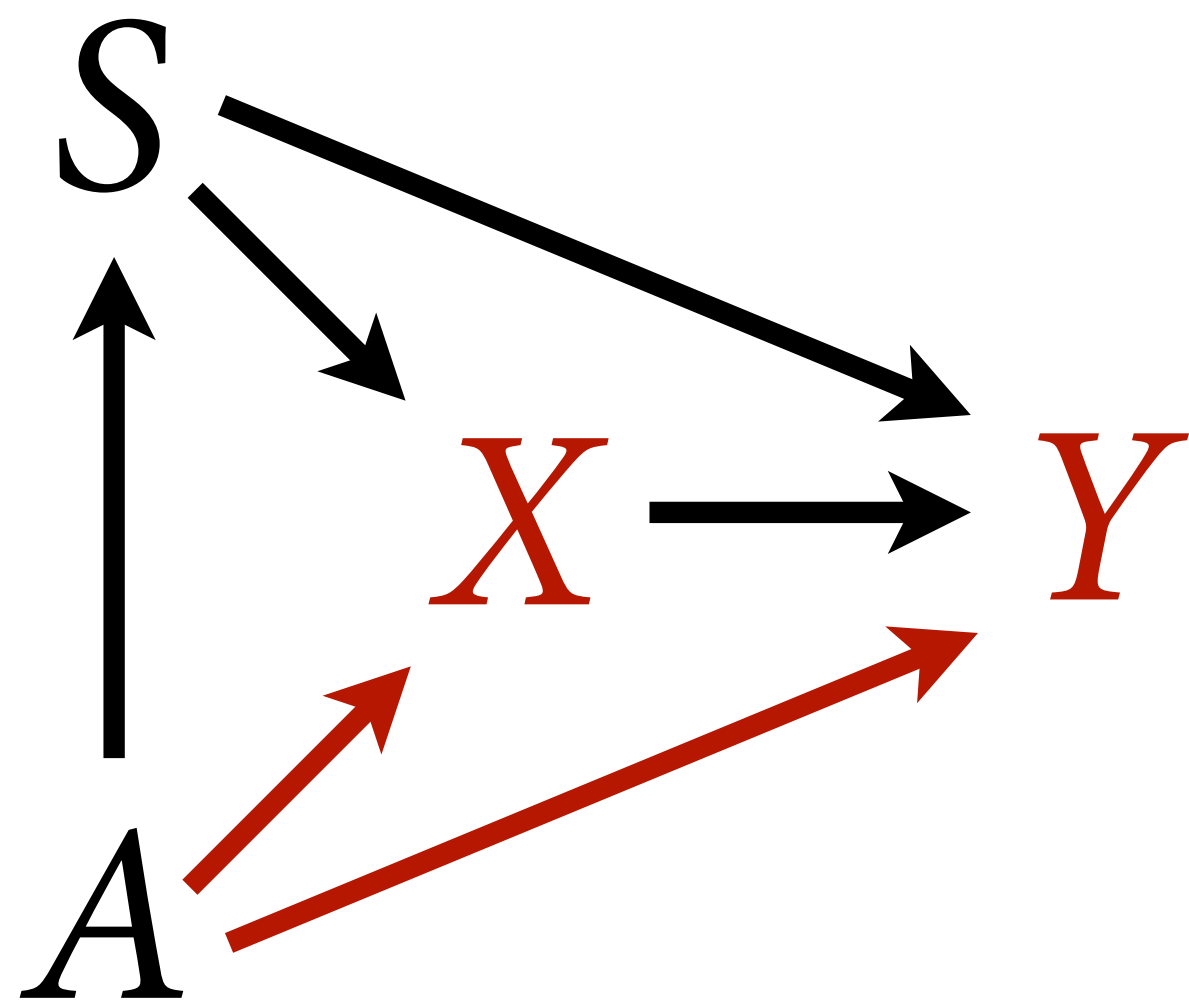
Use Backdoor Criterion



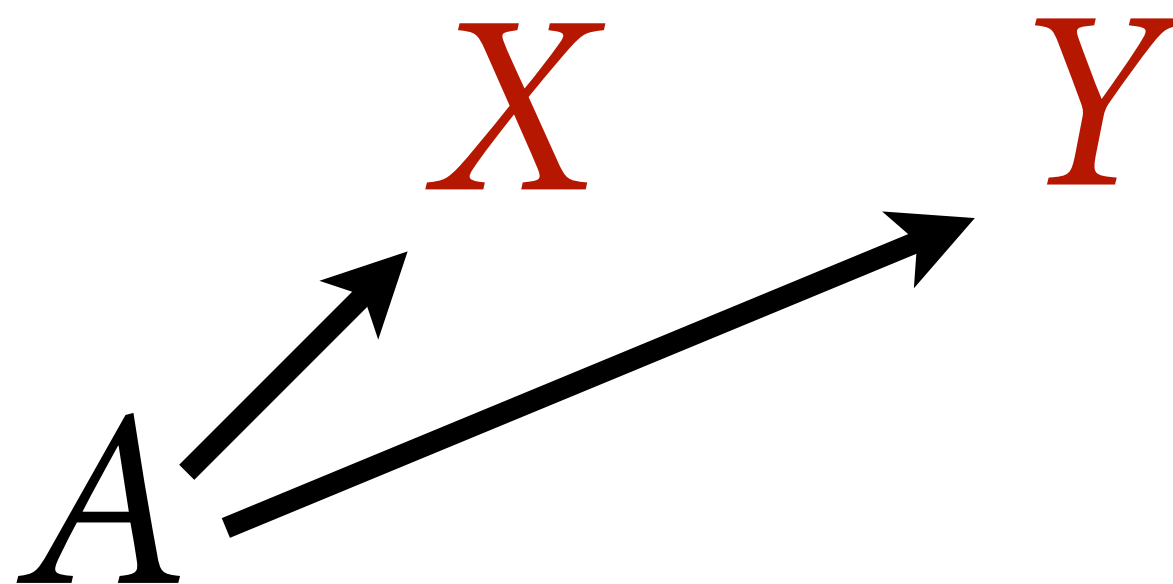
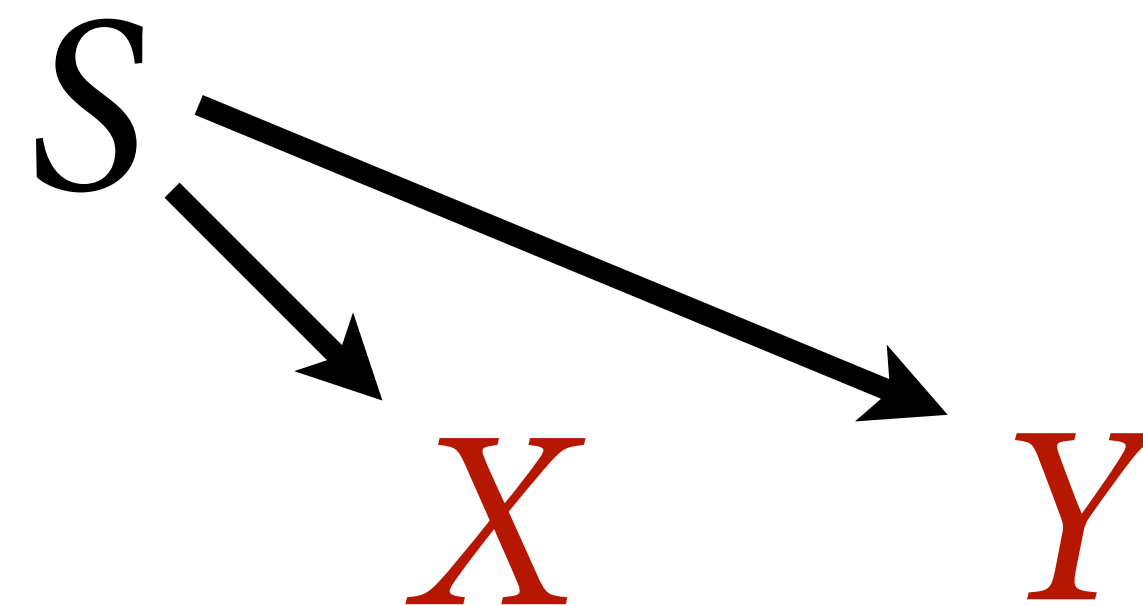
$$X \longrightarrow Y$$



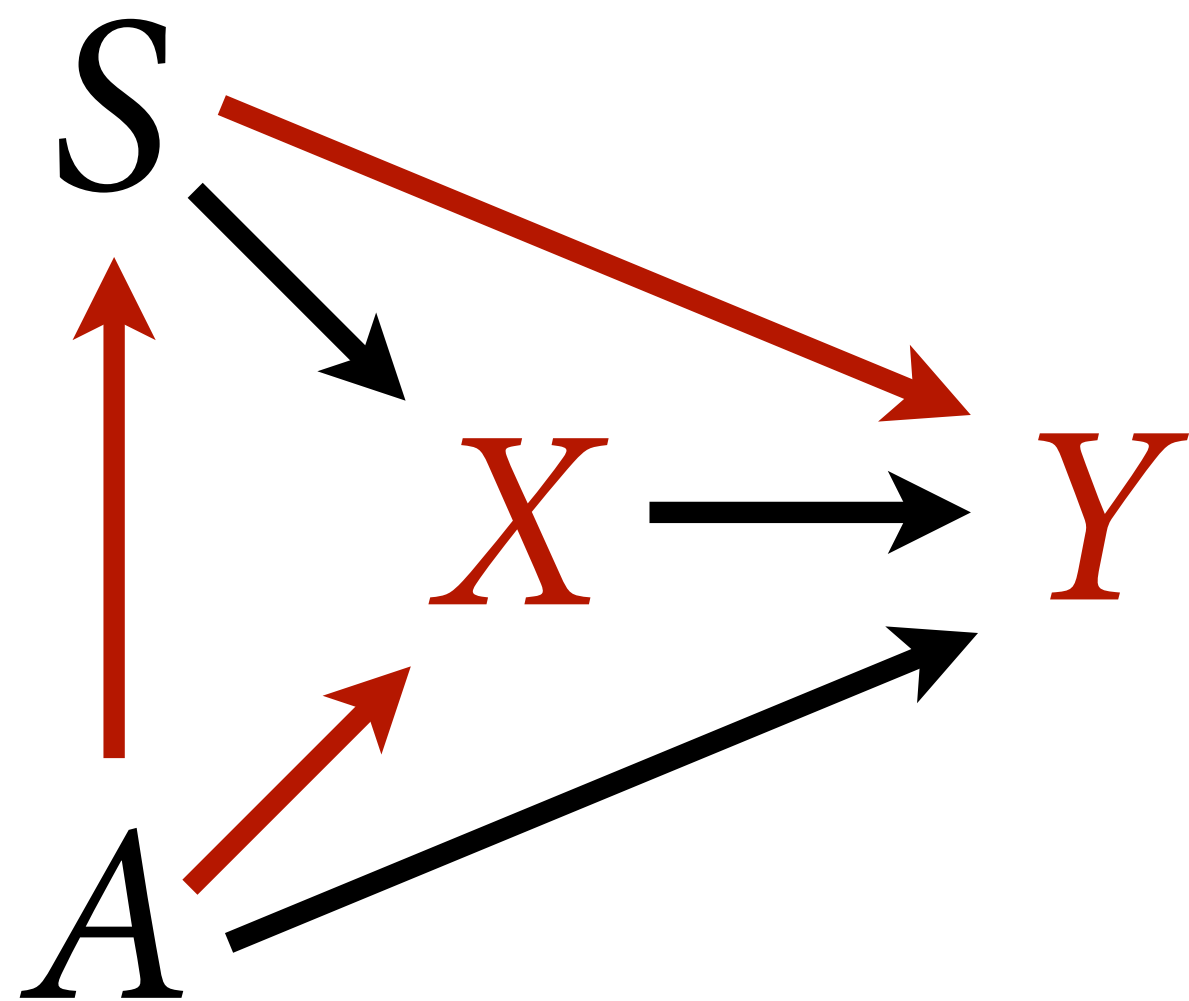
Use Backdoor Criterion



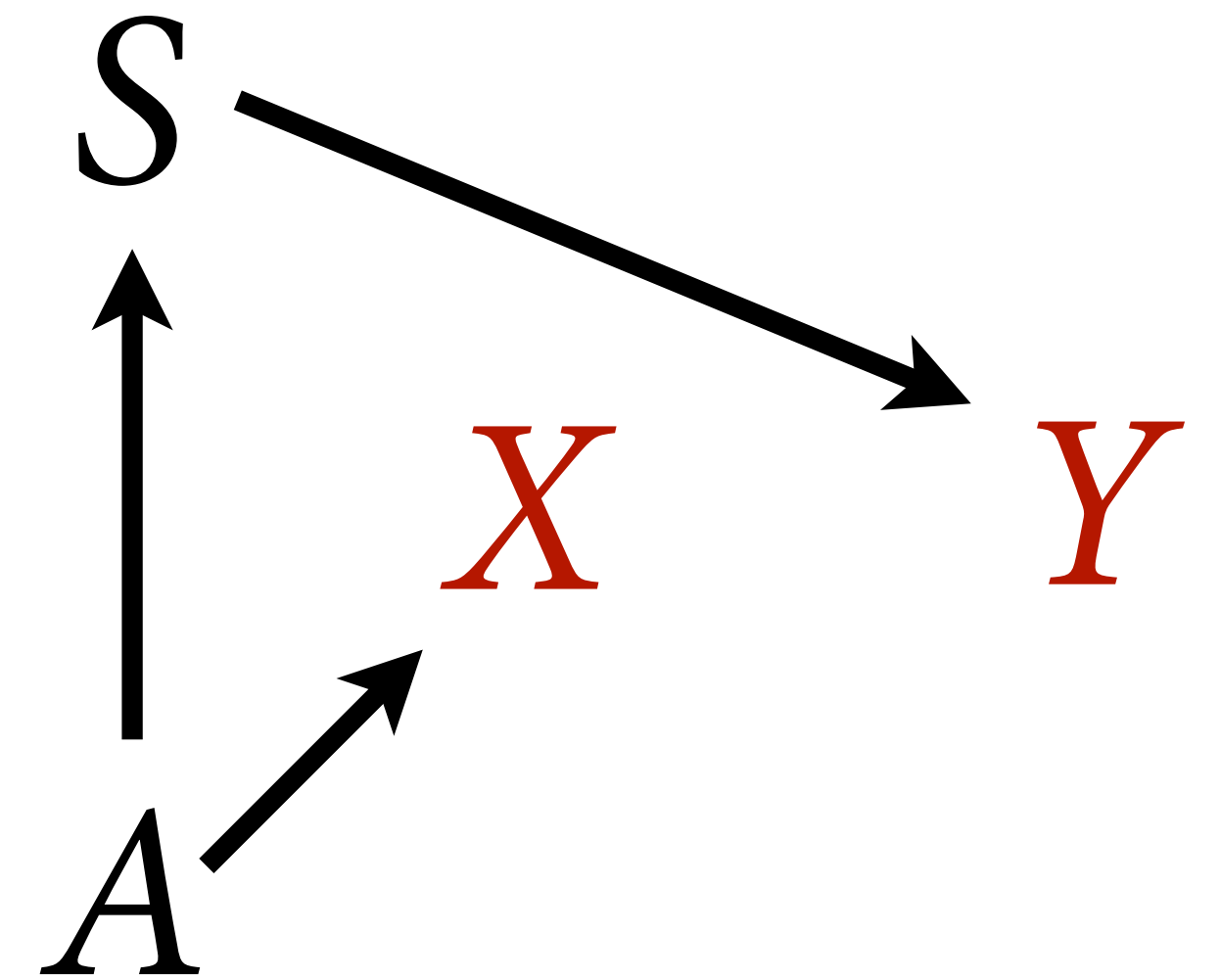
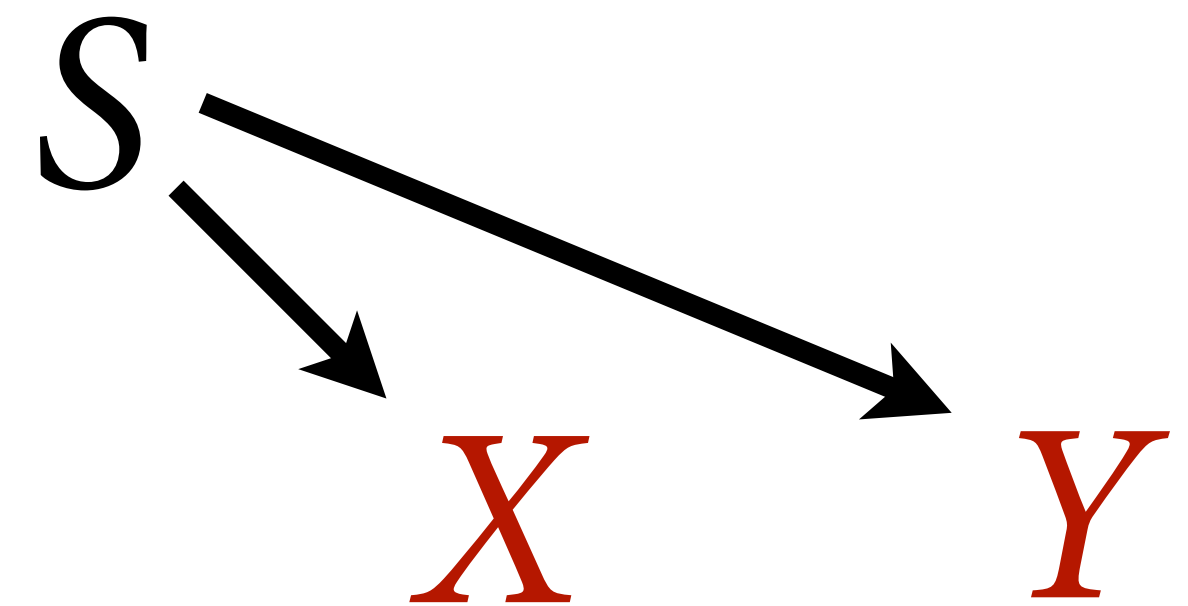
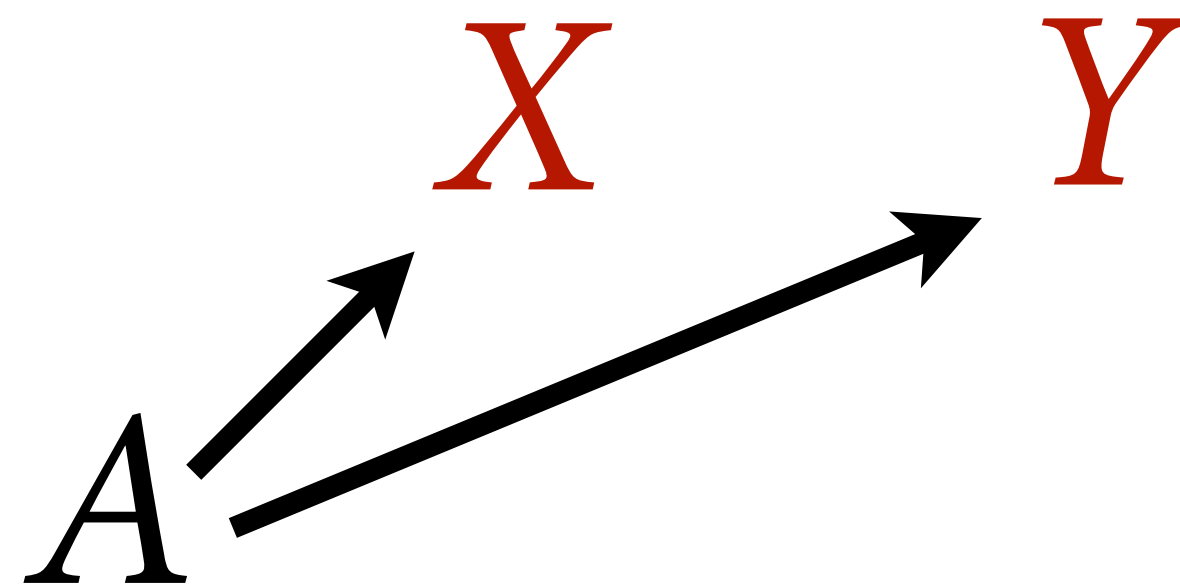
$$X \longrightarrow Y$$



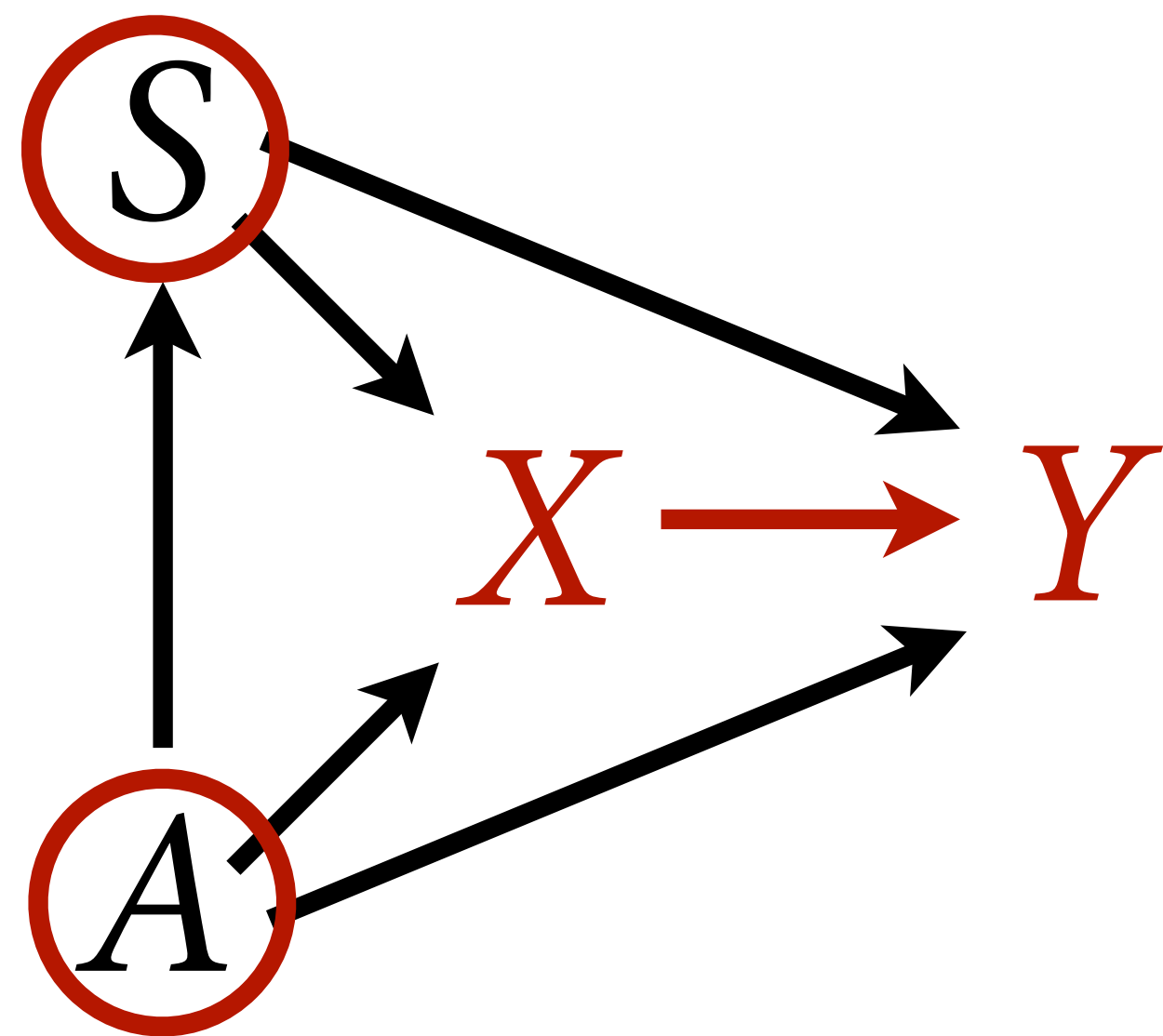
Use Backdoor Criterion



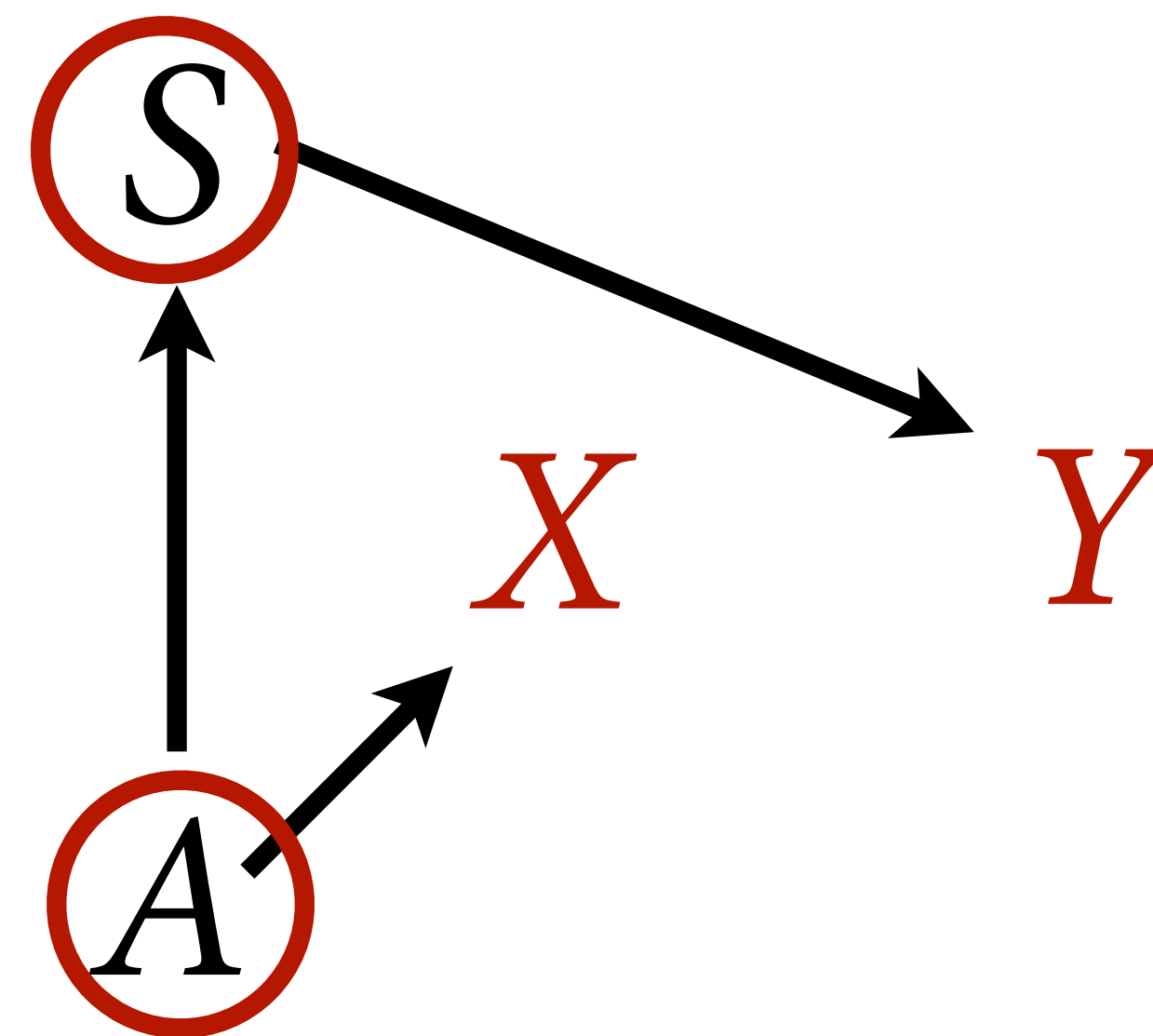
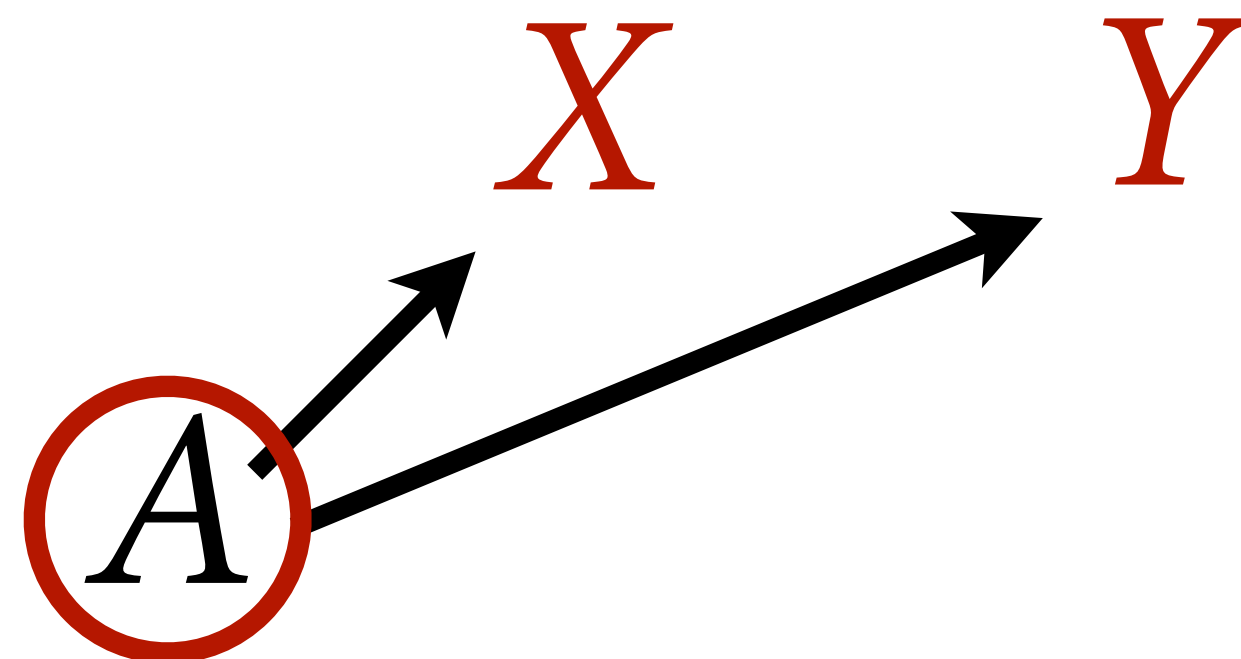
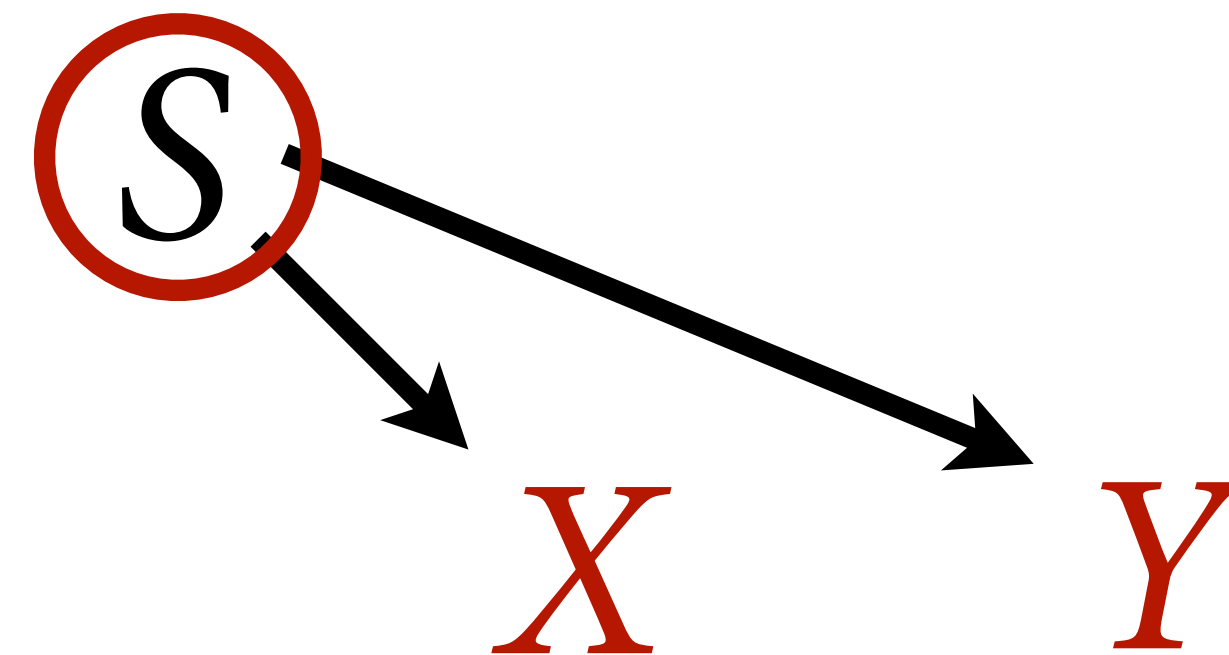
$$X \longrightarrow Y$$

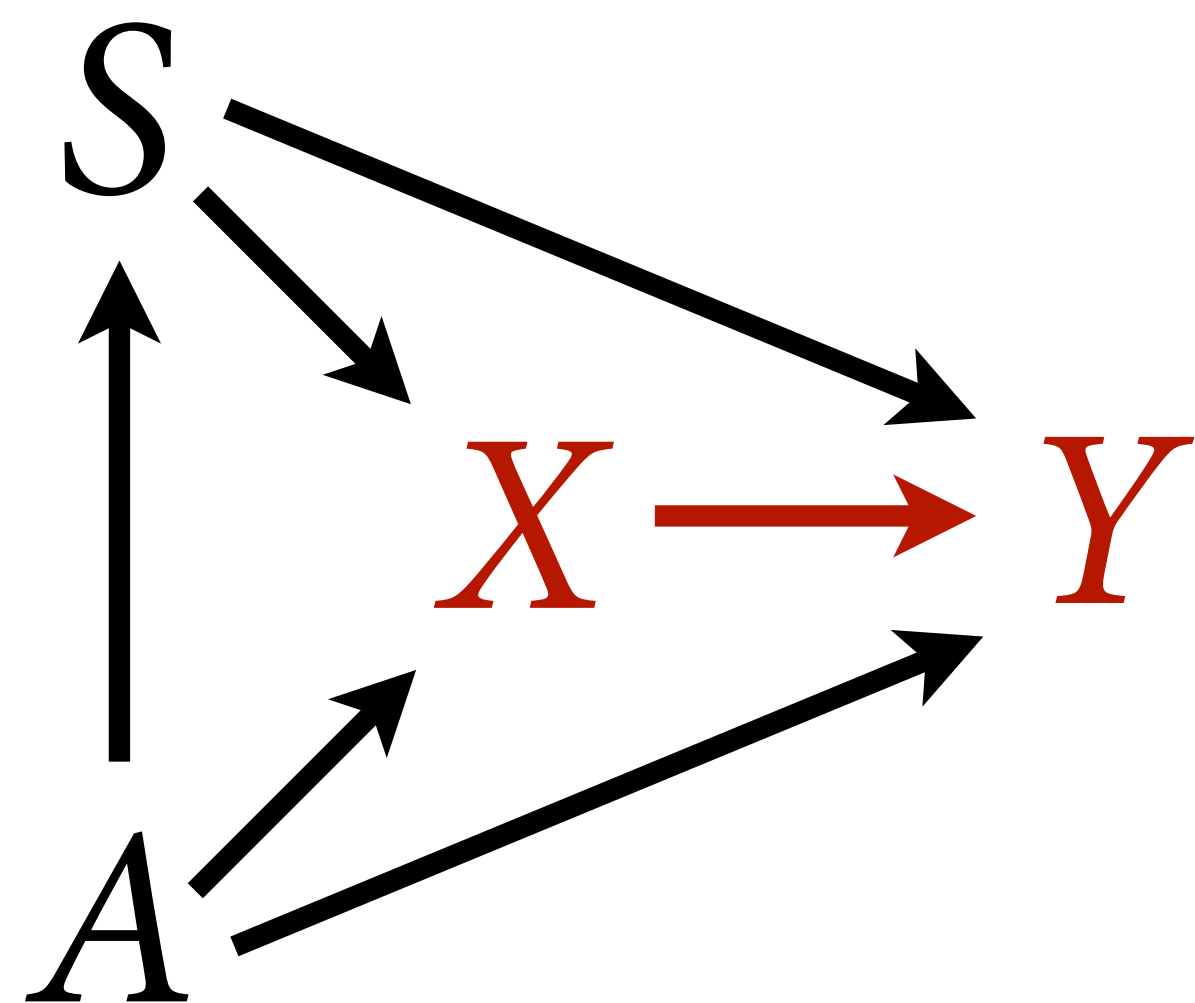


Use Backdoor Criterion



$$X \longrightarrow Y$$



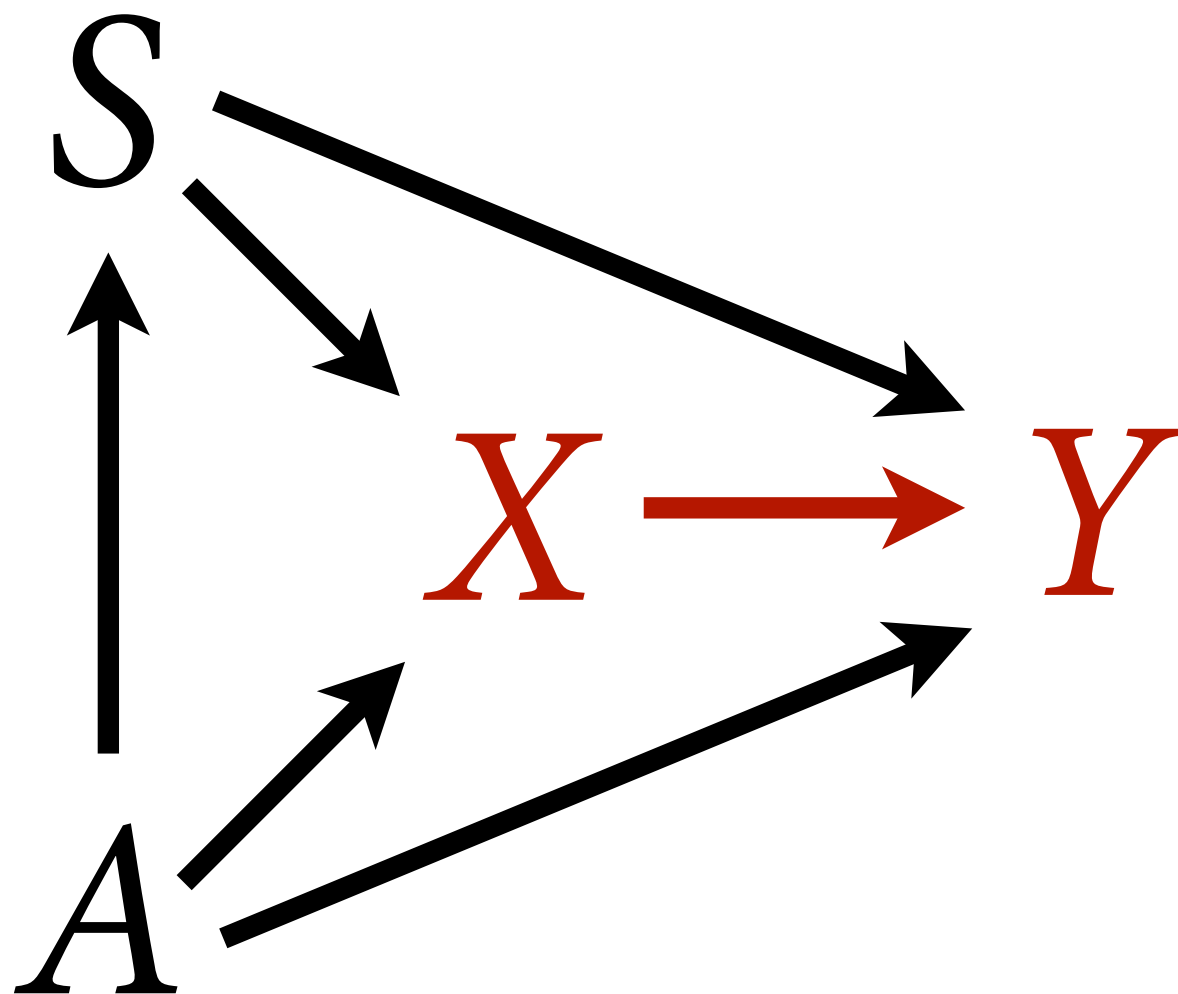


$$Y_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta_X X_i + \beta_S S_i + \beta_A A_i$$

X

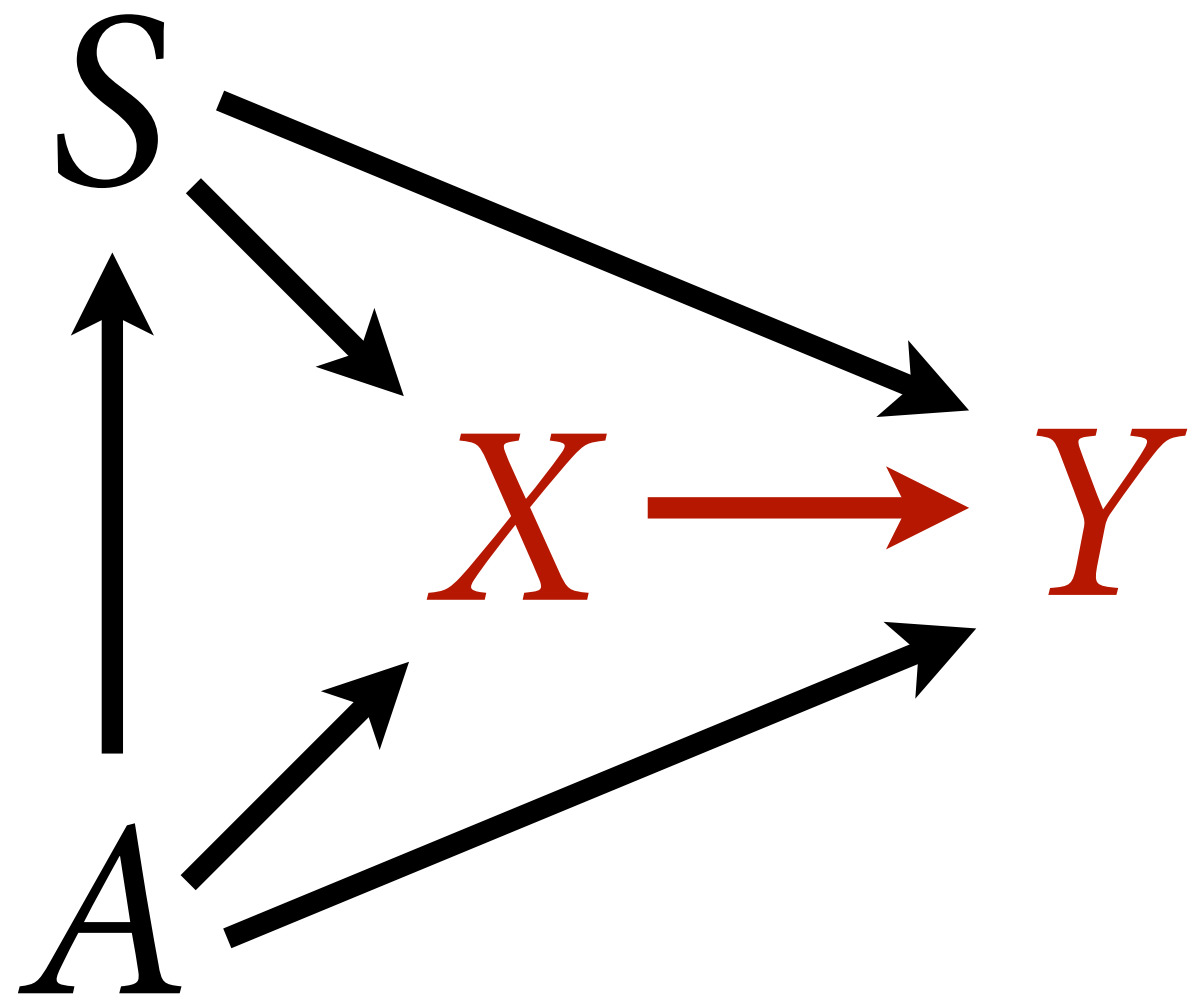
Unconditional



Confounded by *A*
and *S*

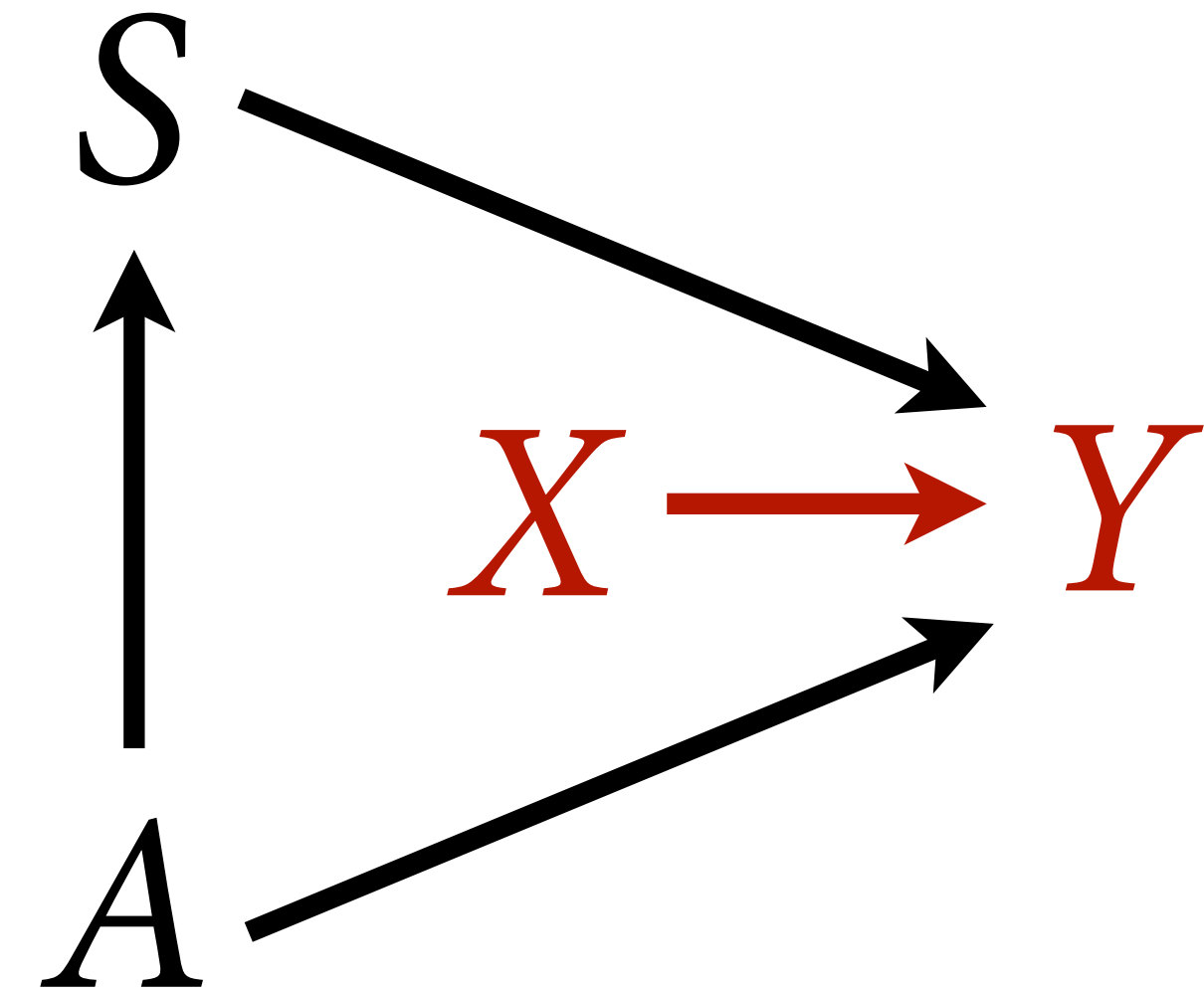
X

Unconditional



Confounded by A
and S

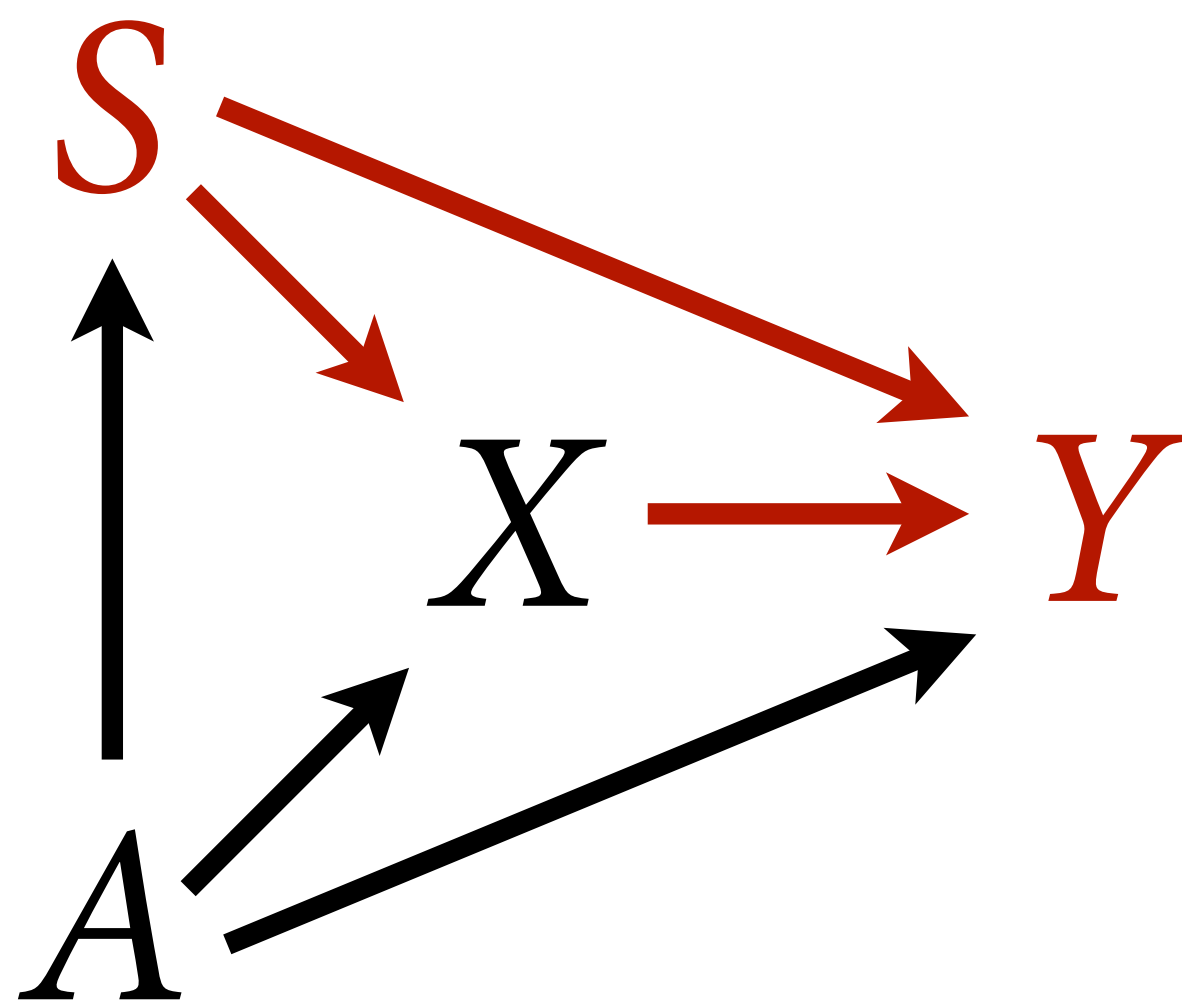
Conditional on A and S



Coefficient for X :
Effect of X on Y
(still must
marginalize!)

S

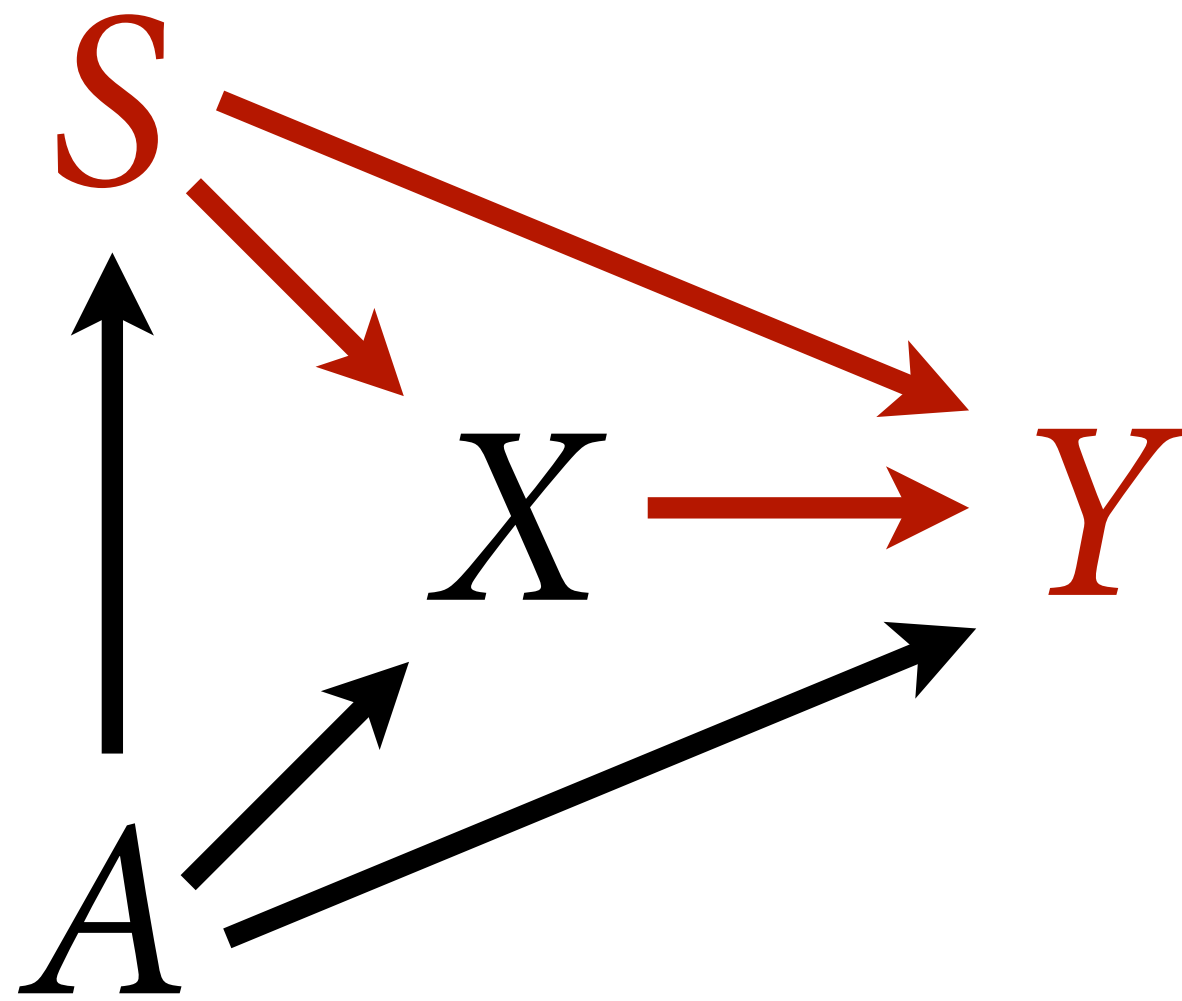
Unconditional



Effect of S
confounded by A

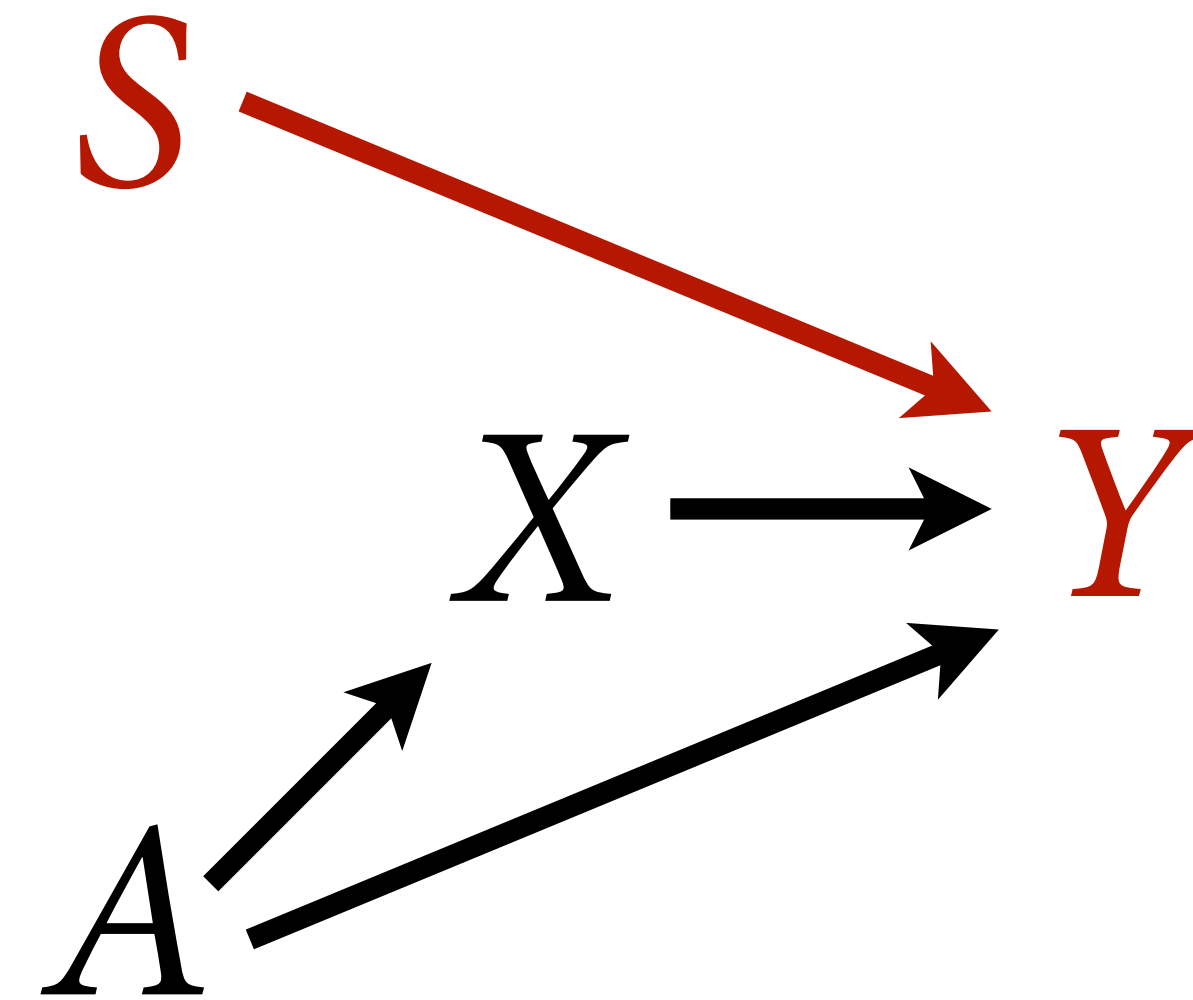
S

Unconditional



Effect of **S**
confounded by **A**

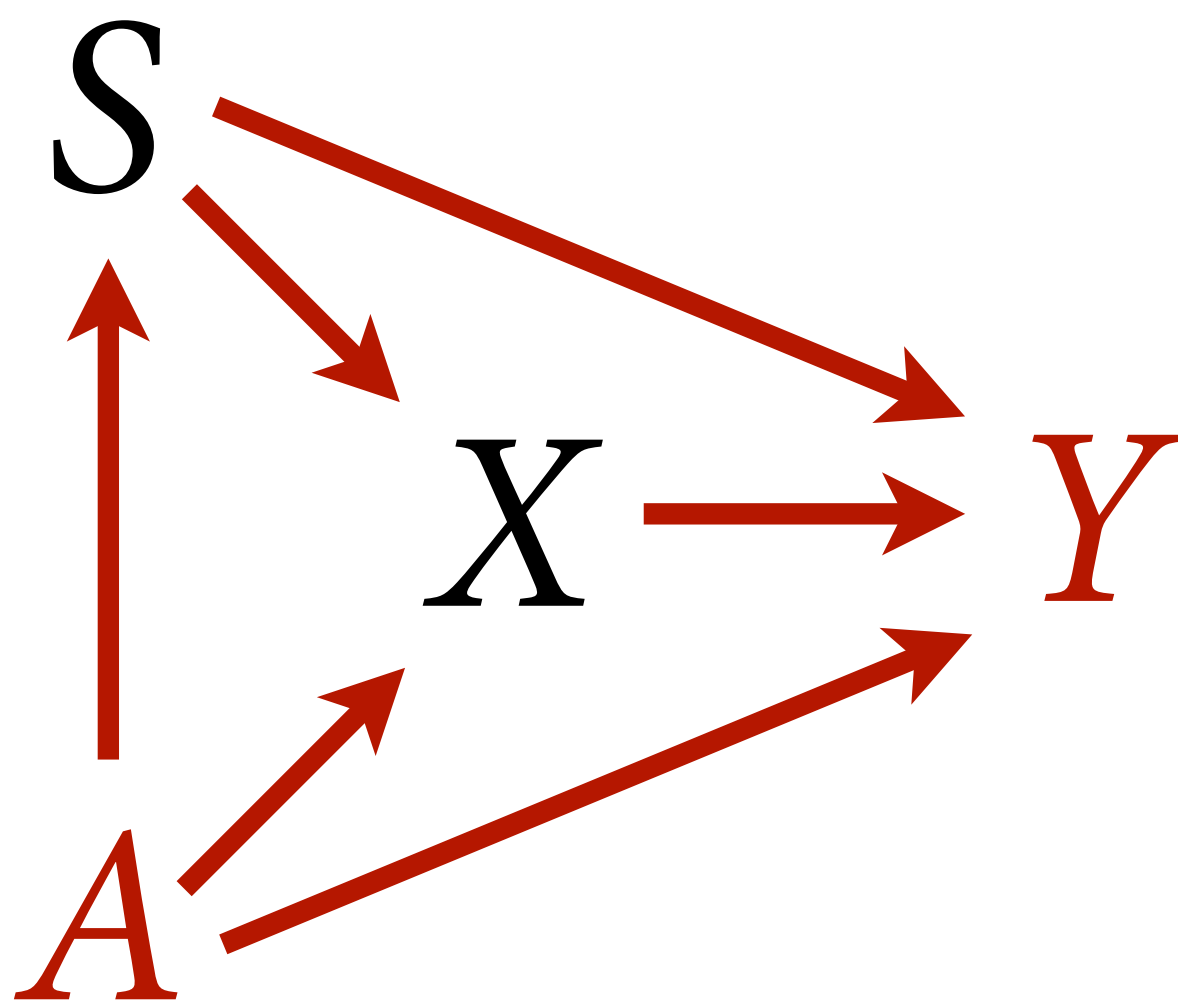
*Conditional on **A** and **X***



Coefficient for **S**:
Direct effect of **S** on **Y**

A

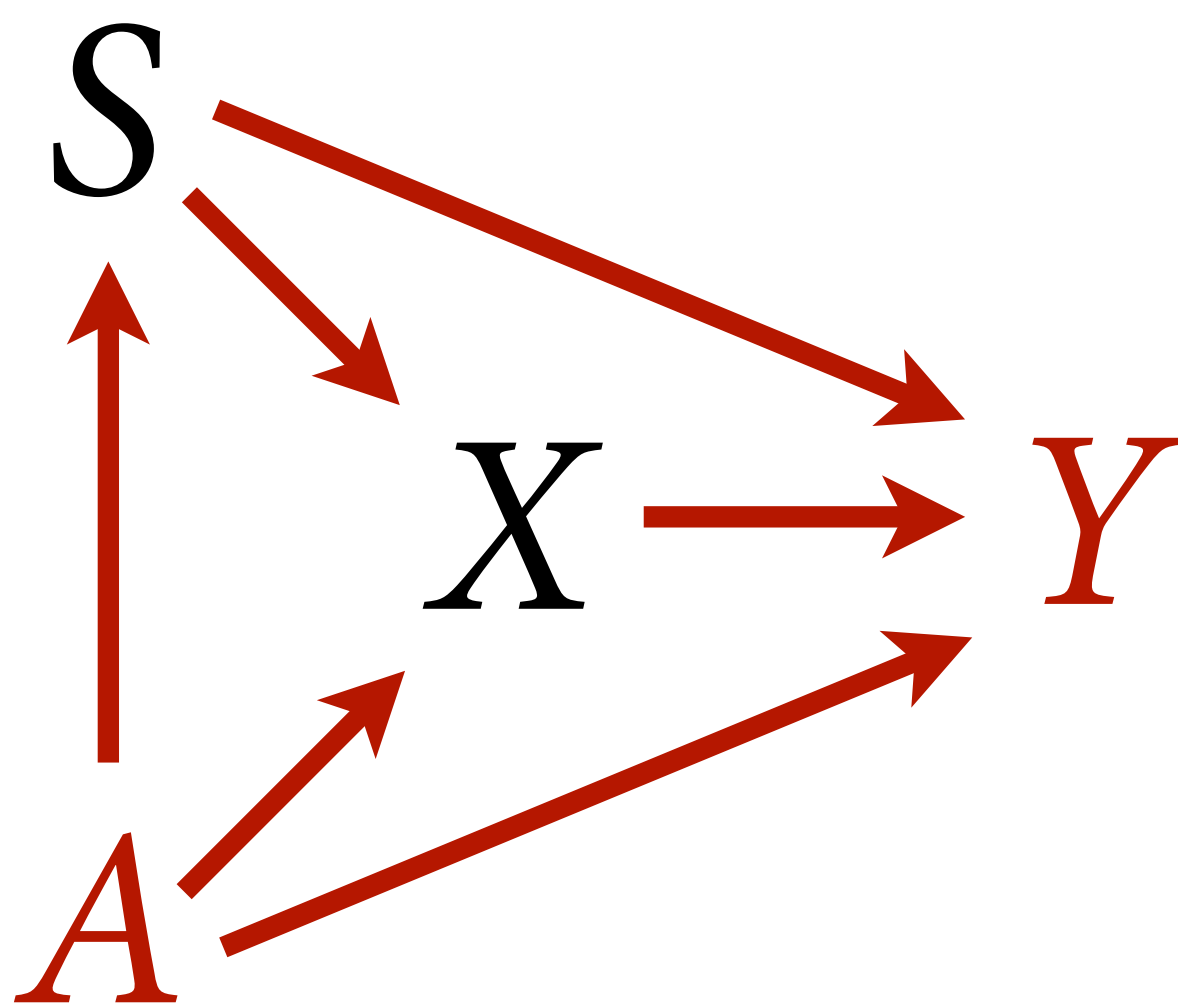
Unconditional



Total causal effect
of *A* on *Y* flows
through all paths

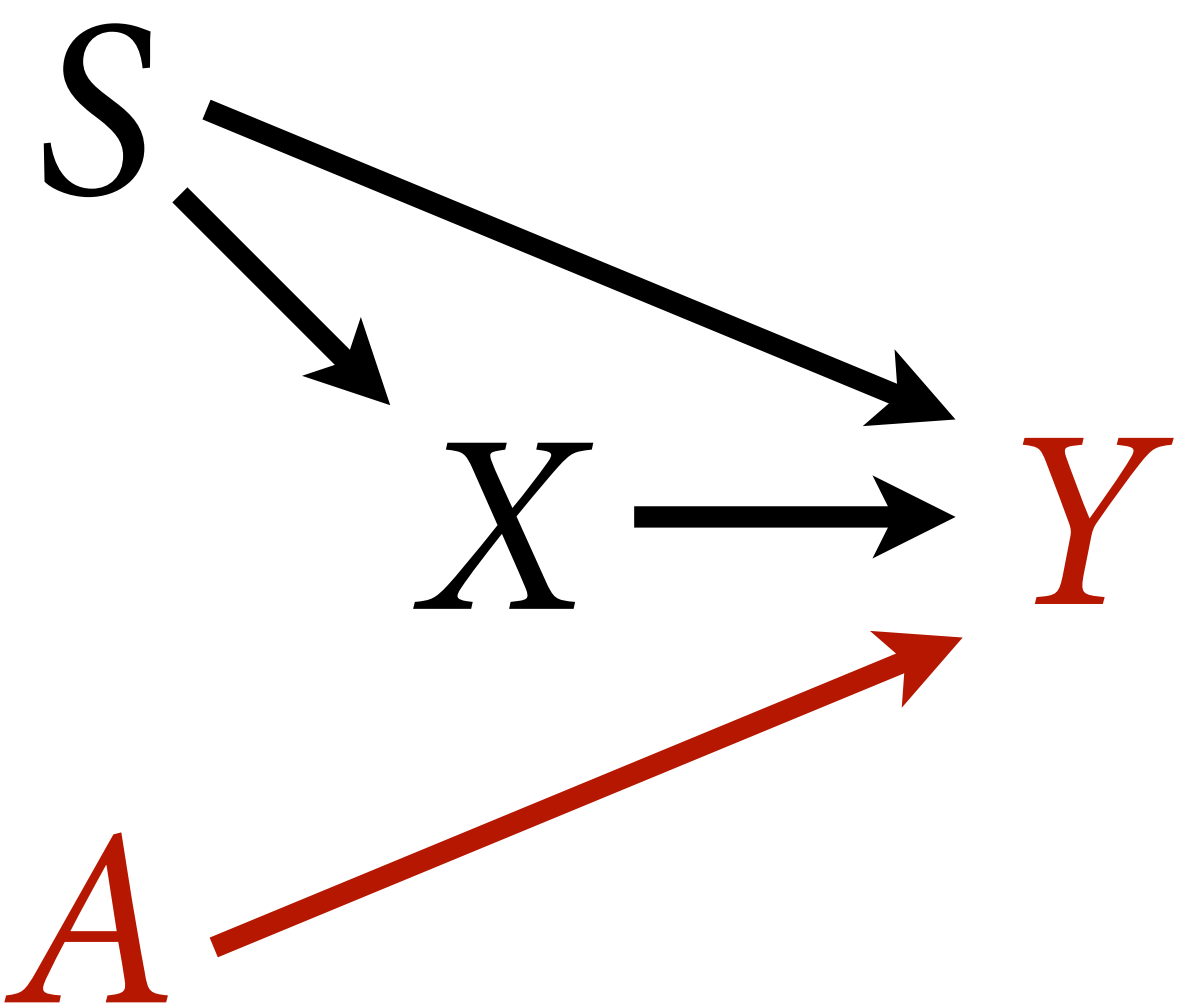
A

Unconditional



Total causal effect
of A on Y flows
through all paths

Conditional on X and S



Coefficient for *A*:
Direct effect of A on Y

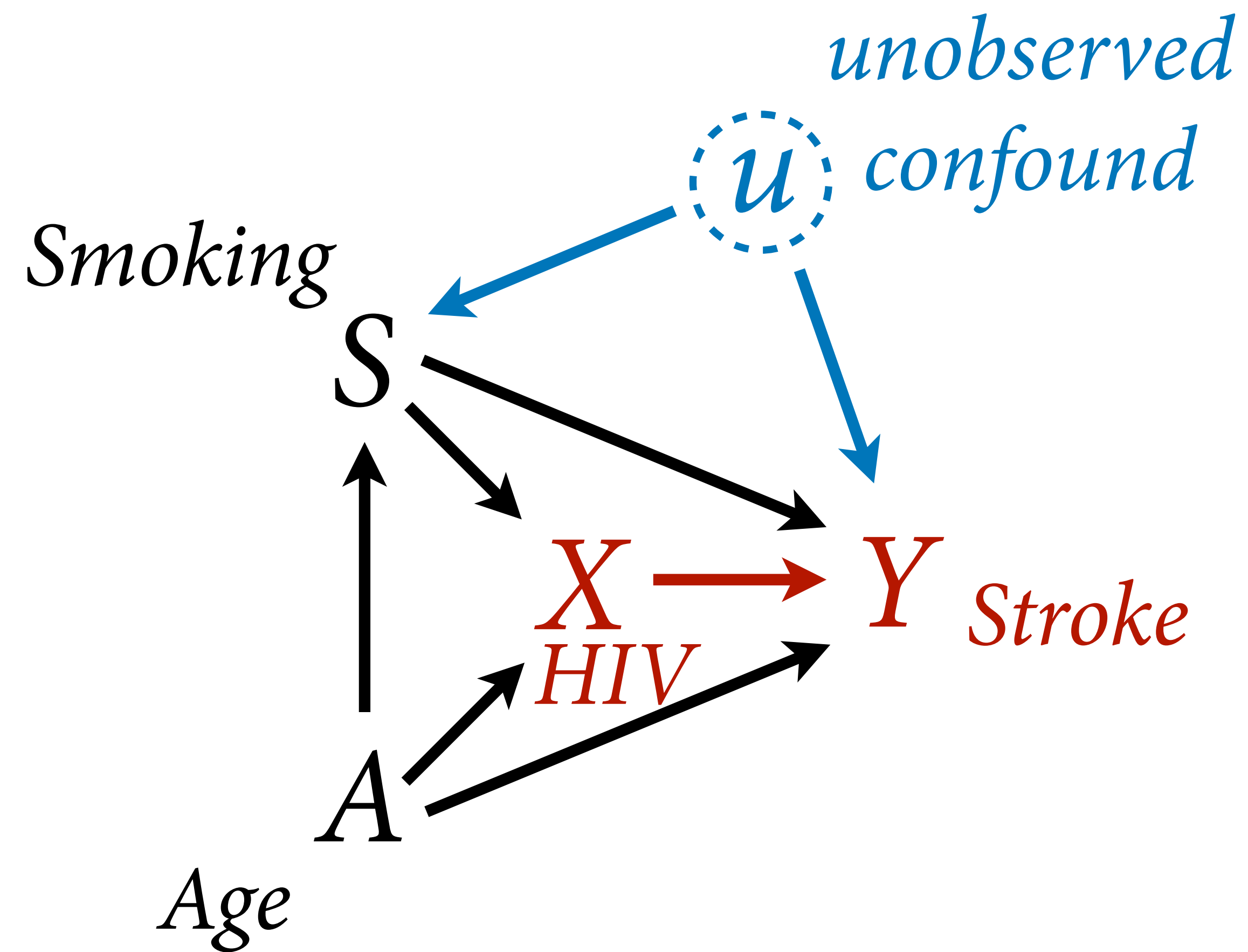


Table 2 Fallacy

Not all coefficients created equal

So do not present them as equal

Options:

Do not present control coefficients

Give explicit interpretation of each

No interpretation without causal
representation

